Algebraic Geometry

— ***EYNTKA*** **Series** —

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Contents

Algebraic Geometry originated as the study of the zero-sets of polynomials, namely it was (and in many ways behind all the abstraction predominantly still is) the solution to a system of polynomials. By explicitly working with the zero sets of polynomials, we can discover a great multitude of geometric results within algebra, however as the field progressed it was discovered zero sets are in many ways insufficient to capture all the geometric data. For some quick examples:

1. it was discovered in the mid 20th century that results in number theory seem eerily similar to the study of 1 dimensional curves.
2. The galois group of a field share many of the same properties of as the fundamental group of a topological space.
3. The fact that a polynomial may have a root of high multiplicity is lost in the classical geometric picture $V(f)$, but this is often key information.
4. Number theorists do not work over algebraically closed fields as they are looking for integer (or integral) solutions to polynomial equations. However, these do not have the same nice geometric correspondence within classical algebraic geometry as the Nullstellensatz works well when the field is algebraically closed.
5. All rings can be represented as the quotient of $\mathbb{Z}[x_i, \mid i \in \mathcal{I}]$ or $k[x_i, \mid i \in \mathcal{I}]$ for some indexing set I, hence all ring (not just reduced finitely generated rings over algebraically closed fields) have some notion of being a function with evaluation
6. When studying a family of varieties (ex. $x^2 + ty^2 = 1$ as t varies) the resulting object will often have "bad" singularities which varieties are poorly equipped to work with.

Such realizations lead to the need to ground algebraic geometry with a new object that encapsulates all of these ideas. this new object is the *scheme*. As compared to varieties that requires an ambient space and working over polynomial rings over fields, schemes are *intrinsic* in that they don't require any ambient space to be defined in, and they generalize classical algebraic geometry to be able to capture the geometry of both Diophantine equations (for number theory) as well as the geometry of (quasi-projective) varieties and other more novel. Perhaps key in their success over other variations is how well they also remember "multiplicity" information of roots of polynomials. This data shall be paramount in many results we shall introduce[1].

This book shall be written with graduate or well-read undergraduates in mind. The reason is to show how inter-connected schemes are to various other fields of mathematics. For example:

1. The geometry of schemes draws many parallels to complex geometry and some results from complex analysis (ex. Hartog's lemma from Complex analysis over several variables shall have a strong appearance when discussing normal schemes).
2. A sheaf draws many similarities to the bundles that are put on manifolds; in fact a manifold can be defined equivalently by using a sheaf of differentiable functions instead of an atlas. Some results such as Poincaré duality shall be proven for schemes (see 5.3)
3. Chapter 5 shall strongly rely on the readers intuitions from algebraic topology and use modern homological algebra.

[1]We shall see for Elliptic Curves that the finite-generation of certain torsion groups shall rely on Schemes in a key way to remember multiplicity

4. Many of the properties in commutative algebra shall become interesting geometric properties in algebraic geometry.

Essentially, anything non-optional material from EYNTKA Undergraduate Algebra ([8]), EYNTKA Differential Geometry ([4]), and EYNTKA Algebraic Topology ([2]) will be considered as assumed prerequisites. Some results from EYNTKA complex analysis ([3]) will be drawn on as well given the strong connection between complex geometry and the geometry of schemes.

All essential category theory will be assumed; it can be reviewed in EYNTKA Undergraduate Algebra.

Throughout this book, we shall use the following terminology:

- Let k be a field, $\overline{k}$ the algebraic closure of k
- Let R an arbitrary ring, $M_n(R)$ the matrix ring over R, $R_{\mathfrak{p}}$ the localization of R by $R \setminus \mathfrak{p}$, R_f the localization of R by $\{f, f^2, ...\}$, $\text{Frac}(R)$ the field of fractions of R is R is an integral domain
- A is an arbitrary R-algebra.

The material covered in the preliminary chapter is at a break-neck pace and is only there for expository purposes for the reader to see if they have the right intuitions to cover algebraic geometry in its greater generality. The first main Part, Scheme Theory, focuses heavily on building up all the necessary theoretical framework to have a comfortable grasp of the material and is heavily inspired by the many famous texts covering the material (ex. Hartshorne, Shafarevich, Vakil, Serre, Eisenbud, and so forth).

Recall Classical Algebraic Geometry

In this chapter, we shall quickly review elementary concepts linking algebra and geometry without going any proofs, that is we are going to explore some classical algebraic geometry. For all the important details, see [8].

0.1 Interpreting Rings as Function Spaces

If A is an R-algebra, then by the universal property of free R-algebras we have if S is a set of generators for A,

$$\begin{tikzcd} R[S] \arrow[r, "\varphi"] & A \\ S \arrow[u, "\iota"] \arrow[ur, "\iota"'] & \end{tikzcd}$$

If S, is finite, then $R[S] \cong R[x_1, x_2, ..., x_n]$ by relabelling S. Since every ring is a $\mathbb{Z}$-algebra, we see that polynomial rings are central to the study of general rings. It is thus fruitful to realize that polynomial rings can naturally be interpreted as functions, that is:

$$f \in R[S],\ R^S \to R[S]^\vee \qquad f \mapsto f(s_i)_{i \in S}$$

In the finite case:

$$f \in R[x_1, x_2, ..., x_n], \qquad R^n \to R[x_1, x_2, ..., x_n]^\vee \qquad (a_1, a_2, ..., a_n) \mapsto (f \mapsto f(a_1, a_2, ..., a_n))$$

We don't want to consider R^S as having any structure besides being an affine space, hence we shall label it as $\mathbb{A}_R^S$ or in the finite case $\mathbb{A}_R^n$. If the ring is evident in context, we shall write $\mathbb{A}^n$. In a number theory context, it shall of be assumed that $\mathbb{A}^n$ implies $\mathbb{A}_{\mathbb{C}}^n$ (or $\mathbb{A}_{\overline{\mathbb{Q}}}^n$ and $\mathbb{A}^n(R) = \mathbb{A}_R^n$; this is because polynomials over $\mathbb{C}$ always have solutions, which is not guaranteed over R.

For now, we shall stick to the finitely generated case. Note that in $\mathbb{A}^n_R$, $(0, ..., 0)$ has no special property, as we are treating this space as simply a collection of points. To emphasize this geometric interpretation, any $p \in \mathbb{A}^n_R$ will be called a *point*, and for any

$$p = (r_1, r_2, ..., r_n)$$

Each r_i are the *coordinates* of p. Hence, we may more accurately any $f \in R[x_1, x_2, ..., x_n]$ as a function as

$$f : \mathbb{A}^n_R \to R \qquad (r_1, r_2, ..., r_n) \mapsto f(r_1, r_2, ..., r_n)$$

The zeros of polynomials form many interesting algebraic shapes, a prototypical example being $x^2 + y^2 - 1$ over $\mathbb{R}$. Hence, we define:

$$Z(S) = \{p \in \mathbb{A}^n_R \mid f(p) = 0, \ \forall f \in S\}$$

A subset $X \subseteq \mathbb{A}^n_R$ is called an *algebraic set* if there exists $S \subseteq R[x_1, x_2, ..., x_n]$ such that $Z(S) = X$. The collection of algebraic sets are closed under finite union and intersection, and the sets $\mathbb{A}^n_R$ and $\emptyset$ are algebraic, hence the collection of algebraic sets form a topology on $\mathbb{A}^n_R$ known as the *Zariski topology*. Most often, we shall have $R = \mathbb{C}$ so that the polynomials always have solutions. Conversely, if we have a subset $S \subseteq \mathbb{A}^n_R$, then

$$\mathcal{I}(S) = \{f \in R[x_1, ..., x_n] \mid f(s) = 0, \ \forall s \in S\}$$

It should be checked that $\mathcal{I}(S) = (\mathcal{I}(S))$, namely it is always an ideal (and the smallest ideal of function that vanish on S). These two almost form a bijective correspondence. The ideals must be limited to radical ideals and the algebraic sets to varieties:

Definition 0.1.1: (classical) Variety

Let $V \subseteq \mathbb{A}^n_k$ be an algebraic subset. Then V is an [affine] *variety* if V is irreducible.

Theorem 0.1.2: Nullstellensatz

Let K be algebraically closed, $I \subseteq K[x_1, ..., x_n]$ a radical ideal, and $\mathcal{I}(-)$, $\mathcal{Z}(-)$ as defined above. Then these two functors form a bijective galois correspondence between radical ideals and varieties.

Proof **:**
[8]

This gives an equivalence of categories:

$$\text{Aff-Var} \underset{\mathcal{Z}}{\overset{\mathcal{I}}{\rightleftarrows}} (\text{Alg}_k^{\text{f.g.}})^{\text{op}}$$

(a lot more words here)

In many ways, the classical Nullstellensatz with finitely generated reduced rings (over algebraically closed fields) shows the symmetries of our "usual" geometry, that is the geometry of zero-sets of polynomials in $\mathbb{R}^n$ or some extension of $\mathbb{Q}^n$. The geometry is naturally trivial in the case of $\mathbb{F} = \mathbb{C}$, however there is a point to only focusing on such rings and studying their properties. Modern algebraic geometry is a departure from this and generalizes the types of rings we use.

0.2 Projective Space

Projective is the natural algebraic setting to solve equations, adding distinct solutions to rational polynomials. (put a summary of basic results from EYNTKA algebra here)

0.3 Noetherian Induction or Well-founded Induction

This is something that will come up in later proofs (for example, see lemma 5.1.14). This is a form of induction that generalizes the usual induction where the well-ordered set is $\mathbb{N}$. Recall that a binary relation R on S is a subset of $S \times S$. For $(a, b) \in R$, we may write $a \preceq b$ or $b \succeq a$. Then the symbol $\preceq$ is sometimes called the binary relation. A binary relation is a strict partial order if it is:

1. Irreflexive: $\forall x \in A, \neg(x < x)$
2. Transitive: $\forall x, y, z \in A, (x < y \wedge y < z) \implies x < z$
3. Asymmetric: $\forall x, y \in A, x < y \implies \neg(y < x)$

Given a set S with a strict partial order $\preceq$, a descending chain is a sequence $a_1, a_2, a_3, \ldots$ of elements in A such that $a_1 \succeq a_2 \succeq a_3 \succeq \ldots$

Definition 0.3.1: Well-founded Relation

A strict partial order $\preceq$ on a set A is well-founded if there exist no infinite descending chains in A with respect to $\preceq$.

Definition 0.3.2: Minimal Element

Given a subset S of a well-founded set A, an element $m \in S$ is minimal if there is no element $x \in S$ such that $x \preceq m$.

Theorem 0.3.3: Noetherian Induction

Let $(A, \preceq)$ be a set with a well-founded relation $\preceq$, and let P be a property defined on A. If for all $x \in A : [\forall y \in A, (y \preceq x \implies P(y))] \implies P(x)$ then $P(x)$ holds for all $x \in A$.

Proof **:**
see cite:HERE.

The following may be an insightful alternative:

Theorem 0.3.4: Noetherian Induction - Alternative Form

Let $(A, \preceq)$ be a set with a well-founded relation $\preceq$. If $S \subseteq A$ is a subset such that for all minimal elements m of $A \backslash S$, we have $m \in S$, then $S = A$.

> ***Proof*** **:**
> see cite:HERE.

The connection between these forms lies in taking S to be the set of elements where P holds, and showing that if S isn't all of A, then any minimal element of $A \backslash S$ must actually be in S, giving a contradiction.

0.4 Categorical Reminder

here

Part I

Scheme Theory

I believe there will be more than just basic alg geo. This part will go over the core theory, Hartshorne, Vakil, Eisenbud, and so forth. Other parts will surely be specializations.

1

Sheaves

Roughly speaking, Sheaf Theory is the theory of augmenting a topological space[1] that adds information to each open set (adds "local" information) which can be "glued" to get information in larger open sets. The prototypical example of a sheaf is the set of smooth functions on a manifold. Smooth functions on a manifold can be restricted to open subsets of a manifold which are homeomorphic to $U \subseteq \mathbb{R}^n$ and if a smooth function on each part of the manifold is defined in a compatible way (i.e. they agree on intersections), they can be glued back together to form a function on the entire manifold. As an open covering of M can always be chosen so that each U is simply connected connected, the non-trivial information is in the ability or choice of gluing of functions.

A key motivator behind sheaves is that they can hold the "geometric" information of a space. The reader may recall that a vector-space and subspaces can be characterized via their dual spaces, that is all functions from the vector space to $\mathbb{F}$. A similar result holds for Varieties with the coordinate ring uniquely characterizing a Variety. This duality was found to exist everywhere in mathematics, to name three more examples:

- A classical example that started the duality between algebra and geometry is that of $C(X)$ and X where X is a compact space and $C(X)$ all continuous function on X. By taking the stone-Cech compactification of a space X, the space can be achieved for any space (see [9, p.144] for details)
- Similarly, a radon measure's can be characterized using the duals of $C_c(X)$ where X is a locally compact space (classical result can be found in EYTNKA at [17])
- Spin manifolds can be characterized by spectral triples (see [1])

[1] more generally, a topoi, which concretely translates in our case to a category whose objects are open sets and morphisms are inclusions

- Sets with set-function can be thought of as the most "abstract" geometric object (they only have information about containment). There is an appropriate dual called the *complete atomic Boolean algebras*. The reader may be interested in knowing that Boolean algebra's characterize logical operations, which gives an interesting geometric interpretation to logic[2].

There is a long list of such examples (the reader may be curious to checkout this link). This gave rise to the idea that a geometric space can be understood by studying some subset of function on this space with certain. In algebraic cases, there are relatively few functions that are defined globally on a space, but many that are defined on local components (the prototypical example of this would be $\mathbb{C}$[3]).[4]

Why do functions capture this geometric information? One can think of functions as tools for representing relations. If our function is of the form $f : X \to \mathbb{R}$, then this is relating X to numbers, i.e. quantification, usually X being a topological space and f being continuous for some notion of closeness. If we have a continuous function of the form $f : X \to \mathbb{R}^n$ (or $\mathbb{C}^n$), with X a topological space, then we may pullback a lot of geometric results from $\mathbb{R}^n$ (or $\mathbb{C}^n$) onto X, depending on how f is defined (ex. it can be shown that f is a diffeomorphism if and only if it pulls back the differential structure of $\mathbb{R}^n$). Very often, it is not possible to define a diffeomorphism $f : X \to \mathbb{R}^n$, however we may define it on components. Functions have the property that we may restrict them or construct them by defining what happening at each point (or in the continuous case, on each open set where they agree on intersections[5]). But you may say that morphisms in a category already capture this intuition. This is indeed the case, what makes functions special is that you can construct larger morphisms from smaller morphisms with an appropriate notion of *gluing* and *restricting*. In particular, the closeness given by the notion of topological spaces and the locality properties inspired by functions shall be the two main properties of sheaves. As is the course of mathematics, it may be asked if these properties of functions can be generalized, and we may ask if we can take special categories which have these features; this is called a *site*. For an introduction to algebraic geometry, limiting to the sites defined on topological spaces is plenty sufficient and shall be the focus of this book.

A particularly important realization will be that commutative rings will in many ways behave like functions on a space, that is if R is a ring, there is a sense in which the elements of R will be the "functions" on a special space (namely, the space $\operatorname{Spec}(R)$). This shall be elaborated on in chapter 2 and was commented on in the front-matter. At this stage it is motivating to know that there is a reason why we may care about sheaves other than the sheaf of functions, namely the interpolation between commutative rings and functions.

Motivating Example from Vakil

Before exploring sheaves in generality, it is useful to have the prototypical example of a sheaf in mind in more detail:

[2] This connection is further refined using Topoi

[3] entire holomorphic are necessarily power series, while holomorphic on (not necessarily simply connected) open sets are analytic and may even have non-trivial Laurent expansion around holes

[4] see this link for more details

[5] by the gluing lemma

Example 1.0.1: Sheaf Of Differentiable Functions On Manifold

Let M be a smooth manifold. Then the we shall constuct the sheaf of differentiable function on M. For each open subset $U \subseteq M$, we shall assocaite to it the set of smooth function $C^\infty(U)$. Denote this collection by $\mathcal{O}(U)$ (this will be common notation for sheaves). Then if we have an open subset of U, $V \subseteq U$, we may restrict the function from U to function on V, namely we have a map $\mathrm{res}_{U,V} : \mathcal{O}(U) \to \mathcal{O}(V)$ mapping $f \mapsto f|_V$. Naturally, if we have $W \hookrightarrow V \hookrightarrow U$, then the following commutative diagram.

$$\begin{tikzcd}
 & V \arrow[dr, "\mathrm{res}_{V,W}"] & \\
U \arrow[ur, "\mathrm{res}_{U,V}"] \arrow[rr, "\mathrm{res}_{U,W}"'] & & W
\end{tikzcd}$$

This collection of maps satisfy two important properties. In the following let and $U = \bigcup_i U_i$.

1. *Defining Locally*: Let's say $f_1, f_2 \in \mathcal{O}(U)$, Then if f_1 and f_2 agree on all restriction, that is $\mathrm{res}_{U,U_i} f_1 = \mathrm{res}_{U,U_j} f_2$, then $f_1 = f_2$
2. *Gluing*: If there exists a collection $f_i \in \mathcal{O}(U_i)$ such that f_i and f_j agree on intersections, that is $\mathrm{res}_{U_i, U_i \cap U_j} f_i = \mathrm{res}_{U_j, U_i \cap U_j} f_j$, then there is a $f \in \mathcal{O}(U)$ such that $\mathrm{res}_{U,U_i} f = f_i$ for all i.

Then a sheaf will respect these properties. Note that in fact, we may define a manifold M to be $(M, \mathcal{O}_M)$ where M Is a topological space and $\mathcal{O}_M$ is a sheaf of smooth functions on M. If instead $\mathcal{O}(U)$ was the collection of continuous functions, we would get a topological manifold. Even more generally, we may let the functions be set functions, for set functions work well under restrictions and we may define set functions locally and construct set function via gluing.

Note that Diffeology can be thought of as a generalization of this result or of a special case of sheaf theory, and we shall explore it briefly at the end of this chapter.

So if even set functions can form a sheaf, then why did I write "nice" function earlier? As it turns out, what we shall do is abstract the notion of function! In particular, we shall look at systems where we may have the behavior of localizing and gluing "functions" on a topological space. Hence, in the following abstraction, we will consider $\mathcal{O}_M$ to be the data that stores a collection of objects that *behave like functions*.

A key property that distinguishes functions from general morphisms is the notion of evaluation (taking $f(x)$ for some element x and function f). Since we are abstracting the notion of function, we shall need a notion of *evaluation*. To motivate this generalization, let's come back to $(M, \mathcal{O}_M)$ and consider some $\mathcal{O}(U) = C^\infty(U)$ for some open $U \subseteq M$ and take $p \in U$. Recall that $\mathcal{O}(U)$ is a ring and that the *germ* at p is collection of equivalence relations:

$$[f] = \{(f, U) \mid p \in U,\ f \in \mathcal{O}(U)\}$$
$$(f, U) \sim (g, V) \iff W \subseteq U, W \subseteq V, p \in W,\ f|_W = g|_W$$

that is, $[f] = [g]$ if and only if there is an open neighborhood $W \subseteq U, V$ containing p ($p \in W$) such that $f|_W = g|_W$ (in the language of sheaf theory, $\mathrm{res}_{U,W} f = \mathrm{res}_{V,W} g$). Denote this collection $\mathcal{O}(U)_p$. The key is that the germ of $\mathcal{O}(U)$ at p is a *local ring* with maximal ideal $\mathfrak{m}_p$ of germs such that $[f](p) = 0$ (or simply $f(p) = 0$ if unambiguous). Then:

$$0 \to \mathfrak{m}_p \to \mathcal{O}(U)_p \xrightarrow{f \mapsto f(p)} \mathbb{R} \to 0$$

and all elements $\mathcal{O}(U) \setminus \mathfrak{m}_p$ are invertible. Hence, we shall see that we may thing of "evalutation" of f at some point $p \in U$ as taking first the germ $\mathcal{O}(U)_p$ and quotient it by the maximal ideal $\mathfrak{m}_p$, $\mathcal{O}(U)_p/\mathfrak{m}_p$, and then taking $\overline{f} \in \mathcal{O}(U)_p/\mathfrak{m}_p$ to be the value of f. Note that the above construction required that $\mathcal{O}(U)$ is a *local ring*. This is evidently the case when $\mathcal{O}(U) = C^\infty(U)$, however in general sheaf theory this ring will not in general be local. In fact, $\mathcal{O}(U)$ will not even need to be a ring! When each $\mathcal{O}(U)$ is a ring, then we shall say that X is a *ringed space*, and when all the germs are local rings, then we shall say that X is a *locally ringed space*.

A final word on generalization: we have been working over a topological space, however we may abstract further and work over a general category; in this way, we would define the notion of a "topology" on a category, known as a *Grothendeick topology* (a category with this structure is called a *site*), and develop sheaf theory accordingly. Though this is fruitful, this generlization shall be ommited for now to concentrate on the geoemtric properties at hand when using a topological space.

1.1 Definitions and Examples

We start by formalizing the above intuitions.

Definition 1.1.1: Presheaf Axiomatic Definition

Let X be a topological space. Then a *presheaf* over a category $\mathbf{C}$ on X, denoted $\mathscr{F}_X$ contains the following data:

1. for each open set $U \subseteq X$, there is an object c of $\mathbf{C}$ and denote it by $\mathscr{F}(U)$. If C is a category whose objects are sets with additioanl structure, then the elements of $\mathscr{F}(U)$ are called the *sections* of $\mathscr{F}$ over U. If we say that s is a section of $\mathscr{F}$, then we mean that s is a section of $\mathscr{F}$ over X.

2. For each inclusion map $U \hookrightarrow V$, there is a morphism $\mathrm{res}_{V,U} : \mathscr{F}(V) \to \mathscr{F}(U)$ called the *restriction* morphism (or *restriction map*) such that

   (a) $\mathrm{res}_{U,U} = \mathrm{id}_U$

   (b) If $W \hookrightarrow V \hookrightarrow U$, then the following diagram commutes

$$\begin{tikzcd} & \mathscr{F}(V) \arrow[dr, "\mathrm{res}_{V,W}"] & \\ F(U) \arrow[ur, "\mathrm{res}_{U,V}"] \arrow[rr, "\mathrm{res}_{U,W}"'] & & F(W) \end{tikzcd}$$

The collection of all presheaves on a topological space X is denoted $\mathbf{PSh}_{\mathbf{C}}(X)$ (or $\mathbf{PSh}(X)$ if unambiguous). If U is clear from context, we shall often write $\mathrm{res}_{U,V} f$ as $f|_V$.

The nomenclature "section" will be made clear in proposition 1.1.11. When it is no ambiguous the image of the restriction morphism for $s \in \mathscr{F}(U)$ shall be denoted:

$$\mathrm{res}_{U,V}(s) = s|_V$$

This notation can be ambiguous as it is not explicit which $U \subseteq X$ is taken from which $s \in \mathscr{F}(U)$, however it will be unambiguous in-context. Notice that it is equivalent to define a presheaf as a *contravariant functor*. Namely

Definition 1.1.2: Presheaf Functorial Definition

Let X be a topological space and let $O(X)$ be the category whose objects are the open sets of X and whose morphism are $\subseteq$ (i.e. the category induced by the partial order given by $\subseteq$). Then a *presheaf* over $\mathbf{C}$ is a contravariant functor from $O(X)$ to $\mathbf{C}$, that is a functor $\mathscr{F} : O(X)^{op} \to \mathbf{C}$.

Usually, our presheaves shall be over **Set**, Ab, $\mathbf{Mod}_k$ or **Ring**[6].

An important notion for geometric objects is their behavior locally. The extreme (or limit) of the notation of locality is the behavior around a point p. This idea originated when studying germs of differential functions in Differential Geometry. These are well-defined on presheaves;

Definition 1.1.3: Stalk as Equivalence Relation

let $(X, \mathcal{O}_X)$ be a presheaf. Then the *stalks* at $p \in X$ is the collection of equivalence class:

$$[f] = \{(f, U) \mid p \in U,\ f \in \mathcal{O}(U)\}$$
$$(f, U) \sim (g, V) \qquad \Leftrightarrow \qquad W \subseteq U, W \subseteq V, p \in W,\ \operatorname{res}_{U,W} f = \operatorname{res}_{V,W} g$$

The stalks at p is usually denoted $\mathcal{O}_{X,p}$ or $\mathcal{O}_p$ if unambiguous in context.

If $\mathbf{C}$ is a category whose objects are augmented sets[a], then if $p \in U$ and $f \in \mathscr{F}(U)$, $[f] \in \mathscr{F}_p$ is called the *germ* of f at p.

[a] ex. **Set**, **Ring**, **Grp**, etc.

Note that in general, the notion of "evaluation" of a stalk/germ as it is usually defined on differential geometry won't make sense; though the symbol f is written the open sets need not contain functions or objects that act like function. We shall later define nice presheaves in which we may recover the notion of evaluation and treat the elements like functions. There is again a categorical way of interpreting stalks that is useful to us:

Definition 1.1.4: Stalks As Colimits

Let $\mathscr{F} \in \mathbf{PSh}_{\mathbf{C}}(X)$ be a presheaf where $\mathbf{C}$ is a category with small filtered colimits. Then a *stalk* at $p \in X$ is:

$$\mathscr{F}_p := \varinjlim \mathscr{F}(U)$$

that is $\mathscr{F}_p$ is the colimit over all $\mathscr{F}(U)$ where $p \in U$.

The adjectives "small filtered colimit" are only there to guarantee that the colimit exists. Most base category that will be worked over shall have the stronger property of bieng *cocompelte* (having all colimits) and often also being *complete* (having all limits). Categories that are complete and cocomplete include:

1. **Set** (Category of Sets)

[6] Technically, **Ab** is $\mathbf{Mod}_{\mathbb{Z}}$, however distinguishing these two is useful as we usually mean k to be a field

2. **Grp** (Category of Groups)
3. **Ab** (Category of Abelian Groups)
4. **Ring** (Category of Rings)
5. R-**Mod** (Category of modules over a ring R)
6. **Top** (Category of Topological Spaces)

Hence, this shows us that as long as the colimit exist in our category, we may define the notion of stalks for presheaves over that category without needing to reference the notion of elements. Note that it is the colimit and not the limit since we need $\mathscr{F}_p$ to "represent" the behaviour around p rather than "embed" germs.

One of the most important properties of stalks is their ability to be defined locally. First, if $U \subseteq X$ is an open set, we can define the sub-presheaf on U by letting $\mathcal{F}|_U(V) = \mathcal{F}(V \cap U)$. The following proposition will be really useful later when we need to work with stalks between larger and smaller (pre)sheaves, and mirrors the property in differential geometry (recall [4, chapter 2.1])

Proposition 1.1.5: Stalk Isomorphism

Let X be a presheaf of sets and $U \subseteq X$ an open set. Then for any $x \in U$:

$$\mathscr{F}_{X,x} \cong_{\mathbf{C}} \mathscr{F}_{U,x}$$

***Proof* :**
We shall prove this in the case where $\mathbf{C} = \mathbf{Set}$. The general case is left as an exercise.

Let us start by defining the map $\varphi : \mathscr{F}_{X,x} \to \mathscr{F}_{U,x}$. Take $[(f, V)] \in \mathcal{F}_{X,x}$ such that $x \in V$. As $x \in U$, $x \in U \cap V$. Hence, notice that:

$$(f, V) \sim (f|_{U\cap V}, U \cap V) \qquad \text{as} \qquad \mathrm{res}_{U,U\cap V} f = \mathrm{res}_{U\cap V,U\cap V} f|_{U\cap V} = f|_{U\cap V}$$

Then define the map between stalk via:

$$\varphi : (f, V) \sim (f|_{U\cap V}, U \cap V) \mapsto (f|_{U\cap V}, U \cap V)$$

This map is well-defined. Take $(f, V_1) \sim (g, V_2)$ where $V_1, V_2 \subseteq X$ so that we have to show that $(f|_{U\cap V_1}, U \cap V_1) \sim (g|_{U\cap V_2}, U \cap V_2)$. As $(f, V_1) \sim (g, V_2)$ there exists $W \subseteq V_1, V_2$ such that

$$f|_W = g|_W$$

Take $U \cap W \subseteq W \subseteq V_1, V_2$. Then by the restriction axioms we have that:

$$f|_{U\cap W} = g|_{U\cap W}$$

So that $(f|_{U\cap W}, U \cap W) \sim (g|_{U\cap W}, U \cap W)$. Then as:

$$\mathrm{res}_{U\cap V_1,U\cap W}(f|_{U\cap V_1}) = \mathrm{res}_{U\cap W,U\cap W}(f|_{U\cap W})$$

(and similarly for V_2), we get:

$$(f|_{U\cap V_1}, U \cap V_1) \sim (f|_{U\cap W}, U \cap W) \sim (g|_{U\cap W}, U \cap W) \sim (g|_{U\cap V_2}, U \cap V_2)$$

and hence the map is well-defined. This map has a natural inverse, namely

$$\varphi^{-1} : \mathscr{F}_{U,x} \to \mathscr{F}_{X,x} \qquad (f, V) \mapsto (f, V)$$

which can be shown to be well-defined and $\varphi \circ \varphi^{-1} = \varphi^{-1} \circ \varphi = \mathrm{id}$, hence these sets are bijective, which are the isomorphisms in set. To generalize to categories that are augmented sets, take advantage of the restriction maps being morphisms.

Now, presheaves allow for the notion of restricting and localizing, but as we shall show soon not all presheaves are amenable to gluing local information to create global information (think of bounded holomorphic functions on open subsets of $\mathbb{C}$). We must add this condition axiomatically:

Definition 1.1.6: Sheaf using Axioms

Let $\mathscr{F} \in \mathbf{PSh}_C(X)$ be a presheaf over a category $\mathbf{C}$ whose objects are sets possibly with additional structure. Then $\mathscr{F}$ is also a sheaf if it satisfies the following two axioms;

1. *Locality Axiom*. Let $U = \bigcup_i U_i$. If $f_1, f_2 \in \mathscr{F}(U)$, then if $\mathrm{res}_{U,U_i} f_1 = \mathrm{res}_{U,U_i} f_2$ for all i, then $f_1 = f_2$
2. *Gluing Axiom*: Let $U = \bigcup_i U_i$. If there exists a collection $f_i \in \mathscr{F}(U_i)$ such that $\mathrm{res}_{U_i, U_i \cap U_j} f_i = \mathrm{res}_{U_j, U_i \cap U_j} f_j$, then there is a $f \in \mathscr{F}(U)$ such that $\mathrm{res}_{U,U_i} f = f_i$ for all i.
3. $\mathscr{F}(\emptyset)$ is associated with the terminal object (if we're working over set, $\mathscr{F}(\emptyset)$ is a singleton).

If only the locality axiom is satisfied, then our presheaf is called a *separated presheaf*. The collection of all sheaves on a topological space X is denoted $\mathbf{Sh}_C(X)$ (or $\mathbf{Sh}(X)$ if unambiguous)

We may think of the locality axiom telling us there is *at most* one way to glue, while the gluing axiom is telling us there is *at least* one way to glue. A separated presheaf is so named as it allows sections to "separate" points (there a cannot be two sections identical on restrictions of an open covering).

1.1.1 Separation and Evaluation Later, the notion of "evaluation" shall be defined for sections where the objects of $\mathbf{C}$ behave like sets of functions; being separated does *not* imply a function is defined by the evaluation at every point.

Sometimes, $\mathscr{F}(\emptyset)$ is specified to be the terminal object (in **Set**, that would be a singleton). The following is sometimes (redundantly) included in the definition of a presheaf:

Lemma 1.1.7: Sheaf and Zero Section

Let $\mathbf{C}$ be a category with the zero object 0 and $\mathscr{F}$ a sheaf (or particularly, a separated presheaf) on X. Let $\bigcup_{i \in I} U_i = X$ be an open cover and $s \in \mathscr{F}(X)$. Then if $s|_{U_i} = 0$ for all i, then $s = 0$.

Proof **:**
exercise.

Notice that we are taking elements of each $\mathscr{F}(U)$ because $\mathscr{F}(U)$ is an object in a category whose objects are sets with possibly additional structure. This is a deviation from the generalization presented in the definition of presheaves as $\mathbf{C}$ may be more general. There is in fact a way to define a sheaf for general categories:

Proposition 1.1.8: Sheaf Categorical Interpretation

Let $\mathscr{F} \in \mathbf{PSh}(X)$ be a presheaf, and let $\cdot$ be a terminal object (in $\mathbf{Set}$, an arbitrary 1-point set). Let $U = \bigcup_i U_i$, and consider:

$$\cdot \to \mathscr{F}(U) \to \prod_{i \in I} \mathscr{F}(U_i) \rightrightarrows \prod_{i,j \in I} \mathscr{F}(U_i \cap U_j)$$

for any indexing set I. Then if the sequence is exact at $\mathscr{F}(U)$, the locality axiom holds, and if it is exact at $\prod \mathscr{F}(U_i)$, the gluing axiom holds.

Proof **:**
This shall be proven when $\mathbf{C} = \mathrm{Ab}$, the general result can be proven by the reader interested in practicing their category theory.

Assume $\mathscr{F}(U) \xrightarrow{\alpha} \prod \mathscr{F}(U_i)$ is injective so that

$$s \hookrightarrow (s|_{U \cap U_i})_i$$

Then by definition of injectivity any s_a, s_b mapping to this collection of restricted sections is equal, $s_a = s_b$, giving the locality.

Now, let $s_i = \mathscr{F}(U_i)$. Take the equalizer:

$$\prod_{i \in I} \mathscr{F}(U_i) \xrightarrow{\beta,\gamma} \prod_{j \in J} \mathscr{F}(U_i \cap U_j)$$

where:

$$\beta((s_i)_i) = \big(s_i|_{U_i \cap U_j}\big)_{i,j \in I} \qquad \text{and} \qquad \gamma((s_i)_i) = \big(s_j|_{U_i \cap U_j}\big)_{i,j \in I}$$

For example Let $I = \{1, 2\}$ and $U_1 \cup U_2 = U$. Let us not put in the "redundant" elements for the pairs $(1, 1), (2, 2)$, and $(2, 1)$; the last pair is not automatically excluded in the definition to avoid putting an unnecessary ordering on I, and the former two are not excluded for the simplicity of notation. As we are working with abelian groups, we can turn the equalizer into:

$$\cdot \to \mathscr{F}(U) \to \prod_{i \in I} \mathscr{F}(U_i) \xrightarrow{s_i|_{U_i \cap U_j} - s_j|_{U_i \cap U_j}} \prod_{i,j \in I} \mathscr{F}(U_i \cap U_j)$$

Thus the kernel of the equalize is when these two are equal, namely:

$$\beta(s_1, s_2) = (s_1|_{U_1 \cap U_2}) \qquad \text{and} \qquad \gamma(s_1, s_2) = (s_2|_{U_1 \cap U_2})$$

are equal:

$$(s_1|_{U_1 \cap U_2}) = \beta(s_1, s_2) = \gamma(s_1, s_2) = (s_2|_{U_1 \cap U_2})$$

Giving the gluing condition. In general, let us denote the kernel of the equalizer $\mathrm{Eq}(\beta, \gamma)$. By exactness we get that:

$$\mathrm{im}(\alpha) = \mathrm{Eq}(\beta, \gamma)$$

Thus, every element in the equalizer comes from a section in U. If α is also a monomorphism, it is unique, as we sought to show.

1.1.2 Foreshadowing Čech Cohomology

What if we extend this to have further equalizers? Sheaves that satisfy higher-order intersection-exactness conditions are a proper subset of sheaves (over non-trivial categories and topological spaces). Such sheaves can be thought of as never having any gluing obstruction, and hence have trivial cohomology. We shall return to these ideas when exploring Čech cohomology in section 5.2.

Example 1.1.9: Presheaves failing exactness (i.e. gluability and locality)

Let X be any topological space and define:

$$\mathcal{F}(X) = \mathbb{Z} \qquad \mathcal{F}(U) = 0 \qquad (\text{for } U \subsetneq X)$$

Then this sheaf is gluable, but not separated. If $U \subsetneq X$ is covered, then it is clear that gluing must hold. If $U = X = \cup_i U_i$, then that restrictions holds as any $z \in \mathbb{Z}$ has $z|_{U_i} = 0$ and so they certainly agree on all intersections. Hence, we see that each $z \in \mathbb{Z}$ is glued by a bunch of 0-elements!

Example 1.1.10: Presheaves And Sheaves

Most of these examples will be on sheaves of sets.

1. Verify that the sheaf of differentiable functions over M, namely $\mathcal{O}_M$, is indeed a sheaf. More generally, if X is any topological space, then the data defined by $\mathscr{F}(U) = \mathrm{Hom}_{\mathbf{C}}(U, \mathbb{R})$ where $\mathbf{C}$ is either **Top** or **Set** is a sheaf.

2. (*Restriction of Sheaves*) Let $\mathscr{F} \in \mathbf{Sh}(X)$ and $U \subseteq X$ an open subset of X. Then the *restriction* of $\mathscr{F}$ to U, denoted $\mathscr{F}|_U$ has as data for each open $V \subseteq U$ $\mathscr{F}|_U(V) = \mathscr{F}(V)$ (note hos it is important that U is open for well-definedness). We shall later see how we may define restrictions of sheaves to arbitrary sets[a]

3. (*Skyscraper Sheaf*) Let X be a topological space, $p \in X$, and S be any set. Let $\iota_p : \{p\} \to X$ be the inclusion map. Then define the sheaf of sets $(\iota_p)_* S$ on X (not on S!) to be:

$$(\iota_p)_* S(U) = \begin{cases} S & p \in U \\ \{e\} & p \notin U \end{cases}$$

   This sheaf is called the *skyscraper sheaf.* The name comes from the fact that the informal picture for this sheaf looks like a skyscraper at p. For sheaves of Abelian groups, $\{e\}$ is the trivial group. More generally, $(\iota_p)_*(U)$ would be the final object if $p \notin U$.

4. (*Constant Presheaf*) Let X be a topological space and S be any set. Define $\underline{S}_{\text{pre}}(U) = S$ for all open $U \subseteq X$ Then it should be verified that $\underline{S}_{\text{pre}}$ forms a presheaf on X, called the *constant presheaf associated to* S. This set is not in general a sheaf if $|S| > 1$: take $X = U_1 \coprod U_2$ to be the disjoint union of two (open) sets, take the constant presheaf so that all restriction maps are constant maps. Chose $s_1 \in S$ and $s_2 \in S$ which are distinct:

$s_1 \neq s_2$. Then by the gluing axiom, there would have to exist a $s \in S$ such that $s|_{U_1} = s_1$ and $s|_{U_2} = s_2$ and their intersection is empty and hence tautologically follow the gluing axiom, however as we have the identity map we have $s_1 = s = s_2$ however $s_1 \neq s_2$.

This is fixed by a process called *sheafification* which will add the appropriate section equivalent to (s_1, s_2) that would restrict to both of these elements.

What makes the sky-scrapper sheaf special is the "distinguishing" of a point making it act like a generalization of an indicator function. In the above example, one of the two sets will not have the point p.

5. (*Constant Sheaf*) Again let X be a topological space and S be any set. For every open $U \subseteq X$, let $\mathscr{F}(U)$ be the collection of all *locally constant* continuous functions from U to S, that is if $f \in \mathscr{F}(U)$, for each $p \in U$ there is an open neighborhood $V \subseteq U$ containing p, $p \in V \subseteq U$, such that $f|_V : V \to S$ is constant (on euclidean space, this is equivalent to having a constant function on each connected component). Then this sheaf is called the *constant sheaf associated to* S, and is denoted $\underline{S}$.

   Another way of describing this sheaf is if we endow S with the discrete topology and let $\mathscr{F}(U)$ be the continuous maps $U \to S$.

6. A great example of a pre-sheaf on $\mathbb{C}$ that is not a sheaf is the sheaf of bounded holomorphic functions on $\mathbb{C}$. Recall that bounded entire holomorphic functions (defined on all of $\mathbb{C}$) must be constant by Louisville's Theorem, however there do exist bounded functions on open subsets of $\mathbb{C}$. For example , on each disk of radius r, $\mathbb{D}^2_r$, the holomorphic function z^2 is bounded but $\mathbb{C} = \cup_r \mathbb{D}^2_r$ has no bounded holomorphic function for which the restriction to any disk $\mathbb{D}^2_r$ is z^2; hence the gluing axiom fails.

   Another example would be the sheaf composed of holomorphic functions admitting holomorphic square roots, as there is no global square-root function on $\mathbb{C}$ due to non-trivial monodromy.

7. (*Sheaf of continuous maps*) Let X and Y be topological spaces. Define $\mathscr{F}(U) = \mathrm{Hom}_{\mathbf{Top}}(U, Y)$. This sheaf is a generalization sheaf of differentiable functions and the constant sheaf, namely

   - The sheaf of differentiable functions has $Y = \mathbb{R}$
   - the constant sheaf has $Y = S$ with the discrete topology.

   If Y is also a topological group, then the continuous maps to Y form a sheaf of groups.

8. (*Sheaf of section of a maps*) This is an important special case of the above sheaf. Let X, Y be topological spaces and let $\mu : Y \to X$ be a continuous map. Then the sections of μ (in the sense of the existence of a map $s : X \to Y$ such that $\mu \circ s = \mathrm{id}$) form a sheaf. In particular, for each open $U \subseteq X$, $\mathscr{F}(U)$ is the set of section maps $s : U \to Y$ (continuous maps $s : U \to Y$ such that $\mu \circ s = \mathrm{id}|_U$). Vector bundles and vector fields are a good example of such a sheaf.

9. The following examples shows how different sheaves can be put on the same object to characterize different type of information. Throughout this example, let $X = \mathbb{A}^1_k$ be the (classical) affine line (see definition 0.1.1). The open subsets in the Zariski topology are all cofinite subsets. If $U \subseteq \mathbb{A}^1_k$ is open let $a_1, ..., a_n$ be the points that are omitted from U.

(a) (Sheaf of regular function) Then letting $f_i = (x - a_i)$ and $F = \{f_1, ..., f_n\}$, define $\mathscr{F}(U) = k[x]_F$ i.e. $\mathscr{F}(U)$ is the ring generated by $k[x]$ and $1/f_i$ for each i. The reader should verify that this is indeed a pre-sheaf, and even a sheaf.

(b) (Sheaf of invertible functions) Let X be as above, and let $\mathscr{F}(U) = (k[x]_F)^*$, that is consider only the invertible functions. Then X is again a sheaf.

(c) (sheaf of roots of unit) Again consider X as above. Then let $\mathscr{F}(U)$ be the subgroup of $k[x]_F^*$ consisting of all roots of unity present in $\mathscr{F}(U)$. This forms a sheaf

(d) (Sheaf of differential forms) If the reader is familiar with differential geometry, let $k = \mathbb{R}$ and let $\mathscr{F}(U)$ be the collection of all closed forms on U (or more generally, all smooth forms on U). Note that only $\mathscr{F}(\mathbb{A}^1_{\mathbb{R}})$ are all the closed forms exact.

[a] see ref:HERE

If $\mathscr{F} \in \mathbf{PSh}(U)$ and $f \in \mathscr{F}(U)$ for some open $U \subseteq X$, then f is called a section. The following proposition motivates this idea

Proposition 1.1.11: Space of Sections of (Pre)Sheaf: étale spaces

Let $\mathscr{F}$ be a presheaf of sets. Then there exists a topological space F and a local homeomorphism $\pi : F \to X$ so that $\mathscr{F}$ can be represented as the sheaf of sections from F to X. The space F is called the *étalé space*. In particular, F is the *étalé space* of $\mathscr{F}$.

Furthermore, there is an equivalence between the category $\mathbf{Sh}(X)$ and the category of étalé spaces over X (with bundle morphisms as morphisms).

The idea is that (pre)sheaves correspond to a space F that is it is locally homeomorphic to X. Thus, geometrically sheaves could be thought of as local homeomorphisms, though the total space F may be rather ugly as shall be shown in example 1.1.12. Note that local homeomorphisms need not be surjective; this is a common misconceptions as many examples of surjective local homeomorphisms are covering spaces, but examples such as the sky-scrapper sheaf should show why this need not be the case.

***Proof* :**
Let F be the disjoint union of the stalks on F.

$$F = \coprod_{x \in X} \mathscr{F}_x$$

Then $\pi : F \to X$ is naturally a set map. Define a basis for F as follows: for each $s \in \mathscr{F}(U)$, take the collection

$$\{[s]_x \in \mathscr{F}_x \mid x \in U\}$$

These form a basis for F (note that each stalk has the discrete topology). With the setup, it is now easy to see the equivalence of categories.

The étale space may seem like it ought to be pretty similar to X being locally homeomorphic, however this can be far from the case:

Example 1.1.12: Non-Hausdorff Étale Space
Take $X = \mathbb{R}$ be the real numbers and take $\mathcal{F}$ to be the smooth functions on $\mathbb{R}$. Take the constant germ around 0 as well as the germ which can be represented by the function:

$$g(x) = \begin{cases} x\sin(1/x) & x\sin(1/x) > 0 \\ 0 & x\sin(1/x) \leq 0 \\ 0 & x < 0 \end{cases}$$

In other words, g is zero when x is negative and when the function $x\sin(1/x)$ would be negative; we want to keep when $x\sin(1/x)$ is positive. Then these two are different points in the étale space, but *cannot* be separated by open sets.

Note that if we were working with $X = \mathbb{C}$ and took analytic or holomorphic functions, then this cannot happen as we have a non-isolated 0 without all of g being zero. This turns out to be the main impingement for the étale space being Hausdorff. In general sheaves of analytic functions shall form Hausdorff étale spaces.

1.1.1 Stalks and Sheaves

As mentioned earlier, presheaves induce stalks. It was mentioned that sheaves but *not* presheaves can be characterized on the level of stalks, meaning the stalks of a sheaf shall give us all the information about the morphism a sheaf. This in fact almost characterizes sheaves from presheaves, as it in fact characterizes *separated presheaves* (recall they satisfy the locality but not gluing axiom). The following shows the value of working with separated presheaves:

Lemma 1.1.13: Sections Determined By Germs

Let $\mathscr{F}$ be a sheaf (more specifically, a separated presheaf). Then the map:

$$\mathscr{F}(U) \to \prod_{p\in U}(\mathscr{F}_p)$$

is injective

This map being injective justifies the term "separated".

Proof **:**
Let f be the map and consider $f(s) = (s_x)_{x\in U}$ Say:

$$(s_x)_{x\in U} = (t_x)_{x\in U} \qquad \text{equiv.} \qquad [(s, V_x)]_{x\in U} = [(t, W_x)]_{x\in U}$$

Then by definition there exists a $C_x \subseteq V_x \cap W_x$ such that

$$s|_{C_x} = t|_{C_x}$$

Now as $x \in C_x$ for each $x \in U$ we have $U = \bigcup_{x\in U} C_x$ and furthermore for for any pair C_x, C_y in the open cover we have:

$$\mathrm{res}_{C_x, C_x\cap C_y} s = \mathrm{res}_{C_y, C_x\cap C_y} t$$

hence, by the gluing axiom $s = t$, as we sought to show.

Note that we can show the locality axiom form this argument:

Corollary 1.1.14: Global Section Over Local Sections

Let $\mathscr{F}$ be a sheaf on X and $U_i \subseteq X$ is an open cover. Then:

$$\mathscr{F}(X) \to \prod_i \mathscr{F}(U_i)$$

is injective

Proof **:**
Map $s \mapsto \left(s|_{U_i}\right)_i$. Consider

$$\left(s|_{U_i}\right)_i = \left(t|_{U_i}\right)_i$$

then s and t agree on intersections of U_i, hence there exists a unique element in $\mathcal{F}(X)$ for both these elements, namely $s = t$

In example 1.1.9, the sheaf $\mathcal{F}(X) = \mathbb{Z}$ and $\mathcal{F}(U) = 0$ for $U \subsetneq X$ was not a separated presheaf, and certainly $\mathcal{F}(X) \to \prod_{x\in X} \mathcal{F}_x$ is *not* injective, as each $\mathcal{F}_x$ consist of a single equivalence class representing the zero element (and hence $\prod_{x\in X} \mathcal{F}_x \equiv \{[0]\}$, and $\mathcal{F}(X) = \mathbb{Z}$. This example shows that we can somehow construct a nonzero object from a bunch of zero-objects. This can be thought of as being the key-blocker, especially in the case of abelian categories where most of the time the pre-image of 0 is only 0. This motivate the following:

Definition 1.1.15: Support of Sections

Let $\mathscr{F}$ be a sheaf of an abelian group on X, and s a global section of $\mathscr{F}$. Then the *support* of s, denoted $\mathrm{Supp}(s)$, is the points $p \in X$ where s has a nonzero germ:

$$\mathrm{Supp}(s) := \{p \in X \mid s_p \neq 0 \text{ in } \mathscr{F}_p\}$$

This notion differs a bit from the usual notion of support in analysis, in particular notice that it is the germ itself that is nonzero and not that the evaluation of the germ s_p at the point p that is nonzero.

Lemma 1.1.16: Support is Closed

Let $\mathscr{F}$ be a sheaf of abelian groups on X and $s \in \mathscr{F}(X)$. Then $\mathrm{Supp}(s)$ is closed

Proof **:**
We'll show $\mathrm{Supp}(s)^c$ is open, that is the set of all points $p \in X$ such that $s_p = 0$ is open. If $s_p = 0$ in the stalk, there exists an open subset $W_p \subseteq X$ such that $s|_{W_p} = 0$. Taking the union of arbitrary open sets is open, and hence taking the union of all W_p cover $\mathrm{Supp}(s)^c$ making $\mathrm{Supp}(s)$ closed, completing the proof.

The equivalent in analysis is the definition of the *closed support* of a function. Due to this, the compliment of the support of an element s is the 0-section. The section can also be defined on presheaves, however there are examples of presheaves where there are nonzero sections that "live nowhere" because it is 0 in every stalk[7]

Let us now take a closer look at $\prod_{p\in U}(\mathscr{F}_p)$. This set is much bigger than sections of a sheaf. We thus take the special sub-collection that associates with $\mathscr{F}(U)$ (essentially characterizing the image of $\mathscr{F}(U)$:

Definition 1.1.17: Compatible Germs

Let $\mathscr{F}$ be a sheaf on X and $U \subseteq X$ an open set. Then we say an elements $(s_p)_{p\in U} \in \prod_{p\in U} \mathscr{F}_p$ consists of *compatible germs* if there exists some representative open sets U_p for s_p:

$$U_p \subseteq U \text{ open}, \ \tilde{s}_p \in \mathscr{F}(U_p)$$

such that the germ of $\tilde{s}_p$ for all $q \in U_p$ is s_q. In other words, if $\{U_i\}$ is an open cover of U where $p \in U_i$ for all i, then s_p is the germ of f_i at p.

Hence, a compatible germ is the collection of germs that are "compatible" when taking subsets, that is *restrictions*. Using the gluing axiom, it is a good exercise to show that any choice of compatible germs for a sheaf $\mathscr{F}$ over U is the image of a section of $\mathscr{F}$ over U, that is we have successfully identified the right hand side of the equation in lemma 1.1.13. Note how important it is that we use sheaves (or at least separated presheaves) for the proof. The proof also motivate the agricultural terminology of "sheaf", for in agriculture a sheaf is a collection of "Stalks" (which is the main stem of a plant).

1.1.3 Remark When working with sheaves of sets or some structures on sets, the elements of the section can literally be thought of as functions using the above definition, namely function of the form $s : U \to \prod_{p\in U} \mathscr{F}_p$. The generalization presented above allows to generalize the notion of "open set U" so that the topological space X is replaced special category called a *site*, see ref:HERE.

1.1.2 Sheaf from a base

When defining a differential structure on a manifold, one of the strategies that is taken is to define pieces of the manifold in a compatible way, or said differently define it on some basis of the manifold. This section endeavours to do this for sheaves.

Let X be a topological space with a sheaf $\mathscr{F}$. Take a basis $\{B_i\}$ for X and take the subset of $\mathscr{F}$ consisting of $\mathscr{F}(B_i)$ along with all the appropriate restriction maps. Then there is enough Stalk information to recover all of $\mathscr{F}$ (show this), and hence we my want to start with such a collection to generate a sheaf. This collection alone will generate a presheaf, we will require the basis to satisfy the locality and gluability axioms:

[7] Take $\mathscr{F}$ to be the sheaf of holomorphic functions on $\mathbb{C}$, $\mathscr{G}$ the sheaf of bounded holomorphic functions. Then take the sheaf $\mathcal{H} = \mathscr{F}/\mathscr{G}$. Then show that $\mathcal{H}_x = 0$, however $\mathcal{H}(\mathbb{C}) = \{\text{entire functions}\}/\mathbb{C}$

Definition 1.1.18: Sheaf On A Base

Let X be a topological space and $\mathcal{B}$ be the collection defined above. Then this collection is caleld a *sheaf on a base* if it satisfies:

1. **Base locality axiom**: If $B = \cup_i B_i$, then if $f, g \in \mathscr{F}(B)$ are such $\text{res}_{B,B_i} f = \text{res}_{B,B_i} g$ for all i, then $f = g$.
2. **Base Gluability**: If $B = \cup_i B_i$, and $f_i \in \mathscr{F}(B_i)$ such that f_i agrees with f_j on any basic open set contained in $B_i \cap B_j$, then there exists $f \in \mathscr{F}(B)$ such that $\text{res}_{B,B_i} f = f_i$ for all i

Theorem 1.1.19: Define Sheaf From Base

Let $\{B_i\}$ be a base for the topological space X and let F be a of sets on the base. Then there is a sheaf $\mathscr{F}$ extending F with isomorphisms $\mathscr{F}(B_i) \cong F(B_i)$ agreeing with restrictions. Furthermore, this sheaf is unique up to unique isomorhpism

***Proof* :**
The sheaf $\mathscr{F}$ will be defined as the sheaf of compatible germs of F. To this end, define the stalk of a presheaf F on a base at $p \in X$ by taking:

$$F_p = \varinjlim F(B_i)$$

Define:

$$\mathscr{F}(U) = \{(f_p \in F_p)_{p \in U} \mid \forall p \in U, \exists B \text{ with } p \in B \subseteq U,\ s \in F(B) \text{ that is compatible: } s_q = f_q\ \forall q \in B\}$$

Then the map $F(B) \to \mathscr{F}(B)$ is an isomorphism.

Lots more to cover here that is worth going over

1.2 Morphisms of (pre)Sheaves and build-up for Cohomology

Now that we have defined (pre)sheaves, it is natural to define morphism between sheaves to show that $\mathbf{PSh}(X)$ and $\mathbf{Sh}(X)$ form a category. Sheaf morphisms shall be over the same object, and hence can be thought of as the equivalent of looking at local homeomorphisms between two étale spaces:

Definition 1.2.1: Morphism Of Sheaves

Let $\mathscr{F}, \mathscr{G} \in \mathbf{PSh}(X)$ (note that $\mathbf{Sh}(X) \subseteq \mathbf{PSh}(X)$). Then a *morphism of (pre)sheaves* (over the chosen category) $\varphi : \mathscr{F} \to \mathscr{G}$ is a natural transformation between the two functors. Concretely, it contains the data $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$ such that if $V \subseteq U$ then the following diagram commutes:

$$\begin{tikzcd} \mathscr{F}(U) \arrow[r, "\varphi(U)"] \arrow[d, "\mathrm{res}_{U,V}"'] & \mathscr{G}(U) \arrow[d, "\mathrm{res}_{U,V}"] \\ \mathscr{F}(V) \arrow[r, "\varphi(V)"'] & \mathscr{G}(V) \end{tikzcd}$$

that is

$$\mathrm{res}_{U,V}(\varphi(U)(-)) = \varphi(V)(\mathrm{res}_{U,V}(-))$$

The collection of morphisms between two sheaves is denoted $\mathrm{Hom}_{\mathbf{Sh}(X)}(\mathscr{F}, \mathscr{G})$ or $\mathrm{Mor}_{\mathbf{Sh}(X)}(\mathscr{F}, \mathscr{G})$

To emphasize that the domain and codomain of a presheaf morphism is a sheaf, the morphism shall be called a *sheaf morphism*. A useful geometric intuition of sheaf morphisms is that they apply a transformation to each map (ex. $\exp(f)(z) = e^{f(z)}$). With the morphisms and objects defined, it is easy to show that they form a category. We shall denote the category of sheaves over a category $\mathbf{C}$ on a space X as $\mathbf{C}_X$. Thus:

$$\mathbf{Set}_X \qquad \mathbf{Ab}_X \qquad \mathbf{Ring}_X$$

are the categories of sheaves over sets, abelian groups, and rings respectively. Since the morphisms are the same, the category $\mathbf{Sh}(X)$ is a full subcategory of $\mathbf{PSh}(X)$. For presheaves, we shall denote the category with an extra superscript:

$$\mathbf{Set}_X^{\mathrm{pre}} \qquad \mathbf{Ab}_X^{\mathrm{pre}} \qquad \mathbf{Ring}_X^{\mathrm{pre}}$$

Example 1.2.2: Morphism of Presheaves and Sheaves

(If the reader is familiar with Riemann surface) Let X be a Riemann surface and $\mathcal{O}_X(-P)$ be the collection of meromorphic function $f \in K^*(X)$ such that $\mathrm{div}\,(f) + (-p) \geq 0$ ($P \in X$ a point). Then if $\mathcal{O}_X$ is the sheaf of holomorphic functions on X there are sheaf morphisms:

$$0 \to \mathcal{O}_K(-P) \to \mathcal{O}_X \to \kappa_P \to 0$$

where κ_P is the sky-scrapper sheaf at P, namely:

$$\kappa_P = \begin{cases} (0) & x \neq P \\ \mathbb{C} & x = P \end{cases}$$

This is even an exact sequence of sheaves, but we need to build-up to show how to properly define exactness in $\mathbf{C}_X$.

From now on, we shall be working with *abelian categories* unless stated otherwise: they all have 0 objects, they are closed under finite products and coproducts, and their kernel/cokernels behave well with epimorphisms and monomorphisms; see section 0.4 for a refresher or [8] for more details. We shall build-up to showing that $\mathbf{C}_X$ and $\mathbf{C}_X^{\mathrm{pre}}$ are abelian categories.

As a presheaf morphism it is a natural transformation, it important to emphasize that it does not make sense to say that a sheaf morphism is injective or surjective. Instead, it can be a *monomorphism* or an *epimorphism* and have corresponding kernel, cokernels, and images. Let us find objects in $\mathbf{PSh}(X)$ and $\mathbf{Sh}(X)$ that can represent these kernel, cokernel, and image. Starting with the kernel of a presheaf morphism $\varphi : \mathscr{F} \to \mathscr{G}$ which is defined to be be object $\mathcal{K}$ with the morphism $K : \mathcal{K} \to \mathcal{F}$ such that $\varphi \circ K = 0$ universally.

Proposition 1.2.3: Kernels in Presheaf Category

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between presheaves. Then $\ker \varphi$ can be defined to be the kernel of each $\varphi(U)$:

$$\mathcal{K}(U) = \ker(\varphi(U))$$

The resulting presheaf is denoted $\ker_{\text{pre}}(\varphi)$. Furthermore, if $\mathscr{F}$ is a sheaf and $\mathscr{G}$ satisfies at least the locality axiom (in particular if it is also a sheaf), then $\mathcal{K}$ is a sheaf.

Proof **:**
chase the diagram, and use a similar separated/locality condition in the case of a sheaf as was done before.

Note that as the representation of the kernel for sheaves is the same as the characterization of presheaves, we can just write $\ker(\varphi)$ without any ambiguity. As the monomorphism condition can be checked on open sets, the following usual definition of injectivity is recovered on the level of open sets:

Definition 1.2.4: Injective Presheaf and Sheaf Morphism

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between shaves or presheaves. Then φ is *injective* if $\ker \varphi(U) = 0$ for all open $U \subseteq X$.

The image of a presheaf morphism $\varphi : \mathscr{F} \to \mathscr{G}$ is defined to be be object $\mathscr{I}$ with the monomorphism $I : \mathscr{G} \to \mathscr{I}$ and a morphism $E : \mathscr{F} \to \mathscr{I}$ such that $\varphi = E \circ I$ universally, namely:

$$\begin{tikzcd}
\mathscr{F} \arrow[rr, "\varphi"] \arrow[dr, "E"] \arrow[ddr, bend right, "E'"'] & & \mathscr{G} \\
& \mathscr{I} \arrow[ur, "I"'] \arrow[d, dashed, "\exists! v"] & \\
& \mathscr{I}' \arrow[uur, bend right, "I'"'] &
\end{tikzcd}$$

The reader familiar with category may be more familiar with images an abelian category:

$$\operatorname{im}(\varphi) = \ker(\operatorname{coker}(\varphi))$$

As we are working with abelian categories, $\ker(\operatorname{coker}(\varphi)) \cong \operatorname{coker}(\ker(\varphi))$, and hence there is no need of defining the coimage.

Proposition 1.2.5: Images in Presheaf and Sheaf Category

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between presheaves. Then $\text{im}_{\text{pre}}\varphi$ can be defined to be the kernel of each $\varphi(U)$:

$$\text{im}_{\text{pre}}\varphi(U) = \text{im}(\varphi(U))$$

and similarly for cokernels:

$$\text{coker}_{\text{pre}}(\varphi(U)) = \frac{\mathscr{G}(U)}{\text{im}(\varphi(U))}$$

The resulting $\text{im}_{\text{pre}}(\varphi)$ and $\text{coker}_{\text{pre}}(\varphi)$ is called the *image presheaf* and *cokernel presheaf* respectively.

This is *not* necessarily the case if $\mathscr{F}$ is a sheaf. Images and cokernel in $\mathbf{C}_X$ will require a bit more work to be characterized.

Proof **:**
chase the diagram.

Definition 1.2.6: Surjective Presheaf Morphism

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a presheaf morphism between presheaves. Then φ is *surjective* if $\text{im}(\varphi(U)) = \mathscr{G}(U)$ for all open $U \subseteq X$.

As noted earlier, images of sheaves need not be sheaves:

Example 1.2.7: Sheaves: difficulty With Cokernels and Images

1. Let $X = \mathbb{C}$ with the usual topology, $\underline{\mathbb{Z}}$ be the constant sheaf on X, $\mathcal{O}_X$ the sheaf of holomorphic functions, and $\mathscr{F}$ the *presheaf* of functions admitting a holomorphic logarithm. Then there is an exact sequence of presheaves on X:

$$0 \to \underline{\mathbb{Z}} \to \mathcal{O}_X \to \mathscr{F} \to 0$$

where $\underline{\mathbb{Z}} \to \mathcal{O}_X$ is the natural inclusion on each open set, and $\mathcal{O}_X \to \mathscr{F}$ is given by $f \mapsto \exp(2\pi i f)$ (again, over each open set). This is indeed a short exact sequence, for if $U \subseteq V \subseteq \mathbb{C}$, then:

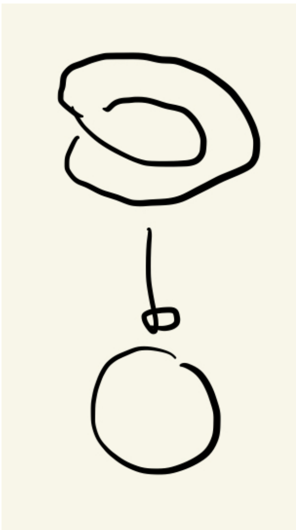
$$\begin{tikzcd}
0 \arrow[r] & \underline{\mathbb{Z}}(V) \arrow[r] \arrow[d, "\text{res}"] & \mathcal{O}_X(V) \arrow[r] \arrow[d, "\text{res}"] & \mathscr{F}(V) \arrow[r] \arrow[d, "\text{res}"] & 0 \\
0 \arrow[r] & \underline{\mathbb{Z}}(U) \arrow[r] & \mathcal{O}_X(U) \arrow[r] & \mathscr{F}(U) \arrow[r] & 0
\end{tikzcd}$$

commutes. Certainly each $\underline{\mathbb{Z}}(V) \to \mathcal{O}_X(V)$ is an inclusion, and $\mathcal{O}_X(V) \to \mathscr{F}(V)$ is surjective with kernel being the constant functions $\mathbb{Z}$ since if f admits a logarithmic inverse, then $f = \log(g)$, so $g \mapsto \exp(2\pi i \log g) = f$.

Then it should be clear that $\mathscr{F}$ is *not* a sheaf: take $U = \mathbb{C}^*$, cover it with $U_1 = \mathbb{C} \setminus \mathbb{R}_{\geq 0}$ and $U_2 = \mathbb{C} \setminus \mathbb{R}_{\leq 0}$, and define a logarithm on both which are compatible (see [3]). If $\mathscr{F}$ were a sheaf, then there should be a function defined on U, that restricts to both of these, but no such function exists (consider that such a function would have to be continuous, but going around the origin introduces non-trivial monodromy).

2. Consider the following similar but subtly different example. Consider $\mathcal{O}_{\mathbb{C}}^*$ the sheaf of all non-vanishing holomorphic functions and consider the following short exact sequence of *sheaves*:
$$0 \to \underline{\mathbb{Z}} \xrightarrow{\iota} \mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^* \xrightarrow{!} 0$$
Notice that on $U = \mathbb{C}^*$, z does not vanish and so $z \in \mathcal{O}_{\mathbb{C}}^*(U)$. However, z does not admit a logarithmic inverse (again see [3]). This would imply that $\exp(\mathbb{C}^*)$ is *not* surjective and so exp is not surjective on all open sets. And yet it is indeed the case that there exists a morphism $\xrightarrow{!}$ making exp a "surjection" on sheaves. This shall be explained by defining sheafification after these examples.

3. For the reader less comfortable with complex analysis, here is a more rudimentary example: take the sheaf of continuous sections of the circle S^1 with the total space for the first sheaf being S^1 and for the second sheaf being the twisted circle, which can be visualized as:



Label this shape S^1_2. Then the first sheaf is all continuous sections on all open subsets of the codomain for the map $\pi : S^1 \to S^1$, while the second is all the continuous sections of all the open subsets of the codomain for the map $\pi : S^1_2 \to S^1$ with a reasonable definition of projection map mirroring the above image. Then there is certainly a surjective continuous map φ satisfying the commutative diagram:
$$\begin{tikzcd} S^1_2 \arrow[rr, "\varphi"] \arrow[dr] & & S^1 \arrow[dl] \\ & S^1 & \end{tikzcd}$$
However, there is no surjective maps:
$$\mathcal{F}_2(X) \to \mathcal{F}_1(X)$$

namely since there no continuous section from S^1 to S^1_2 that would satisfy the projection condition.

Just like we can form an abelian group from an abelian semi-group via groupification, we can from a sheaf from a presheaf via sheafification. As is usual, this is done in a universal way that is adjoint to the forgetful functor. This sheafification shall be the solution to defining right notion of surjection for sheaves:

Theorem 1.2.8: Sheafification

Let $\mathscr{F}$ be a presheaf on X. Then there exists a sheaf $\mathscr{F}^{\text{sh}}$ and a map $\text{sh} : \mathscr{F} \to \mathscr{F}^{\text{sh}}$ such that for any sheaf $\mathscr{G}$ and any presheaf morphism $g : \mathscr{F} \to \mathscr{G}$, there exists a unique morphism of sheaves $f : \mathscr{F}^{\text{sh}} \to \mathscr{G}$ such that the following diagram commutes:

$$\begin{tikzcd} & \mathscr{F}^{\text{sh}} \arrow[d, "g"] \\ \mathscr{F} \arrow[ur, "\text{sh}"] \arrow[r, "f"'] & \mathscr{G} \end{tikzcd}$$

The sheafification of the presheaf $\mathscr{F}$ is often denoted $\mathscr{F}^+$.

The idea will be to remove all section that violate locality, and add sections to respect gluability. Another good take is the following post here.

***Proof* :**
Let $\mathscr{F}$ be the presheaf in the question. Define $\mathscr{F}^{\text{sh}}$ as the set of compatible germs of the presheaf $\mathcal{F}$ over U, namely:

$$\mathscr{F}^{\text{sh}}(U) := \{(f_p \in \mathscr{F}_p)_{p \in U} \mid \forall p \in U, \exists p \in V \subseteq U \text{ open and } s \in \mathscr{F}(V) \text{ such that } s_q = f_q, \forall q \in V\}$$

Note this resemble to the étale space. Verify that this is a sheaf, that $\mathscr{F} \to \mathscr{F}^{\text{sh}}$ induces an isomorphism so stalks, and that it satisfies the free-forgetful adjoint.

The "failure" of a presheaf being a sheaf can be measured as the lack of exactness of at the $\prod_i F(U_i)$ in proposition 1.1.8.

Example 1.2.9: Sheafification
As practice, show the following sheafification:

1. Show that $(\underline{S}_{\text{pre}})^{\text{sh}} = \underline{S}$, that is the presheaf of constant function sheafifies to the sheaf of constant functions.
2. Let $\mathscr{F}$ on $\mathbb{R}$ be all bounded continuous functions on each open set. Show that $\mathscr{F}^{\text{sh}}$ is *all* continuous functions on each open subset.
3. Show that $\mathcal{H}$ on $\mathbb{C}$ of non-vanishing holomorphic functions that admit a logarithm sheafify to all non-vanishing holomorphic functions.
4. Let $\mathscr{F}$ be a non-trivial presheaf where all stalks are trivial (see exercise ref:HERE). Show that $\mathscr{F}^{\text{sh}}$ is trivial.

Corollary 1.2.10: Sheafification and Stalks

Let $\mathcal{F} \in \mathbf{PSh}(X)$ be a presheaf. Then for all $p \in X$

$$\mathcal{F}_p \cong \mathcal{F}_p^{\mathrm{sh}}$$

Proof **:**
Use something similar given in the proof of proposition 1.1.5

Let us now put together the universal properties we've shown so far to get that the category of sheaves also have cokernels:

Proposition 1.2.11: Sheaf Morphisms Have Cokernels

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a sheaf morphism. Then it has a cokernel and it is the sheafification of the cokernel-presheaf.

Proof **:**
Recall that the universal property of the cokernel can be seen as the colimit of the following diagram:

$$\begin{tikzcd} \mathcal{F} \arrow[r, "\varphi"] \arrow[d] & \mathcal{G} \\ 0 & \end{tikzcd}$$

As we know, the cokernel forms a presheaf $\mathcal{H}_{\mathrm{pre}}$. By sheafification we have a unique map $\mathrm{sh} : \mathcal{H}_{\mathrm{pre}} \to \mathcal{H}$. Now let $\mathcal{A}$ be any sheaf and consider the following diagram:

$$\begin{tikzcd} \mathcal{F} \arrow[r, "\varphi"] \arrow[d] & \mathcal{G} \arrow[d] \arrow[rrd, bend left] & & \\ 0 \arrow[r] \arrow[rrrd] & \mathcal{H}_{\mathrm{pre}} \arrow[r, "\mathrm{sh}"] & \mathcal{H} \arrow[rd, dashed, "\exists!"] & \\ & & & \mathcal{A} \end{tikzcd}$$

By the universal property of cokernels for presheaves, we have a unique map $\mathcal{H}_{\mathrm{pre}} \to \mathcal{A}$. By the universal property of sheafification, we will get a unique map composed from the two unique universal maps giving us $\mathcal{H} \to \mathcal{A}$, as we sought to show.

Corollary 1.2.12: Image and Sheafification

Let $\varphi \in \mathbf{Sh}(X)$ be a morphism of sheaves over an abelian category. Then the image of φ is the sheafification of the image presheaf.

***Proof* :**
exercise

Definition 1.2.13: Surjective Sheaf Morphism

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of sheaves and $\mathrm{im}\varphi$ the image presheaf. Then the cokernel and image of φ is the sheafification of the cokernel-presheaf and image-presheaf respectively. The morphism φ is *surjective* if

$$(\mathrm{im}_{\mathrm{pre}}\varphi)^{\mathrm{sh}} = \mathscr{G}$$

Using this, we see that the map $\mathcal{O}_{\mathbb{C}} \xrightarrow{\exp} \mathcal{O}_{\mathbb{C}}^*$ is *surjective* in the category of sheaves, even though it is not surjective in the category of presheaves. Hence, exactness in presheaves and sheaves is *different*. The following example using simpler categories should demonstrate why this is not so strange:

Example 1.2.14: Ab Vs TF-Ab

Let Ab be the category of abelian groups and TF-Ab be the category of torsion-free abelian groups. Consider the morphism $\varphi : \mathbb{Z} \to \mathbb{Z}$, $\varphi(n) = 2n$. This morphism exists in both categories. The cokernel of φ in Ab is $\mathbb{Z}/2\mathbb{Z}$, which is torsion. To find the cokernel in TF-AB, the "best" torsion-free group must be found so that its composition with φ creates is zero. It isn't hard to see this is 0, so:

$$\mathrm{coker}_{\mathrm{Ab}}(\varphi) = \mathbb{Z}/2\mathbb{Z} \neq 0 = \mathrm{coker}_{\text{TF-Ab}}(\varphi)$$

Later in this section, an easier way to show that a sheaf-morphism is surjective shall be shown.

Functors from (pre)sheaves: simplifying checking exactness

As presheaves and sheaves store all local information, we may want that there are some natural functors that preserve some exactness to extract that information. The two natural candidates for functors to would be $-(U)$ taking a (pre)sheaf to its sections over an open set U, and $(-)_p$ the functor taking a (pre)sheaf to it's stalk at p.

It is clear that $-(U) : \mathbf{C}_X \to \mathbf{C}$ is a functor for:

$$\mathscr{F} \mapsto \mathscr{F}(U) \qquad (\varphi : \mathscr{F} \to \mathscr{G}) \mapsto (\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U))$$

preserves composition's an identities. Let us show $(-)_p$ is also a functor:

Proposition 1.2.15: Morphism Of (Pre)Sheaves Induces Morphism On Stalks

Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a morphism of (pre)sheaves over any category $\mathbf{C}$ having colimits (so that stalks are well-defined). Then there is an induced morphism on stalks $\varphi_p : \mathscr{F}_p \to \mathscr{G}_p$, that is, it induced a functor $\mathbf{C}_X \to \mathbf{C}$.

***Proof* :**
This is definition pushing: For each open $U \subseteq X$ we have the morphism $\varphi(U) : \mathscr{F}(U) \to \mathscr{G}(U)$. Define the map:

$$[(f, U)]_p \mapsto [(\varphi(U)(f), U]_p$$

For this map to be well-defined if $(f,U) \sim (f',V)$ then we need:

$$[(f',V)]_p \mapsto [(\varphi(V)(f'),V]_p = [(\varphi(U)(f),U]_p$$

Hence we need that $((\varphi(V)(f'),V) \sim ((\varphi(U)(f),U)$. As $(f,U) \sim (f',V)$ we have there exists $W \subseteq U,V$ such that

$$\text{res}_{U,W} f = \text{res}_{V,W} f' = g$$

Then as φ is a pre-sheaf morphism we have:

$$\begin{tikzcd} \varphi(U) : f \arrow[r, mapsto] \arrow[d, "\text{res}_{U,W}"] & \varphi(U)(f) \arrow[d, "\text{res}_{U,W}"] \\ \varphi(W) : \text{res}_{U,W}(f) = g \arrow[r, mapsto] & \varphi(W)(g) \end{tikzcd} \qquad \begin{tikzcd} \varphi(V) : f' \arrow[r, mapsto] \arrow[d, "\text{res}_{V,W}"] & \varphi(U)(f') \arrow[d, "\text{res}_{V,W}"] \\ \varphi(W) : \text{res}_{V,W}(f) = g \arrow[r, mapsto] & \varphi(W)(g) \end{tikzcd}$$

which implies

$$\text{res}_{U,W}\varphi(U)(f) = \text{res}_{V,W}\varphi(V)(f')$$

But that means they are similar, $((\varphi(V)(f'),V) \sim ((\varphi(U)(f),U)$, as we sought to show

Let us now see how these functors preserve monomorphisms/epimorphisms (i.e. injectivity and surjectivity on opens/stalks). Starting with open subsets:

Proposition 1.2.16: On Presheaf: Open Set Functor is Exact

The functor $-(U) : \mathbf{PSh_C}(X) \to \mathbf{C}$ (where $\mathbf{C}$ is abelian) is an exact functor

On **Sh**, the functor $-(U)$ is left-exact.

***Proof* :**
Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a monomorphism so that if $f : \mathcal{H} \to \mathcal{F}$ and $g : \mathcal{H} \to \mathcal{F}$ are morphisms such that $\varphi \circ f = \varphi \circ g$ then $f = g$. Then after applying the functor we get:

$$\varphi(U) \circ f(U) = \varphi(U) \circ g(U) \qquad \text{implies} \qquad f(U) = g(U)$$

Note that any element $f(U), g(U)$ is associated to a presheaf an "indicator sheaf" which is the sheaf where if $V \subseteq U$ then $\mathcal{F}(U) = \{e\}$ and if $V \not\subseteq U$ then $\mathcal{F}(U) = \emptyset$, and so we have that this holds for every possible element (or function) $f(U), g(U)$, giving they are monomorphism, or in the right category injective. A similar argument can be done for epimorphisms/surjective maps using the sky-scrapper sheaf.

The argument for left-exactness works just as well for the sheaf case, and example 1.2.7 shows why it cannot be extended to be right-exact in general.

Proposition 1.2.17: Exactness of Presheaves On The Level of Open Sets

An exact sequence of presheaves $0 \to \mathscr{F}_1 \to \mathscr{F}_2 \to \cdots \to \mathscr{F}_n \to 0$ is exact if and only if the sequence $0 \to \mathscr{F}_1(U) \to \mathscr{F}_2(U) \to \cdots \to \mathscr{F}_n(U) \to 0$ is exact for all U.

***Proof* :**
exercise.

Corollary 1.2.18: Exactness of Stalk For Presheaves

Let $0 \to \mathscr{F} \to \mathscr{G} \to \mathscr{H} \to 0$ be an exact sequence of presheaves. Then if filtered colimits exists (so that stalks

$$0 \to \mathscr{F}_p \to \mathscr{G}_g \to \mathscr{H}_p \to 0$$

is exact.

***Proof* :**
Use that each functor $-(U)$ is exact for each U.

Theorem 1.2.19: Presheaves Over Ab Category Form Ab Category

The category $\mathbf{PSh_C}(X)$ over an abelian category C is an abelian category

***Proof* :**
The main thing that wasn't checked was how to define addition of morphism. By the above work, this can be done on each open set: if $\varphi, \psi : \mathcal{F} \to \mathcal{G}$ then $\varphi + \psi$ defined via open sets as:

$$\varphi + \psi(U) := \varphi(U) + \psi(U)$$

is a map of presheaf (it respects restriction by the commutativity of both natural transformation), and so the hom-sets form an abelian group. Then as we also have a 0-object, and we have finite products (again, something the reader could verify), we have an additive category.

To show kernels and cokernels exist, by proposition 1.2.17 it suffices to check on the level of open sets. Then we define:

$$(\ker_{\text{pre}} \varphi)(U) = \ker \varphi(U)$$

The key is to consider the following diagram where $U \hookrightarrow V$:

$$\begin{tikzcd}
0 \arrow[r] & \ker_{\text{pre}}(\varphi(V)) \arrow[r] \arrow[d, dashed, "\exists!"] & \mathcal{F}(V) \arrow[r] \arrow[d, "\text{res}_{V,U}"] & \mathcal{G}(V) \arrow[d, "\text{res}_{V,U}"] \\
0 \arrow[r] & (\ker_{\text{pre}} \varphi)(U) \arrow[r] & \mathcal{F}(U) \arrow[r] & \mathcal{G}(U)
\end{tikzcd}$$

which shows that this is indeed well-defined. The reader can further check that this does indeed satisfy the universal property of the kernel. By dualizing the argument, we can do the same for the cokernel, and hence the additive category of presheaves has kernels and cokernels, making it abelian.

To keep track of all morphisms between open sets, we box the following definition:

Definition 1.2.20: Sheaf Hom

Let $\mathscr{F}, \mathscr{G} \in \mathbf{Sh}_{\mathbf{C}}(X)$. Then the *sheaf hom* is defined to be:

$$\mathcal{H}om(\mathscr{F}, \mathscr{G})(U) := \mathrm{Mor}(\mathscr{F}_U, \mathscr{G}|_U)$$

where $\mathrm{Mor}(\mathscr{F}|_U, \mathscr{G}|_U)$ are all morphism of (pre)sheaves. The dual of $\mathscr{F}$ is:

$$\mathscr{F}^\vee = \mathrm{Hom}_{\mathbf{Mod}_{\mathcal{O}_X}}(\mathscr{F}, \mathcal{O}_X)$$

Just like Hom, $\mathcal{H}om$ is a (co/contra)variant functor, depending on which position you choose.

The situation is not so clean for $(-)_p$, namely the functor on $\mathbf{Sh}_{\mathbf{C}}(X)$ is not always exact for *any* abelian category; it need not be the case that filtered colimits preserve exactness, and hence that $(-)_p$ preserve exactness (think of the category of finite abelian groups which doesn't have $\varinjlim_p \mathbb{Z}/p^n\mathbb{Z}$). Grothendieck axiomatically added this condition to abelian categories (such categories are called *Grothendieck categories*). Fortunately for us, this will never be an issue as all the categories of interest to us shall have this property, namely:

1. abelian groups, more generally R-**Mod**
2. Sheaves of abelian groups (more generally $\mathcal{O}_X$-modules which are defined later)
3. the product of Grothendieck categories
4. In ref:HERE, it shall be important to have internal Hom's: given a small category $\mathbf{C}$ and a Grothendieck category $\mathbf{G}$, all functors from $\mathbf{C}$ to $\mathbf{G}$ (with morphism being natural transformation) form Grothendieck categories.

Proposition 1.2.21: Sheaves and exactness For Stalk

Let $\mathbf{Sh}(X)$ be the category of sheaves over any R-**Mod**. Then $(-)_p : \mathbf{Sh}(X) \to R\text{-}\mathbf{Mod}$ is exact.

Proof **:**
exercise.

With the right notion of compatible stalks defined, we may show how to determine morphisms of sheaves at the level of stalks:

Proposition 1.2.22: Morphisms Determined By Stalks

Let φ_1, φ_2 be morphisms from a presheaf of sets $\mathscr{F}$ to a *sheaf* of sets $\mathscr{G}$ that induce the same maps on each stalk. Then $\varphi_1 = \varphi_2$

***Proof* :**
Consider the diagram:

$$\begin{tikzcd} \mathscr{F}(U) \arrow[r] \arrow[d, hook] & \mathscr{G}(U) \arrow[d, hook] \\ \prod_{p\in U} \mathscr{F}_p \arrow[r] & \prod_{p\in U} \mathscr{G}_p \end{tikzcd}$$

Proposition 1.2.23: Isomorphisms Determined By Stalks

Let φ be a morphism of sheaves. Then φ is an isomorphism if and only if it induces an isomorphism for all stalks.

***Proof* :**
Let $\varphi : \mathscr{F} \to \mathscr{G}$ be a sheaf morphism. If φ is an isomorphism, then certainly each φ_p is an isomorphism, hence say each φ_p is an isomorphism. Then it must be shown that each $\varphi(U)$ is an isomorphism. For injectivity, say for $s \in \mathscr{F}(U)$ $\varphi(s) = 0 \in \mathscr{G}(U)$. Then for each $p \in U$, $\varphi(s)_p = 0 \in \mathscr{G}_p$. As φ_p is injective at each p, we have $s_p = 0 \in \mathscr{F}_p$. Using the definition of stalk, we have that s is zero on some open set, namely $s|_{W_p} = 0$ for $W_p \subseteq U$. As U is covered by such neighborhoods, by the gluing axiom we have $s = 0$

For surjectivity, let $t \in \mathscr{G}(U)$. Consider $t_p \in \mathscr{G}_p$. As φ_p is surjective, we have $s_p \in \mathscr{F}_F$ such that $\varphi_p(s_p) = t_p$. Now, let $(s(p), V_p)$ be a representative of s_p on the set V_p containing the point p. Then $\varphi(s(p))$ and $t|_{V_p}$ are two elements of $\mathscr{G}(V_p)$ whose germs at p are the same. Thus, replacing V_p by a smaller neighborhood if necessary, we have $\varphi(s(p)) = t|_{V_p} \in \mathscr{G}(V_p)$.

Now, U is covered by open sets V_p where on each V_p we have $s(p) \in \mathscr{F}(V_p)$ and $\varphi(s(p)) = t|_{V_p}$. If p, q are two points, then $s(p)|_{V_p\cap V_q}$ and $s(q)|_{V_p\cap V_q}$ are two sections of $\mathscr{F}(V_p \cap V_q)$ which are both sent to $t|_{V_p\cap V_q}$ via φ. As φ was shown to be injective, they are equal. then by the gluing axiom, there exists an $s \in \mathscr{F}(U)$ such that

$$s|_{V_p} = s(p) \qquad \forall p \in U$$

Finally, to show that $\varphi(s) = t$, notice that $\varphi(s)$ and t are two section of $\mathscr{F}(U)$ where for each p $\varphi(s)|_{V_p} = t|_{V_p}$, hence they glue to create $\varphi(s) = t$, as we sought to show.

This does not imply that if two sheaves have isomorphic stalks, then they are isomorphic! You need to already have a sheaf homomorphism which in turn induces the isomorphic stalks. Conceptually, this is saying that being locally isomorphic does not imply it is globally isomorphic, which there are plenty concrete example form differential geometry to choose from.

Example 1.2.24: Isomorphic Stalks, Not Isomorphic Sheafs

1. Let X be a Hausdorff space. Consider the constant sheaf $\underline{\mathbb{Z}}$ and the sky-scrapper sheaf $\iota_*(\mathbb{Z})$. Show that the stalks at each point is $\mathbb{Z}$, however these two sheaves are not isomorphic.
2. The following example will rely on Spec for full understanding that is covered in the next chapter, however the reader may rely on their intuitions from [8]. A more geometric example

is the following: Take
$$A = \mathbb{R}[x, y, z]/(x^2 + y^2 + z^2 - 1)$$
which is the 2-sphere in $\mathbb{R}^3$. Take the tangent bundle, i.e. A-module, of all derivations $E = \mathrm{Der}_{\mathbb{R}}(A)$. Then E is a rank 2 locally trivial module since for any open subset or distinguished open subset, $E_x \cong A_x^2$, showing each stalk is isomorphic. However, $E \not\cong A^2$.

3. Another famous example would be the trivial line bundle on S^1 vs. the möbius bundle.

It is still useful to know under what conditions do the stalk's of a map induce a sheaf morphism:

Proposition 1.2.25: Characterizing Stalks Inducing Sheaf Morphism

Let X be a topological space with two sheaves $\mathscr{F}, \mathscr{G}$. Let $f_p : \mathscr{F}_p \to \mathscr{G}_p$ be a collection of stalk morphisms for all $p \in X$. Then this gives a map of sheaves $f : \mathscr{F} \to \mathscr{G}$ if and only if for all open subsets $U \subseteq X$, for all sections $s \in \mathscr{F}(U)$, there is an open cover $U = \cup_i U_i$ such that
$$f_p(s_p) = (t_i)_p$$
for all $p \in U_i$ and all i

***Proof* :**
exercise.

Summarizing the results for morphisms of sheaves:

Theorem 1.2.26: Mono- and Epimorhpisms of Sheaves Equivalence

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves on X (over **Set**, or an abelian category). Then the following are equivalent and characterize kernels:

1. φ is a monomorphism in the category of sheaves
2. φ is injective on the level of stalks: $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$ for all $p \in X$
3. φ is injective on the level of pen sets: $\varphi(U) : \mathcal{F}(U) \to \mathcal{F}(U)$ for all $U \subseteq X$

Furthermore, the following are equivalent and characterize cokernels:

1. φ is an epimorphism in the category of sheaves
2. φ is surjective on the level of stalks: $\varphi_p : \mathcal{F}_p \to \mathcal{G}_p$ for all $p \in X$

Furthermore, if φ is surjective on open sets, it implies the two above conditions (but not the other way around).

***Proof* :**
This is combining the results from earlier. In particular, apply the results just proven about stalks to the proof of proposition 1.2.16.

Theorem 1.2.27: Sheaves Over Ab Category Form Ab Category

The category $\mathbf{Sh_C}(X)$ over an abelian category C is an abelian category

Proof **:**
exercise.

Morphisms and Stalks

Corollary 1.2.28: Commuting Kernel and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then the stalk of the kernel is he kernel of the stalk:

$$(\ker(\mathcal{F} \to \mathcal{G}))_p \cong \ker(\mathcal{F}_p \to \mathcal{G}_p)$$

Proof **:**
exercise.

Corollary 1.2.29: Commuting Cokernel and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then the stalk of the kernel is he kernel of the stalk:

$$(\operatorname{coker}(\mathcal{F} \to \mathcal{G}))_p \cong \operatorname{coker}(\mathcal{F}_p \to \mathcal{G}_p)$$

Proof **:**
exercise.

Note how due to the sheafification, the surjectivity of a sheaf is no longer necessarily conditional on the surjectivity of on all open sets, even on $U = X$

Example 1.2.30: Surjective, But Not On Open Sets

Take $X = \mathbb{C} \setminus \{0\}$, $\mathscr{F}$ the collection of meromorphic functions, and $\mathscr{G}$ the collection of differential 1-forms. Take $\varphi : \mathscr{F} \to \mathscr{G}$ given by $f \mapsto df$. This map is surjective on stalks: indeed locally, each differential 1-form $\omega = df$ for appropriate f (namely as around every point there is a simply-connected open neighborhood). However, on all of X, the form dz/z has nothing mapping to it: indeed the only possible candidate is $\log(z)$ which is not a meromorphic function on $\mathbb{C} \setminus \{0\}$.

Thus, $\varphi(X) : \mathscr{F}(X) \to \mathscr{G}(X)$ is *not* surjective, even though $\varphi : \mathscr{F} \to \mathscr{G}$ *is* surjective!

For a purely algebraic examples, take $X = \mathbb{R}$, $\mathscr{F} = \underline{\mathbb{Z}}$, and $\mathscr{G} = \iota_p(\mathbb{Z}) \oplus \iota_q(\mathbb{Z})$ (i.e. the direct sum of two skyscraper sheaves at two distinct points p, q). Take the natural restriction map φ, show it is surjective on each stalk, but $\operatorname{im}(\mathscr{F}(X)) = \{(n, n) \mid n \in \mathbb{Z}\} \neq \mathbb{Z} \oplus \mathbb{Z}$.

Corollary 1.2.31: Commuting Image and Stalk

Let $\varphi : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves. Then the stalk of the image is he image of the stalk:

$$(\mathrm{im}(\varphi))_p \cong \mathrm{im}(\varphi_p)$$

This last corollary can be interpreted as saying that we can take the stalk of the presheaf image or the stalk of the sheaf image, and get the same result.

Overall, the reason why presheaves are an abelian category is because of the computation "open set by open set", and sheaves are abelian categories because all abelian-categorical notions make sense "stalk by stalk".

Exercise 1.2.1

1. Show that $\varphi : \mathscr{F} \to \mathscr{G}$ is surjective if and only if for every $s \in \mathscr{G}(U)$, there exists an open covering $\bigcup_i U_i = U$ and elements $t_i \in \mathscr{F}(U_i)$ such that $\varphi(t_i) = s|_{U_i}$ (hint: recall φ is surjective if and only if it is surjective on stalks).

   This exercise will come back in proposition 4.1.15

2. Let $(X, \mathcal{O}_X)$ be a ringed space and $U \subseteq X$ an open subset. Show that we can define non-vanishing distinguished open subset by:

$$\begin{array}{ccc} U_f & \longrightarrow & \mathcal{O}_X^* \\ \downarrow & & \downarrow \\ U & \xrightarrow{f} & \mathcal{O}_X \end{array}$$

   that is, $U_f = f^{-1}(\mathcal{O}_X^*) = \left\{ x \in U \;\middle|\; f(x) \in \mathcal{O}_{X,x}^* \right\}$. Show these form a sheaf.

3. Define a *local ringed space* to be a ringed space $(X, \mathcal{O}_X)$ such that for all open $U \subseteq X$ and all $f \in \mathcal{O}_X(U)$, $U = U_f \cup U_{1-f}$, namely for all $x \in U$, f or $1 - f$ is invertible.

   a) Show that if X, $\mathrm{mSpec}\, \mathcal{O}_X(X)$ is a singleton.

   b) Let $C(X)$ be all continuous real function on X. Show that $(X, \mathbb{C}(X))$ is local.

4. Let $(X, \mathcal{O}_X)$ is a local ringed space, $U \subseteq X$ open, and $f \in \mathcal{O}_X(U)$. Then if $f_1 + \cdots + f_r = 1$, then $\bigcup U_i = U$.

5. Let $\mathscr{F}$ be a sheaf of sets, show that $\mathcal{H}om(\underline{\{p\}}, \mathscr{F}) \cong \mathscr{F}$

6. Let $\mathscr{F}$ be a sheaf of groups, show that $\mathcal{H}om(\underline{\mathbb{Z}}, \mathscr{F}) \cong \mathscr{F}$

7. It is tempting to then think that $\mathcal{H}om(\mathscr{F}, \mathscr{G})_p \cong \mathrm{Hom}(\mathscr{F}_p, \mathscr{G}_p)$. However this is *not* the case: think of a counter-example. There is a map from one of these sets to another, what is it?

1.3 Pushforward and Pullback

The next important question is if we can pushforward or pullback a sheaf given a continuous map $f : X \to Y$. If $U \subseteq V \subseteq Y$ and $f : X \to Y$, then $f^{-1}(U) \subseteq f^{-1}(V)$. Hence, f^{-1} can be thought of as a contravariant functor preserving $\subseteq$. In fact, more is true: f^{-1} also preserves the properties in proposition 1.1.8, and hence works on sheaves as well:

Definition 1.3.1: Pushforward Of (Pre)sheaf

Let $f : X \to Y$ be a continuous map and $\mathscr{F} \in \mathbf{PSh}(X)$. Then the *pushforward* or *direct image* of $\mathscr{F}$ through f is defined on every open set $U \subseteq Y$ to be

$$f_*\mathscr{F}(U) = \mathscr{F}(f^{-1}(U))$$

and induces a (per)sheaf on Y. The pushforward defines a [covariant] functor $f_* : \mathbf{Sh}(X) \to \mathbf{Sh}(Y)$ sending:

$$f_* : \mathscr{F} \to f_*(\mathscr{F})$$

If working with a particular category, for example **Set**, we can write $f_* : \mathbf{Set}_X \to \mathbf{Set}_Y$.

The term "pushforward" comes from the fact that we are taking a sheaf on X and defining a sheaf on Y through the map $f : X \to Y$. The pullback shall do the same thing in reverse.

It is perhaps unsurprising that f^{-1} preserves pre-sheaves, however it is a little less obvious that it preserves sheaves as well; it is worth thinking this through. To me, this shows that sheaves are a very natural concept, as the usually topological function f^{-1} naturally preserves them. This also motivates the Grothendieck topology that will be introduced much later. We have already seen an example of a pushforward: the skyscraper sheaf is induced that pushing forward along the inclusion map $\iota_p : \{p\} \to X$ the constant sheaf $\underline{S}$.

Proposition 1.3.2: Pushforward Induces Map On Stalks

Let $f : X \to Y$ be a continuous map and $\mathscr{F} \in \mathbf{Sh}(X)$. Then for each $\pi(p) = q$, there exists a map

$$(f_*\mathscr{F})_q \to \mathscr{F}_p$$

between the stalks of $f_*\mathscr{F}$ and $\mathscr{F}$.

Proof **:**
exercise. This can be verified using the equivalence class definition or the universal property. Doing it using equivalence classes, define the map

$$[s]_q \mapsto [f^{-1}(s)]_p$$

Pullback

So far, most of the discussion of sheaves where with the assumption of a topological X. In example 1.1.10, there were examples of sheaves being "pushed-forward" (Definition 1.3.1) via a continuous map $\pi : X \to Y$. Namely there is a sheaf $\pi_*\mathscr{F}$ on Y defined via:

$$\pi_*\mathscr{F}(U) = \mathscr{F}\left(f^{-1}(U)\right)$$

There is an adjoint to it known as the inverse image sheaf[8]. It is not as clear as the case for the push-forward as the image of an open set need not be open, hence the existence is more subtle (similar to the existence of cokernels for groups). The adjoint will be important not only for having more than one perspective on sheaves, but for definition morphisms of schemes.

Definition 1.3.3: Inverse Image Sheaf

Let $f : X \to Y$ be a function function. and $\mathscr{F}, \mathscr{G}$ be sheaves on X and Y respectively. Then the *inverse image sheaf* or *pullback* $f^*\mathscr{G}$ on X is the sheaf associated to the presheaf:

$$f^*_{\text{pre}}\mathscr{G}(U) = \varinjlim_{V \supseteq f(U)} \mathscr{G}(V)$$

or more concisely

$$f^*\mathscr{G} = (f^*_{\text{pre}}\mathscr{G})^{\text{sh}}$$

Giving us a map $f^* : \mathbf{Sh}(Y) \to \mathbf{Sh}(X)$.

The construction of the sheaf is done in the usual quotient way: the elements of $f^*_{\text{pre}}\mathscr{F}(U)$ are equivalent classes $(s, U), (t, V)$ where $(s, U) \sim (t, V)$ if there exists a $W \subseteq U \cap V$ where $f(U) \subseteq W$ and the restriction of s, t are equal on W.

The sheafification is crucial:

Example 1.3.4: Presheaf Inverse Image

Let $f : Y \to X$ where $Y = X \coprod X$ and f is the natural projection map and a sheaf $\mathscr{G}$ defined on X. Then by definition, we would get $f^*_{\text{pre}}\mathscr{G}(U) = \mathscr{G}(U)$, however the sheafification will get:

$$f^*\mathscr{G}(U) = \mathscr{G}(U) \times \mathscr{G}(U)$$

showing the two are in general not equal.

Proposition 1.3.5: push-forward Sheaf Adjoint

Let $f : X \to Y$ be a continuous functions and $\mathscr{F}, \mathscr{G}$ be sheaves on X and Y respectfully. Then there is a bijection:

$$\text{Mor}_X(f^*\mathscr{G}, \mathscr{G}) \leftrightarrow \text{Mor}_Y(\mathscr{G}, f_*\mathscr{F})$$

That is, they are adjoints

***Proof* :**
exercise.

One advantage of the inverse image sheaf is the computation of stalk

[8] This is not the pullback, as this will be for a concept with quasi-coherent sheaves

Proposition 1.3.6: Stalks Via Inverse Image Sheaf

Let $f : X \to Y$ be a continuous function with $\mathscr{G}$ a sheaf on Y. Then:

$$(f^*\mathscr{G})_p = \mathscr{F}_{f(p)}$$

Proof **:**
exercise, follows more naturally via the definition.

In fact, the adjoint of this map is the *skyscraper sheaf*, namely given $f : \{p\} \to X$ where $p \in X$, then $f^*\mathscr{G} = \mathscr{G}_p$. Conversely, if we define the sheaf $\mathscr{F}(\{p\}) = S$, then $f_*\mathscr{F}(U)$ is the skyscraper sheaf.

will get back to this when it is more needed

Exercise 1.3.1

1. Let $U \subseteq Y$ open and $\iota : U \hookrightarrow Y$ be the inclusion. Show that $\iota^*\mathscr{G} \cong \mathscr{G}|_U$.
2. Show that f^* is an exact functor (naturally on abelian categories, however is just as good to show it for any ${}_R\mathbf{Mod}$)
3. Let $U \subseteq X$ be an open subset of a topological space so that $V = X \setminus U$ is closed. Let $\mathscr{F}$ be a sheaf on V, and $\iota : V \hookrightarrow X$ the inclusion.
   a) Show that $(\iota_*\mathscr{F})_p \cong \mathscr{F}_p$ if $p \in V$ and 0 otherwise.
   b) Let $j_!(\mathscr{F})$ be a sheaf on X given by
   $$W \mapsto \mathscr{F}(W) \qquad W \mapsto 0$$
   where the first happens when $W \subseteq V$, and the second in all other cases. Show that $(j_!\mathscr{F})_p \cong \mathscr{F}_p$ if $p \in U$ and 0 otherwise. .
   c) conclude there is a short exact sequence $0 \to j_!(\mathscr{F}|_U) \to \mathscr{F} \to \iota_*(\mathscr{F}|_V) \to 0$.

1.4 Ringed Space and Locally Ringed Space

Having defined the morphisms in $\mathbf{Sh}(X)$, an important class of sheaves to distinguish are sheaves over rings, as one of the ultimate goals in this Part of the textbook is the generalize the Nullstellensatz to arbitrary (commutative) rings:

Theorem 1.4.1: (Commutative) Ringed Objects in Sheaf

let $\mathbf{Sh}_{\mathbf{C}}(X)$ be the category of sheaves over a category $\mathbf{C}$ that has finite products and a terminal object. Then commutative ringed object exists, and the subcategory $\mathcal{R} \subseteq \mathbf{Sh}_{\mathbf{C}}(X)$ of commutative ringed object is equivalent to the category of sheaves $\mathbf{Sh}_{\mathscr{R}(C)}(X)$ on the *commutative ringed objects* of $\mathbf{C}$.

Proof **:**
This is a lot of diagram chasing. With some of the proofs later in this section, this result can be proved more easily on the level of stalks.

We shall almost always work in the case of $\mathbf{C} = \mathbf{Set}$ so that $\mathscr{R}(\mathbf{C}) = \mathbf{Ring}$:

Definition 1.4.2: Ringed Space And Structure Sheaf

Let $\mathbf{Sh}_{\mathbf{Ring}}(X)$ be the category of sheaves over $\mathbf{Ring}$ on X. Then given a sheaf of [commutative] rings over X, denote it $\mathcal{O}_X$, the tuple $(X, \mathcal{O}_X)$ is called a *ringed space*. The sheaf $\mathcal{O}_X$ is called the *structure sheaf*, and the sections of the structure sheaf $\mathcal{O}_X$ over an open subset U are called *[regular] functions on U*.

If it is unambiguous, the $\mathcal{O}_X$ is dropped and X is simply called the ringed space.

1.4.1 Warning Though these are called functions, they are elements of an arbitrary [commutative] ring. The reader should have regular function from [8, chatper 22.3] in mind.

Definition 1.4.3: Ringed Space Morphism

A map $(f, f^{\#}) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_X)$ is a *ringed space morphism* if $f : X \to Y$ is a continuous map and $f^{\#} : \mathcal{O}_Y \to f_*(\mathcal{O}_X)$ is a sheaf morphism.

It is easy to that these form a category and shall often be denoted **RingedSp**.

Example 1.4.4: Ringed Space

1. The simplest and prototypical ringed space is again the sheaf of [real/complex] continuous/smooth/holomorphic functions on a manifold/variety with component-wise addition and multiplication.
2. Schemes in the next part shall form a large class of ringed spaces
3. vector fields on a manifold do not form a ringed space as they are not closed under multiplication (they are closed under the lie bracket operation). Hoover, they shall form a $\mathcal{O}_X$-module over manifold, as shall be explored in chapter 4.

An important sheaf that shall often be used is the sheaf of units:

Definition 1.4.5: Unit Sheaf

Let $(X, \mathcal{O}_X)$ be a ringed space. Then $\mathcal{O}_X^*$ is the sheaf given by the fibered product:

$$\begin{tikzcd} \mathcal{O}_X^* \arrow[r] \arrow[d] & \{1\} \arrow[d] \\ \mathcal{O}_X \times \mathcal{O}_X \arrow[r, "\cdot"] & \mathcal{O}_X \end{tikzcd}$$

We shall almost never refer to the categories interpretations of the sheaf of rings in this Part of the book (it shall be important in ref:HERE), however ringed spaces are the central object of study in algebraic geometry.

As elements the sections in $\mathcal{O}_X(U)$ are called functions, there should be a notion of evaluation. When working over coordinate rings, evaluation of f at a point a is equivalent to modding f by the (maximal) ideal $(x - a)$. This works over coordinate rings since by the Nullstellensatz the maximal ideals correspond to points. For a general (commutative) ring, there can be many more pathological behaviours, namely it is not clear how to associate to each point $x \in X$ a maximal ideal $\mathfrak{m} \subseteq R$.

The first natural place to consider is the behaviour at the stalk, namely each point x is associated to the ring $\mathcal{O}_{X,x}$. Is there a natural choice of ideal in this ring? In this case, a natural choice would mean there is only *one* choice of ideal, in which case any $f \in \mathcal{O}_X(U)$ where $x \in U$ maps into $\mathcal{O}_{X,x}$ and can be modded by this maximal ideal. In general the answer is no:

Example 1.4.6: Stalk that is not a Local Ring

Take $X = \{p, q\}$ with the topology given by $\mathcal{T}_X = \{\emptyset, \{p\}, X\}$. Let:

$$\mathcal{O}_X(\emptyset) = \mathbf{0} \qquad \mathcal{O}_X(\{p\}) = k \qquad \mathcal{O}_X(X) = k \times k$$

where k is a field with the only non-trivial restriction map being

$$\mathcal{O}_X(X) \to \mathcal{O}_X(\{p\}) \qquad (a, b) \mapsto a$$

Then the stalk at p and q are:

$$\mathcal{O}_{X,p} = k \qquad \text{and} \qquad \mathcal{O}_{X,q} = k \times k$$

showing that the stalk at q has more than one maximal ideal (namely $k \times \{0\}$ and $\{0\} \times k$).

When working over coordinate rings of varieties, all stalks will be local rings (exercise 1.4.1(1)). In fact, the same holds for (smooth, complex, topological) manifolds [4, section 2.1]. This motivate focusing on ringed spaces whose stalks are local rings:

Definition 1.4.7: Locally Ringed Space

Let X be a ringed space. Then X is a *locally ringed space* if its stalks are local rings.

Definition 1.4.8: Evaluation on Locally Ringed Space

Let $(X, \mathcal{O}_X)$ be a locally ringed space. Then as every stalk is a local ring, it has a unique maximal ideal, hence a unique field

$$\mathcal{O}_{X,x}/\mathfrak{m}_x = \kappa(x)$$

called the *residue field*. Define *evaluation* of an element $f \in \mathcal{O}_X(U)$ where $x \in U$ by defining:

$$f(x) := \overline{f} \bmod \mathfrak{m}_x$$

where $\overline{f}$ is the image of f in $\mathcal{O}_X(U) \to \mathcal{O}_{X,x}$.

In general, ringed space morphisms *do not* preserve the local ring structure of the stalks (an example is worked through at the beginning of chapter 3, see example 3.1.3). This condition must be imposed:

Definition 1.4.9: Locally Ringed Space Morphism

Let X, Y be locally ringed spaces. Then a morphism of locally ringed spaces $(\pi, \pi^{\#}) : X \to Y$ is a morphism of ringed spaces such that the induced map of stalks $\pi^{\sharp} : \mathcal{O}_{Y,q} \to \pi_* \mathcal{O}_{X,p}$ (where $\pi(p) = q$) are local ring homomorphisms (the pre-image of the maximal ideal is contained in the maximal ideal).

It can be checked that locally ringed spaces with locally ringed morphisms form a category which shall often be denoted **LocRingedSp**.

Example 1.4.10: Manifolds and Locally ringed sheaves

The two basic examples are varieties and manifolds. Abstracting, topological spaces with continuous function form locally ringed spaces. Complex analytic space (which generalize varieties by taking the vanishing set of holomorphic functions) also form locally ringed spaces. p-adic spaces (central to number theory and modern representation theory) form locally ringed spaces. More generally, Algebraic Spaces and Stacks (defined in Part ref:HERE) form locally ringed spaces.

Overall, locally ringed spaces are the "right" over-arching category when considering the duality between algebra and geometry.

Definition 1.4.11: Isomorphism Of Ringed Spaces

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be two ringed spaces. Then an isomorphism or ringed space is a map $\pi : X \to Y$ where:

1. π is a homeomorphism
2. There is an isomorphism of sheaves Between $\mathcal{O}_X$ and $\mathcal{O}_Y$ where two two sheaves are given via the pushforward π_* (i.e. $\mathcal{O}_Y \cong \pi_* \mathcal{O}_X$ on Y)[a]

[a] By proposition 1.2.23, it is enough to check this isomorphism on stalks

Exercise 1.4.1

1. Show that the stalks of coordinate rings at maximal ideals are all local rings

1.5 *Diffeology

(would love to add a word here)

2

Schemes I: Basics

Recall the following correspondence:

$$\left\{\text{Affine Varieties}\right\} \underset{\mathcal{Z}(S)}{\overset{\mathcal{I}(V)}{\rightleftarrows}} \left\{\begin{matrix}\text{Finitely Generated,}\\ \text{Reduced rings}\\ \text{over algebraically closed fields}\end{matrix}\right\} \tag{2.1}$$

This allows for a certain subset of commutative rings to be interpreted functions on geometric objects, namely zero sets of polynomials, and vice-versa to interpret affine varieties as being determined by the set of functions on them. The question naturally arises then if this duality can be extended to general commutative rings. In particular, does there exist some topological space such that:

$$\left\{\begin{matrix}\\ ???\\ \\ \end{matrix}\right\} \underset{\mathcal{Z}(S)}{\overset{\mathcal{I}(V)}{\rightleftarrows}} \left\{\begin{matrix}\\ \text{Commutative Ring}\\ \\ \end{matrix}\right\}$$

The necessity of commutativity stems from the need of prime ideals being well-defined[1]. For a more geometric motivation, smooth/holomorphic functions over a real/complex manifold $\mathbb{C}^{\infty}(M)/\mathcal{H}(M)$ form a commutative ring since they map into $\mathbb{R}$ or $\mathbb{C}$, and hence we copy this behaviour.

Why do this? This abstraction is not simply for abstractions-sake. It is because the dictionary between algebra and geometry that has been built up to is incredibly useful, and will fix some of the problems of algebraic varieties, for example:

1. There are many algebras over more general rings that have geometric interpretations. For a rather trivial but illustrative example, the coordinate ring of a line whose points are in some

[1] The theory of "non-commutative geometry" using non-commutative rings can be explored in *Spectral Theory* in a class on functional analysis (see [5])

arbitrary (commutative) ring R (the coordinate ring being $R[x]$) currently does not have a good representation since the Nullstellensatz requires k to be an algebraically closed fields. Another example that may at first seem "un-geometric" but turns out to have many strong geometric properties are rings such as $\mathbb{Z}$, $\mathbb{Z}[i]$ (the Gaussian integers), and more generally *algebraic number rings* (see [8, chapter 22]). Mathematicians have found that these corresponds naturally to looking for integer solutions (or integral solutions when working over an algebraic ring) of a curves (see [8, chapter 22]). In particular, the element of $\mathbb{Z}[i] = \mathbb{Z}[x]/(x^2+1)$ should be functions that "evaluate" at points that would find information mod $\mathfrak{p}$ to solutions of x^2+1.

2. As we saw in front-matter, localization gives us powerful tools to explore the behavior of a variety at a point. In the theoretical framework given by the Nullstellensatz, we cannot consider such rings, for example, if we take the coordinate ring of an affine line, $k[x]$, takes the closure $k(x)$, or study $k[x]_{(x)}$, we loose finite generation. However, as we saw these certainly still have interpretations as functions on a geometric space (the former being the rational functions on a variety, the latter being the "germs" around $0 \in \mathbb{A}^1_k$).

3. Varieties loose multiplicity information. Take for example two varieties over $\mathbb{C}$ defined by the equations
$$y = x^2 \qquad \text{and} \qquad y = 0$$
then if we take their intersection, we get a point $(0,0)$. The corresponding affine variety is given by taking the union of quotient of the two coordinate rings, namely:
$$\frac{k[x,y]}{(y, x^2-y)} = \frac{k[x]}{(x^2)}$$
Usually, we would omit the x^2 term and just use $k[x]/(x)$ for the coordinate ring, however this eliminate the multiplicity information, namely that the variety $y = x^2$ "doubly intersects" the variety $y = 0$. The ring $k[x]/(x^2)$ keeps track that we have a "double-point", while $k[x]/(x)$ does not. This is a achieved by the fact that the ring has nilpotent elements.

4. (If you know differential geometry) A geometrical object that has many algebraic parallels is the vector bundle. Most vector bundles are very algebraic, however many are not projective (cannot be embedded into $\mathbb{P}^n_k$, or even $\mathbb{P}^n_R$ for some ring R). We thus need need to generalize better our notion of projective space. This shall lead to different sheaves on spectra and will be studied in chapter 4.

This build-up will introduce to us the *affine scheme*. Many familiar geometric objects shall look like affine schemes, but not all familiar geometric objects shall look like them (notably, projective space). We will need to generalize a bit more and define a *scheme* which is locally an affine scheme. This shall complete the liaison that algebraic geometry promises between algebra and geometry.

We shall then focus on how what properties of commutative rings translate to geometric properties on a scheme, and conversely when imposing desirable geometric properties how does this translate as algebraic properties. Properties such as being an integral domain, PID, UFD, Noetherian, reduced, and so forth will have geometric interpretations, and conversely notions like dimension, smoothness, open/closed embeddings, and so forth shall have algebraic counter-parts.

2.1 Defining Spec

Recall that the maximal ideals of $k[x_1, x_2, \cdots, x_n]/V$ correspond to points in a variety V (theorem 0.1.2). It may then be natural to say that the collection of maximal ideals should be considered the "points" of an arbitrary ring R. These collection were called the spectrum (plural spectra). For such an association to work, there would need to be a functor translating maps between rings to maps between the collection of maximal ideals. In particular, we would want that the functor translating a ring homomorphism $\varphi : R \to S$ to the associated continuous function mapping the maximal ideal $\mathfrak{m} \subseteq S$ to a maximal ideal $\varphi^{-1}(\mathfrak{m})$ (recall that regular function corresponds contra-variantly to k-algebra homomorphisms), However, the pre-image of a maximal ideal need not be maximal breaking functoriality (ex. take $\mathbb{Z} \hookrightarrow \mathbb{Q}$. Then (0) is maximal in $\mathbb{Q}$, but certainly not maximal in $\mathbb{Z}$).

When the theory was originally being developed within spectral theory from functional analysis (see [5]), the spaces were compact Hausdorff spaces, in which case it would have sufficed to work with the maximal ideals as within this setting the pre-image of maximal ideals (of the ring of continuous functions) is usually maximal. However, as demonstrated in the example above this doesn't hold for general rings.

Fortunately, if we loosen our requirement a bit and consider prime ideals, then it is indeed the case that $\varphi^{-1}(P)$ is prime, making it the more natural choice of "points" for functorial purposes, even though it shall produce some more "complicated" points (see example 2.1.3). Working with primes also brings the added benefit of incurring a lot more structure which shall turn out to be very insightful and make the addition of primes not only necessary for the definition to work, but to explain a lot of phenomena[2].

Definition 2.1.1: Spectrum

Let R be an [arbitrary] commutative ring. Then $\operatorname{Spec} R$ is the collection of prime ideals of R. Let $\operatorname{mSpec}(R)$ represent the collection of maximal ideals of R.

We call an element of $\operatorname{Spec} R$ a *point*, and we call an element of R a *regular functions*

In the special case where R is a coordinate ring $k[x_1, ..., x_n]/I(V)$, $\operatorname{Spec} R$ corresponds to points of an affine variety as well as "extra" points associated to primes that are not maximal (ex. $(y-x^2) \subseteq k[x, y]$) along with the zero ideal which is prime in integral coordinate rings; the intuition behind having these extra points shall be elucidated throughout this section.

To avoid confusion between points of $\operatorname{Spec} R$ and ideals of R, we shall write points in square brackets, for example $[\mathfrak{p}] \in \operatorname{Spec} R$ for $\mathfrak{p} \subseteq R$. There is some merit in writing it in the same notation as the equivalence relation notation, as points on in the topology are defined up to the radical of an ideal, for example for $(x^2) \subseteq \mathbb{C}[x]$, $V(\{(x^2)\}) = [(x)]$ (the $V(-)$ function will be defined in section 2.1.1, or see [8, chapter 21]). As we shall see, once a structure sheaf is introduced, the sets $\operatorname{Spec} \mathbb{C}[x]/(x^2)$ and $\operatorname{Spec} \mathbb{C}[x]/(x)$ which each have one element shall not be have the same structure sheaf added to them. We shall see that in most cases, we can interpret the maximal ideals as points and can write an ideal of the form $[(x_1 - a_1, ..., x_n - a_n)]$ as $(a_1, ..., a_n)$ without ambiguity, however this is not always the case (we shall define non-trivial schemes that shall have no such point[3]). In example 2.1.3, we shall

[2]For example, recall that if k was not algebraically closed, there exists regular functions that are not polynomials. Another is the distinct characterization of divisors in number theory which unifies describing the same cohomological properties of schemes

[3]In particular, there shall be no closed points

see that we can write any point associated to a maximal ideal from a coordinate ring in this way.

Another advantage of working with spectra is that there is no need to reference an ambient space for the algebraic subsets, the information is "within" the space! In particular, every quasi-projective variety in [8] was a subset of $\mathbb{P}^n_k$, while a spectra need not be a subset of $\mathbb{P}^n_k$! In fact, there shall be schemes that are not embed-able in projective space, though they are few and far between[4]. We shall later see under which condition we can embed a scheme into $\mathbb{P}^n_R$ for a ring R, and naturally this shall always be be the case for schemes associated to quasi-projective varieties (covered in section 4.4).

When working with coordinate rings R of a variety V, we may consider R to be the set of functions on V, since we may have a notion of "evaluation" for each element of R. We can do the same for a general commutative ring R, motivating the notation f for a general element of a commutative ring R ($f \in R$) and for calling such elements *regular functions.* Notice that if $p(x) \in k[x]$, then we may find the value of the evaluation $p(a)$ for some $a \in k$ by doing

$$\frac{p(x)}{(x-a)} = p(a) \bmod (x-a)$$

To represent the above more formally, every element $f \in k[x]$ (for some algebraically closed field k) is associated to an element of $\mathrm{Hom}_{\mathbf{Set}}\left(\mathrm{Spec}\,(k[x]), \coprod_{(x-a)\in \mathrm{Spec}\,(k[x])} k[x]/(x-a)\right)$, in particular:

$$k[x] \to \mathrm{Hom}_{\mathbf{Set}}\left(\mathrm{Spec}\,(k[x]), \coprod_{(x-a)\in \mathrm{Spec}\,(k[x])} \frac{k[x]}{(x-a)}\right)$$

$$f \mapsto ((x-a) \mapsto f \bmod (x-a))$$

Stare at the above until it makes sense; this is essentially a result of the Nullstellensatz ([8]). Notably, we can think of a function as holding local information (indeed, the "most local" as it gives point-wise information rather than open-set local, component local, germ local, and so forth) at each point of the spectrum[5]. We extract this point-wise information by evaluating, or in the case of spectra by modding.

Recall that if we take subsets of the classical affine space, we can often define more regular functions[6], for example by taking the (classical) affine line minus the origin, $\mathbb{A}^1_k \setminus \{0\}$, we have $f(x) = 1/x$ is a well-defined regular function in the classical sense. We shall thus want to focus on all possible functions from $\mathrm{Frac}(R)$[7] that are regular at a point $\mathfrak{p} \in \mathrm{Spec}\, R$; we saw in [8, chapter 21.6] that this collection is given by localizing at the prime: $R_{\mathfrak{p}}$ (for $R = k[x]$ with $\mathfrak{p} = (x-a)$ we had $k[x]_{(x-a)}$). This collection is called the *stalk* at the point $\mathfrak{p}$. Notice that the codomain of the hom-set can be thought of as the disjoint union of field of fractions of the germs $k[x]_{(x-a)}$ as

$$\frac{k[x]_{(x-a)}}{\mathfrak{m}_{(x-a)}} \cong \mathrm{Frac}\left(\frac{k[x]}{(x-a)}\right) \cong \mathrm{Frac}(k) = k$$

which show that we recover the same notion of evaluation when working over the localization. This greater generality shall be important when we define the sheaf on $\mathrm{Spec}\, R$ namely since the ring $R_{\mathfrak{p}}$

[4] We shall see in ref:HERE that all proper scheme are covered by a projective scheme, hence we shall require some "awkward blow-ups" to forcefully construct a scheme that cannot be embedded in projective space

[5] or more generally, at each point of a geometric space, like a manifold

[6] though not always, for example the punctured plane has the same algebraic functions as the plane, see example 2.2.20

[7] Technically, it is required that R be an integral domain as this corresponds to irreducible subsets. Each irreducible subsets will have its own $\mathrm{Frac}(R_i)$ of possible regular functions

shall be a *local ring* that has only a single maximal ideal $\mathfrak{m}_\mathfrak{p}$ making it unambiguous which ideal to mod by (as explored in section 1.4), and so we shall use it for our definition:

Definition 2.1.2: Value Of Regular Function

Let $f \in R$. Then the *value* of f at the point $\mathfrak{p} \in \operatorname{Spec}(R)$ is:

$$f(\mathfrak{p}) := \overline{f} \bmod \mathfrak{m}_\mathfrak{p} \in\in R_\mathfrak{p}/\mathfrak{m}_\mathfrak{p}$$

Let us now give multiple examples to get some grasp on spec and the functions on Spec

Example 2.1.3: Spectra and [regular] functions

This is a long set of examples, the reader should cover as many as they need to feel comfortable with the notion of spectrum; they may return here later for further elucidation on spectra.

1. Take $\operatorname{Spec}(\mathbb{C}[x])$ which is all prime (even maximal) ideal $(x-a)$ for $a \in \mathbb{C}$ and the prime ideal (0). This spectrum is labeled $\mathbb{A}^1_\mathbb{C}$; more generally the spectrum of $R[x_1, ..., x_n]$ is labeled $\mathbb{A}^n_R$. As $\mathbb{C}[x]$ is a finitely generated reduced ring over an algebraically closed field, the maximal ideals correspond exactly to the classical points of $\mathbb{A}_\mathbb{C}$ by the Nullstellensatz, with the exception of the special point (0) called the *generic point* which we shall return too soon. To visualize this set, imagine a real line (to remember that it is a 1-dimensional object over $\mathbb{C}$):

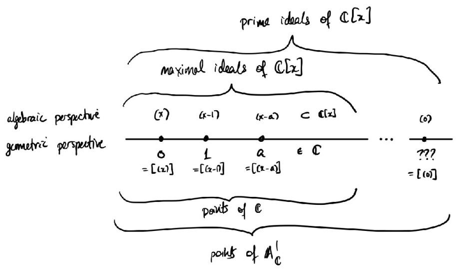


Figure 2.1: Visual Picture of $\operatorname{Spec}(\mathbb{C}[x])$

The generic point was located "away" from the line, but that is not in fact accurate. The problem is that it is a single point that is near every other point, while remaining to be a single point. What it means for points to be near shall be defined shortly when the Zariski topology on Spectra is introduced, but for now you can imagine that this causes issues when trying to draw out such point on a euclidean piece of paper. There are many ways to visualize it, some write it as a cloud of points in the corner, others at some edge of the visual, and eventually some omit the point when it is clear.

Let us evaluate some regular functions $f \in \mathbb{C}[x]$. Let's say $f = x^2 + 2x + 1$ and input input $[(x-3)]$. Then the answer is

$$16 \bmod (x-3) = f(2) \bmod (x-3)$$

which can be thought of as our usual answer $f(2)$! Notice that $\mathbb{C}[x]/(x-a) \cong \mathbb{C}$, and hence the codomain when evaluating at each point is the same. We shall see this is usually not the case soon. We can think of the fact that all the codomain of each evaluation as being a special symmetry for abstract varieties (that the evaluation information locally can all be embedded into a single global function-space).

Let us now plug in the generic point (0) into our function f (i.e. the point that isn't in our usual visualization of the complex line). We get:

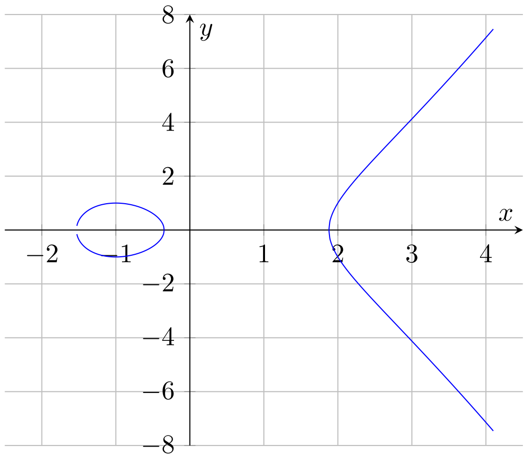
$$f \bmod 0 \in \mathbb{C}[x]$$

that it, it gives back the function itself, and is the only point for which the codomain of f is not isomorphic to $\mathbb{C}$. You can think of 0 as a "1" [complex] dimensional point. When we have spectrum's of higher dimension, say n, then the point will be "n" dimensional. We can't yet define dimension as we are still working generically with sets, this shall be rectified in section 2.6.

More generally, this entire discussion works just as well over algebraically closed fields $k = \overline{k}$, as the completeness of $\mathbb{C}$ did not factor into the above results.

Often when working with $\overline{k}$-polynomials over several variables, we shall write the maximal ideals as tuples of point, ex $[(x-2, y-3)] \in \mathbb{A}^2_{\overline{k}}$ will be represented as $(2,3) \in \mathbb{A}^2_{\overline{k}}$. We shall often then denote the generic point $[(0)]$ as η to distinguish it from the point 0 which is associated to the ideal (x).

2. Take $y^2 - x^3 + 3x - 1 \in \mathbb{C}[x,y]$ and consider Spec $\frac{\mathbb{C}[x,y]}{(y^2-x^3+3x-1)}$. This this spectrum corresponds to the elliptic curve $y^2 = y^3 - 3x + 1$. The spectrum will contain all the maximal ideals of the ring $\frac{\mathbb{C}[x,y]}{(y^2-x^3+3x-1)}$ as well as a generic point η corresponding to $[(0)]$ representing the entire space, and hence can be visualized as:



More generally, all the points of an affine algebraic sets correspond to the maximal ideals of a spectrum, and so a subset of the geometry of spectra's includes the classical geometry of curves, surfaces, and so forth!

3. Here is a Spectra that has no corresponding algebraic set in the classical sense: $\text{Spec}(\mathbb{Z}) = \{(2),(5),(7),\cdots,(p),...,(0)\}$. In other words, $\text{Spec}(\mathbb{Z}) = \{(p) \mid p \text{ is prime}\} \cup \{0\}$ and $\text{mSpec}(\mathbb{Z}) = \text{Spec}(\mathbb{Z}) \setminus \{(0)\}$.

Let us evaluate some of the regular function $n \in \mathbb{Z}$ on the points $\operatorname{Spec}(\mathbb{Z})$. The regular function 15 at the point $[(7)] \in \operatorname{Spec}(\mathbb{Z})$ is $15 \bmod (7) = 1 \bmod (7)$. The regular function 9 evaluated at $[(3)]$ is $0 \bmod (3)$. We may even say it has a double-zero at 3.

Hence, functions on $\operatorname{Spec}(\mathbb{Z})$ return some information about divisibility by primes.

4. For any field k, $\operatorname{Spec} k = \operatorname{mSpec} k = \{(0)\}$ (or $\operatorname{Spec} k = \{\eta\}$. For example, $\operatorname{Spec} \mathbb{Z}/3\mathbb{Z} = \{0\}$. Thus, fields are like "points" within algebraic geometry. An advantage of adding a structure sheaf will be to remember the original field k: we shall see that as schemes $\operatorname{Spec} K \ncong \operatorname{Spec} L$ as their sheaves will differ. Though the geometric picture as Spectra is trivial, they shall still hold valuable non-trivial information.

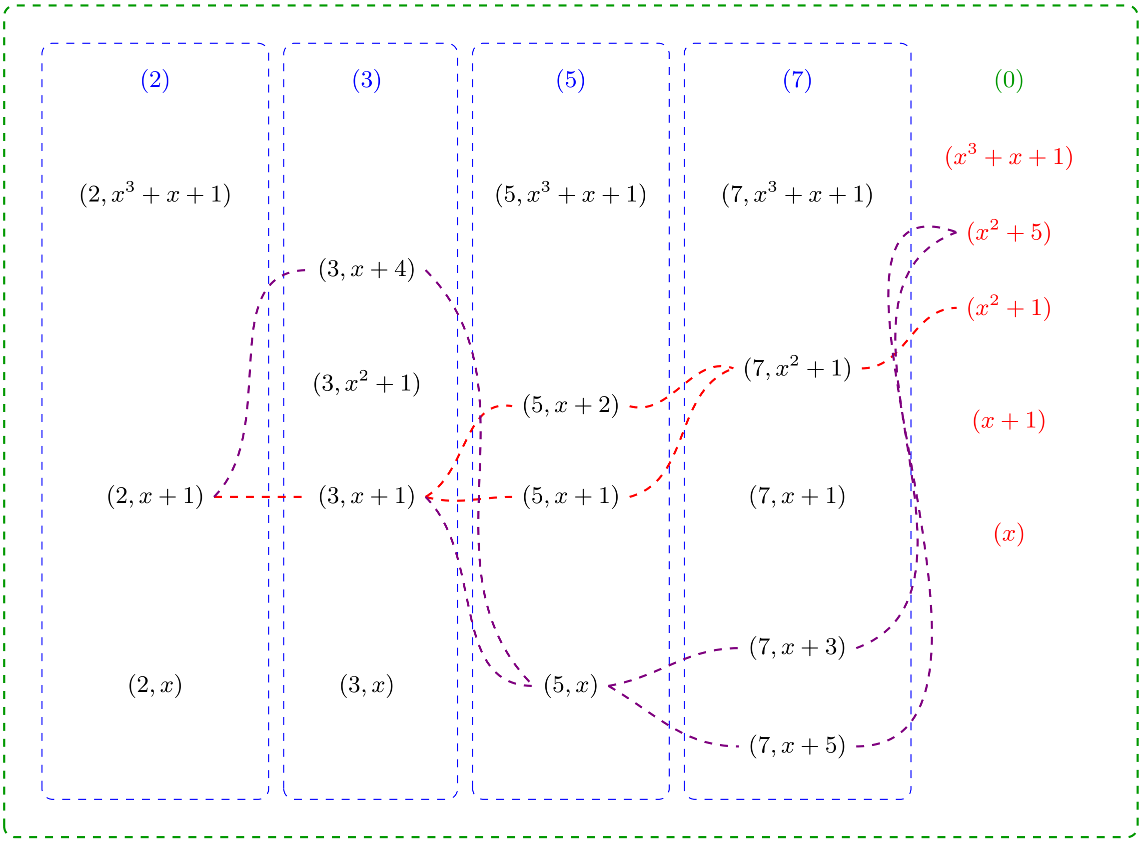
   The evaluation at of each function at the singular point shall just return the function, namely it shall be a "constant"
   $$3 \bmod 0 = 3 \in k$$

5. Take $\operatorname{Spec} \mathbb{R}[x]$. This consists of $[(0)]$, all prime (even maximal) ideals of the form $[(x-a)]$, and all prime (even maximal) ideals that can be represented by an irreducible 2nd degree polynomial $\left[\left(x^2+bx+c\right)\right]$, for example $\left[\left(x^2+1\right)\right]$. Evaluating at any of these points (besides $[(0)]$) will give a field: if evaluating points of the form $(x-a)$ the field will be isomorphic to $\mathbb{R}$, and it evaluating points of the form $\left(x^2+ax+b\right)$, the field will be isomorphic to $\mathbb{C}$. As every irreducible quadratic splits into conjugate pairs of linear terms over $\mathbb{C}$, we can think of $\operatorname{Spec}(\mathbb{R}[x])$ as being a copy of $\mathbb{C}$ folded over itself attaching the conjugate pairs (don't continue reading until this is clear)
   Let us take the function x^3-1 and evaluate it as some point. At the point $(x-2)$ we get $7 \bmod (x-2)$, or in our classical interpretation, $f(2)$. At the point $\left(x^2+1\right)$, we get:
   $$x^3-1 \equiv -x-1 \bmod \left(x^2+1\right)$$
   which we can think of as being $-i-1$. Hence, the Spectra of polynomials over fields that are not algebraically closed will behave more like algebraically closed fields as the extra non-maximal primes shall play the role of the required "additional points"! Hence, one piece of the puzzle for the "additional points" in a spectra is that they allow us to think of the spectrum as being "algebraically closed"!

6. Generalizing the above result, show that the points of $\operatorname{Spec} \mathbb{Q}[x]$ (minus the generic point 0) are in bijective correspondence with the points of $\overline{\mathbb{Q}}$, the algebraic closure of $\mathbb{Q}$, where the points of $\overline{\mathbb{Q}}$ are glue to their galois-conjugates: $\operatorname{Gal}_{\mathbb{Q}}(\overline{\mathbb{Q}})$.

7. Consider $\operatorname{Spec} \mathbb{Z}[x]$. Unlike $\mathbb{Q}[x]$, the ring $\mathbb{Z}[x]$ is not a PID, and has prime ideals of the form $(p, f(x))$ where $f(x)$ and $f(x) \bmod p$ is irreducible. This is the first example where the irreducible functions of a polynomial ring are not in one-to-one correspondence with the points, there are more points than there are irreducible functions[a]! The maximal ideals are of the form
   $$(p, f(x))$$
   while the prime ideals are of the form:
   $$(p) \quad (f(x)) \quad (p, f(x))$$
   There are now non-maximal prime ideals, namely the principal ideals (p) for prime p and principal ideals $(f(x))$ where $f(x)$ is irreducible over $\mathbb{Z}$ (or equivalently by Gauss's lemma

over $\mathbb{Q}$). To understand this spectrum better, the following is a part of a visual of Spec $\mathbb{Z}[x]$:



Let us parse this image: the maximal points are all of the form $(p, f(x))$ and they are the black points. The prime (p) are contained in each maximal ideal $(p, f(x))$. They are like a more specialized generic point, where they are not near all the points, but near every point for which the ideal representing the point contains (p). Due to this, each column is surround by dotted blue lines that indicating the generic location of each prime (p). The entire image is surround by a green dotted line which represents the generic point $[(0)]$ and how it is close to every point. Notice that there are gaps in the columns exactly where the irreducible polynomial $f(x)$ is reducible mod$p$. For example $x^2+1 \bmod 2$ is reducible with solution $x=1$, and so $\left(2, x^2+1\right)$ is not prime, namely given $\mathbb{Z}[x]/\left(2, x^2+1\right) = (\mathbb{Z}/2\mathbb{Z})[\omega]$ we have

$$(\omega+1)(\omega+1) = \omega^2 + 2\omega + 1 = 1 + 1 = 0$$

showing that $\left(2, x^2+1\right)$ is not prime, and hence not a point.
Next, each point $(f(x))$ is also a generic point, however their "shape" is not a simple as the points (p). The simplest case is when a point $(p, g(x))$ contains $f(x)$, namely $f(x) = g(x)$. In the image, we see that $\left[\left(3, x^2+1\right)\right]$ is a point and hence $\left[\left(x^2+1\right)\right]$ is near this point. On the other hand, both $(5, x+2), (5, x+1)$ are contained in $\left(x^2+1\right)$, precisely because x^2+1 factors into a product of those two linear terms mod 5 . Hence $\left[\left(x^2+1\right)\right]$ is near $[(5, x+2)]$ and $[(5, x+1)]$. This closeness is shown via a red-line passing near these points and

represents the fact that $\left[\left(x^2+1\right)\right]$ is near each of these points. Essentially, this represents the quadratic residues of x^2+1 !
The same thing has been done for the point $\left[\left(x^2+5\right)\right]$. Some extra points have been added to the image to demonstrate more clearly the branching behavior. Notice that $\left[\left(x^2+5\right)\right]$ and $\left[\left(x^2+1\right)\right]$ "intersect", for example at $[(2, x+1)]$ and at $[(13, x+5)]$. These give us some number-theoretic information about these equations!
The way I like to interpret these "fuzzy" points corresponding to the irreducible polynomials from some intuition given by complex analysis. This is not unreasonable as the spectrum in many ways mirror the behavior of algebraically closed fields and complex analysis is the analysis of functions that are locally infinite polynomials, and indeed these two fields are very closed linked, something we shall cover in ref:HERE.
For now, consider the following: if we have a "germ" around a single point of a holomorphic function (i.e. the information given by its power-series representation around an arbitrarily small neighborhood around the point), we in fact have information about the entire holomorphic function. From this perspective, it doesn't matter at which point we choose to take the power-series representation, it will return equivalent information. Thus, the information does not live at any point point in particular but "generically" at all these points. If we choose to "specialize" and specify which data we want, we shall evaluate further to a single maximal ideal.
Notice that $\mathbb{Z}[x]$ is seen on a two-dimensional grid. We shall later see that $\operatorname{Spec} \mathbb{Z}[x]$ is naturally seen as a 2 -dimensional scheme, while $\operatorname{Spec} \mathbb{Q}[x]$ shall be 1-dimensional. Furthermore, the red-lines and blue dotted lines will be seen as 1-dimensional points, the generic point will be seen as a 2 -dimensional point, while all the maximal ideals will be 0 -dimensional points.

Finally, let us consider evaluation of functions $f \in \mathbb{Z}[x]$ at points. At maximal points, we shall get values in the field $\mathbb{F}_p(\omega)$[b]where ω is a root of the irreducible polynomial. For example, take $\left[\left(7, x^3+x+1\right)\right] \in \operatorname{Spec} \mathbb{Z}[x]$ and $x^2-4 \in \mathbb{Z}[x]$. Then the value will be in $(\mathbb{Z}/7\mathbb{Z})(\omega)$ where ω is an element such that $\omega^3+\omega+1=0$. Then we would get that the value of x^2-4 is $\omega^2+3 \in \mathbb{F}_7(\omega)$. Another example would be evaluating at the point $\left[\left(3, x^2+1\right)\right]$ the function x^4-1. In this case, we get

$$x^4+2 \bmod x^2+1 = x^2+2 \bmod x^2+1 = 1 \bmod x^2+1$$

In other words, $f\left(\left[\left(3, x^2+1\right)\right]\right) = 1 \in \mathbb{F}_3(i)$. As we are evaluating at maximal ideals, we shall always get a "number" as the output, in particular since we can always embed fields into an algebraic closure, the output of function at points corresponding to maximal ideals is away within an algebraically closed field.

Let us now evaluate at point that corresponds to a non-maximal prime ideal, say at 3. The output of each function $f \in \mathbb{Z}[x]$ will then be:

$$\bar{f} \in (\mathbb{Z}/3\mathbb{Z})[x] = \mathbb{F}_3[x]$$

The way I like to think about these output is that we are asking for some of the information stored in f, namely it shall return enough information about f to determine the generic behavior of that function around $[(3)]$. If we wanted the generic behavior around a different generic point, for example $\left[\left(x^2+1\right)\right]$, then we would get:

$$f(i) \in \mathbb{Z}[i]$$

Namely, if we choose any irreducible polynomial, we are choosing which new element of an integral extension we are adding that we then evaluate the function f at. We may then extract further information by modding by different primes. It shall often be the case that evaluating at a generic point corresponds to reducing the information of a function down to a smaller geometric object or to modding information.

8. If you know some number theory, $\operatorname{Spec}\mathbb{Z}[\sqrt{-5}]$ is the quintessential example of a ring that has primes that are *not* principle, for example $(2, 1+\sqrt{-5})$ is prime but is provably not principal (see [8, chapter 22]). Evaluating at (2) and $(1+\sqrt{-5})$ will return 0, however the zero set of both of these function's shall be larger than $(2, 1+\sqrt{-5})$; we shall require the vanishing set to contain both of these.

9. any field k has a spectrum with a single element (0).

10. The following is another example of a spectrum with a single element. Take $\operatorname{Spec}\left(k[x]/\left(x^2\right)\right)$, which the spectrum of is a ring with nilpotent elements. This ring can be thought of as an extension of k by an element $\epsilon = x \bmod x^2$ that is thought to be "infinitely small". It has the interesting property that $\epsilon^2 = 0$, even though ϵ itself is nonzero. As the spectrum only contains one point[c], namely ϵ, this can be though of as a "thicker" point (as shall be explored in greater detail in section 2.2.2). The ring $k[x]/\left(x^2\right)$ is called the *dual numbers* and shall be a recurring ring to study the behaviour of nilpotent elements. This ring has the interesting property of having the function $x \in k(\epsilon)$ which always evaluates to 0 even though it is not zero! This gives us our first example of a function[d] that is not defined by what it does at every point, namely because the functions 0 and x have the same outputs.

    More generally, $k[x]/(x^n)$ has a single element. As k and $k[x]/(x^n)$ have the same spectrum, what shall distinguish them will be the sheaf structure associated to these two spectra (they shall not be isomorphic as schemes).

11. The above ring only had one point in its spectrum. There are many spectrum's with only two points: a generic point and a non-generic point. A large class of such rings shall be *local rings*. For example $k[x]_{(x-a)}$ will have (0) and $(x-a)$ as the two points of the spectrum. Another example example of a spectrum with only two points is $k[[x]]$ which has (x) and (0) as its two points. As spectra, $k[x]_{(x)}$ and $k[[x]]$ are indistinguishable. What shall distinguish these two spectra shall be the sheaf-structure associated to these two spectra.

12. Let us take a look at the number theoretic equivalent of localization: take $\mathbb{Z}_{(p)}$, the localization of $\mathbb{Z}$ at $\mathbb{Z}\backslash(p)$. This spectrum contains two points, $[(0)]$ and $[(p)]$. In particular, $\operatorname{Spec}\mathbb{Z}_{(p)}$ is not a single point like $\operatorname{Spec}\mathbb{F}_p$, it somehow has some more "fuzz" around it's single point $[(p)]$. This can be thought of as $\operatorname{Spec}\mathbb{Z}_{(p)}$ "remembering" that there is some arbitrarily small neighborhood around the point $[(p)]$ that it must keep track of. Once we introduce a topology on spectra's, we'll see that the closure of $[(p)]$ is itself (it is a closed point), while the closure of $[(0)]$ is $\{[(0)], [(p)]\}$, motivating the idea that $[(0)]$ "remembers" that there is some more information outside of just the single point! This shall be even more clear once we introduce the notion of dimension: the point $[(p)]$ will be zero-dimensional, while $[(0)]$ will be one-dimensional!

13. Let $R = k[x] \times k[x]$. What is $\operatorname{Spec} R$? Notice that $\operatorname{Spec} R$ does not contain just one generic point, but two! Importantly, $[(0)]$ is not prime as $(1,0)\cdot(0,1) \in (0)$ but $(1,0),(0,1) \notin (0)$.

14. Take Spec $\mathbb{F}_p[x] = \mathbb{A}^1_{\mathbb{F}_p}$. As it is a euclidean domain, the points are (0) and $(f(x))$ for all irreducible polynomials over $\mathbb{F}_p$. One of the interesting facets of $\mathbb{A}^1_{\mathbb{F}_p}$ is that there are infinitely many points that correspond to each point in $\mathbb{F}_p$ (one for each possible extension $\mathbb{F}_q/\mathbb{F}_p$). This richness in points will allow us to distinguish functions on $\mathbb{F}_p$ by the evaluation, namely recall that a polynomial is not distinguished by evaluation at all points in $\mathbb{F}_p$, however a polynomial will be distinguished by the evaluation of all points in Spec $(\mathbb{F}_p[x])$, as we shall show using proposition 2.1.4

15. Take $\operatorname{Spec}(\bar{k}[x,y]) = \mathbb{A}^2_{\bar{k}}$(ex. $\bar{k} = \mathbb{C}$). Then it should be proven that the points in the set are

$$(0) \quad (x-a, x-b) \quad (f(x,y))$$

where $f(x,y)$ is an irreducible polynomial in $\bar{k}[x,y]$. Visually, all the primes of the form $(x-a, x-b)$ correspond to the usual points in $\mathbb{C}^2$. The points $(f(x,y))$ can be thought of as tracing out their curve in $\mathbb{C}^2$, for example $(y-x^2)$ is the point that is the curve $y = x^2$. Notice how The maximal ideal correspond to " 0 " dimensional points, while the ideal $(y-x^2)$ correspond to " 1 " dimensional points. Similarly, (0) is a "2" dimensional point. We shall come back to that through krull dimensions in section ref:HERE.

More generally, by the Nullstellensatz the maximal points of $\bar{k}[x_1, \ldots, x_n]$ are the ideals of the form:

$$(x_1 - a_1, x_2 - a_2, \ldots, x_n - a_n)$$

The prime ideals of this ring don't lend themselves to a nice classification. Each prime ideal corresponds to an irreducible variety of a certain dimension, and hence the classification of the prime ideals is equivalent to the classification of irreducible varieties, which is a rather difficult task that is carried out in special cases, and more generally leads to the theory of moduli spaces. Some context on this classification in a classical setting was covered in [8].

16. Find the points of Spec $(k[x_1, \ldots, x_n])$ where k is not algebraically closed, ex. $\mathbb{Q}$; generalize example 5[e].

[a]This sort of exploration on the correspondence between objects that "principal" (or locally principal) and how much they are not will be the focus of section 4.3

[b]Recall that $k[\omega] = k(\omega)$, or re-prove this

[c]Prove that (0) is not a prime ideal

[d]granted, a generalization of the notion of function

[e]If you solved it, note that $(\sqrt{2}, \sqrt{2})$ is glued to $(-\sqrt{2}, -\sqrt{2})$ but *not* to $(\sqrt{2}, -\sqrt{2})$

2.1.1 Zariski Topology

Having defined the set that represents the space, the next thing to do is give a topology to this set. This topology would have to be a generalization of the Zariski topology defined on $k[x_1, ..., x_n]$ using the definition of regular function that was just introduced:

Definition 2.1.4: Zariski Topology

Let R be a commutative ring and $\mathrm{Spec}\,(R)$ the collection of prime ideals of R. Then define the *closed sets* for the Zariski topology on $\mathrm{Spec}\,(R)$ to be:

$$V(S) = \{x \in \mathrm{Spec}\,(R) \mid f(x) = 0,\ \forall f \in S\} = \{[\mathfrak{p}] \in \mathrm{Spec}\,(R) \mid S \subseteq \mathfrak{p}\}$$

Take a moment to see how this parallels the condition that all the functions in the coordinate ring must equal zero when evaluated at $\mathfrak{p}$. It is an exercise in commutative algebra to check that this is indeed a topology, namely:

Lemma 2.1.5: Zariski-closed Sets Form Topology

Let $\mathrm{Spec}\,(R)$ be the collection of primes and let:

$$\mathcal{T} = \{V(S) \mid S \subseteq R\}$$

Then:

1. $\mathcal{T}$ contain the empty set and $\mathrm{Spec}\,(R)$ (these are both the vanishing set of single elements)
2. $\mathcal{T}$ is closed under arbitrary unions and finite intersections, namely:

   $$\cap_i V(I_i) = V\left(\sum_i I_i\right)$$

   for arbitrary indexing set, and that

   $$V(I_1) \cup V(I_2) = V(I_1 I_2)$$

   see [8, chapter 21.1].

Proof **:**
exercise.

Note that the vanishing set $V(-)$ defined for the spectra of rings usually have more points than their counterpart for coordinate rings in classical algebraic geometry as they will contains primes that are not maximal ideals (particularly for rings of "dimension" ≥ 2 as we have briefly discussed), and these primes will be within $V(S)$ for appropriate S.

The reader should check how this topology compares to the topology for classical algebraic sets. For example, as there exist generic points in most spectra we have that this space is no longer T_1 (it shall soon be shown that generic points are not closed). If R is Noetherian, then $\mathrm{Spec}\, R$ is a Noetherian topological space; see exercise 2.1.1 and section 2.1.3 for more comparisons.

Just like for varieties, we may define the inverse function $I(-)$, that is for [nonempty] $S \subseteq \mathrm{Spec}\,(R)$,

$$I(S) \subseteq R \qquad I(S) = \bigcap_{[\mathfrak{p}] \in S} \mathfrak{p}$$

Hence $I(S)$ is a radical ideal (recall the intersection of prime ideals is a radical ideal, think $2\mathbb{Z} \cap 3\mathbb{Z} = 6\mathbb{Z}$). Just like in classical algebraic geometry:

Proposition 2.1.6: Properties of $I(-)$ and $V(-)$

1. $I(S)$ is an ideal of R, and furthermore it is always *radical*.
2. $I(-)$ is inclusion-reversing
3. $I(A \cup B) = I(A) \cap I(B)$, and $I(A \cap B) = \sqrt{I(A) + I(B)}$.
4. $I(\overline{S}) = I(S)$
5. $V(I(S)) = \overline{S}$ and $I(V(J)) = \sqrt{J}$.
6. If S is closed, $V(I(S)) = S$ is S. If I is radical then $I(V(J)) = J$.
7. $I(-)$ and $V(-)$ are an inclusion-reversing bijection between closed subsets of Spec (R) and radical ideals of R.

Hence:

$$\text{closed subset} \subseteq \operatorname{Spec}(R) \qquad \Leftrightarrow \qquad \text{radical ideal of } R$$

Proof **:**
exercise.

Thus, we get the bijective correspondence:

$$\{\text{Radical ideals of } R\} \underset{V(-)}{\overset{I(-)}{\rightleftarrows}} \{\text{closed sets of Spec}(R)\} \tag{2.2}$$

Notice that prime ideals are radical and hence are part of this association. This correspondence does not extend to all ideals until a sheaf, the structure sheaf, is introduced (see section 2.2.2).

As we see, the Zariski topology on Spec (R) is a generalization of the Zariski topology on $k[x_1, ..., x_n]$. The closed sets of Spec (R) will contain the "traditional" points as well as the new points, for example $V(xy, yz)$ will contain all the "traditional" points where $y = 0$ or $x = z = 0$, as well as the generic point that vanish on $V(xy)$ and $V(yz)$ (i.e. $[(y)]$ vanishes on this set). Just like for affine varieties, Spec (R) is almost never Hausdorff. This is even clearer in Spec (R), for if there are primes that are not maximal, these are *not* closed points (in a Hausdorff or T_1 space, every point is closed)

Example 2.1.7: Non-closed Points

Take $\mathbb{A}^2_{\mathbb{C}}$ with $[(y^2 - x)]$ and $[(x-2, y-4)]$. The point $[(y^2 - x)]$ is not closed. Notice that:

$$[(y-4, x-2)] \in V(I([(y^2-x)])) = V(y^2-x) \overset{\text{pr. } 2.1.6}{=} \overline{\{[(y^2-x)]\}}$$

Showing that the point is *not* closed, namely points associated to non-maximal prime ideals are not closed. However, notice the set $V(I([(y^2 - x)]))$ is closed by definition. In a moment, the point $[(y^2 - x)]$ will be seen as the generic point of the above closed set.

In fact, due to Hilbert's Nullstellensatz, we know that in $\mathbb{A}^n_k$ the maximal ideals correspond to the "traditional points", that is the closed points. From here on, we shall call the "traditional points" (more generally points associated to maximal ideals) the *closed points*, and the "extra" or "bonus" points that were being referred to shall be called the *non-closed points* or *generic points*.

With a topology, we can now link points like $[(x-2, y-4)]$ to points like $[(y-x^2)]$. As these are point, it doesn't make sense to say one point contains another. However we we be able to link these points using closure:

Definition 2.1.8: Specialization And Generization

Let $x, y \in X$ be two points in a topological space. Then we say that x is a *specialization* of y and y a *generization* of x if

$$x \in \overline{\{y\}}$$

Algebraically, $[\mathfrak{q}]$ is a specialization of $[\mathfrak{p}]$ if and only if $\mathfrak{p} \subseteq \mathfrak{q}$, hence $V(\mathfrak{p}) = \overline{\{[\mathfrak{p}]\}}$.

We can now define generic point more precisely:

Definition 2.1.9: Generic Point

Let X be a topological space. Then $p \in X$ is a *generic point* for a closed subset K iff

$$\overline{\{p\}} = K$$

By this definition, closed points are technically generic points associated the closet set of a maximal idea; we shall only use the term generic point for closed sets that are not given by a maximal ideal. With generic points and the Zariski topology defined, the idea that generic points of a closed set (including all of $\operatorname{Spec}(R)$) are close to all the points in the closed subset (it is contained in all the open subsets (or all the open subsets of $V(S) \cap U$).

Let us end with upgrading proposition proposition 2.2.7

Proposition 2.1.10: Nilpotent Specturm Homeomorphism

Let R be a ring and I an ideal of nilpotent elements. Then

$$\operatorname{Spec} R/I \to \operatorname{Spec} R$$

is a homeomorphism.

***Proof* :**
The map is a bijective continuous map. Show the inverse is continuous.

Non-closed points shall have their use in maps such as $k[x, y] \to k[x](y)$ as well as in intersection theory where the intersection or "creases" of schemes shall often be generic points. We shall see that there is a "local-ring" associated to both closed and non-closed points alike and their reduction behaviour is the same (though naturally being in different dimensions means that the behaviour may differ in exactitude's).

2.1.2 (Topological) Basis: Distinguished Open Sets

As the topology of $|\mathrm{Spec}\,(R)|$ is generate by the closed sets $V(S)$, and these sets are the intersection of:

$$V(S) = \bigcap_{f \in S} V(f)$$

the sets $V(f)$ generate the topology of $\mathrm{Spec}\,(R)$. We may want to call these sets "distinguished closed sets", however we shall favor their open counter-part due to there more simple behaviour when introducing sheaf theory to spectra[8]:

Definition 2.1.11: Distinguished Open Sets

The sets $D(f)$ are called the *distinguished open sets*, *basic open sets*, or *principal open sets*, and are defined to be

$$D(f) = X_f = |\mathrm{Spec}\,(R)| \setminus V(f) = \{\mathfrak{p} \in \mathrm{Spec}\,(R) \mid f \notin \mathfrak{p}\}$$

that is, it is the collection of prime ideals of R that do not contain f. Any open set U can be written as:

$$U = \mathrm{Spec}\,(R) \setminus V(S) = \mathrm{Spec}\,(R) \setminus \bigcap_{f \in S} V(f) = \bigcup_{f \in S} D(f)$$

Writing it as $D(f)$ may be helpful, Prof. Ravi Vakil calls these "doesn't-vanish sets" which may serve as a helpful mnemonic. These sets are usually not closed (especially if $\mathrm{Spec}\,(R)$ is connected, in which case unless $D(f) = 0$ it cannot be), and hence they are not the vanishing set for any functions in R. However, they are still isomorphic to a spectrum:

Proposition 2.1.12: Correspondence: Distinguished Open Sets and Localization

Let $D(f) \subseteq \mathrm{Spec}\,(R)$ be a distinguished open set. Then:

$$D(f) \cong_{\mathbf{Top}} \mathrm{Spec}\,(R_f)$$

Justifying the X_f notation. Furthermore, if $g = f_1 f_2 \cdots f_n$, then

$$\bigcap_{i=1,\cdots,n} (\mathrm{Spec}\,R)_{f_i} = (\mathrm{Spec}\,R)_g$$

Proof **:**
Recall that $R_f = R[f^{-1}]$, and that $\mathfrak{p} \subseteq R[f^{-1}]$ iff "$f \notin \mathfrak{p}$ and $\mathfrak{p} \in R$". This is exactly the points of $D(f)$, it is left to check that the map is bi-continuous. The second part is left as an exercise.

A mistake the reader may do is believe that each $D(f)$ can be thought of as the spectrum minus some points, but this need not be the case:

[8] There shall be a natural unique open subscheme structure up to isomorphism, but infinitely many natural closed subscheme structure up to isomorphism due to the possibility of nilpotent

Example 2.1.13: Spectrum And Distinguished Open Sets
Take $X = \operatorname{Spec}(k[x,y])$ and take $U = X \setminus \{[(x,y)]\}$, i.e. X minus the origin (which is a closed set as it is a maximal ideal). Show that $U \ncong D(f)$ for any f (hint: notice that (x,y) is not principle). This shows an important distinction between points and functions.

Show further that U can be represented as the union of two distinguished open sets.

Another result we may want to conclude is that each open set $U \subseteq \operatorname{Spec}(R)$ corresponds to $\operatorname{Spec}(R_S)$ for an appropriate S, i.e. $U \cong_{\mathbf{Top}} \operatorname{Spec}(R_S)$. This shall be shown to be *false*, the curious reader may jump to example 2.2.20 for a counter-example.

2.1.3 Topological Properties

Let us see what classical topological properties applies to Spec R. An important consequence is that for any open cover of $\operatorname{Spec}(R)$ given by distinguished open sets, there exists a finite open sub-cover. To see this, notice that $\cup_{i\in I} D(f_i) = \operatorname{Spec}(R)$ For some indexing set I if and only if $(\{f_i\}_{i\in I}) = R$ that is the ideal generated by the set $\{f_i\}_{i\in I}$. Then 1 must be inside that ideal, and must be a finite linear combination, and the result follows by the algebraic/geometric correspondence. This shows that $\operatorname{Spec}(R)$ is always compact in the topological sense. We may wish that this implies that $\operatorname{Spec}(R)$ has the usual nice properties of compactness, but this is very much not the case (notice this implies $\mathbb{A}^n_{\mathbb{C}}$ is compact). This turns out to be a consequence of lacking Hausdroffness. For this reason, a different name is given to compactness:

Definition 2.1.14: Quasicompact

Let X be a topological space. Then if X is compact and non-Hausdorff, X is *quasicompact*

As an exercise, show that if R is Noetherian, every subset of $\operatorname{Spec}(R)$ is quasicompact. In section 3.4.2, we shall show how the notion of proper maps from analysis is the right way of defining the notion of compactness which recovers many of the nice results expected from compactness.

The companion property of [quasi]compactness is connectedness. Recall a set is connected if it cannot be written as a disjoint of two nonempty open sets. Disconnectedness of Spec is equivalent to a ring representable as the product of two rings, $R = R_1 \times R_2$, or if the reader remembers the brief discussion in Representation theory(see [8, chapter 23.2]), if R has idempotent elements

Proposition 2.1.15: Spectrum Connectedness And Idempotent Elements

IF $R = R_1 \times \cdots \times R_n = \prod_i^n R_i$ is a ring, then show that

$$\operatorname{Spec}\left(\prod_i^n R_i\right) = \coprod_i \operatorname{Spec}(R_i)$$

More generally, show that $\operatorname{Spec}(R)$ is not connected if R is isomorphic to the product of two zero-rings, that is there exists $a_1, a_2 \in R$ st $a_1^2 = a_2, a_2^2 = a_2, a_1 + a_2 = 1$.

Proof **:**
good exercise.

Note that for the homeomorphism, the product must be finite, if the product is infinite then Spec $(\prod_i R_i)$ is larger than the right hand side. In fact, an infinite disjoint union of Spec (R_i) cannot be represented as Spec (R) for an appropriate R; it can be seen as the "simplest" example of a scheme that is not an affine scheme (see example 2.2.10(6)).

A notion similar to connectedness is *irreducibility*. Recall that a topological space is said to be *irreducible* if it is nonempty and it is not the union of two proper closed subsets. Written differently, X is irreducible if whenever $X = Y \cup Z$ with Y, Z closed in X, then $Y = X$ or $Z = X$[9]. Note that Y, Z may have nonempty intersection. In the euclidean topology, this notion is much less important (think $\mathbb{C}$), however in the Zariski topology we shall see that many interesting sets are irreducible. The intuition stems from traditional varieties where irreducibility corresponds to irreducible polynomials, and indeed this intuition when properly generalized translates over to irreducible [closed] sets in the Zariski topology of Spectra.

Let $X = Y \cup Z$ for closed sets Y, Z. Then there are radical ideals J_X, J_Y, J_Z such that $V(J_X) = V(J_Y) \cup V(J_Z) = V(J_Y J_Z) = V(J_Y \cap J_Z)$. As these are all radical ideals, $J_X = J_Y \cap J_Z$. Then if J_X is prime, it is an exercise in commutative algebra to show that either $J_Y \subseteq J_Z$ or $J_Z \subseteq J_Y$. Hence, irreducibility is in correspondence with prime ideals:

$$\{\text{prime ideals of } R\} \underset{V(-)}{\overset{I(-)}{\rightleftarrows}} \{\text{closed irreducible sets of Spec}\,(R)\} \tag{2.3}$$

If we look at irreducible closed sets, then $V(-)$ and $I(-)$ gives a bijection between the irreducible closed subsets of Spec (R) and the prime ideals of R, in other words there is a bijection between the *points* of Spec (R) and the irreducible closed subsets of Spec (R). Since it is a bijection, every irreducible closed subset of Spec (R) has *exactly* one generic point associated to it.

As usual, we define the Noetherian property on Spec (R) to mean that the descending chain condition for closed sets is satisfied (to mirror the ascending chain of ideals in R). In the Noetherian case, all closed sets can be decomposed into irreducibles:

Proposition 2.1.16: Noetherian Decomposition

Let X be a Noetherian topological space. Then every nonempty closed subset Z can be expressed uniquely as a finite union

$$Z = Z_1 \cup \cdots \cup Z_n$$

of irreducible closed subsets, none contained in another.That is , any closed subset Z has a finite number of pieces.

Proof **:**
see [8].

[9]The finiteness is important, for there may be irreducible sets that contain within them reducible sets of lower dimension

The reader should check the equivalent result for prime ideals

2.1.4 Ring to Top Functor: Induced Spectra Maps

As spectra are given by rings, we shall want to translate over ring homomorphisms to maps between spectra:

Definition 2.1.17: Induced Spectra Map

let $Y = \operatorname{Spec}(S)$ and $X = \operatorname{Spec}(R)$ be two spectrum. Then given the ring homomorphism: $\varphi : R \to S$, the induced *spectrum map* $\varphi^a : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ is given by:

$$\varphi^a(\mathfrak{p}) = \varphi^{-1}(\mathfrak{p})$$

The reader ought to verify that if $\varphi : R \to S$ is a ring homomorphism, then $\varphi^a : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ sending $\mathfrak{p} \mapsto \varphi^{-1}(\mathfrak{p})$ is functorial (recall or reprove that the pre-image of a prime ideal is prime). We now have two functions happening: φ^a given by ring homomorphisms, and regular functions given by ring elements. Often, φ^a is called a map, while the elements of R are called functions (following a similar convention from differential geometry).

Example 2.1.18: Maps between Spectra: φ^a

1. Show that the surjective map $R \to R/I$ induce an injective map $\operatorname{Spec}(R/I) \to \operatorname{Spec}(R)$. Algebraically, the surjective map adds relations. Geometrically, we are taking the points with the extra algebraic relations given by I and associating them to the points in R.

2. Show that the injective map $R \to S^{-1}R$ induce a injective map $\operatorname{Spec}(S^{-1}R) \to \operatorname{Spec}(R)$. Algebraically, the map $R \to S^{-1}R$ takes (integral) elements[a] and maps it to units.

3. Here is a concrete example generalizing the usual notion of a regular morphism as introduced in[8]: Take $\mathbb{A}^1_{\mathbb{C}} = \operatorname{Spec}\mathbb{C}[x]$ which by Nullstellensatz we can think of as the traditional points $\mathbb{C}$ along with the generic point (0). Consider the ring homomorphism between these rings defined by:
$$\varphi : \mathbb{C}[y] \to \mathbb{C}[x] \qquad y \mapsto x^2$$
Note this is not the squaring map, as that is not a ring morphism, but rather the map that maps the variable y to x^2 and then extends linearly. Then the map on the sets Spec's is given by:
$$\varphi^a : \operatorname{Spec}(\mathbb{C}[x]) \to \operatorname{Spec}(\mathbb{C}[y])$$
$$\begin{aligned}\varphi^a(\mathfrak{p}) &= \varphi^{-1}((x-a)) \\ &= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; f(x^2) \in (x-a) \right\} \\ &= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; x-a \middle| f(x^2) \right\} \\ &= \left\{ f(y) \in \mathbb{C}[y] \;\middle|\; f(x^2) = (x-a)q(x) \right\} \\ &= (y-a^2)\end{aligned}$$
In one line:
$$\varphi^a([x-a]) = [(y-a^2)]$$

verify that

$$(\varphi^a)^{-1}((y-a)) = \{(x-\sqrt{a}), (x+\sqrt{a})\}$$

Perhaps more intuitively, this is the Spec-equivalent of the more usual map $\mathbb{C} \to \mathbb{C}$ given by $z \to z^2$ which can be thought of as the double-covering of $\mathbb{C}$ with branching point 0. Note the importance of visualizing schemes using complex numbers, or more generally an algebraically closed field, even if we replaced $\mathbb{C}$ with $\mathbb{R}$; for example the image of (x^2+1) is $(y+1)$ (i.e. $-i^2=-1$) and $(\varphi^a)^{-1}\left((y^2+1)\right) = \{(x^2+1)\}$ where the information about the 2 points is captured in the degree of the (irreducible) polynomial in the pre-image (recall they are glued together in $\operatorname{Spec}\mathbb{R}[x]$).

4. More generally, any collection of regular functions can produce a spectra map, a "regular map" (in particular, this mirror's regular maps between affine varieties). Let k be any field and take polynomial $f_1, ..., f_n \in k[x_1, ..., x_m]$. Then define:

$$\varphi : k[y_1, ..., y_n] \to k[x_1, ..., x_m] \qquad y_i \to f_i$$

Let A represent the domain and B the codomain. Then show that for any ideal $I \subseteq A$, $J \subseteq B$ such that $\varphi(I) \subseteq J$, φ induces a map of sets:

$$\operatorname{Spec}(k[x_1, ..., x_m]/J) \to \operatorname{Spec}(k[y_1, ..., y_n]/I)$$

More particularly, show that this maps sends the point $[(x_1-a_1, ..., x_m-a_m)]$ (traditionally, $(a_1, ..., a_m) \in k^m$ to the point traditionally represented as:

$$(f_1(a_1, ..., a_m), ..., f_n(a_1, ..., a_m)) \in k^n$$

From now on, we shall sometimes take liberties in spectra maps between spectra resembling varieties by defining polynomial maps on the maximal ideals (this shall pose no problems with extending the definition to the prime ideals or to the later sheaf structure that shall be added by theorem 3.1.19).

5. The map $\mathbb{A}^1_k \to \mathbb{A}^1_k$ given by $a \mapsto p(a)$ can always be represented as a map to the graph:

$$\mathbb{A}^1_k \to \mathbb{A}^2_k \qquad a \mapsto (a, p(a))$$

Furthermore, by the projection map $\pi : \mathbb{A}^2_k \to \mathbb{A}^1_k$ given by $(a,b) \mapsto a$, we shall see that this maps $\mathbb{A}^1_k$ isomorphic to all of its graphs (this works even for higher dimensions). Note that this does not extend if we have more variable where more than one term is not the identity, for example:

$$\mathbb{A}^1_k \to \mathbb{A}^3_k \qquad t \mapsto (t, t^2, t^3)$$

maps to the point of $\mathbb{A}^1_k$ to the associated to the variety $\mathcal{Z}(x-y^2, x-z^3)$ which in [8] was shown to be rational but not isomorphic. Another example that isn't even rational is a map from $\mathbb{P}^1_k$ to an elliptic curve E.

6. This example gives a way of thinking of $\mathbb{A}^n_{\mathbb{Z}} = \operatorname{Spec}(\mathbb{Z}[x_1, ..., x_n])$ as an "$\mathbb{A}^n$-bundle" on $\operatorname{Spec}(\mathbb{Z})$ and is an important example for number theory. Define the map $\pi : \mathbb{A}^n_{\mathbb{Z}} \to \operatorname{Spec}(\mathbb{Z})$ given by the inclusion $\mathbb{Z} \hookrightarrow \mathbb{Z}[x_1, ..., x_n]$. Then show there is a bijection between $\pi^{-1}([(p)])$ where $(p) \neq (0)$and $\mathbb{A}^n_{\mathbb{F}_p}$! If $(p) = (0)$ then the pre-image is $\mathbb{A}^n_k$. The picture of this bundle can be seen as follows:

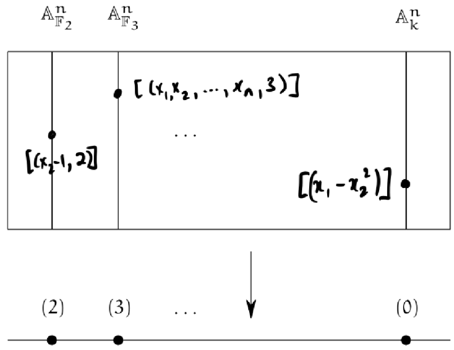


Figure 2.2: Bundle on $\operatorname{Spec}\mathbb{Z}$

7. Consider $\varphi : \mathbb{Z} \hookrightarrow k[x]$. Then $\varphi^a : \operatorname{Spec}\mathbb{C}[x] \to \operatorname{Spec}\mathbb{Z}$ is the zero map, namely $\varphi^a(\operatorname{Spec} k[x]) = \{[(0)]\}$. From this perspective, we can see that classical field theory has trivial number theory as it has no interesting fibers over $\operatorname{Spec}\mathbb{Z}$.

8. Let $\varphi : R \to S$ be an integral extension. Show that $\operatorname{Spec}(S) \to \operatorname{Spec}(R)$ is surjective (hint: Lying Over). This results also motivates the terminology "lying over" geometrically. Recall that is R or S is a field, the other must be too. Show that S being a field and the map being surjective can be used to conclude R is a field. What is the geometric argument if R was a field?

9. Let us see why it is important to work with primes instead of maximal ideals. Consider Spec $k(x)[y] \to$ Spec $k[x]$ induced by $k[x] \to k(x)[y]$ where $k[x]$ maps to itself (notably, all elements become units). Then as $k(x)[y]$ is a PID, all of its primes are maximal and are of the form $(y - f(x))$ for some $f(x) \in k(x)$. Then the pre-image of this ideal under the inclusion map must be the ideal (0), which is certainly not maximal (in fact it is the generic point). Hence the map Spec $k(x)[y] \to$ Spec $k[x]$ will map $[(y - f(x))] \mapsto \eta$, notably a 0 -dimensional point mapped to a 1-dimensional point!

10. Let R be a ring with nilpotence and $I = \sqrt{(0)}$ the nilradical. Take $\varphi : R \to R/I$ to be the usual projection. Then φ^a is a bijection; the key is to recall that the nilradical intersection of all prime ideals, and hence any prime $\overline{\mathfrak{p}} \subseteq R/I$ which is of the for $\mathfrak{p} + I$ must in fact equal itself: $\mathfrak{p} + I = \mathfrak{p}$ (as $I \subseteq \mathfrak{p}$). The example to keep in mind should be $R = \mathbb{C}[x]/(x^2)$ with $I = (x)$.

[a] Note that if an element is nilpotent, localizing it will map it to zero

Lemma 2.1.19: Induced Sepctra Map Is Continuous

Let $\varphi : R \to S$ be a ring homomorphism. Then φ^a is a continuous map

Proof **:**
It suffices to show the pre-image of a closed set is closed, so take $V(S) \subseteq \text{Spec}(R)$ and consider $(\varphi^a)^{-1}(V(I))$. Show that this is equal to $V(\varphi(I))$, showing it is closed.

The map φ^a given by the natural inverting function is a naturally continuous map, that is φ^a is a continuous function between the Spec of two rings with the Zariski topology. Hence Spec is a [contravariant] functor from **CRing** to **Top**. Furthermore, continuity can be shown on distinguished open sets, namely as they form a basis we can show that:

$$(\varphi^a)^{-1}(D([\mathfrak{p}]) = D(\varphi([\mathfrak{p}]))$$

Note that there will be continuous maps between the spectrum of rings that are not φ^a,

Example 2.1.20: Continuous Not Induced By Ring Homomorphism
The map from $\mathbb{A}^1_{\mathbb{C}}$ to itself that is the identity on all points except $[(x)]$ and $[(x-1)]$ (which will be represented with 0 and 1) where it swaps these two values is *not* a polynomial and hence cannot have been induced by a ring homomorphism, however it *is* continuous. If $0, 1 \notin V(S)$, then $\varphi^{-1}(V(S)) = V(S)$. Similarly if both values are inside. If 0 or 1 is in $V(S)$, say $0 \in V(S)$, then $\varphi^{-1}(V(S)) = V(S) \setminus \{0\} \cup \{1\}$ which is closed as all finite subsets of $\mathbb{A}^1_{\mathbb{C}}$ are closed.

Can you now find two ring homomorphisms that induce the same Spectra map? See exercise 2.1.1(13)

Exercise 2.1.1(14) goes over some of the properties preserved by continuous maps. Most of the properties that we shall be want for schemes to have (ex. separatedness, properness, smoothness, multiplicity information, finiteness conditions) shall require more than continuity.

The correspondence is terse as Spec does not carry enough information to fully and faithfully correspond to ring homomorphism. In chapter 3 it shall be shown that scheme morphisms are the right object to carry this information (namely, they shall insure that around every point, that is each stalk, the spectra map behaves like a polynomial).

2.1.5 *Subsets of Spectra via quotient and Localization

Two common constructions that came up in a few example is the process of taking the quotient and the localization of a ring R. These form important subsets of $\text{Spec}(R)$, namely by the 4th isomorphism theorem and the prime-correspondence theorem for prime ideals in localization (see [8, chapter 9.3] and [8, chapter 20.5]). By these theorems, $\text{Spec}(R/I), \text{Spec}(S^{-1}R) \subseteq \text{Spec}(R)$. If $I = \mathfrak{p}$ and $S = R \setminus \mathfrak{p}$,the first corresponds to all prime ideals containing $\mathfrak{p}$, while the second corresponds to all prime ideals that are contained in the prime ideal $\mathfrak{p}$[10] These two sets need not union to the entire $\text{Spec}(R)$ (take $\mathbb{Z}$ and 5), however they do correspond to two "opposite" ways of carving out a part of $\text{Spec}(R)$. In the following visual is a very common illustration:

[10]More generally it is all prime ideals not having trivial intersection with the multiplicative set.

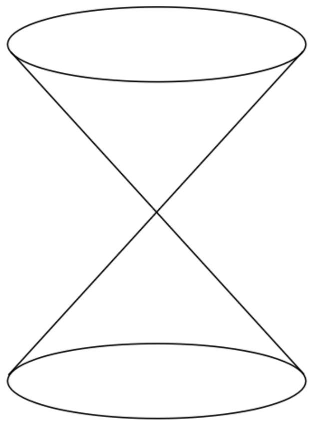


Figure 2.3: partial decomposition of Spec (R)

Geometrically, Spec (R/I) can be thought of as "carving out" a part of Spec (R). For example, take $R = \mathbb{C}[x, y, z]$ and $\mathfrak{p} = (x^2 + y^2 - z)$. If we were working with affine complex varieties, then we would say this is the 3D-parabola in $\mathbb{C}^3$, however as we are on Spec $(R/\mathfrak{p})$ this would come with some extra points (which ones?). For Spec $(S^{-1}R)$, we may think it as the opposite (we shall cover the special case of $R_\mathfrak{p}$ afterwards). For example let's say $S = \{1, f, f^2, ...\}$ and take $S^{-1}R$. Then the primes of Spec $(S^{-1}R)$ would be all primes that do not containing f, that is all points where f doesn't vanish! As an illustration, the set Spec $(\mathbb{C}[x, y]_{y-x^2})$ are the points in the affine plane, minus all the points on the parabola $y - x^2$, namely all the singletons (a, a^2) for $a \in \mathbb{C}$ (so $[(x - a, y - a^2)]$), as well as the point $[(y - x^2)]$.

Proposition 2.1.21: Closed And Open Subsets

Let R be a commutative ring, $I \subseteq R$ an ideal, $f \in R$, $\mathfrak{p} \subseteq R$ prime, and S a multiplicative set. Then:

1. Spec $(R/I) \subseteq$ Spec (R) is a closed subset
2. Spec (R_f) is an open subset
3. Spec $\left(S^{-1}R\right) \subseteq$ Spec(R) need not be open or closed in general. In particular Spec $(R_\mathfrak{p}) \subseteq$ Spec(R) need not be open or closed. However, it is an embedding, namely $\iota : \operatorname{Spec} S^{-1}R \hookrightarrow \operatorname{Spec} R$ is bijective, continuous, and closed.

Proof **:**

1. Think of $V(I)$ (induce the topology on Spec (R/I) via the inclusion map)
2. Think Spec $(R) \setminus V(f)$

3. think of Spec $(\mathbb{Q}) \subseteq$ Spec $(\mathbb{Z})$ and Spec $(\mathbb{Z}_{\mathfrak{p}}) \to$ Spec $(\mathbb{Z})$ induced by $\mathbb{Z} \to \mathbb{Z}_{\mathfrak{p}}$ (this shows that Spec $(R_{\mathfrak{p}})$ need not be open.

For the case of $S = R - \mathfrak{p}$, we can think of Spec $(R_{\mathfrak{p}})$ as keeping near the $\mathfrak{p}$, namely since the prime ideals away from $\mathfrak{p}$ will be turned into units, the primes intersecting $R \setminus \mathfrak{p}$ will be eliminated. For example, if $R = \mathbb{C}[x, y]$ and $\mathfrak{p} = (x, y)$, then we keep all the points corresponding to being "near" the origin, that is the point $[(x, y)]$, i.e. $(0, 0)$, and the 1-dimensional points $[(f(x, y))]$ where $f(0, 0) = 0$ (i.e. where $f \equiv 0 \bmod (x, y)$) that is irreducible curves that pass through the origin. In the special case where $R = k[x]$, then the localization will always only have two points, usually called the classical point and the generic point. These shall eventually be seen as geometric interpretations of DVR.

Using these geometric results, they can lend intuition towards algebraic manipulation:

Example 2.1.22: Illustration Of Quotient And Localization

Using the above intuitions, we can quickly come up with some intuition isomorphisms. Take $R = \mathbb{C}[x, y]/(xy)$. Visually, this is the union of the y and x-axis. If we localize this at x, then by our above intuition this should eliminated all points where the "function" x vanishes. The function x vanishes at the y-axis, and so we must be left with the x-axis minus the origin. This happens to be Spec $\mathbb{C}[x]_x$. In fact, these two rings are isomorphic!

$$\left(\frac{\mathbb{C}[x, y]}{(xy)}\right)_x \cong \mathbb{C}[x]_x$$

which the reader should check.

Exercise 2.1.1

1. Show that $D(f) \cap D(g) = D(fg)$.
2. $D(f) \subseteq D(g)$ iff $f^n \in (g)$ iff g is invertible in R_f. How would you prove this via "geometric explanation"? For an algebraic explanation, change to $V(f) \supseteq V(g)$ and take $I(-)$ to inverse and conclude.
3. $D(f) = \emptyset$ iff f is in the Nilradical.
4. Show that any connected component is the union of irreducible components (think of the + shape, i.e. Spec $(\mathbb{C}[x, y]/(xy))$).
5. Show that $\mathbb{A}^1_{\mathbb{C}}$ is irreducible.
6. Show that in an irreducible topological space, any nonempty open set is dense.
7. If X is a topological and Z an irreducible subspace, then $\overline{Z}$ is irreducible as well
8. Show that an irreducible space is connected.
9. If A is a integral domain, Spec (A) is irreducible.
10. Show that Spec $(\mathbb{C}[x, y]/(xy))$ is connected but reducible.
11. Show that there is a natural bijection irreducible components (maximal irreducible subsets) of Spec (R) and the *minimal* prime ideals of R.

12. these next two exercises show us how the theory of classical varieties persists with Schemes
    a) Let A be a finitely generated k-algebra. Then show that the closed points of $\mathrm{Spec}\,(A)$ are dense; show that if $f \in A$ and $D(f)$ is nonempty, then $D(f)$ contains a closed point of $\mathrm{Spec}\,(A)$[11]
    b) Let $A = \overline{k}[x_1, ..., x_n]/I$ where the Nilradical is trivial, and let $X = \mathrm{Spec}\,(A) \subseteq \mathbb{A}^n_{\overline{k}}$. Show that [regular] functions on X are determined by their valued on the closed points. Hint: if f, g are different [regular] functions on X, then $f - g$ is nowhere zero on an open subset of X. Use the above result.
13. Find two ring homomorphism that induce the same spectra map.
14. Let $f : X \to Y$ be a continuous map.
    a) Show that irreducible sets map to irreducible sets to irreducible sets.
    b) Show that if X is Noetherian, then the image of f is Noetherian
    c) Show that the dimension of X is greater than or equal to the dimension of the image of Y (also see 2.6).

2.2 Schemes

The spectrum of a ring with the Zariski is just the underlying set. It does not contain any extra information, namely the equivalent of the coordinate ring. The *structure sheaf* will be added to $\mathrm{Spec}\,R$ in order to add this extra information (namely, we shall have the collection of regular function defined on each open set as part of the data we are tracking). The resulting object is called the *affine scheme*.

After defining the affine scheme, we shall require one more crucial abstraction that links it to differential geometry: we shall want our object to the glued-together open sets. In differential geometry, these would all be open subsets of euclidean space, usually simply connected and all isomorphic. A key distinguishing feature in algebraic geometry is that the sets that can be glued can be very different, namely $\mathrm{Spec}\,R$ for all commutative rings R will be the sets from which we can select, and as the reader has already seen these sets can have very different topologies.

2.2.1 The Structure Sheaf: Affine Scheme

When working with geometric objects (varieties, real and complex manifolds, etc), we study the local behaviour of the space, namely by studying some given information on open sets or germs/stalks. In the theory of differential manifold, the information at each local was the "linear" equation (recall that one way of defining the tangent bundle is by taking all the derivations of smooth functions, which can be shown to "preserve" the linear part of each smooth function). For an , we are interested in the functions functions are *regular* on open sets. By section 2.1.2, it is enough to define the regular functions on distinguished open sets. For those motivated by analytic parallels, Complex manifold have a sheaf of meromorphic functions which are defined on the open subsets of the complex manifold, and we usually have holomorphic functions on the entire manifold.

[11] Hint: A_f is also a finitely generated k-algebra use generalized Nullstellensatz. Show the closed points of the Spec of a finitely generated k-algebra are those for which the residue field is a finite extension of k

To motivate the following sheaf, recall that on $\mathbb{C}^n$, if we take open subsets there are many more holomorphic functions that could be defined on that open subset, namely that rational functions are well-defined when the roots of the denominator (which are not cancel by the numerator) are outside the open set. For example, on $\mathbb{C}^2 \setminus \{x\text{-axis} \cup y\text{-axis}\}$, the function

$$\frac{x^2 + 4x + y + 1}{x^5 y^4}$$

is well-defined. Moving this to Spec, in $\mathbb{A}^2_{\mathbb{C}}$, we ought to have the above function to be well-defined on $D(xy)$. Furthermore, we want the collection of global sections, $\mathcal{O}_X$, to be R, mirroring the "global section" on $\mathbb{C}^2$ being the (non-rational) holomorphic functions in two variables.

As we saw in theorem 1.1.19, it suffice to define a sheaf on a base of $\operatorname{Spec}(R)$, namely on the distinguished open sets. Thus, we may define on $D(f) \subseteq \operatorname{Spec}(R)$ the ring that is the localization of R at the multiplicative subset of all function that do not vanish on $V(f)$ (note how this is subtly different than functions that do not vanish on f, what if f is nilpotent?). This can be represented as:

$$\mathcal{O}_X(D(f)) = \mathcal{O}_X(X_f) \cong R_f$$

First off, this is a presheaf: if $D(f) \supseteq D(g)$, by exercise 2.1.1 $g^n \in (f)$ so $rf = g^n$. Then we may define the restriction map:

$$\operatorname{res}_{X_f, X_g} : R_f \to R_{fg} = R_g \qquad \frac{s}{f^m} \mapsto \frac{r^m s}{r^m f^m} = \frac{r^m s}{g^{nm}}$$

which shows we have defined a presheaf on a base. We next must show that it satisfies the sheaf on base axioms.

Proposition 2.2.1: Checking Sheaf Axioms For Spec R

Let $X = \operatorname{Spec} R$, and suppose that X_f is covered by distinguished open sets $X_{f_a} \subseteq X_f$. Then:

1. If $g, h \in R_f$ are equal in F_{f_a} when passing through the restriction map, they are equal
2. If for all $a \in R$, there exists a $g_a \in R_{f_a}$ such that for each pair $a, b \in R$, the images of g_a and g_b in $R_{f_a f_b}$ are equal, then there is an element $g \in R_f$ whose image in R_{f_a} is g_a.

In other words, we define a $\mathcal{B}$-sheaf, and by theorem 1.1.19, $\mathcal{O}_X$ extends to a sheaf on X.

***Proof* :**
Let $\operatorname{Spec}(R) = \cup_{i \in I} D(f_i)$ where i indexes over the elements of R, or alternatively I is the ideal generated by the f_i that creates the entire ring R. What we shall be showing is the locality and gluability Axiom, or in categorical terms the exactness of:

$$0 \longrightarrow R \longrightarrow \prod_{i \in I} R_{f_i} \longrightarrow \prod_{i \neq j \in I} R_{f_i f_j}$$

Starting with the locality. As $s_1 = s_2$ is equivalent to $s_1 - s_2 = 0$, it suffices to show the result in the case where $s = 0$. By quasicompactness we know there is some finite subset of I where

$$\operatorname{Spec}(R) = \bigcup_{i=1}^{n} D(f_i) \qquad (f_1, ..., f_n) = A$$

Let's first suppose that for some $s \in R$, $\text{res}_{\text{Spec}\,(R),D(f_i)} s = 0$. It has to be shown that $s = 0$. By definition of R_{f_i}

$$s|_{D(f_i)} = 0|_{D(f_i)} \qquad \text{if and only if} \qquad (f_i)^m(s-0) = f_i^{m_i} s = 0$$

and by the finite number of f_i, there must exist some $m \geq 0$ such that

$$f_i^m s = 0 \qquad \forall i \in \{1, ..., n\}$$

Now, as $D(f_i) = D(f_i^m)$, $(f_1^m, ..., f_n^m) = R$. Thus, for appropriates choices $r_i \in R$ where $\sum_i^n r_i f_i^m = 1$, we get:

$$s = \left(\sum_i r_i f_i^m \right) s = \sum_i r_i (f_i^m s) = 0$$

For general $D(f)$, do the above argument with minor modifications changing R to R_f (why can the sum still be finite? How do you adjust when working with s/f?)

We next show Gluability[a]. Starting with the case of $\text{Spec}\,(R) = \cup_i D(f_i)$, let's say that there are elements in each R_{f_i} that agree on overlaps $R_{f_i f_j}$ (note that intersections of distinguished open sets are distinguished open sets). As there may be infinitely many sets on which there are overlaps, it cannot be assumed that I is finite. However, the case where I is finite is a good start, say $I = \{1, ..., n\}$. Then we have element $\frac{a_i}{f_i^{l_i}} \in R_{f_i}$ agreeing on the overlaps $R_{f_i f_j}$, meaning:

$$(f_i^{l_i} f_j^{l_j})^{m_{ij}} (f_j^{l_j} a_i - f_i^{l_i} a_j) = 0$$

simplifying notation by writing $f_i^{l_i} = g_i$:

$$(g_i g_j)^{m_{ij}} (g_j a_i - g_i a_j) = 0$$

As there are finitely many m_{ij}, take $m = \max_{i,j}(m_{ij})$:

$$(g_i g_j)^m (g_j a_i - g_i a_j) = 0$$

Next, we'll convert to a different set of distinguished open bases: take $b_i = a_i g_i^m$ and $h_i = g_i^{m+1}$ so that $D(h_i) = D(g_i)$. Then on each $D(h_i)$, we have the function $b_i/h - i$, and the overlap condition is:

$$h_j b_i = h_i b_j$$

Noting that $\cup_i D(h_i) = \text{Spec}\,(R)$ giving $1 = \sum_i^n r_i h_i$ for appropriate i, replace h_i with b_i to get :

$$r = \sum_i r_i b_i$$

Then r is the element of R that restricts to each b_j/h_j:

$$r h_j = \sum_i r_i b_i h_j = \sum_i r_i h_i b_j = b_j$$

showing Gluability in the finite case. For the infinite case, we will use the previously shown base-locality axiom. By quasi-compactness, to choose appropriate f_i so that $(f_1, ..., f_n) = R$. Construct r as was just shown. Now, it shall be shown that for any $z \in I \setminus \{1, ..., n\}$, r restricts to the desired element a_z of A_{f_z}. To do this, repeat the argument above on $\{1, ..., n, z, \}$ to get

$r' \in R$ which restricts to a_i for $i \in \{1, ..., n, z\}$. But then by the base locality, $r' = r$. Hence, r restricts to $a_z/f_z^{l_z}$.

To finish of the proof, we must do the same for an arbitrary $D(f)$, which shall then be what we wanted to show.

[a]Vakil pointed out that Serre called this as "partition of unity argument", which is the local-to-global principle physically manifested in differential geometry, which is a rather good perspective on what is about to happen

It may be tempting to say that $\mathcal{O}_{\text{Spec}\,(R)}(U)$ for any open subset U is the localization of R at the multiplicative set of all functions that do not vanish at any point of U, however this is not true:

Example 2.2.2: Subtle Local Section Of Affine Scheme

Let Spec (R) be two copies of $\mathbb{A}_k^2$ glues together at their origins (it shall later be shown this is the spectrum of the ring $R = k[w, x, y, z]/(wy, wz, xy, xz)$). Let U be the compliment of the origin. Then there is a function which is 1 on the first copy and 0 on the second copy (minus the origin). This function will not be of the aforementioned described form.

This is because each ring of section's is given by:

$$\mathcal{O}(U) = \varinjlim_{D(f) \subseteq U} \mathcal{O}(D(f))$$

which does not preserve the localization structure in general.

More explicitly, given an arbitrary open subset U, we can cover it by affine opens $\{U_i\}_{i \in I}$ and cover the intersections $U_i \cap U_j$ by affine opens $\{U_{ijk}\}_{k \in K_{ij}}$. Then the sheaf condition gives that $\mathcal{O}_X(U)$ is the equalizer of:

$$\prod_{i \in I} \mathcal{O}_X(U_i) \rightrightarrows \prod_{i,j \in I, k \in K_{ij}} \mathcal{O}_X(U_{ijk})$$

The general idea is that $\mathcal{O}_X(U)$ will contain rational functions f/g where g is nonzero on U. In practice, we shall often only require the local sections on distinguished open sets or on affine open sets, and many computations reduce to these two cases.

Adding this information to Spec (R) gives us our first definition we need:

Definition 2.2.3: Affine Scheme

Let $X = \text{Spec}\,(R)$ be the spectrum of the ring R with the Zariski topology, and let $\mathcal{O}_{\text{Spec}\,(R)}$ be the structure sheaf. Then $(\text{Spec}\,(R), \mathcal{O}_{\text{Spec}\,(R)})$ is called an *affine scheme.*

With the sheaf defined, we have now "justified" thinking of elements on an arbitrary ring R as functions, as

$$\mathcal{O}_{\text{Spec}\,(R)}(\text{Spec}\,(R)) = \mathcal{O}_{\text{Spec}\,(R)}(\text{Spec}\,(1)) = R_1 = R$$

Even more is true, due to the section's being constructed through localization of R, they may be ratio's of functions, and we give elements of section a particular name:

Definition 2.2.4: Regular Functions

Let $X = \operatorname{Spec} R$ be an and $U \subseteq X$ an open subset. Then the set $\mathcal{O}_X(U)$ is called the set of *regular functions* on U.

Note that like complex manifolds (and unlike general smooth manifolds), there are in general relatively few functions in the global section of an as compared to the global section (in the global section of an affine scheme, all rational functions are regular, that is they are R). On the other hand, $\mathcal{O}_X(U)$ may have many interesting rational functions. In fact, in many cases all open subsets of X are dense meaning the local behavior can give a lot of fruitful information on the entire scheme (as shall be the case for abstract varieties, see ref:HERE)

Example 2.2.5: Affine Schemes

All spectra can be equipped with a structure sheaf by definition and so they are all affine schemes. For less obvious examples:

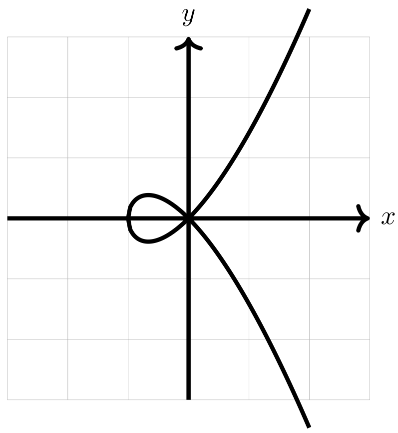
1. If $k \ncong k'$ are distinct fields fields, then as schemes $\operatorname{Spec}(k) \ncong \operatorname{Spec}(k')$ as their sheaves are different. Since there is only one point $[(0)]$ an all such Spectra, the functions (elements of the sheaf) will all return themselves, which the reader can think of as there only being constant functions, where each sheaf have different constant functions.

2. Let $R = k[x]_{(x)}$. Then $\operatorname{Spec} R$ contains two points, $[(x)]$ and $[(0)]$ and has 3 open sets, namely
$$\emptyset \quad \{[(0)]\} \quad \{[(0)], [(x)]\} = \operatorname{Spec} R$$
Note that U and $\emptyset$ are distinguished open sets (since $\{(0)\} = X_x$). The corresponding structure sheaf is:
$$\mathcal{O}_X(X) = R = k[x]_{(x)}$$
$$\mathcal{O}_X(U) = k(x)$$
The restriction map is the natural inclusion. This schemes is an example of a *local scheme*.

3. Let $R = k[[x]]$, the completion of $k[x]$ at x. The only points of Spec R are again $[(0)]$ and $[(x)]$, and so it has the same open sets. The corresponding structure sheaf is:
$$\mathcal{O}_X(X) = R = k[[x]]$$
$$\mathcal{O}_X(U) = \operatorname{Frac}(k[[x]])$$
Thus, we see that there can be schemes with very similar sheaves and identical spectra, but are distinct.

   This scheme is considered to be *even more local* than localization. Consider $\mathbb{A}^2_k$. Then the sequence of maps:
$$k[x,y] \hookrightarrow k[x,y]_{(x,y)} \hookrightarrow k[[x,y]]$$
induces a sequence of inclusion spectra maps $Y \to X \to \mathbb{A}^2_k$, showing how the completion at the point (x, y) is "smaller" than the localization at that point. To demonstrate this, assume

k does not have characteristic 2 or 3 and consider the affine scheme given by $y^2 = x^2 + x^3$:



Note that $y^2 - x^2 - x^3$ is irreducible, and localizing at (x, y) keeps the corresponding $(\frac{k[x,y]}{(y^2-x^2-x^3)})_{(x,y)}$ with only two points. However, in

$$\frac{k[[x, y]]}{(y^2 - x^2 - x^3)}$$

we have the access to power series expansion around $(0, 0)$, and we have that the power series:

$$x + \frac{1}{2}x^2 - \frac{1}{8}x^3 + \cdots$$

belongs to this ring, and notably this power series is the square-root of $x^2 + x^3$. Thus, we may write:

$$y^2 - x^3 - x^2 = (y - \sqrt{x^3 + x^2})(y + \sqrt{x^3 + x^2})$$

and so we get that $\frac{k[[x,y]]}{(y^2-x^3-x^2)}$ is *not* an integral domain, and in particular we shall see that $\operatorname{Spec} \frac{k[[x,y]]}{(y^2-x^3-x^2)}$ has *two* irreducible components corresponding to the two branches at $(0, 0)$.

4. The reader may recall that most power series do not satisfy an algebraic equation. For this reason, the above ring is often considered too big, and instead we take an intermediary ring H which contains all power series that satisfy algebraic equations over $\operatorname{Frac}(R)$. This is called the *Henselization* of $\operatorname{Spec} R$ at a point $\mathfrak{p}$.

5. If X, Y are then the product of X and Y are:

$$\operatorname{Spec}(R) \times \operatorname{Spec}(S) = \operatorname{Spec}(R \otimes_{\mathbb{Z}} S) = \operatorname{Spec}(R \otimes S)$$

where the tensor product with no base ring by default assume it is over the initial ring. The generalization of this concept shall bee shown in section 3.2 .

2.2.2 Ideal Correspondence in Affine Schemes

So far, the dictionary between certain types of ideals of R and the geometric equivalent in Spec (R) has been establish, namely through $V(-)$ and $I(-)$:

$$\begin{aligned} \text{maximal ideal of } R &\quad\Leftrightarrow\quad \text{closed points of Spec}\,(R) \\ \text{minimal prime ideals of } R &\quad\Leftrightarrow\quad \text{irreducible components of Spec}\,(R) \\ \text{prime ideal of } R &\quad\Leftrightarrow\quad \text{irreducible closed subset of Spec}\,(R) \\ \text{radical ideals of } R &\quad\Leftrightarrow\quad \text{closed subsets of Spec}\,(R) \end{aligned}$$

What of a general ideal $J \subseteq R$? In section 2.1.1, this information was "lost" as $I(V(J)) = \sqrt{J}$. However, a sheaf would capture this information!

Let's start with the simplest case of $R = \mathbb{C}[x]/(x^2)$ with $X = \text{Spec}\,(R)$ that was mentioned in example 2.1.3(6). If we consider the set Spec (R), then this is just the origin, i.e. $[(x)]$. Namely, by proposition 2.1.10[12],

$$\text{Spec}\,(\mathbb{C}[x]/(x^2)) \cong \text{Spec}\,(\mathbb{C}[x]/(x)) \cong \{\overline{(x)}\} \equiv \{0\}$$

Showing that topologically, these two spaces are the same (i.e. they are a single point). To show that the $\left(\text{Spec}\left(\mathbb{C}[x]/\left(x^2\right)\right), \mathcal{O}_{\text{Spec}(\mathbb{C}[x]/(x^2))}\right)$ is different from $\left(\text{Spec}(\mathbb{C}[x]/(x)), \mathcal{O}_{\text{Spec }(\mathbb{C}[x]/(x)))}\right)$, let us first label:

$$(X, \mathcal{O}_X) = \left(\text{Spec}\,\frac{\mathbb{C}[x]}{(x^2)}, \mathcal{O}_{\text{Spec}\,\mathbb{C}[x]/(x^2)}\right)$$

Consider the image of a polynomial $f(x)$ (i.e. an element in the global section) in $\mathbb{C}[x]/\left(x^2\right)$. Then we shall see that its evaluation is no longer an element in the field! For example if $\overline{f(x)} = \overline{(x-a)^2} \in \mathbb{C}[x]/\left(x^2\right)$, then:

$$\bar{f} = \overline{(x-a)^2} = \overline{x^2 - 2ax + a^2} = \overline{-2ax + a^2} \in \mathbb{C}[x]/\left(x^2\right) = \mathcal{O}_X(X)$$

evaluating at $\epsilon \in \mathbb{C}(\epsilon)$ we get:

$$\bar{f}(\epsilon) = a^2$$

One very important algebraic fact that is "obvious" for traditional functions but is not necessarily the case for ring with nilpotents is that a function that is zero everywhere will have to be zero:

Lemma 2.2.6: Nilpotence and Nonzero "Zero" Functions

Let R be a ring with nilpotence (i.e. a nonreduced ring). Then $\mathcal{O}_{\text{Spec}\,(R)}(\text{Spec}\,(R))$ contains a function that evaluate to 0 everywhere but is not zero.

Proof **:**
Let $f \in R$ be a nilpotence element so that $f \neq 0$ but $f^n = 0$. Then f is contained in each prime ideal, for $ff^{n-1} = f^n = 0 \in \mathfrak{p}$, and so either $f, f^{n-1} \in \mathfrak{p}$, which we can then reduce to having

[12] Note that this is a topological isomorphism, these are not scheme isomorphic as we shall see in section 3.1.1!

$f \in \mathfrak{p}$. Hence it is in the Nilradical, and so as a function

$$f(\mathfrak{p}) = 0$$

However $f \neq 0$, as we sought to show.

For completeness, we register this result whose proof was outlined in example 2.1.18[8]. Recall that there exists functions on a Spectrum of a ring that are not determined by their value at every-point (recall Spec $(k[\epsilon]/(\epsilon^2)$ with the regular function ϵ). As it turns out, this behaviour can completely be put at the feet of nilpotents:

Proposition 2.2.7: Spectrum And Nilpotent Elements

Let R be a ring and I an ideal of nilpotent elements. Then

$$\operatorname{Spec} R/I \to \operatorname{Spec} R$$

is a homeomorphism. As a consequence, two regular functions (elements of R) have the same value if they differ by a nilpotent element.

Thus, as topological spaces, nilpotents have no effect. Their presence is felt within the sheaf we put on spectra (see section 2.2.1)

***Proof* :**
Hint: recall that the nilradical is equal to the intersection of all prime elements (see [8, chapter 22]).

Hence if the nilradical is zero, $\sqrt{(0)} = (0)$, functions (in global sections of s) are determined by their points[13]! In the new language we developed, this say that a function vanishes at every point if and only if it is nilpotent! Note how this contrasts to the case where we required k to be algebraically closed for the same result, showing the greater generality achieved by having the extra prime points!

How to then visualize the Spec (R) along with its structure sheaf? It is usually done by picturing a "cloud" or "fuzz" around the point:

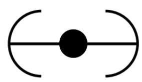


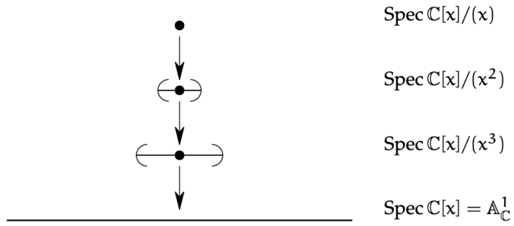
Figure 2.4: Picture of $(X, \mathcal{O}_X)$, not just $X = \operatorname{Spec}(R)$

The sequence of restrictions $\mathbb{C}[x] \to \mathbb{C}[x]/(x^2) \to \mathbb{C}[x]/(x)$ should be thought of as inching closer to $f(0)$:

[13]Proposition 2.4.5 shall push this result to its maximal generalization on quasicompact schemes

$$\begin{tikzcd}
\mathbb{C}[x] \arrow[r, two heads] & \mathbb{C}[x]/(x^2) \arrow[r, two heads] & \mathbb{C}[x]/(x) \\
f(x) \arrow[rr, mapsto] & & f(0),
\end{tikzcd}$$

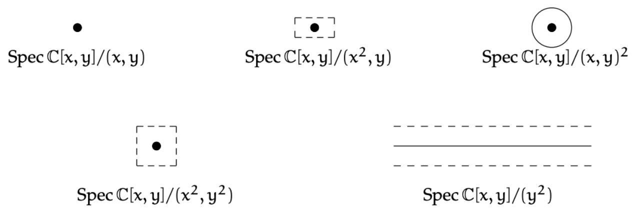
If we take, $\mathbb{C}[x]/(x^3)$, then then the spectrum will again be a single point, and when finding an element from the global section, the value, 1st derivative, and 2nd derivative will be remembered. This can be thought of a larger fuzz. We can have the following chain of inclusions:



Going to higher dimensions and considering:

$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{(x,y)^2}\right)$$

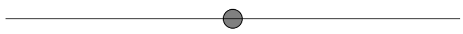
this can be thought of as remembering the derivative in all directions, and can be represented by a circle. The reader should now stare at this image to make sure it makes sense:



Next consider,

$$\operatorname{Spec}\left(\frac{\mathbb{C}[x,y]}{(y^2,xy)}\right)$$

Then this can be thought of as the intersection of a thickened x-axis and the "+" created by xy, that is this can be thought of as the x-axis as well as the first-order differential information around the origin.



Once we introduce stalks in section 2.2.6, these points will be more detectable. Notice that $[(x)]$ is *no longer* a generic point since $y^2 = 0 \in \overline{(x)}$ but $y \notin \overline{(x)}$, and hence the ideal is not prime, thus no longer a point. On the other hand, the function y when evaluating at the points $[\overline{(x-a, y-b)}] = [\overline{(x-a, y)}]$ is 0 at each point

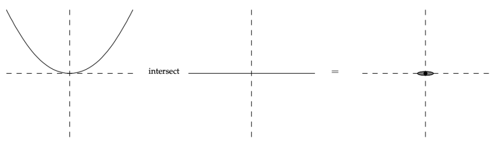
Intersection Theory and Family of Schemes

When introducing intersection theory for curves in [8, chapter 21.6], to remember the multiplicity information of a curve we had to define an equivalence class elements of $k[x,y]$ up to scaling. Schemes naturally capture multiplicity information.

For example, consider the following:

$$\begin{aligned}\operatorname{Spec}\left(\mathbb{C}[x/y]/(y-x^2)\right) \cap \operatorname{Spec}\left(\mathbb{C}[x,y]/(y)\right) &= \operatorname{Spec}\left(\mathbb{C}[x,y]/(y-x^2, y)\right)\\ &= \operatorname{Spec}\left(\mathbb{C}[x,y]/(y, x^2)\right)\end{aligned}$$

That is, intersecting the parabola $y = x^2$ with the y axis should produce produce a fuzzy point to remember the multiplicity:



Let us push this idea further. Let us see what would be the local subscheme for a curve passing through the origin. To start, consider $R = k[\epsilon]/(\epsilon^2)$ and take the surjection:

$$\varphi : k[x,y] \to R$$

mapping x, y to ϵ. This corresponds to a scheme morphism mapping the unique maximal ideal $[(\epsilon)]$ is $(x,y) \subseteq k[x,y]$. As $(\epsilon)^2 = 0$, it must be that φ vanishes on $(x,y)^2 = (x^2, xy, y^2)$, and hence φ must factor through:

$$\overline{\varphi} : \frac{k[x,y]}{(x^2, xy, y^2)} \to R$$

As R is 2 dimensional while $k[x,y]/(x^2,xy,y^2)$ is 3-dimensional, it follows the $\overline{\varphi}$ contains a non-trivial kernel. It is not hard to see with some algebraic manipulation that the kernel must be of the form $(ax + by)$, and hence we get the isomorphism:

$$X_{a,b} = \operatorname{Spec} \frac{k[x,y]}{(x^2, xy, y^2, ax + by)} \cong \operatorname{Spec} R$$

How can we interpret this scheme?

- Algebraically, we can interpret $X_{a,b} \subseteq \mathbb{A}^2_k$ as the local subscheme associated to the ideal containing all function $f \in k[x,y]$ that vanish at the origin *and* have partial derivatives satisfying:
$$b\frac{\partial f}{\partial x} - a\frac{\partial f}{\partial y} = 0$$
Note that this implies that $f = c(ax+by) + g(x)$ where $g(x)$ is the higher-order terms.
- geometrically, we can consider $X_{a,b}$ as the image of the inclusion map $\mathbb{A}^1_k \to \mathbb{A}^2_k$ given by
$$x \mapsto (bx, -ax)$$

We interpret $X_{a,b}$ as containing the point $(0,0)$ as well as remembering the infinitely near information in the direction specified by the line $ax + by = 0$ (the "tangent-line" information!) Such a scheme arises naturally when we would intersect two curves non-transversally (the example with the parabola and the line is the simplest such example).

Another place where such curves naturally arise when when looking at the *limit of schemes.* To understand, let us take the case of two point $(0,0), (a,b) \in \mathbb{A}^2_k$. These two points together form a scheme:

$$X_1 = \operatorname{Spec} R = \operatorname{Spec} \frac{k[x,y]}{(x,y) \cap (x-a, y-b)} = \operatorname{Spec} \frac{k[x,y]}{(x^2-ax, xy-bx, xy-ay, y^2-by)}$$

It should not be hard to see (ex. via the Chinese Remainder Theorem) that $R \cong k \times k$.

Now, suppose (a,b) moves towards $(0,0)$ along a polynomial curve. In particular, say there are two polynomial functions $a(t), b(t)$ such that $(a(0), b(0)) = (0,0)$. Let $X_t = \{(0,0), (a(t), b(t))\}$. Then what scheme would be expect to get when $t \to 0$? The advantage we have outline above in using schemes with intersection theory suggests that this limit should remember what direction the points came from, and indeed the limit will. Even more than this, the limit shall still contain two points which are arbitrarily close to each other!

Let us elucidate the details. Take

$$I_t = (x,y) \cap (x - a(t), y - b(t)) = (x^2 - a(t)x, xy - b(t)x, xy - a(t)y, y^2 - b(t)y)$$

Then as $t \to 0$, we see that we must contain in the ideal I_0 the polynomials x^2, xy, and y^2. Furthermore, notice that I_i always contains a linear form:

$$a(t)y - b(t)x = (xy - b(t)x) - (xy - a(t)y)$$

Thus, when $t \neq 0$, we must have that the ideal I_t always contains:

$$\frac{a(t)y - b(t)x}{t} = a - 1y - b_1x + t \cdot (g(x,y))$$

Thus, we have that:
$$(x^2, xy, y^2, a_1y - b_1x) \subseteq I_0$$

These are in fact equal. Indeed, we first notice that $I_t \subseteq k[x]$ is a k-vector space of codimension 2. By continuity (in the usual analysis sense), we can see that the limit will be codimension 2[14]. As the

[14]This argument is more subtle if we did not have finite-codimension which shall be the case when we cover this in fuller generality in section 5.4

left hand side of the above equation is codimension 2, it follows that we have an equality. Labelling $\alpha = a_1$, $\beta = b_1$ and $X_0 = X_{\alpha,\beta}$, we may write:

$$\lim_{t \to 0} X_t = X_{\alpha,\beta}$$

In higher-dimensions, things get more complicated, and when our schemes are on longer local schemes even more so. We shall see in section 5.4 that we have been looking at a *flat family* of schemes. Being "flat" shall give good conditions for paramaterized over.

Exercise 2.2.1

1. Show that for any polynomial $f \in k[x]$, if $a + b\epsilon \in \mathbb{C}(\epsilon)$, then

$$f(a + b\epsilon) = f(a) + bf'(a)\epsilon$$

2. let k be algebraically closed, R be a local k-algebra of dimension 2 with maximal ideal $\mathfrak{m}$. Show that $R \cong k[x]/(x^2)$ (hint: use some dimension argument and Nakayama's lemma to conclude $\mathfrak{m}^2 = 0$).
3. Let k be algebraically closed. Find two non-isomorphic local k-algebras of dimension 3. Geometrically, we can think of this as representing the extra flexibility in directions or local schemes can capture.
4. Let k be algebraically closed and $\operatorname{Spec} R = \operatorname{Spec} k[x_1, ..., x_n]/I \subseteq \mathbb{A}^n_k$ be a local subscheme of dimension 0 and degree 3 (R has dimension 3 over k) supported at the origin. Then $\operatorname{Spec} R$ is either isomorphic to $\operatorname{Spec} k[x]/(x^3)$ or $\operatorname{Spec} k[x, y]/(x^2, xy, y^2)$, and these two schemes are not isomorphic to each other.
5. Let R be a ring that is not a field. Show that the restriction map on the structure sheaf is almost never surjective. Hence, there are more local functions than global function.
6. Let R be a reduced ring (contains no nilpotent elements). Show that the restriction maps are injective, that is each function on a larger open set can only restrict to one function. Give a counter-example for a ring with nilpotent elements.

2.2.3 Schemes

The last step to define schemes in general is by noticing that many geometric objects are not the spectra of a ring, but are locally the spectrum of a ring, as shall be demonstrated after the definition:

Definition 2.2.8: Scheme

Let X be a topological space. Then X is a *scheme* if it comes with a sheaf of rings $\mathcal{O}_X$ such that the ringed space $(X, \mathcal{O}_X)$ is locally affine, that is, X is covered in open sets U_i

$$X = \bigcup_i U_i$$

such that there exists a ring R_i and a homeomorphism $U_i \cong_{\mathbf{Top}} |\operatorname{Spec}(R_i)|$ such that

$$\mathcal{O}_X|_{U_i} \cong_{\mathbf{Sh}} \mathcal{O}_{\operatorname{Spec} R_i}$$

Given a scheme X, we can always find an open covering by taking the collection of distinguished open subsets of the corresponding $\operatorname{Spec} R_i$, which are still open in X as $\mathcal{O}_X|_{U_i}$ is homeomorphic topologically to $\operatorname{Spec} R_i$. The way to interpret this is that locally a scheme respects the *algebraic relations* imposed by a ring R, and the agreement on intersections (if the intersection is nonempty) corresponds to the algebraic relations between two rings R_i, R_j being compatible.

To compare to real manifolds, the patches all have only "linear smooth" information (and in fact, the category of piecewise-linear manifolds of dimension ≤ 3 is equivalent to topological manifolds of dimension ≤ 3 and smooth manifolds of dimension ≤ 3, showing how "linear" these categories are in at least low dimensions[15]).

2.2.1 Lack of $V(-)$ and $I(-)$? The topology of affine schemes are characterized by the ideals of the global ring. Schemes are no longer tethered to the global sections (as shall be momentarily demonstrated). However, there is a generalization of $V(-)$ and $I(-)$ that work with the (quasi-coherent) *ideal sheaves* of a scheme X; the details are covered in section 4.1.3. From this perspective, the structure sheaf $\mathcal{O}_X$ can be thought of as generalizing the notion of a ring, remembering local information. This shall be of central importance in ref:HERE[a].

[a] related to derived scheme theory

The following will define a scheme isomorphism, but *not* a scheme morphism. This is because not all stalks of ringed-space morphisms will not preserve local rings (see example 3.1.3).

Definition 2.2.9: Isomorphism of Schemes

Let $(X, \mathcal{O}_X)$ and $(Y, \mathcal{O}_Y)$ be two schemes. Then an *isomorphism of schemes* is an isomorphism of ringed space $(f, f^\sharp) : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$, namely it is a homeomorphism $f : X \to Y$ and a sheaf isomorphism $f^\sharp : f_*\mathcal{O}_Y \to \mathcal{O}_X$ (or equiv. $f^\sharp : \mathcal{O}_Y \to f^{-1}\mathcal{O}_X$)

Example 2.2.10: Schemes

1. Naturally, all affine schemes are schemes covered by one open set
2. The key scheme to consider is projective space $\mathbb{P}^n_k$. We shall cover the construction of a general projective scheme over any ring, $\mathbb{P}^n_R$ in section 2.3; however the reader can already verify that the usual projective space as defined in [8] is a scheme. Furthermore, any sub-

[15] this becomes much more complex in higher dimensions, see this link for more details

variety of $\mathbb{P}_k^n$ (with the appropriate sheaf associated to it that we shall define in section 2.3) shall also be a scheme that isn't always isomorphic to an affine scheme.

3. The following is the smallest non-affine scheme. Take
$$X = \{p, q_1, q_2\}$$
Define the topology by making
$$X_1 = \{p, q_1\}, \; X_2 = \{p, q_2\}$$
be a sub-base. Define the presheaf $\mathcal{O}$ of rings on X by lettings
$$\mathcal{O}(X) = \mathcal{O}(X_1) = \mathcal{O}(X_2) = K[x]_{(x)} \qquad \mathcal{O}(\{p\}) = K(x)$$
(this is the *completion* at (x), not the localization at x) with the restriction map $\mathcal{O}(X) \to \mathcal{O}(X_i)$ the identify and $\mathcal{O}(X_i) \to \mathcal{O}(\{p\})$ the natural inclusion. Then this presheaf is also a sheaf, and in fact $(X, \mathcal{O})$ is a scheme, but *not* an (see exercise ref:HERE). In the geometric context, this is thought of as the "germ of doubled points".
4. By using proposition 2.1.15, the finite disjoint union of schemes is a scheme. Note that an arbitrary disjoint union of s is not an affine scheme (notice it is not quasicompact).
5. If the reader has already encountered blow-ups of a point, then the schemes associated to blow up are usually not affine (consider the example in [8] of a proper scheme that is not projective).
6. Let $X = \{*\}$ be a singleton and $\mathcal{O}_X(X) = R$ be any (commutative) ring. Is $(X, \mathcal{O}_X)$ a scheme?

In general, it is not immediate that a scheme candidate $(X, \mathcal{O}_X)$ is or is not covered by a single , since there are so many types of rings to choose from. For example, the space $\mathbb{A}_{\mathbb{C}}^1 \backslash \{([x])\}$ may seem like it cannot be a non-affine scheme as it is not an affine variety in the classical sense, however space is an affine scheme, namely it is isomorphic to $\mathbb{C}\left[x, x^{-1}\right]$. However, in example 2.11 we shall see that this $\mathbb{A}_{\mathbb{C}}^2 \backslash \{[(x, y)]\}$ is not affine. We shall soon establish some way of detecting whether a ringed space is affine or not.

With the extra structure, we shall now denote $|X|$ to represent the topological space of X rather than the underlying set. The elements of $\mathcal{O}_X(U)$ are naturally called sections, and sections of $\mathcal{O}_X(X)$ are called *global sections*. Section can be variously denoted as:
$$\mathcal{O}_X(U) \qquad \Gamma(U, \mathcal{O}_X) \qquad H^0(U, \mathcal{O}_X)$$
depending on the use-case[16]. Since stalks of schemes are isomorphic to the stalk in the , we call stalks at a point $x \in X$ a *local ring*, and quotient field $\mathcal{O}_{X,x}$ is called the residue field, which is denoted by $\kappa(x)$. We may ask what ringed spaces are isomorphic to affine schemes. The following addresses this question:

[16] the first is useful if there is only the structure sheaf, the second will be useful when we will introduce multiple sheaves on a scheme, and the third hints at how the global section is a functor $\Gamma(X, -)$ which shall in general not be exact

Proposition 2.2.11: Locally Ringed Spaces And Affine Schemes

Let $(X, \mathcal{O}_X)$ be a ringed spaced such that $R = \mathcal{O}(X)$. Then if:

1. $\mathcal{O}(U_f) = R[f^{-1}]$
2. The stalks of $\mathcal{O}_x$ of $\mathcal{O}$ are local rings
3. The map $X \to |\text{Spec}\,(\mathcal{O}(X))\,|$ is a homeomorphism

Then $(X, \mathcal{O}) \cong (|\text{Spec}\,R|, \mathcal{O}_{\text{Spec}\,R})$, and we call X a *affine space.* A ring satisfying the first two conditions is called a *locally ringed space.*

Though all these properties are important, the key new property we haven't mentioned before is that the stalks must be *local rings.*

***Proof* :**
exercise.

The following is put down here for the reader to see that schemes are indeed the geometric notion of rings, however the proof shall be relegated as the corollary of a more general result (or the reader can try to prove it themselves, as it is doable)

Theorem 2.2.12: Correspondence: Ring and Affine Scheme Isomorphisms

Let R, S be two rings. Then if the ringed-space isomorphism $\pi : \text{Spec}\,(R) \to \text{Spec}\,(S)$ induces an isomorphism $\pi^\sharp : S \to R$, which induces a ringed-space isomorphism $\rho : \text{Spec}\,(R) \to \text{Spec}\,(S)$,

$$\rho = \pi$$

***Proof* :**
corollary of theorem 3.1.6.

The following are useful results for later proofs

Proposition 2.2.13: Distinguished Open Sets And Schemes

Let $f \in R$. Then:

$$(D(f), \mathcal{O}_{\text{Spec}\,(R)|_{D(f)}}) \cong (\text{Spec}\,(R_f), \mathcal{O}_{\text{Spec}\,(R_f)})$$

If $f = 1$, then:

$$(D(1), \mathcal{O}_{\text{Spec}\,(R)|_{D(1)}}) \cong (\text{Spec}\,(R), \mathcal{O}_{\text{Spec}\,(R)})$$

Thus, the natural set-inclusion map $\text{Spec}\,(R_f) \hookrightarrow \text{Spec}\,(R)$ gives a restriction from the Zariski topology on $\text{Spec}\,(R)$ to the Zariski topology on $\text{Spec}\,(R_f)$ and the structure sheaf restricts to the structure sheaf on $\text{Spec}\,(R_f)$.

Furthermore, if X be an *affine* scheme, meaning $(X, \mathcal{O}_X) \cong (\text{Spec}\,(R), \mathcal{O}_{\text{Spec}\,(R)})$ for some R not necessarily known. Then we can recover R through the global sections of X.

Proof **:**
As X is an affine scheme, it is generated by the distinguished open sets for some (not yet known but existing) ring R. Then as $1 \in R$, $X = D(1)$. Then:

$$\mathcal{O}_X(X) = \mathcal{O}_X(\mathrm{Spec}\,(R)) = \mathcal{O}_X(D(1)) \overset{!}{=} R_1 \cong R$$

Where the $\overset{!}{=}$ inequality came from the definition of the structure sheaf, namely localizing R at 1, but 1 is already a unit and so we get:

$$\mathcal{O}_X(X) \cong R$$

From this, show that we have an isomorphism π

$$(\mathrm{Spec}\,(R), \mathcal{O}_{\mathrm{Spec}\,(R)}) = (\mathrm{Spec}\,(\mathcal{O}_X(X)), \mathcal{O}_{\mathrm{Spec}\,(\mathcal{O}_X(X))}) \xrightarrow{\sim} (X, \mathcal{O}_X)$$

The motivation behind this proposition can come if the reader recalls that it wasn't immediate what the global sections of projective space should be, while it was in a sense intrinsic in the definition of an affine variety.

In algebra, we saw that every ring is a $\mathbb{Z}$-algebra, and we can put other actions on a ring, which naturally gives the collection R-algebras. The scheme-equivalent is the following:

Definition 2.2.14: R-scheme and Finite Type Schemes

Let X be a scheme. Then it is an *R-scheme* (or a *scheme over R*) if all the ring of sections are R-algebras, and all restriction maps are R-algebra homomorphisms. Namely, there is a scheme morphism $X \to \mathrm{Spec}\, R$.

Furthermore, if an R-scheme X can be covered by $\mathrm{Spec}\,(S_i)$ where S_i is a finitely generated R-algebra, we say that X is *locally finite type over R*. If X is quasicompact, then X is said to be *finite type over R*.

Example 2.2.15: Finite type R-scheme

1. Show that $\mathbb{A}^n_k$ is a k-scheme of finite type.
2. Show that $\mathrm{Spec}\, R[x_1, ..., x_n]/(f)$ for some polynomial $f \in R[x_1, ..., x_n]$ is a finite type R-scheme. More generally, for a finite collection of polynomials $f_1, ..., f_m \in R[x_1, ..., x_m]$, $\mathrm{Spec}\, \mathbb{R}[x_1, ..., x_n]/(f_1, ..., f_n)$ is of finite type. Hence, most intuitions from algebraic sets are about $\overline{k}$-schemes that are finite type over $\overline{k}$ (abstract varieties shall require a couple more, see section 3.5)
3. Show that if R is a local ring given by some polynomial ring over k, then $\mathrm{Spec}\, R$ is not of finite type over k.
4. Show that if the fibers of $X \to \mathrm{Spec}\, R$ are infinite dimensional then X is not finite type.

2.2.4 Comparing Schemes and Real/Complex Manifolds

Before proceeding to further explore properties of schemes, it is worth taking a moment to show how schemes and real smooth manifolds have some different fundamental geometric properties:

1. The scheme given by $V(x) \cup V(yz) \subseteq \mathbb{A}^3_k$ has two irreducible components of different dimensions (as we shall more rigorously show in section 2.6). A manifold (differential and even topological) will always be of a single dimension.

2. A manifold cannot self-intersect[17], while schemes (even affine schemes, even varieties) can: consider $V(y^2 - x^3 - x^2) \subseteq \mathbb{A}^2_k$.

3. The patches of manifold are very regular, being diffeomorphic to $\mathbb{R}^n$, however patches of a scheme may be "pathological", for example if the ring is not Noetherian then there may be uncountably many connected components. This reflects how affine schemes represent "local algebraic information" which is more complicated than "local smooth linear information" that real manifolds work with. Some of the conditions and properties we shall introduce later will add some regularity to the components.

4. In Manifold Theory, the notion of irreducible component is trivial as only points are irreducible. From an analytic perspective, this can be thought of as the lack of separability of points by functions for schemes; part of this consequence is the existence of "large" points for schemes. More generally the correspondence between irreducible sets and non-closed points (in at least Noetherian schemes) shows that schemes have a lot more "global" information on its codimension subsets as compared to real smooth manifolds.

5. If a scheme X is irreducible, all its open sets are dense. Many of the schemes we shall work with will be the union of irreducible subschemes, where for each irreducible subscheme all affine open subsets are dense in the irreducible subscheme; having dense open subsets is far from being the case for most manifolds.

6. More generally, the behaviour of schemes matches more closely with the behaviour of complex geometry (we shall draw such analogies often, for example *Hartog's Extension Lemma* ([3, chapter 8]) shall have parallels with normal schemes, the existence of certain holomorphic/meromorphic functions shall tie into divisor theory, compact complex Riemann surfaces shall correspond to projective curves, and so forth).

7. Any smooth real manifold M can be embedded into $\mathbb{R}^N$ for appropriate N. For classical affine (resp. projective) varieties of dimension n, they essentially by definition embed into $\mathbb{A}^n_k$ (resp. $\mathbb{P}^n_k$). For complex manifolds, it is *not* the case they all embed in $\mathbb{C}^N$[18], and no compact complex scheme embeds into $\mathbb{C}^N$.This is because complex manifolds have much more rigidity that $\mathbb{C}^n$ lacks [6][19]. Schemes follow in this complexity.

   By the Kodaira Embedding Theorem [6], a "positive Khäler manifold" can embed into projective space . Proper schemes (the equivalent of compact complex manifolds) with an *ample line bundle* (the equivalent of positive Khäler structure) shall embedded into projective space. In fact, this means that positive Khäler manifolds are isomorphic to a proper scheme with an

[17] Naturally, a submersion of a manifold may have intersections, however (naturally) the immersion is not a manifold

[18] If it does embed, such complex manifold are called *stein manifolds* and they share many properties of quasi-projective varieties

[19] consider that there is a unique complex structure on the complex torus for each complex number

ample bundle!. This is the closest to an embedding theorem that is widely used and will be proven in section 4.4.

For a closed embeddings $X \hookrightarrow \mathbb{A}^n_R$ so that $X \cong V \subseteq \mathbb{A}^n_R$ where V is closed, this means that X is topologically sub-variety, though it may have a more complicated sheaf structure (for more details see section 3.5).

For an open embedding $X \hookrightarrow \mathbb{A}^n_R$ so that $X \cong U \subseteq A^n_{\mathbb{R}}$ where U is open, this means that X is a *quasi-affine subscheme*, which by definition is an open subscheme of affine space. Their characterization is more complicated: the map $X \to \operatorname{Spec}(\mathcal{O}_X(X))$ must be an open embedding and, as will be discussed in section 4.1.1, all quasi-coherent sheaves of $\mathcal{O}_X$-modules must be generated by global sections. To check that each $\mathcal{O}_X$-module is generated by global sections is difficult, there is a simpler characterization by the Goodman-Hartshorne Theorem [10], but it is still a complicated characterization showing the greater complexities that arise when working with non-proper schemes.

More generally, much later in ref:HERE, there shall be other interesting spaces to embed certain types of schemes into (ex. into toric varieties, Grassmannians, flag varieties, etc.) which shall have different purposes depending on what Moduli problem is being studied.

8. Schemes will be able to have "double points", recall the line with two origin is a popular counter for a manifold as it is not Hausdorff at the origin.

9. A scheme remembers multiplicity. For example, we can define a lot of functions on a single point, however scheme functions (i.e. regular functions) can hold multiplicity information (ex. $\operatorname{Spec} k[x]/(x)^2$). A smooth manifold $(M, \mathcal{A})$ with maximal atlas does not hold this information.

10. There can be multiple structure sheaves on the same set defining different schemes (think of the different fields that can form the sheaf over a singleton). Manifolds of dimension ≤ 3 have unique differential structures up to diffeomorphism.

11. Though there wasn't an atlas defined for schemes, proposition 2.2.37 will show that schemes can be constructed via gluing affine just like an atlas on a manifold. Exercise ref:HERE goes over the construction of a maximal affine atlas and how the category is equivalent (want to say why not as useful here).

12. the natural additional structure on a smooth manifold is a *tangent bundle* with transition mappings of the form $D(\mathbf{y} \circ \mathbf{x}^{-1})$, namely they are not just smooth transitions between open sets and fibers, but need to respect a $\mathrm{GL}_n(\mathbb{R})$-action given by the derivation D. We shall define the equivalent for schemes in chapter 4 section 4.5, but more care must be done as schemes are not "locally linear" everywhere in the way that smooth manifolds must be. Much of the intuition of Poincaré's duality (see [2, section 4.3]) will influence the generalization.

13. Schemes shall be able to better remember different levels of *local* behaviour. For example, a germ is not a smooth manifold on a point, but the equivalent for a scheme shall be a scheme, known as a *local scheme*.

14. Scheme functions are more "rigid"; while on a manifold we may have partitions of unity due to the flexibility of euclidean topology, no exact equivalent can be defined for schemes. This rigidity will also add structure and make it easier to defined some concepts (ex. closed subschemes, introduced in section 2.2.5, shall be a rather "natural" concept as compared to the work that goes into ensuring closed submanifolds are locally isomorphic to $\mathbb{R}^k$).

As smooth functions on each $U \subseteq M$ from a sheaf, sheaves that admit a "partition of unity" will be called *fine sheaves*. A weaker notion of *soft sheaf* will be introduced in section 5.1 that will be the closest equivalent to having the partition of unity property.

15. Schemes unify "arithmetic" and "geometry" in a way manifold theory is poorly equipped for. As an example, there is no equivalent of $\operatorname{Spec}\mathbb{Z}$, $\operatorname{Spec}\mathcal{O}_K$ or $\operatorname{Spec}\mathbb{Z}[t]$ where $\mathcal{O}_K$ is a ring of integers. In this way, scheme theory also allows for a new type of object that is not tractable in differential geometry. Often, to deal with more "geometric" schemes, one works with $\overline{k}$-schemes where $\overline{k}$ denotes the algebraic closure of a characteristic 0 field. In fact, when defining Z-points in section 3.1.3 where Z is a scheme, if $Z = \operatorname{Spec}\overline{k}$ then such points shall be called *geometric points*. Working over a non-algebraically closed field often implies the schemes contains some arithmetic information.

Though this is a long list of differences, very often in practice (especially in the setting of k-schemes or when studying curves, surfaces, moduli spaces, etc.) we work with schemes that most of the time are proper (the correct generalization of compact manifolds), and have many methods to "normalize" singularities (ex. Intersections, cusps, etc) to make these scheme regular[20]. In fact, as we shall see in section 5.6, a result known as *GAGA* shows when proper schemes with appropriate coherent sheaves are equivalent to a compact complex manifolds. Naturally, arithmetic schemes will not be part of this manifold-scheme correspondence.

2.2.5 Subschemes: Open, Closed, and Neither

An overarching driver with the definition of affine schemes is that it ought to be the dual category to the category of rings. Thus, we would like it if the sub-objects in the category of schemes correspond to the quotient objects in the category of rings, and indeed given $R \to R/I$, by proposition 2.1.21 the subset $\operatorname{Spec} R/I \subseteq \operatorname{Spec} R$ is a closed subset and there exists a continuous inclusion map that is also a map of locally-ringed spaces on the sheaves. Intuitively, $R \to R/I$ trivializes some relations, and hence shrinks the given affine scheme. For a general scheme, this condition shall be generalized to requiring that the map on stalks be surjective (as we will see in definition 2.2.21).

Though categorically correct, it must be pointed out that there can be more interesting classes of subschemes. We may generally ask the question "if $Y \subseteq X$ is a subset and $(X, \mathcal{O}_X)$ is a scheme, is there a sheaf-structure we can put on Y that is naturally induced by X via the pullback". For one, open subsets shall have a natural structure sheaf induced by the original scheme. For another, we shall have inclusions $\operatorname{Spec} X \hookrightarrow \operatorname{Spec} Y$ where $\operatorname{Spec} X$ is neither open nor closed, but it is homeomorphic onto its image and the induced sheaf map is an isomorphism (in particular, this shall be true for *local schemes*). We thus see that monomorphisms are too restrictive to fully capture all the behaviour we may wish out of a subscheme.

Nevertheless, the monomorphisms which are given as open subschemes or locally closed subschemes will create a great class of schemes that we shall encounter very often and deserve to be singled out. Other notions of subschemes shall appear when we need them.

[20] Though as we'll see, there is no universal way of doing this with the exception of some low-dimension that have been completely proved, and sometimes this singularity information is important.

Open Subschemes

An easy way to get new schemes from known ones is to take open and closed subsets. It is easy to see that it is a topological subspace, The important thing to see is that the structure-sheaf is well-defined. Let us start with open subsets as they are a bit easier:

Proposition 2.2.16: Open Subscheme

Let X be a scheme and $U \subseteq X$ an open subset. Then $(U, \mathcal{O}_X|_U)$ is a scheme.

Proof **:**
U is certainly a subspace and a subsheaf with local rings as stalks. To show it is locally affine, for any $x \in U$ choose affine open $V \subseteq X$ where $x \in U \cap V$ and $V \cong \operatorname{Spec} R$. As $D(f)$ forms a basis for $\operatorname{Spec} R$, there exists an f_0 such that $x \in D(f_0) \subseteq U \cap V \subseteq U$, showing that U is covered in affine, completing the proof.

Definition 2.2.17: Open Subscheme

Let X be a scheme. Then $(U, \mathcal{O}_X|_U)$ is called an *open subscheme*. If U is also an affine scheme, then U is often said to be an *affine open subscheme* or *affine open subset*.

If a scheme is isomorphic to an open subscheme, we shall say that the isomorphism map (onto the image) is an open embedding:

Definition 2.2.18: Open Embedding

A map of schemes $\pi : X \to Y$ is an open embedding if:

$$(X, \mathcal{O}_X) \xrightarrow{\sim} (U, \mathcal{O}_Y|_U) \hookrightarrow (Y, \mathcal{O}_Y)$$

2.2.2 Remark about Hartshorne's Definition of Open Subscheme Note that in Hartshorne, an open subscheme was technically not defined as $(U, \mathcal{O}_X|_U)$ but as $(U, [\mathcal{O}_X|_U])$ where $[\mathcal{O}_X|_U]$ is the equivalence class of all isomorphic sheaves. This is in fact not correct: isomorphic sheaves shall give technically different open subschemes (which are of course isomorphic). Hence, Hartshorne claims that an open subscheme has a unique subscheme structure, not unique up to isomorphism.

Corollary 2.2.19: Subspace and Zariski Topology On Open Subscheme

Let $U = \operatorname{Spec}(R) \subseteq X$ be affine an open subset and take $(U, \mathcal{O}_X|_U)$. Then the Zariski topology on U is the subspace topology on U with respect to X

Proof **:**
exercise.

The prototypical example of an open subscheme would be the open sets $D_+(x_i) \subseteq \mathbb{P}^n_k$ of projective space or $D(f) \subseteq \operatorname{Spec} R$ of affine space. Note that an open sub-scheme of an affine sub-scheme need not be affine:

Example 2.2.20: Open Sub-Scheme need not be Affine

Let $R = k[x, y]$, $X = \mathbb{A}^2_k = \operatorname{Spec}(k[x, y])$, and consider $U = \mathbb{A}^2_k \setminus \{(x, y)\}$ which may be considered the plane without the origin. Before seeing why this is a scheme, try to show that U is indeed an open subset (it is the compliment of intersection of certain closed subsets, equivalently the union of certain open sets, there is more than one way of creating it).

Now, there does not exist an R such that $\operatorname{Spec}(R) \cong U$. It may be tempting to say that there ought to be a distinguished open set of $\mathbb{A}^2_k$ that produces U, but the reader should think through why this is not the case (hint: (x, y) is not principle). However, U can certainly be described as the union of two distinguished open sets, namely:

$$U = D(x) \cup D(y)$$

What is the global section on the open sub-scheme U? This would be all functions in $D(x) \cup D(y)$ that agree on the intersection, $D(x) \cap D(y) = D(xy) = R_{xy} = k[x, y]_{xy}$. Computing the local sections, as $D(x)$ is all points in $\mathbb{A}^2_k$ that do not vanish in the function x, and $D(y)$ is all the points that do not vanish in the function y, we get:

$$\mathcal{O}_U(D(x)) = R_x = k\left[x, y, \frac{1}{x}\right] \qquad \mathcal{O}_U(D(y)) = R_y = k\left[x, y, \frac{1}{y}\right]$$

Now consider the functions that agree on the intersection: $R_x \cap R_y \subseteq R_{xy}$. These are all rationals functions with only powers of x in the denominator, and only powers of y in the denominator (as it must be in the intersection). But this leaves only the elements of R: $R_x \cap R_y = R$ (where R is identified with it's canonical embedding in R_{xy}). Thus:

$$\mathcal{O}_{\mathbb{A}^2_k}(U) = \mathcal{O}_U(U) = k[x, y]$$

in other words, the global section of U is identical to the global section of $\mathbb{A}^2_k$. Now, if U was affine, then by proposition 2.2.13, we would have that the is determined by the global ring, that is:

$$U \cong \mathbb{A}^2_k$$

that is, this is a scheme isomorphism, and hence also a homeomorphic map between prime ideals that must respect $V(-)$ and $I(-)$. But in U we have:

$$V(x) \cap V(y) = \emptyset$$

while this is certainly not the case in $\mathbb{A}^2_k$ (as it gives $[(x - 0, y - 0)]$). Hence U is *not* an affine scheme!

The main idea is that $D(f)$ misses a codimension 1 subset if R is Noetherian integral domain (see exercise 1), where Noetherian is to avoid infinite pathologies and integral domain is to insure there is only one irreducible component (see section 2.4). More generally, given R is Noetherian and an integral domain, $\operatorname{Spec} R_S$ for some $S \subseteq R$ not containing units misses a (pure) codimension 1 subset. The reason that higher codimensional sets "don't matter" when considering global sections very reminiscent of complex geometry, namely this is an algebraic version of *Hartog's Lemma* (see [3, chapter 8]).

Closed Subschemes

When working with closed subschemes, we have to be a little more careful as there can be multiple scheme structure, namely since we can have varying multiplicity of the zero sets, for example Spec $k[x]/(x)$, Spec $k[x]/\left(x^2\right)$, Spec $k[x]/\left(x^3\right)$, and so forth. This comes down to the very elementary fact that $0^n = 0$ and $n \cdot 0 = 0$. For any non-zero number a, it is not the case that both $a^n \neq a$ and $n \cdot a = a$ (for all $n \in \mathbb{Z}$) be true unless $a = 0$. Hence, up to isomorphism, open subschemes have a unique compatible sheaf-structure with the parent scheme, while closed subscheme can have multiple:

Definition 2.2.21: Closed Embedding, immersion, and Closed subschemes

Let $f : Y \to X$ be a spectrum map along with $f^\# : \mathcal{O}_X \to f_*\mathcal{O}_Y$. Then:

1. if f induces a homeomorphism from Y onto it's image which is closed in X, and $f^\# : \mathcal{O}_X \to f_*\mathcal{O}_Y$ is surjective, then f is said to be a closed embedding.
2. If f is injective and closed and $f^\#$ is surjective, then f is a *closed immersion*.
3. A closed subscheme $Y \subseteq X$ is a closed subset where the structure sheaf on Y makes id : $Y \hookrightarrow X$ is a closed embedding.

The main difference between a closed subscheme and closed embedding is that a closed embedding comes with an isomorphism, while a closed subscheme would just be Y along with an appropriate structure sheaf. Note that the compliment of a closed subscheme is a unique open subscheme, but the compliment of an open subscheme has multiple closed subscheme, usually infinitely many!

2.2.3 Nuance in Terminology Many textbooks use the term closed immersion and closed embedding interchangeably or define a closed immersion as how closed embeddings where defined here. For this textbook when a scheme is *reduced* then the two definition match (see exercise 2.4.2(6)).

Example 2.2.22: Closed Subscheme

1. Take $\{0\} \subseteq \mathbb{A}^1_k$. This point corresponds to the Spec $k[x]/(x) \cong \operatorname{Spec} k$. Then f : Spec $k \to$ Spec $k[x]$ mapping $\eta \mapsto 0$ is a closed embedding: its image being closed, the map $f^\#$: $\mathcal{O}_{\operatorname{Spec} k[x]}(\operatorname{Spec} k[x]) \to \mathcal{O}_{\operatorname{Spec} k(x)}(\operatorname{Spec} k(x))$ is the map

$$k[x] \to k$$

   is naturally surjective, hence all its localizations are surjective, hence we have a closed subscheme.

2. Now consider Spec $k[x]/\left(x^2\right) \to \operatorname{Spec} k[x]$. The image of f is the same as the above and so is still closed, and the sheaf morphism:

$$k[x] \to k[x]/\left(x^2\right)$$

is surjective, hence this is also a closed subscheme. More generally, $\operatorname{Spec} k[x]/(x^n) \to \operatorname{Spec} k[x]$ is a closed subscheme. This example shows that the sheaf of a subscheme is *not* a sub-sheaf of the original scheme.

Furthermore, shows that the single point $\{0\}$ has countably many closed subscheme associated to it, depending how much "infinitesimal data" is preserved!

3. Let $I \subseteq R$ be an ideal. Then $\operatorname{Spec} R/I \to \operatorname{Spec} R$ is a closed embedding, it's image being closed as it is equal to $V(I)$, and the maps $R \to R/I$ are all surjective.

Lemma 2.2.23: Properties of Closed Embedding

1. f is a closed embedding if and only if for every affine fine open subset $\operatorname{Spec}(S) \subseteq Y$ where $f^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R), S \to R$ is surjective.
2. Every surjective ring homomorphism $S \to R$ induces a closed embedding $\operatorname{Spec}(R) \to \operatorname{Spec}(S)$
3. The composition two closed embeddings is a closed embedding.

***Proof* :**

1. Let f be a closed embedding. Then $f^{\#} : \mathcal{O}_X \to f_*\mathcal{O}_X$ is surjective (which implies it is surjective on stalks), which implies it is surjective on affine local subsets, which gives the result one way. For the other way, a map is surjective if it is surjective on open subsets that cover the space, which we get by taking the open subsets $D(\varphi)$ and taking their pre-image.
2. exercise
3. Use the two above parts to conclude the result by using the fact that the composition of surjective maps is surjective.

As closed subschemes are given by ideals for each open subsets, it would be good to describe all possible closed subschemes of a scheme by looking at an "ideal sub-sheaf" of the structure sheaf. This requires some more elaboration on the different types of sheaves that can be put on a scheme, which will be covered in section 4.1.2.

Locally Closed Subschemes

(worth defining since schemes are inherently local defined, and we would computationally often want to define a subset that is locally closed in some choice of affine open subsets)

Local Subscheme

These are subschemes which are are not sub-objects in the category **Sch**, however they still serve as an important class of schemes which we want to consider subschemes.

Definition 2.2.24: Local Subscheme

Let $(X, \mathcal{O}_X)$ be a scheme. Then if $Y \subseteq X$ is a subset such that $f : Y \hookrightarrow X$ is an embedding and $\mathcal{O}_X \to f_*(Y)$ is an isomorphism

Complete this section later.

2.2.6 Evaluation and Stalks on Schemes

In differential geometry, the stalks of a manifold can be used define the evaluation of a function. This same trick is endeavored for scheme:

Proposition 2.2.25: Stalks of Schemes

Let $\operatorname{Spec}(R)$ be a scheme. Then the stalk at $\mathcal{O}_{\operatorname{Spec}(R)}$ at the point $[\mathfrak{p}]$ is isomorphic to the local ring $R_\mathfrak{p}$.

$$\mathcal{O}_{\operatorname{Spec}(R),\mathfrak{p}} \cong R_\mathfrak{p}$$

Furthermore, for any $[\mathfrak{p}] \in X$, as $[\mathfrak{p}] \in \operatorname{Spec}(R_i)$ for appropriate R_i, we have

$$\mathcal{O}_{X,[\mathfrak{p}]} \cong \mathcal{O}_{\operatorname{Spec}(R_i),[\mathfrak{p}]}$$

***Proof* :**
For the first part, prove that $R_\mathfrak{p}$ is the colimit of R_f where $f \notin \mathfrak{p}$. The second part comes from proposition 1.1.5.

Definition 2.2.26: Local Scheme

Let X be a scheme. Then the scheme $(\operatorname{Spec} \mathcal{O}_{\operatorname{Spec} R,\mathfrak{p}}, \mathcal{O}_{\operatorname{Spec} R,\mathfrak{p}})$ is called a *local scheme* at the point $\mathfrak{p}$.

The reader should show that the local scheme at the point $\mathfrak{p}$ is the intersection of all open subsets containing $\mathfrak{p}$. Hence, local schemes are the germs of schemes. Note that as there are "points within points", namely as there are non-closed points in schemes, then there is also stalks that are subsets of stalks, in particular if $\mathfrak{p}, \mathfrak{m}$ are a prime and maximal ideal respectively where $\mathfrak{p} \subsetneq \mathfrak{m} \in \operatorname{Spec} R$ so that $[\mathfrak{m}] \in \overline{[\mathfrak{p}]}$, then $\mathcal{O}_{X,\mathfrak{p}} \cong R_\mathfrak{p}$ is a localization of $\mathcal{O}_{X,\mathfrak{m}} \cong R_\mathfrak{m}$. Due to this, notice that the local schemes in higher dimensions (section 2.6) can have local schemes contained within them. For example, the scheme $\operatorname{Spec} k[x,y]_{(x,y)}$ is the local scheme at the origin, it contain a single closed but, and a non-closed point for every irreducible curve in $\mathbb{A}^2_k$. Notice too how the local schemes are still schemes, as compared to the collection of germs at a point along with the point is not considered to be a manifold.

The main idea is that the single maximal ideal represents that there is one point at which we can evaluate for each germ. As all local rings have a maximal ideal, this motivates remembering the following notion form commutative algebra:

Definition 2.2.27: Residue Field

Let R be a commutative ring and $\mathfrak{p} \subseteq R$ a prime ideal. Then the ring of fractions of the integral domain domain $R/\mathfrak{p}$ is called the *residue field*, i.e. $\mathrm{Frac}(R/\mathfrak{p})$ is the residue field of $\mathfrak{p}$ (with respect to R). This field is often denoted $\kappa(\mathfrak{p})$.

The use of kappa in $\kappa(\mathfrak{p})$ is due to the fact that there can be many different residue fields at different points. For example in $\operatorname{Spec} \mathbb{R}[x]$, all residue fields are isomorphic to either $\mathbb{R}$ or $\mathbb{C}$ (depending if the localization is being taken at a non-maximal or maximal prime ideal).

Using residue fields, we can generalize the evaluation of a regular function (i.e. the evaluation of any element in the sections of $\mathcal{O}_X$). Recall that:

$$R_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}} \cong \mathrm{Frac}(R/\mathfrak{p})$$

as localization and taking the quotient commutes (see [8, section 20.5]). This makes it much easier to evaluate a function if we can determine R. Then as the valuation of a function is usually in $R/\mathfrak{p}$, as rational functions are now permitted on (proper) open subsets of X, the evaluation will be in $\mathrm{Frac}(R/\mathfrak{p})$. For a general point $x \in X$ in a scheme, we may diminish to the to define evaluation.

Definition 2.2.28: Evaluation of Function [on a Scheme]

Let X be a scheme, $U \subseteq X$ an open subset, and $f \in \mathcal{O}_X(U)$ a section. Then the value of f at the point $[\mathfrak{p}] \in U$ is the element:

$$f(\mathfrak{p}) := \bar{f} \in \mathcal{O}_{X,\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}} \cong \mathrm{Frac}(R/\mathfrak{p}) = \kappa(\mathfrak{p})$$

where $\mathrm{Spec}(R)$ is some affine local neighborhood of $\mathfrak{p}$.

As the quotient of any ring by a prime ideal is an integral domain, by taking its field of fraction we are forcing the codomain of each function to be a disjoint union of fields:

$$f : U \to \coprod_{\mathfrak{p}} \mathcal{O}_{U,\mathfrak{p}} \quad \text{(and in affine case)} \quad f : U \to \coprod_{\mathfrak{p} \in U} \mathrm{Frac}(R/\mathfrak{p})$$

If $\mathfrak{p}$ is maximal, then $\mathrm{Frac}(R/\mathfrak{p}) = R/\mathfrak{m}$. For certain rings, it suffices sections only over maximal points (namely Jacobson Schemes). In the case of affine Jacobson schemes, we can consider:

$$f : U \to \coprod_{\mathfrak{p} \in U} R/\mathfrak{m}$$

Varieties will be an example of Jacobson schemes. If $R/\mathfrak{m}_i \cong R/\mathfrak{m}_j$ for all maximal ideals (which shall be shown to be the case for varieties), then we can simply write:

$$f : U \to R/\mathfrak{m}$$

which recovers the notion of evaluation from classical algebraic geometry!

Example 2.2.29: Valuation On Affine Schemes
The key idea is that the stalk $R_\mathfrak{p}$ gives all function's that will not vanish at $\mathfrak{p}$, and hence can divide by the function at that point, and then taking the quotient by the maximal ideal evaluate the function at that point, equivalently considering the value of the rational function in $\mathrm{Frac}(R/\mathfrak{p})$:

1. Take $X = \mathbb{C}[x]$. Then each $f/g \in \mathcal{O}_X(U)$ is a rational polynomial where $g(\mathfrak{p}) \neq 0$ for all $\mathfrak{p} \in U$. For example if $U = D(x)$, then $h = \frac{x+1}{x} \in \mathcal{O}_X(D(x))$ and:

$$h(3) = \frac{3+1}{3} = \frac{4}{3} \in \mathbb{C} = \frac{\mathbb{C}[x]}{(x-3)} \cong \mathcal{O}_{X,3}/(x-3)$$
$$h(-1) = \frac{-1+1}{-1} = 0 \in \mathbb{C} = \frac{\mathbb{C}[x]}{(x+1)} \cong \mathcal{O}_{X,-1}(x+1)$$

As the points of $\mathbb{C}[x]$ correspond to the variety, we see that the evaluation will always be in $\mathbb{C}$, and there is no need to take the field of fractions as $\mathrm{Frac}(\mathbb{C}) = \mathbb{C}$.
Another important example, take (want the vanishing at 0 , but something feels off)

2. Take $\mathbb{A}^2_k$ for a field k where char $\neq 2$. Then in the open set U away from the y-axis and the curve $y^2 - x^5$ has in it's local section $\mathcal{O}_{\mathbb{A}^2_k}(U)$ the rational function $\left(x^2+y^2\right)/x\left(y^2-x^5\right) \in \mathcal{O}_{\mathbb{A}^2_k}(U)$. It's value at the point $(2,4)$ (i.e. $[(x-2,y-4)]$) give a values in $\mathrm{Frac}(R/(x-2,y-4)) = \mathrm{Frac}(k[x,y]/(x-2,y-4)) \cong R_{(x-2,y-4)}/\overline{(x-2,y-4)}$. Then:

$$\frac{x^2+y^2}{x\left(y^2-x^5\right)} \bmod (x-2,y-4) = \frac{2^2+4^2}{2\left(4^2-2^5\right)} \in \mathbb{C} = \frac{\mathbb{C}[x,y]}{(x-2,y-4)} \cong \mathcal{O}_{\mathbb{A}^2_k,(2,4)}$$

The value at the point $[(y)]$, which is the generic point of the x axis, is

$$\frac{x^2+y^2}{x\left(y^2-x^5\right)} \bmod (y) = \frac{x^2}{-x^6} = -1\frac{1}{x^4} \in \mathbb{C}(x) \cong \mathrm{Frac}(\mathbb{C}[x,y]/(y)) \cong \mathcal{O}_{\mathbb{A}^2_k,[(y)]}$$

3. Take $X = \mathrm{Spec}(\mathbb{Z})$ and take the open set $U = \mathrm{Spec}(\mathbb{Z})\backslash\{(2),(7)\}$. This open set has many rational function, but let's take $27/4$. Then evaluating this function at different points of U we get:

$$(5): 2/(-1) \equiv -2 \equiv 3 \bmod 5$$
$$(0): 27/4$$
$$(3): 0/1 = 0 \bmod 3$$

Notice that at 3, the function became 0. A (rational) function whose output is 0 is said to *vanish* at that point. If $\mathfrak{m}_\mathfrak{p}$ represents the maximal ideal of $\mathcal{O}_{X,\mathfrak{p}}$, then functions on an open subset U (i.e. section in $\mathscr{F}(U)$) have values at each point of U whose value lies in $\kappa(\mathfrak{p})$. A function *vanishes* at $\mathfrak{p}$ if its value at $\mathfrak{p}$ is 0. This idea is not well-defined on ringed spaces in general! There is furthermore the following properties mirroring manifold theory:

Proposition 2.2.30: Properties of Functions on Locally Ringed Spaces

Let f be function on a locally ringed space X. Then:

1. The subset of X where f vanishes is closed
2. if f vanishes nowhere, then f is invertible

Note that the property "if f vanishes everywhere, then $f = 0$ " is not one of the properties. This will be addressed in proposition 2.4 .10 .

Proof **:**

1. the subset of X where the germ of f is invertible is open
2. If $f \not\equiv 0 \bmod \mathfrak{p}$, then in particular $f \not\equiv 0 \bmod \mathfrak{m}$ for all maximal ideal, meaning f is contained in no maximal ideal. But then f must be a unit. If f where not a unit, then there would exist an $\mathfrak{m}$ that f belongs to, namely the maximal ideal containing f. But only units are not contained in any maximal ideal.

Note how this is is *not* true on a variety defined in the classical sense:

Example 2.2.31: On Variety: Vanishes Nowhere, Not Invertible
Take

$$\overline{x - 100} \in \frac{\mathbb{R}[x,y]}{(x^2 + y^2 - 1)}$$

The reader should check that this function vanishes nowhere on $V(x^2 + y^2 - 1)$ as a variety, but it is *not* in invertible (i.e it is not a unit). However, $\overline{x - 100}$ *does* vanish on Spec $\mathbb{R}[x,y]/(x^2+y^2-1)$; can the reader find what (non-maximal) prime ideal it vanishes on?

This ties in with the intuition that the spectrum of $\mathbb{R}[x,y]/(x^2 + y^2 - 1)$ would be some "gluing" of $\mathbb{C}[x,y]/(x^2 + y^2 - 1)$ via galois conjugate.

Finally, a moment should be taken to interpret evaluating at generic points. Let us consider $X = \mathbb{A}^1_k$ and let η be the generic point. Then looking at the fiber at η, we see that:

$$\mathcal{O}_{\mathbb{A}^1_k,\eta} = (k[x])_{(0)} = k(x)$$

that is, everything is inverted. The earlier intuition about $R_\mathfrak{p}$ is that we only evaluate functions *near* the point $\mathfrak{p}$ and every other $\mathfrak{q} \in \operatorname{Spec} R$ can be inverted, so this means that η is near *no* points as we can invert every point! This contrasts with the intuition that $\overline{\eta} = \mathbb{A}^1_k$ which seems to suggest that it is *close* to every point. This is where the nomenclature *generic* can be of use: it doesn't "see" what global sections capture, and can be thought of as "ignoring" any constraints given by points. This also explains the evaluation at the generic point, it returns the function itself motivating how it is not "particularly" extracting any information.

2.2.7 Independence of Affine Covering

If X is a scheme, it can be covered by affine schemes. Like manifolds, there can be multiple different coverings of X. When working with manifolds of dimension ≤ 3, every topological manifold admits a unique smooth structure (Moise's Theorem, [14]) We shall define dimensions for schemes in section 2.6, but the reader should already recall from [8] the notion of Krull dimension for a topological space and transcendental dimension for a variety. Using these, we can see even in 0 dimensions a scheme may have multiple non-isomorphic atlases (the topological space of a field is always a single closed point, but we may have multiple different sheaves on it depending on our choice of field). It thus become even more important for schemes to know when a problem can be reduced to a choice of open covering. We also don't expect to get a result as powerful as Moise's Theorem as gluing provides algebraic relations on the sheaf on X[21].

Translating between these different equivalent covering shall be given by the *Affine Communication Lemma*. Given this lemma, there shall be properties for schemes that need only be checked on some affine cover, giving rise to affine-local properties. There shall be times where this is not the case, and it will often be that the sheaf under consideration holds some arithmetic information. The affine communication lemma can be thought of as singling out properties that hold "geometric" information. In these cases, this will allow us to talk about schemes without references the particular open cover.

Lemma 2.2.32: Build-up Affine Communication

Let $\text{Spec}(R), \text{Spec}(S)$ be two affine sub-schemes of a scheme X. Then $(\text{Spec}(R)) \cap (\text{Spec}(S))$ is the union of open sets that are distinguished open sets for both $\text{Spec}(R)$ and $\text{Spec}(S)$.

Proof **:**
Let $[\mathfrak{p}] \in \text{Spec}(R) \cap \text{Spec}(S)$. As these are affine schemes, their intersection is the finite union of affine schemes, namely they are given through the appropriate distinguished open sets (recall the proof of proposition 2.2.16). Then there exits an $f \in R$ and $g \in S$ such that $[\mathfrak{p}] \in \text{Spec}(R_f), \text{Spec}(S_g) \subseteq \text{Spec}(R) \cap \text{Spec}(S)$. Then $g \in \mathcal{O}_X(\text{Spec}(S))$ restricts to an element $g' \in \mathcal{O}_X(\text{Spec}(R_f))$.

Now, the points of $\text{Spec}(R_f)$ where g vanishes are precisely the points of $\text{Spec}(R_f)$ where g' vanishes, hence:

$$\begin{aligned}\text{Spec}(S_g) &= \text{Spec}(R_f) \setminus \{[\mathfrak{p}] \mid g' \in \mathfrak{p}\} \\ &= \text{Spec}\left((R_f)_{g'}\right)\end{aligned}$$

Defining g'' where $g' = g'/f^n$ (note $g'' \in R$), then

$$\text{Spec}\left((R_f)_{g'}\right) = \text{Spec}(R_{fg''})$$

which gives the desired set.

[21] when we introduce other sheaves on X, we will run into the same situation. However the affine gluing lemma is not dependent on the structure sheaf

Theorem 2.2.33: Affine Communication Lemma

Let X be a scheme, and let P be a property that is satisfied by some affine open subsets of X set, say $\mathrm{Spec}\,(R^{(i)})$ for some index i, where $X = \cup_i \mathrm{Spec}\,(R^{(i)})$. Then if

1. for each $f \in R^{(i)}$, $\mathrm{Spec}\,(R^{(i)}_f) \hookrightarrow X$ satisfies the property too
2. If $R^{(i)} = (r_1, ..., r_n)$ and $\mathrm{Spec}\,(R^{(i)}_{r_i}) \to X$ satisfies the property, then so does $\mathrm{Spec}\,(R^{(i)}) \to X$

Then X satisfies this property.

***Proof* :**
Let $\mathrm{Spec}\,(R) \subseteq X$ be an affine subscheme. As $\mathrm{Spec}\,(R)$ is quasi-compact, by lemma 2.2.32 cover it with a finite number of distinguished open sets $\mathrm{Spec}\,(R_{f_i})$, each of which is distinguished in some $\mathrm{Spec}\,(R_i)$. By (1), each $\mathrm{Spec}\,(R_{f_i})$ has property P, and by (2) $\mathrm{Spec}\,(R)$ has property P, as we sought to show.

Definition 2.2.34: Afine Local and Stalk Local

Let X be a scheme. Then if a collection of open sets satisfy theorem theorem 2.4.2, the property is said to be *affine local*

Another similar concept that is similar to differential geometry is being able to check result "stalk-locally" (ex. local diffeomorhpisms). As mentioned in chapter 1, some conditions can be defined via stalks, others not. We formalize the notion of being stalk local for future reference:

Definition 2.2.35: Stalk-local

If a property P can be checked for X by checking it on each stalk, then the property is said to be *stalk-local*. In particular, a property P is stalk-local when X satisfies P if and only if all the stalks of X satisfies P.

Affine local does not imply stalk local:

Example 2.2.36: Affine Local But Not Stalk Local
Recall that If R is a Noetherian ring, then it's localization and extension (via exact sequence) is Noetherian. Conversely, if $R = (f_1, ..., f_n)$ and each R_{f_i} is Noetherian, so is R.

Take $X = \mathrm{Spec}\,(\prod_i^\infty \mathbb{Z}/2\mathbb{Z})$. This scheme is stalk-locally Noetherian, as the only prime ideals are those for which when taking the quotient the result is an integral domain (in this case, the quotient will always be isomorphic to a field, and hence every prime must be maximal). However, the ring is *not* Noetherian as can readily be identified, an distinguished open subsets are also not Noetherian. Hence, it is not locally-affine Noetherian.

2.2.8 Gluing Schemes

In Differential Geometry, one of the most common way of constructing a manifold, and indeed what is at the heart of geometry, is the process of gluing pieces together with some amount of consistency (see [4, chapter 1]). More generally, gluing manifolds, and gluing schemes, is a special case of a pushout (or fibered coproduct): $X \cup_Z Y$. The fibered product can be thought of as taking a product of spaces and doing some gluing (i.e. adding an appropriate equivalence-relationship), while the fibered coproduct can be thought of as taking the disjoint union of spaces and doing some gluing. Just like for manifolds, we can't glue schemes any way we want and get another scheme, we must be a bit more careful. Example 3.10 we shall see how we can "badly" glue schemes[22]. The main added condition we need is a consistency in the maps on local sections:

Proposition 2.2.37: Constructing Schemes Via Gluing

Let $\{X_i\}_{i \in I}$ be a collection of schemes. Then if there exists:

1. A collection of open subscheme, not necessarily an open cover, $X_{ij} \subseteq X_i$ (where $X_{ii} = X_i$)
2. a collection of isomorphisms $f_{ij} : X_{ij} \to X_{ji}$ (where f_{ii} would be the identity)

such that the isomorphisms agree on triple intersection[a]:

$$f_{ik}|_{X_{ij} \cap X_{ik}} = f_{jk}|_{X_{ji} \cap X_{jk}} \circ f_{ij}|_{X_{ij} \cap X_{ik}}$$

then there is a unique scheme X (up to unique isomorphism) with open subsets isomorphic to X_i respecting the above gluing conditions.

[a] this is called the *cocycle condition*

This is a translation of the techniques from differential geometry (this condition may look more familiar if written as $\mathbf{y} \circ \mathbf{x}^{-1} : \mathbf{x}(U \cap V) \to \mathbf{y}(U \cap V)$). The cocycle condition can be visualized like so:

[22] seethis link for more details

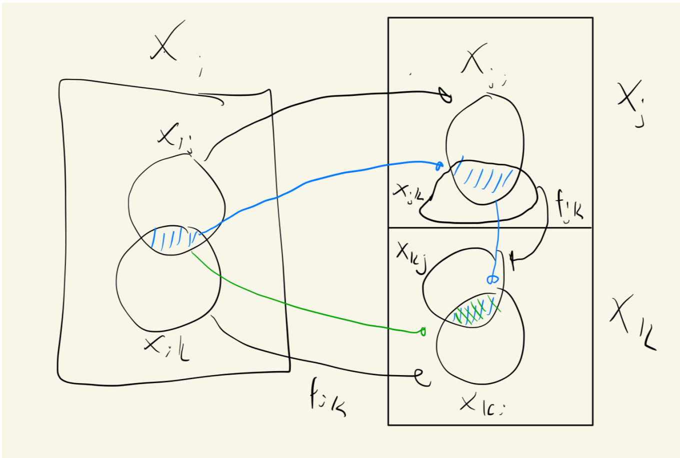


Figure 2.5: Green map must match blue map

Proof **:**
We shall present the high-level idea. Take the following topological space with the quotient topology and quotient-sheaf:

$$X = \frac{\coprod X_i}{(X_{ij} \ni x \sim f_{ij}(x) \in X_{ji})}$$

In particular, we are gluing parts of the scheme together. This gluing is well-defined precisely because of the cocycle condition, it guarantee's transitivity.

This will produce a new topological space; what needs to be shown is that this space is a Scheme. First, let $X_i = \text{Spec}\,(R_i)$ be s so that

$$X = \frac{\coprod \text{Spec}\,(R_i)}{(X_{ij} \ni x \sim f_{ij}(x) \in X_{ji})}$$

Then notice that $\pi_i : X \to \text{Spec}\,(R_i)$, $[x_i] \mapsto x_i$ is a well-defined surjective continuous map; since if $x_i \in U_{ij}$ for any j, as f_{ij} (resp. f_{ji}) is an isomorphism, there is only one point in $\text{Spec}\,(R_i)$ representing it, making the map well-defined. It is certainly surjective. For continuity, by the cocycle condition $\pi^{-1}(D(f_i))$ will be open if it intersects any X_{ij}. Then the sets U_i cover X, and I claim that $U_i \cong_{\mathbf{Sch}} \text{Spec}\,(R_i)$, proving X is a scheme in the affine case. The map is bijective as the fibers of the map π_i over each point $x_i \in \text{Spec}\,(R_i)$ have size 1. The inverse map $\iota : \text{Spec}\,(R_i) \to U_i$ mapping $x_i \mapsto [x_i]$ is also continuous since if $V_i \subseteq U_i$ is open, it is representable by points in a set $\cup_{ij} D(f_{ij})$, and so we get a homeomorphism. Furthermore, by it's construction it is naturally also a ringed-space morphism through careful application of sheaf quotients and the cocycle condition. This will give our covering.

Finally, in the general case, we take projection $\pi_i : X \to X_i$ where as $X_i = \bigcup_j \text{Spec}\,(R_{ij})$, we take the covering $\pi_i^{-1}(\text{Spec}\,(R_{ij})$, and proceed in a similar manner, which shall give the desired result.

What would the resulting global section's of a glued scheme look like? Let us do so for the case of two schemes X_1, X_2 being glued on X_{12} and X_{21} with inclusion maps $\iota_i : X_i \to X$. Then we would get:

$$\mathcal{O}_X(U) = \left\{(s_1, s_2) \;\middle|\; s_1 \in \mathcal{O}_{X_1}(\iota_1^{-1}(U)),\ s_1 \in \mathcal{O}_{X_1}(\iota_1^{-1}(U)) \text{ such that } \varphi(s_1|_{\iota_1^{-1}(V)\cap U_1}) = s_1|_{\iota_1^{-1}(V)\cap U_2}\right\}$$

In this case, we have that the global section could be represented via an appropriate tensor product (note the tensor product is the coproduct in R-algebras, where in this case R would be the ring of the intersection). In general, the global sections will be the limit of the diagram of all the inclusion maps along with all the intersection.

When thinking of the global sections of glued schemes, I like to think of the gluing maps as adding extra relations. In general, the global sections of a scheme shall be more constraint as they shall be all elements of the global sections of the glued subject to the relation imposed by the gluing map between open sets.

Example 2.2.38: Gluing [affine] Schemes

Let $X = \operatorname{Spec}(k[x])$ and $Y = \operatorname{Spec}(k[y])$ Take $D(x) \subseteq X$ and $D(y) \subseteq Y$, which is the collection of all points that do not vanish at x and y respectively. As shown in proposition ref:HERE, these subsets are isomorphic to:

$$D(x) \cong \operatorname{Spec}\left(k\left[x, \frac{1}{x}\right]\right) \qquad D(y) \cong \operatorname{Spec}\left(k\left[y, \frac{1}{y}\right]\right)$$

As a diagram, we have:

$$\begin{tikzcd} \operatorname{Spec}(k[x]) & & \operatorname{Spec}(k[x]) \\ & \operatorname{Spec}(k[x,1/x]) \arrow[ul] \arrow[ur] & \end{tikzcd} \quad \text{and} \quad \begin{tikzcd} k[x] \arrow[dr] & & k[x] \arrow[dl] \\ & k[x,1/x] & \end{tikzcd}$$

or it may be more helpful to see this as:

$$\begin{tikzcd} \operatorname{Spec} k[x] & & \operatorname{Spec} k[y] \\ & \operatorname{Spec} k[x,1/x] \cong \operatorname{Spec} k[y,1/y] \arrow[ul] \arrow[ur] & \end{tikzcd}$$

and

$$\begin{tikzcd} k[x] \arrow[dr] & & k[y] \arrow[dl] \\ & k[x,1/x] \cong k[y,1/y] & \end{tikzcd}$$

where the cocycle condition must be satisfied by the isomorphisms. Notice how these are similar to restriction conditions: we are "manually" figuring out how the restrictions should work so that by uniqueness of gluing local sections on a sheaf we can get the global sections. These two open subsets can be glued in at least two different ways to produce two non-s that will be familiar to the reader: familiar to the reader:

1. (Spaces with double-origins). Define the isomorphism $D(x) \cong D(y)$ via the isomorphism $k[x, 1/x] \cong k[y, 1/y]$ with defined by mapping $x \to y$. This is a scheme by proposition 2.2.37. Label this scheme S_1. This is *not* an affine scheme. Let us show that the global sections are identical to those of $\mathbb{A}_k^1$, and so by proposition 2.2.13 we would $\mathbb{A}_k^1 \cong S_1$, but that is impossible as can be seen by a topological argument.

   Let's find the global sections. Take $p(y) \in k[y]$. Then the restriction map $\operatorname{res}_{\mathbb{A}_k^1, D(y)}$ maps it to $\frac{p(y)}{1}$. The gluing isomorphism maps it to $\frac{p(x)}{1}$. Hence, we have the relationship $p(x) = p(x)$. But this relationship is "trivial"; it adds no extra restrictions. We conclude that the global sections are just $k[x]$. The same can done in higher dimensions (ex. plane with double-origin, more generally $\mathbb{A}_k^n$ with double origin).

   In differential geometry, such spaces are example of non-Hausdorff spaces, however all [non-trivial] schemes are not Hausdorff. This behaviour is not due to a lack of Hausdroffness, but due to the lack of a different condition called *separatedness*. Spaces with double-origins are not separated, while $\mathbb{A}_k^n$ will be an example of a separated space (as a hint: show that in the spaces with double origin, there exists two affine spaces whose intersection is not affine, it is a space that was already covered)

   For a more classical proof (if you're comfortable with Hartog's lemma, or a continuity argument), is that any collection of polynomials that vanish every except maybe at $(x, y) = 0$ must in fact also vanish at 0.

2. (projective space) Define the isomorphism $D(x) \cong D(y)$ via the isomorphism $k[x, 1/x] \cong k[y, 1/y]$ with defined by mapping $x \to 1/y$. The reader will recognize this as $\mathbb{P}_k^1$. When $k \in \{\mathbb{R}, \mathbb{C}\}$, this shape is isomorphic to a circle and a sphere, while in higher-dimension, say n it is an example of a space that cannot be embedded in $\mathbb{R}^n$ or $\mathbb{C}^n$ respectively. For schemes, we must be more prudent given the generic zero point (0) and the generic points given by non-linear irreducible polynomials (recall that the linear terms that are galois conjugates are glued together).

   Let us check that $\mathbb{P}_k^1$ is not affine. Consider the global section $\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1)$. By construction, these are the global sections over X and global sections over Y that agree on the overlap. The sections on X and Y are the polynomials $f(x)$ and $g(y)$ respectively. For them to agree, we would need that, after passing through the isomorphism, $f(x) = g(1/x)$. But such polynomials are only the constant polynomial. Hence:

$$\mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1) = k$$

   Now by proposition 2.2.13 we have

$$\operatorname{Spec} \mathcal{O}_{\mathbb{P}^1}(\mathbb{P}^1) = \operatorname{Spec}(k) = \{0\}$$

   However, the projective line certainly has more than one point, completing the proof[a].

[a] Those familiar with complex analysis may recall that the projective space ha only constant holomorphic functions defined on it

Exercise 2.2.2

1. Let X be the line with two origins, $Y = \mathbb{A}_k^1$ the affine line, and $\pi : X \to Y$ the natural projection. Compute $f_*(\mathcal{O}_X)_0$, the talk of the pushforward at 0.

2. Let X be a scheme. Show that the affine open subsets form a base.
3. A subset Z of a scheme X is closed if and only if there exists an open cover of X by affine subschemes, $\{U_i = \mathrm{Spec}(A_i)\}_{i \in I}$, such that for every $i \in I$, the intersection $Z \cap U_i$ is a closed subset of the affine scheme U_i. In other words, a closed subset of X is characterized as being locally closed on an affine open covering.

2.3 Projective Schemes

Recall projective space from [4]. In [8] a few key properties of projective space, namely:

1. Projective space was constructed by gluing together affine spaces. This construction satisfies the co-cycle condition quickly when $k = \overline{k}$ is algebraically closed, but we must (1) be more careful in the general case of a ring and (2) try to construct it in a more "natural" way without any choice of coordinates.
2. There shall be an equivalent of a homogeneous R from which we can construct the scheme $\mathrm{Proj}\, R$. Just like in the classical setting, R shall not be the set of regular functions on R, the global sections of $\mathrm{Proj}\, R$ shall be constant, and $\mathrm{Proj}\, R$ shall behave like the "compact" sets in algebraic geometry (namely the proper sets, as mentioned in [8]).

   We shall show that R will be associated to a different sheaf over $\mathrm{Proj}\, R$ that is not the structure sheaf, see section 4.2.
3. The properties of Bézout's theorem shall turn out to be the special case of the theory of *divisors* on projective space. See section 4.3 for more details.

Classically, n-projective space $\mathbb{R}\mathrm{P}^n$ or $\mathbb{C}\mathrm{P}^n$ is defined on $\mathbb{R}^{n+1}$ or $\mathbb{C}^{n+1}$ by taking an equivalence class:

$$(x_1, ..., x_{n+1}) \sim (y_1, ..., y_{n+1}) \qquad \Leftrightarrow \qquad (x_1, ..., x_{n+1}) = (\lambda x_1, ..., \lambda, y_{n+1}) \qquad \lambda \neq 0$$

This produces an n-dimensional smooth manifold with the usual patches defined as:

$$U_i = D(x_i) = \{[x] \in \mathbb{P}^n \mid x_i \neq 0\}$$

where the gluing was defined as

$$\begin{aligned} \varphi_i \circ \varphi_j^{-1}(x^1, x^2, ..., x^n) &= \varphi_i(x^1 : \cdots x^{j-1} : 1 : x^{j+1} : \cdots : x^n) \\ &= \left(\frac{x^0}{x^i}, ..., \frac{x^{j-1}}{x^i}, \frac{1}{x^i} \frac{x^{j+1}}{x^i}, ..., \frac{x^{i-1}}{x^i}, \frac{x^{i+1}}{x^i}, ..., \frac{x^n}{x^i}\right) \end{aligned}$$

In [8], we saw we can do the same for any field k, namely since inverses exist. Let us do the same but with a general graded $\mathbb{Z}$-ring R and $S = R[x_0, x_1, \ldots, x_n]$ and $S_+ \subseteq S$ being the positive-degree elements. Note that we are taking a $\mathbb{Z}$-grading instead of an $\mathbb{N}_{\geq 0}$ grading like in [8] as we shall often be working with rational functions that have negative degrees[23]

[23] More general gradings are sometimes useful, however we shall not need this either in this book or for awhile, I'll get back to this later

As a set, define:

Definition 2.3.1: Projective Space – as a Set

$$\text{Proj } S = \{[\mathfrak{p}] \big| \mathfrak{p} \subseteq S \text{ is a homogeneous prime not containing an irrelevant ideal } \}$$

Recall an irrelevant ideal is an ideal $I_s = (x_0^s, \ldots, x_n^s)$ for $s > 0$, as it is the zero set of $[0 : \cdots : 0]$ which doesn't exist in projective space. Then Proj S has the Zariski topology induced by homogeneous functions of positive degree:

$$V_+(T) = \{[\mathfrak{p}] \in \text{Proj } S \big| T \subseteq \mathfrak{p}\}$$

The basis for the topology can be given by the positive degree homogeneous elements:

$$D_+(f) = \{[\mathfrak{p}] \in \text{Proj } S \big| f \notin \mathfrak{p}\}$$

If $f = x_i$, we would like this to correspond to the usual affine space $\mathbb{A}_R^n$. In the classical case, we got this isomorphism by doing:

$$\begin{aligned}
\{[x_0 : \cdots : x_n] \mid x_i \neq 0\} &= \left\{ \left[\frac{x_0}{x_i} : \ldots : 1 : \frac{x_n}{x_i}\right] \middle| \; x_i \neq 0 \right\} \\
&\cong k\left[\frac{x_0}{x_i}, \ldots, \frac{x_n}{x_i}\right] \\
&= k\left[x_{0,i}, \ldots, x_{n,i}\right] && \text{relabeling} \\
&= \mathbb{A}_k^n
\end{aligned}$$

Notice how these are all degree 0 in the grading of S. These sets are not quite the same as Spec R_f, however it turns out that choosing $D_+(f)$ and working with graded rings give us exactly the desired result:

Lemma 2.3.2: Characterizing Basic Open Sets For Projective Space

$$D_+(f) \cong \text{Spec}\left(S_{(f)}\right)_0$$

Proof **:**
Let us establish a bijection between these two sets, and then show they are homeomorphic. First, there is a bijective correspondence between the relevant homogeneous $\mathfrak{p} \in S$ where $f \notin \mathfrak{p}$, and $\mathfrak{p} \subseteq S_{(f)}$ (see [8]). Let us now show there is a bijective correspondence between the primes $S_{(f)} = A$ and $\left(S_{(f)}\right)_0 = A_0$. First, the map $A_0 \to A$ injectively corresponds the prime ideals of A_0 with A via $\mathfrak{p} \mapsto \mathfrak{p}^e$ (the extension of primes). To show the map is surjective, for each $\mathfrak{p}_0 \subseteq A_0$ define $\mathfrak{p} \subseteq A$ as:

$$\mathfrak{p} = \bigoplus_i \mathfrak{q}_i \quad \text{where} \quad \mathfrak{q}_i \subseteq A_i, a \in \mathfrak{q}_i \text{ if and only if } \frac{a^{\deg f}}{f^i} \in \mathfrak{p}_0$$

In the case where $\deg f = 1$, we simply get $\frac{a}{f^i} \in \mathfrak{p}_0$. Note that $\mathfrak{q}_0 = \mathfrak{p}_0$, and hence if we show that $\mathfrak{p}$ is a homogeneous ideal in A, we have our desired bijective correspondence. But this almost immediately comes by construction. If $a \in \mathfrak{q}_i$, then $a^2 \in \mathfrak{q}_{2i}$, and if $a, b \in \mathfrak{q}_i$, then $(a+b)^2 = a^2 + 2ab + b^2 \in \mathfrak{q}_{2i}$, and so $a + b \in \mathfrak{q}_i$, showing it is closed under addition. A similar reasoning shows it is closed under scalar multiplication, giving us an ideal. If $ab \in \mathfrak{p}$, then $ab \in \mathfrak{q}_i$ for some i, which implies

$$\frac{(ab)^{\deg f}}{f^i} \in \mathfrak{p}_0 \quad \Rightarrow \quad \frac{ab}{f^i} \in \mathfrak{p}_0$$

and using that $\mathfrak{p}_0$ is prime, $a, b \in \mathfrak{p}$. We thus have our bijective correspondence, as we sought to show.

Let us apply this to $f = x_i$:

$$\begin{aligned} D_+(x_i) &= \left\{[\mathfrak{p}] \in \operatorname{Proj} S \middle| x_i \notin \mathfrak{p}\right\} \\ &= \left\{[\mathfrak{p}] \in \operatorname{Proj} S \middle| x_i \notin \mathfrak{p} \cap S_0\right\} \\ &\cong \operatorname{Spec}\left(S_{(x_i)}\right)_0 \\ &\cong \operatorname{Spec} \text{degree } 0 \text{ elements of } R\left[x_0, \ldots, x_n, 1/x_i\right] \\ &\stackrel{!}{=} \operatorname{Spec} R\left[\frac{x_0}{x_i}, \ldots, \frac{x_i}{x_i}, \ldots, \frac{x_n}{x_i}\right] \\ &\cong \operatorname{Spec} \frac{R\left[x_{0,i}, \ldots, x_{n,i}\right]}{(x_{i,i} - 1)} \\ &\cong \mathbb{A}^n_R \end{aligned}$$

where $\stackrel{!}{=}$ comes from the fact that $1/x_i$ is now a unit and a polynomial of degree -1, hence x_j/x_j is of degree 0, and these elements generate $\left(S_{(x_i)}\right)_0$ over R. Relabeling x_j/x_i as $x_{j,i}$ and "omitting" $x_{i,i}$ as this is equal to one (equivalently, we "set" it to one by taking the quotient, which is similar to de-homogenizing), we see that it is isomorphic to the usual $\mathbb{A}^n_R$, and hence we recover the usual affine cover! Note the important of taking the degree 0 component, for

$$\operatorname{Spec} R\left[x_0, \ldots, x_n, 1/x_i\right] \cong \mathbb{A}^{n+1}_R \setminus \{x_i = 0\}$$

Let us now construct the "structure sheaf" on Proj S. We shall see that Proj S is not affine (we hope not as the classical projective variety is not affine!), and so it isn't exactly correct to say it is the structure sheaf, however we see that this scheme very naturally behaves a lot like s, justifying the terminology. As we just saw, each $D_+(f)$ is affine and hence we certainly have an affine cover. Let us ensure that the gluing satisfies proposition 2.3.19, namely that all intersections glue appropriately to form a scheme.

Theorem 2.3.3: Structure Sheaf On Proj

Take the Zariski topology on Proj S given by the basis $D_+(f) \cong \operatorname{Spec}\left(S_{(f)}\right)_0$. Then Proj S this is a scheme.

***Proof* :**
Let $f, g \in S$ be homogeneous polynomials and Lake $D_+(f), D_+(g)$ with non-trivial overlap. It is not difficult to show that $D_+(f) \cap D_+(g) = D_+(fg)$. We shall show that the cocycle condition given by proposition proposition 2.2.37. Take

$$D(g^{\deg(f)}/f^{\deg g}) \subseteq D_+(f) \qquad \text{and} \qquad D(f^{\deg(g)}/g^{\deg f}) \subseteq D_+(g)$$

We shall show that:

$$((S_\bullet)_f)_0 \left[\frac{1}{(g^{\deg f}/f^{\deg g})}\right] \cong ((S_\bullet)_{fg})_0 \cong ((S_\bullet)_g)_0 \left[\frac{1}{(f^{\deg g}/g^{\deg f})}\right]$$

This will induce the desired isomorphism on spectra. First off, there is a natural map:

$$\varphi : ((S_\bullet)_f)_0 \to ((S_\bullet)_{fg})_0$$

mapping $h/f^k \mapsto hg^k/(fg)^k$. Taking $U = \left\{(g^{\deg(f)}/f^{\deg(g)})^k \;\middle|\; k \in \mathbb{N}\right\}$, notice that $\varphi(U)$ are units, thus there is a natural map

$$\widetilde{\varphi} : ((S_\bullet)_f)_0 \left[\frac{1}{(g^{\deg f}/f^{\deg g})}\right] ((S_\bullet)_{fg})_0$$

It is straight-forward to verify that this map is in fact an isomorphism (injectivity is easy, and the degree's matching is enough for surjectivity). The same argument works in the other direction, giving the desired result. Thus, we have established the desired ring isomorphism, which induces the required isomorphism of schemes, completing the proof.

Definition 2.3.4: Projective Scheme

Let $S_\bullet$ be a $\mathbb{Z}$-graded ring where $S_0 = R$. Then $\operatorname{Proj} S_\bullet$ is called the projective scheme associated to $S_\bullet$. If $S = R[x_0, \ldots, x_n$, then the above construction defines the classical projective n-scheme or prototypical projective scheme over R, and is denoted $\mathbb{P}^n_R$ or $\mathbb{P}^n$ if unambiguous.

A quasiprojective R-scheme is open subscheme of a projective scheme.
Notice that it the scalars can be defined for any ring, which generalizes the result from classical algebraic geometry. If $R = k$ is an algebraically closed field, the closed points of $\mathbb{P}^n_k$ will behave late the traditional points of projective space defined in classical algebraic geometry.

Proposition 2.3.5: Global Sections of Proj $S\bullet$

Let $S_\bullet$ be a $\mathbb{Z}$-graded ring where $S_0 = R$, and let $X = \operatorname{Proj} S_\bullet$. Then:

$$\mathcal{O}_X(X) = R$$

Hence $\mathbb{P}^n_R$ is not an affine scheme (the case for $n = 1$ was given in example 2.2.38)

***Proof* :**
exercise

Recall from [8] that homogeneous polynomials cut out sets from $\mathbb{P}^n$. This is because if $a \in R^{n+1}$ is the root of a homogeneous polynomial, so are all multiples of a. In modern algebraic geometry, we consider the global sections to be the "functions" that are defined on all of $\mathbb{P}^n_R$. However, $\mathcal{O}_{\mathbb{P}^n_R}(\mathbb{P}^n_R) = R$. So how do we interpret S? We shall see in proposition 4.2.7 that S form a different sheaf on $\mathbb{P}^n_R$, in particular it will be a *line bundle* on $\mathbb{P}^n_R$.

The reader ought to remember that if $f(x_{0,i}, ..., x_{n,i})$ is a homogeneous polynomial on $D(x_i)$, then when converting coordinates to $D(x_j)$ we get:

$$f(x_{0,i}, ..., x_{n,j}) = x_{j,i}^{\deg f} f(x_{0,j}, ..., x_{n,j})$$

Example 2.3.6: Projective Scheme Homogeneous Polynomial Cut
There are many example in [8], and hence we shall just put a few as a refresher.

1. Take $C = V\left(x^2 + y^2 - z^2\right) \subseteq \mathbb{P}^2_k$ for an algebraically closed field k. Then if $z = 0$, we would get $[(0,0,0)] \in C$, however this point doesn't exist in $\mathbb{P}^n_k$. Hence consider $z \neq 0$. Then for each solution $[(a,b,c)] = [(\lambda a, \lambda b, \lambda c)] \in C$. In k^3, we have that we get a cone of points that are solutions. Loosely visualizing $\mathbb{P}^2_k$ as a sphere (we should glue the dichotomous points) we see that this resemble the image:

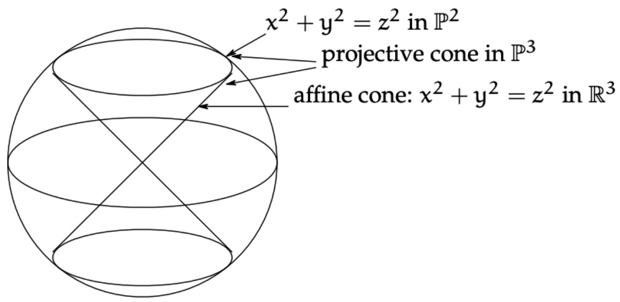


Figure 2.6: Visualization of $V(x^2 + y^2 - z^2)$

such vanishing sets are called *vanishing cones*.

2. $\mathbb{P}^0_R = \operatorname{Proj} R[x_0] \cong \operatorname{Spec} R$, hence all are 0-dimensional projective schemes. Note then that $\mathbb{P}^0_k = \{\overline{0}\}$ since $\operatorname{Spec} k$ is a singleton; or this can be verified directly by noticing that the only non-irrelevant homogeneous ideal of $k[x]$ is (0), for all other homogeneous prime ideals are of the form (x^n).

3. For an example that is not a hypersurface, take $V(wz - xy, x^2 - wy, y^2 - xz)$. This is called the *twisted cubic curve*. This ideal is prime, and the variety is isomorphic to $\mathbb{P}^1_k$ (hint: consider $k[w,x,y,z] \to k[s,t]$, $(w,x,y,z) \mapsto (s^3, s^2t, st^2, t^3)$.

4. In an affine scheme, two polynomials over a field cut out the same shape if they they differ by a scalar. This is not the case for homogeneous polynomials cutting out parts of $\mathbb{P}^n$,

consider $x^2y = 0$ and $xy^2 = 0$. the problem is that these two are homogenizations of different polynomials that produce the same set. However, as we know from section 2.2.2 schemes remember multiplicity. This behaviour shall later be properly taken into account by saying that two polynomials shall cut out different *closed subscheme* (rather than the same *closed subset*). With this addendum, it shall once again be the case that two polynomials cut out the same closed subscheme if and only if they differ by a scalar multiple, as closed schemes shall essentially correspond to polynomial cut outs (as sets) that have sheaves that remember multiplicity information.

Proposition 2.3.7: Irreducibility of classical projective scheme

The scheme $\mathbb{P}^n_R$ is irreducible and is of finite type.

As irreducible spaces are connected, $\mathbb{P}^n_k$ is connected.

Proof **:**
Recall that the affine schemes U_i are irreducible. Notice the argument is invariant under gluing.

To conclude this section, let us formally define the affine and projective cone:

Definition 2.3.8: Affine Cone

Let $S_\bullet$ be a graded ring. Then the affine cone of $\operatorname{Proj} S_\bullet$ is $\operatorname{Spec}(S_\bullet)$.

Note that $\operatorname{Spec}(S_\bullet)$ is dependent on $S_\bullet$, not just $\operatorname{Proj}(S_\bullet)$. Think of the affine cone as eliminating the points at infinite and adding the origin. In fact, given a field k, there is a natural map Spec $S_\bullet \backslash V(S_+) \to \operatorname{Proj} S_\bullet$ (more generally if $S_\bullet$ is a finitely generated graded ring over a base ring R, there is a natural morphism).

Definition 2.3.9: Projective Cone

Take Proj $S_\bullet$. Then projective cone or projective completion is $\operatorname{Proj}(S_\bullet[t])$ for some indeterminate t of degree 1 .

Take for example $\operatorname{Proj}\left(k[x,y,z]/\left(z^2 - x^2 - y^2\right)\right.$. Then the projective cone is $\operatorname{Proj}\left(k[x,y,z,t]/\left(z^2 - x^2 - y^2\right)\right.$. Taking $t = 0$ returns the original projective scheme, and taking the compliment (the distinguished set $D(t)$) gives $\operatorname{Spec}(S_\bullet)$. This construction can be thought of as the opposite of the affine cone, where we attach the points at infinity of a conic curve.

2.4 Properties of Schemes

In this section, we shall cover various different properties a scheme may have. In the process, we shall see if these properties can be checked affine-locally or stalk-locally, that is whether it suffices to check on an open cover, or if it suffices to check on stalks.
Many of the topological properties inquired about s can be asked about for general schemes. For

example, general schemes need not be quasicompact, while all s are quasicompact. Properties that have been already are:

1. (dis)connected: In the affine case, this corresponds to the existence of non-trivial idempotent elements, for if $e \in R$ is such an element then $R \cong eR \times (1-e)R$, and so the spectrum is a disjoint union. For the general case of schemes, it is usually given by the disjoint union of known schemes.

2. (ir)reducible: In the affine case, this corresponds to the ring having *no zero divisors*, i.e. an integral domain. Proposition 2.4.19 shall characterize irreducible schemes[24].

   Recall that an equivalent conditions to irreducibility is that every non-empty open subset is dense (as the reader should verify).

3. quasicompact: Essentially all schemes we shall work with will be quasicompact (with the "trivial" exception being the infinite disjoint union of schemes). Most of the time, non-quasicompact schemes will come from some infinite gluing.

Finiteness Conditions

The most common types of rings in commutative algebra are Noetherian rings. The geometric equivalent is the following:

Definition 2.4.1: Noetherian Scheme

Let X be a scheme. If X can be covered by affine open sets where each ring R_i of $\operatorname{Spec}(R_i)$ is Noetherian, then X is said to be a *locally Noetherian Scheme*. If X is also quasicompact, then X is said to be a *Noetherian Scheme*.

All coordinate rings are Noetherian, and hence we may think of varieties as examples of Noetherian schemes[25]. We can think of the Noetherian condition as insuring that the open affine don't have "difficult" infinite behavior. Most examples which involve non-Noetherian rings involve infinitely many variables (ex. $k[x_1, x_2, ...]$, $k[x, y, y/x, y/x^2, ...]$ and so forth). In part I; examples of non-Noetherian topological spaces we shall encounter will be the fibered product of Noetherian rings (see ref:HERE), and the special product topology put on the ring of adèles. In general, we shall see that there are many results that do not require finite generation, and hence it is worth keeping track of it when this condition is necessary.

One important consequence of being Noetherian is the existence of finitely many irreducible components:

Lemma 2.4.2: Finitely Many Components

Let X be a Noetherian scheme. Then X has finitely many irreducible components.

[24] with the reducedness condition insuring no "fuzz" in the sheaf
[25] Once we introduce theorem 3.1.19 this will be made precise

***Proof* :**
As X is Noetherian, it is locally Noetherian, so let it be covered be Noetherian affine open schemes. As it is quasicompact, we may take a finite subcover so $X = \bigcup_i^n \operatorname{Spec} R_i$. As each R_i is Noetherian, it can have only finitely many irreducible components (check this), and hence their union can have only finitely many irreducible components, completing the proof.

This is not true if the ring is not Noetherian: the spectrum of the ring

$$\varinjlim_{n\to\infty} \frac{k[x_i \,|\, 1 \leq i \leq n]}{\prod_i^n x_i}$$

has infinitely many components, but it is a quasicompact scheme (as it is affine).

Let us next turn to an important property of quasicompact schemes. When working with affine schemes, we always have that there exists closed points corresponding to the maximal ideals of the ring. When working over general schemes, there may be no closed points[26]. Fortunately, all quasicompact scheme have closed points, and even better:

Proposition 2.4.3: Quasicompact Schemes and Closed Points

Let X be a quasicompact scheme. Then X contains closed points. Furthermore, the closure of any non-closed point contains closed points.

***Proof* :**
First, if X is affine, it certainly contains closed points as $\mathcal{O}_X(X) = R$ contain at least one maximal ideal. Furthermore, the closure of any non-closed point would contain maximal ideal as any prime ideal is contained in a maximal ideal.

Now, let X be some quasicompact scheme. Let $X = \bigcup_i^n U_i$ be a finite covering of affine open subsets. Take any closed point $\mathfrak{p}_1 \in U_1$. If it is closed in X we're done. If not, take its closure $\overline{\mathfrak{p}_1}$ in X. As $\mathfrak{p}_1$ is not closed in X, its closure contains more than $\mathfrak{p}_1$, call this point $\mathfrak{p}_2$, and without loss of generality say it is in U_2. If $\mathfrak{p}_2$ is closed in X, we are done. If not, take $\overline{\mathfrak{p}_2}$ in X. Its closure must contain a point $\mathfrak{p}_3$ that is not in U_2 or U_1 (as $\mathfrak{p}_3$ would be in the closure of $\mathfrak{p}_1$ or $\mathfrak{p}_2$ in U_1 and U_2 respectively). Continue this process until a point is closed, or if we get to $\mathfrak{p}_n \in U_n$, then the closure of this point can't contain any other point in $U_1, \ldots, U_n$, making $\mathfrak{p}_n$ closed and completing the proof.
Note that $\mathfrak{p}_n$ is in the closure of $\mathfrak{p}_1$ in X : by a topological argument it is easy to show that:

$$\overline{\mathfrak{p}_1} \supseteq \overline{\mathfrak{p}_2} \supseteq \cdots \supseteq \overline{\mathfrak{p}_n} = \mathfrak{p}_n$$

and hence the closure of any non-closed point contains a closed point.

Note that this does not mean that the closed points are dense, that is it does not imply that the closure of the collection of closed point is the entire space. Such schemes are called Jacobson schemes and are related to Jacobson rings and Jacobson radicals (see [8]). Once we introduce (abstract) varieties, we shall see that the closed points are dense for varieties. The following gives a good set of (affine) schemes whose closed points are dense:

[26]see https://math.stanford.edu/~vakil/files/schwede03.pdf

Lemma 2.4.4: Dense Closed Points

Let A be a finitely generated k-algebra. Then the closed points of $\operatorname{Spec} A$ are dense

Proof **:**
It suffices show that for each $f \in A$ where $D(f) \neq \emptyset$ (i.e. f is not nilpotent, see exercise ref:HERE), $D(f)$ $\operatorname{Spec}(A_f) \cong D(f)$, and A_f certainly contains maximal ideals, there is a maximal ideal $\mathfrak{p}_f \subseteq A_f$ which corresponds to a (at least) prime ideal $\mathfrak{p} \subseteq A$ not containing f. If we show that $A/\mathfrak{p}$ is a field, we're done. By Zariski's lemma, F is a finitely generated k-algebra if it is a finitely generated k-module, hence it suffice to show that $A/\mathfrak{p}$ is a finitely generated k-vector space. As $(A/\mathfrak{p})_f \cong A_f/\mathfrak{p}_f$, if we show that $A_f/\mathfrak{p}_f$ is a finite dimensional k-vector space, we can lift to a finite basis for $A/\mathfrak{p}$.

Now, as $A_f = A[1/f]$, it is a finitely generated k-algebra and $\mathfrak{p}_f$ is maximal, $A_f/\mathfrak{p}_f$ is a finite dimensional k-vector space by Zariski. But then we get the desired conclusion

We can also show it in the following more "straightforward" with a more subtle application of Nullstellensatz. For the sake of contradiction, suppose $D(f)$ has no maximal ideals. Then for each maximal ideal $\mathfrak{m} \subseteq A$, $f \in \mathfrak{m}$. Thus:

$$f \in \bigcap_{\mathfrak{m} \subseteq A} \mathfrak{m} \overset{!}{=} \sqrt{(0)}$$

where $\overset{!}{=}$ comes as a consequence of the Nullstellensatz (key here is the finitely generation as a k-algebra). But then f is nilpotent, and so $D(f) = \emptyset$. Hence, any nonempty set contains a closed point, completing the proof.

Recall in proposition 2.3.18 it was mentioned that invertibility can be determined if a function is non-zero at every point, however a function being zero at all points requires more care to study. Part of this is because this is true for quasicompact schemes

Proposition 2.4.5: Quasicompact Schemes and Vanishing Function

Let X be a quasicompact scheme. Then if $f \in \mathcal{O}_X(X)$ vanishes at all points in X, then there exists an n such that $f^n = 0$.

Proof **:**
Say $f([\mathfrak{p}]) = 0$ so that $f \in \mathfrak{p}$ for all $\mathfrak{p}$. Then on $U_i \cong \operatorname{Spec} R_i \subseteq X, f \in \sqrt{R_i}$, implying $f^{n_i} = 0$ for some n. Cover X with these open sets, and by Quasicompactness we only require finitely many such sets, where in each set we have an f_i and n_i such that $f_i^{n_i} = 0$. As X is a scheme, on intersection $U_i \cap U_j$ we have $g_{i,j} = f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ and as ring homomorphisms must map 0 to $0, g_{i,j}^{\max n_i, n_j} = 0$. Take $n = \max_i n_i$ so that for each intersection $U_i \cap U_j$:

$$f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j} = g_{i,j} \quad g_{i,j}^n = 0$$

Now, as these agree on intersections and cover X, by definition there must exist a function $f \in \mathcal{O}_X(X)$ such that $f|_{U_i} = f_i$. Then as $\operatorname{res}_{X,U_i}(f^n) = \operatorname{res}_{X,U_i}(f)^n = f_i^n f_i^{n_i n'} = 0 f^{n'} = 0$, we have:

$$(f^n)|_{U_i} = \left(f|_{U_i}\right)^n = 0$$

which implies $(f^n)|_{U_i} = 0 \in R_i$. Then by guability, it must be that $f^n = 0$, completing the proof.

This is not necessarily true for a general scheme, namely since we can have nilpotence of every degree:

Example 2.4.6: Non-vanishing Function On Scheme

Take

$$X = \coprod_{n \geq 1} \operatorname{Spec}\left(k[x]/\left(x^n\right)\right)$$

Then the global function ϵ is zero every-where, namely there is only one point to evaluate to, but essentially by definition there cannot exist an n such that $\epsilon^n = 0$.

Lemma 2.4.7: Projective Scheme is Noetherian

Let $X = \mathbb{P}^n_{\mathbb{Z}}$ or $\mathbb{P}^n_k$. Then X is Noetherian.

Proof **:**
exercise.

To conclude this section, we shall define one more type of scheme here due to the fact that all affine and projective schemes have this property, however it shall not be of great use to us until chapter ref:HERE. The following establishes some finiteness conditions on the intersection of important open subsets:

Definition 2.4.8: Quasi-separated

Let X be a topological space. Then X is *quasiseparated* if the intersection of any two quasicompact open subsets is quasicompact.

Proposition 2.4.9: Characterizing Quasi-separated Schemes

Let X be scheme. Then X is quasi-separated if and only if the intersection of any two affine open subsets is a finite union of affine open subsets

Proof **:**
This is very similar to a compactness argument and is left as an exercise

Proposition 2.4.10: Affine Scheme and Quasiseparatedned

Let $X = \operatorname{Spec} R$ be an affine scheme. Then X is quasiseparated.

Proof **:**
The intersection of affine open subsets is affine (example ref:HERE shows that $U \cap V \cong \operatorname{Spec}(S \otimes_R S')$ for $U, V \subseteq \operatorname{Spec} R$ and $U \cong \operatorname{Spec} S$, $V \cong \operatorname{Spec} S'$. Then by quasi-compactness this can be covered by finitely many distinguished open subsets.

From this, we can reduce to the case of affine opens and the rest follows quickly.

A large family of schemes that are quasi-separated and quasicompact are projective R-schemes, and as we shall see many schemes we shall be working with can be embedded into a projective R-scheme, and hence these two finiteness properties will be very common. There do exist non quasi-separated schemes:

Example 2.4.11: Non-quasiseparated Scheme
Let $X = \operatorname{Spec}(k[x_1, x_2, \ldots])$ and let $U = X \backslash [\mathfrak{m}]$ where $\mathfrak{m}$ is the maximal ideal $(x_1, x_2, \ldots)$. Take two copies of X and glue along U to create a double-origin scheme, namely the infinite version of the double-origin scheme seen in example 2.15. Show this is not quasiseparated.

Lemma 2.4.12: Projective Scheme Quasicompact And Quasiseperated

Let X be a projective R-scheme. Then X is quasicompact and quasiseparated

Note that R need not be Noetherian, hence any projective $\mathbb{P}^n_R$ scheme is quasicompact.

Proof **:**
hint: affine communication lemma.

Exercise 2.4.1

1. Let X be a scheme. Show that X is a locally Noetherian scheme if and only if every open affine subset corresponds to a Noetherian ring.
2. Show that $\prod_i^\infty \operatorname{Spec} R_i$ properly embeds in $\operatorname{Spec}\left(\prod_i^\infty R_i\right)$. More precisely, the right hand side is (significantly) larger than the left hand side.

2.4.1 Reducedness and Integrality

The next important property that general rings have that coordinate rings for varieties don't have is the presence of zero-divisors and nilpotence. A ring with zero-divisors can be decomposed into a product to two rings and corresponds to a function vanishing on separate irreducible components (there must be more than 1), while a ring with nilpotence have effect on the sheaf of a ring as discussed in section 2.2.2. A ring with no nilpotent elements is said to be *reduced* (note that it may have zero-divisors[27]). A scheme that has this property of being reduced for every ring of sections is given a name:

Definition 2.4.13: Reduced Scheme

Let X be a scheme. Then X is a *reduced scheme* if $\mathcal{O}_X(U)$ is reduced for every open set U of X

Naturally, on a reduced scheme by proposition 2.4.5 a global section that evaluates to zero at every point will be the zero-section.

Proposition 2.4.14: Reducedness Is Stalk-local

Let X be a scheme. Then X is reduced if and only if non of the stalks have nonzero nilpotents. As a consequence, if $f, g \in \mathcal{O}_X(U)$ are two functions and agree on all points, then $f = g$

***Proof* :**
if X is a reduced scheme, then the localization will preserve reducedness, and hence each stalk is reduced. For the converse, we can show the contrapositive and assume there exists a $f \in \mathcal{O}_X(U)$ such that $f^n = 0$. Note that f cannot be a unit (as the reader could verify). Then some affine open subset $\operatorname{Spec} R \subseteq U$ we have $f|_U^n = f^n|_U = 0|_U$, showing that f is still nilpotent in $\mathcal{O}_X(\operatorname{Spec} R)$. Then it is a non-unit element in $\operatorname{Spec} R$, and hence belongs in some maximal ideal $f \in \mathfrak{m}$. Then localizing at $[\mathfrak{m}]$, it is easy to see that f remains to be a nilpotent element, showing that not all stalks are reduced.

For the next assertion, this will be a simpler version of proposition 2.4.5 as we do not need to keep track of any exponents, and hence the Quasicompactness condition is not needed! Assume $f, g \in \mathcal{O}_X(U)$ such that $f(x) = g(x)$ for all $x \in U$. We want to show that $f = g$. Take $h = f - g$. Now, by definition $h(x) = f(x) - g(x) = 0$, so h must be contained in every prime ideal on every affine open subset of U. As $\mathcal{O}_X(U)$ is reduced, $\sqrt{(0)} = (0)$ on each affine open, and so $h|_{\operatorname{Spec} R} = 0$ for each element in the affine open covering, which by the gluging gives us $h = 0$, completing the proof.

Lemma 2.4.15: Reducedness For Affine And Projective Schemes

Let R be a reduced ring. Then $\operatorname{Spec}(R)$ is reduced. Hence, $\mathbb{A}^n$ and $\mathbb{P}^n$ are reduced.

***Proof* :**
exercise.

Proposition 2.4.16: Reducedness On Quasicompact Schemes

Let X be a quasicompact scheme. Then X is reduced if and only if it is reduced at closed points

[27] if D is the set of all zero-divisors of a reduced ring, then it is the union of all minimal ideals

Proof **:**
Assume it is reduced at all closed points. As X is quasi-compact, the closure of each non-closed point contains a closed point; say $Y \in X$ is a non-closed and $x \in X$ is a closed point contained in its closure. Then $\mathcal{O}_{X,Y}$ is the localization of $\mathcal{O}_{X,x}$. As localization preserves reducedness, $\mathcal{O}_{X,Y}$ must be reduced, completing the proof.

The above showed that reducedness for quasicompact schemes is a "closed-point" local condition. It is not an open set local condition. Furthermore, it is not enough to check the global sections:

Example 2.4.17: Reducedness Edge Case

1. The scheme given by:
$$X = \operatorname{Spec} \frac{\mathbb{C}[x, y_1, y_2, \ldots]}{(y_1^2, y_2^2, \ldots, (x-1)y - 1, (x-2)y_2, \ldots)}$$
is nonreduced at the closed points corresponding to the positive integers. The complement of this set is not Zariski open.
The key is that if X is also locally Noetherian, then reducedness is a locally open condition.
2. Another mistake that is easy to make: let X be scheme where the global sections is reduced. This doesn't imply X is reduced (take the scheme cut out by $x^2 = 0$ in $\mathbb{P}_k^2$; note that the global section will be k, but X will be non-reduced).

One of the most important type of reduced ring is an integral domain. If a scheme is an integral domain for each section, it is given a name:

Definition 2.4.18: Integral Scheme

Let X be a scheme. Then X is *integral* if it is nonempty and $\mathcal{O}_X(U)$ is an integral domain for every nonempty open subset $U \subseteq X$.

Recall that the product of integral domain is not an integral domain, or geometrically, the disjoint union of integral scheme need not be integral. This global behavior cannot be captured through stalks, as the reader should verify, and hence integrality is not stalk-local. Integral schemes can be thought of as schemes that are in "one piece" and have "no fuzz":

Proposition 2.4.19: Integral, then Irreducible and Reduced

Let X be a scheme. Then if X is an integral scheme, it is irreducible and reduced

It is in fact an if and only if, which we shall show in proposition 2.4.25.

Proof **:**
If X is integral, it is certainly reduced as each ring of local sections is integral (and hence reduced). To show it is irreducible, assume $X = X_1 \cup X_2$, X_1, X_2 closed. For the sake of contradiction, let's say these sets don't contain each other. Then there exists open subsets $U \subseteq X_1, V \subseteq X_2$ such that

$U \cap V = \emptyset$. Given $X_1 \not\subseteq X_2$, then here exists a point $p \in X_1 \setminus X_2$. Let

$$U = (X \setminus X_2) \cap X_1$$

Then U is open in X_1, and hence by construction in X. Similarly, define $V = (X \setminus X_1) \cap X_2$. Then $U \cap V = \emptyset$.

Then, by exercise ref:HERE, we have by an almost direct consequence of the sheaf axioms that:

$$\mathcal{O}_X(U \coprod V) \cong \mathcal{O}_X(U) \times \mathcal{O}_X(V)$$

But then $\mathcal{O}_X(U \coprod V)$ certainly contains zero-divisors, and hence cannot be integral.

An important property that integral domains is the ability to localize to obtain a fractional field. Localizing at an integral schemes generic point shall give us a "global" function field in which all local sections can embed, similarity to varieties! We first prove that an integral scheme has a unique generic point. Note the proof can be done if we replace integrality with irreducibility, that is we don't need every ring of local sections to be integral, see exercise ref:HERE.

Lemma 2.4.20: Irreducible Schemes and Generic Points

Let X be an integral scheme. Then for every irreducible closed subset $Z \subseteq X$, there exists a unique generic point η as $\overline{\eta} = Z$. In particular, as X is irreducible, then there exists a point η such that $\overline{\eta} = X$.

***Proof* :**
Let $Z \subseteq X$ be an irreducible subscheme. Then $Z = \bigcup_i U_i$ is an affine open cover. As $U_i = \operatorname{Spec} R_i$ is open and Z is irreducible, $\overline{U_i} = Z$ (exercise in topology). Now, as X is integral, (0_i) is the generic point as $[(0_i)]$ is prime. Let $\eta_i = [(0_i)]$. Then:

$$\overline{\eta_i} = Z$$

This shows a generic point exists, let's shows the generic point is unique. Say η_1, η_2 are generic points of Z so that $\overline{\eta_1} = \overline{\eta_2} = Z$. Take any affine open $U_i \subseteq Z$. Then we must have $\eta_1, \eta_2 \in U_i$. If they weren't, then $\eta_i \in Z \setminus U_i$ would be a closed and hence there exists a smaller closed set containing η_i which would imply $\overline{\eta_i} = Z \setminus U_i$, a contradiction. Thus, we get $\eta_1, \eta_2 \in U_i$. Hence, in the subspace topology

$$\overline{\eta_1} = \overline{\eta_2} = U_i$$

Showing that these are also the generic points of U_i. But as U_i is affine and integral, there can only exist a single generic point, hence $\eta_1 = \eta_2$.

In the special case where X is itself irreducible, we can conclude it has a unique generic point η, as we sought to show.

This is a purely topological fact, and hence can also be proven using only irreducibility, see exercise ref:HERE.

The existence was given by reducing to an affine open and "using" its generic point. Using this, we can quickly calculate the stalk at the generic point of an integral scheme. We give this stalk a name:

Definition 2.4.21: Rational Field

Let X be an integral scheme. Then the function field is the collection

$$\mathcal{O}_{X,\eta} = R_{(0)} = \mathrm{Frac}(R)$$

for any R given by $\mathrm{Spec}(R) \subseteq X$, and it is often denoted $\kappa(X)$. If X is a k-scheme, then it is denoted $k(X)$.

Thus, integral schemes have a similar property to that of classical affine and projective varieties, namely that their sections over different open sets can be considered as subsets of a single ring! Gluing is easy as we can see that two elements are glued if and only if they are the same element of $\kappa(X)$.

Proposition 2.4.22: Integral Scheme: Restriction Maps An Function Field

Let X be a scheme that is irreducible and reduced. If $U \subseteq V$ where $V \neq \emptyset$, then

$$\mathrm{res}_{V,U} : \mathcal{O}_X(V) \hookrightarrow \mathcal{O}_X(U)$$

is an inclusion.

Proof **:**
As X is reduced, its sections are all determined by their value on the points, and hence can be treated as function (see example 2.4.23 on why this proof fails otherwise). Take $U \subseteq V \subseteq X$, $f, g \in \mathcal{O}_X(U)$ and assume $f|_U = g|_U$. We want to show that $f|_V = g|_V$. Define:

$$Z = \{x \in V \mid f(x) = g(x)\}$$

Z is closed in V as it is the zero set of $f - g$. By definition $U \subseteq Z$, $Z \subseteq V$, and U is open in V. As X is integral, the open subset V is irreducible. But then U is *dense* in V. As Z is closed in V and contains U, it must be that $Z = V$, but then

$$f|_V = g|_V$$

as we sought to show

Note that we require *both* the irreducibility and the reducedness, having *only* irreducibility is not enough

Example 2.4.23: Non-injective Non-reduced Scheme

Let $R = \frac{k[x,y]}{(xy,y^2)}$ so that $\mathrm{Spec}\, R$ is the x axis there is a bit of fluff in direction of the y-axis at the point 0. Then given $D(x) \subseteq \mathrm{Spec}\, R$, the map $\mathrm{res} : \mathcal{O}_{\mathrm{Spec}\, R}(\mathrm{Spec}\, R) \to \mathcal{O}_{\mathrm{Spec}\, R}(D(x))$ maps $y \mapsto 0$ as

$$y \mapsto \frac{y}{1} = \frac{xy}{x} = \frac{0}{x} = 0$$

Corollary 2.4.24: Integral Scheme: Generic Point

Let X be an integral scheme. Then there exists a generic point η such that given any choice of $\mathrm{Spec}(R) \subseteq X$, there is always an inclusion:

$$\mathcal{O}_X(U) \hookrightarrow \mathcal{O}_{X,\eta} \cong \kappa(R)$$

Proof **:**
By definition:

$$\mathcal{O}_{X,\eta} = \varinjlim_{\eta \in U} \mathcal{O}_X(U)$$

and as we saw lemma 2.4.20 the generic point is in each open set. in Then in the commutative diagram, all these maps are injective, and hence hence the map $\mathcal{O}_X(U)$ is injective into $\mathcal{O}_{X,\eta}$.

Proposition 2.4.25: Irreducible and Reduced, then Integral

Let X be a scheme. If X is irreducible and reduced, it is integral

Proof **:**
Let X be irreducible and reduced. Then any open subscheme is irreducible and reduced (the irreducibility is proved topologically, and the reducedness can be deduced by the stalks). In particular, any affine open subscheme $\mathrm{Spec}\, R$ must be irreducible and reduced. Certainly R must then be reduced, hence to show that $\mathrm{Spec}\, R$ is integral consider $xy \in R$ such that $xy = 0$. Then

$$\mathrm{Spec}\, R = V(0) = V(xy) = V(x) \cup V(y)$$

But that contradicts irreducedness. Hence, all ring of sections of affine open subsets are integral. For general open $U \subseteq X$, we must have an affine open $\mathrm{Spec}\, R \subseteq U$, and hence a restriction $\mathcal{O}_X(U) \to \mathcal{O}_X(\mathrm{Spec}\, R)$. By proposition 2.4.22, this map is injective, and hence $\mathcal{O}_X(U)$ is a subring of an integral domain, making it an integral domain. Hence all rings of local sections are integral domains, completing the proof.

If we have the added condition of being Noetherian, then we only need to check that it is connected and that all its stalks are integral domains:

Proposition 2.4.26: Characterizing Integrality in Noetherian Schemes

Let X be a Noetherian Scheme. Then X is integral if and only if X is nonempty, connected, and all stalks $\mathcal{O}_{X,\mathfrak{p}}$ are integral domains.

Intuitively, the stalk at the intersection of irreducible components will “remember” the generic points of the respective irreducible component.

Proof **:**
By proposition 2.4.19 and proposition 2.4.25, X is integral if and only if it is reduced and irreducible. Then X is certainly non-empty, connected, and the stalks are integral (the localization of integral domains is integral).

Conversely, assume X is nonempty, connected, with integral stalks. First of all, as the stalks are integral, they are reduced. We must show that X is irreducible. As X is Noetherian, by lemma 2.4.2 X can be decomposed into finitely many irreducible components:

$$X = X_1 \cup X_2 \cup \cdots \cup X_n$$

As X is connected, for each i, j we must have $X_i \cap X_j \neq \emptyset$. Take an affine open subset $\operatorname{Spec} R \subseteq X$ and:

$$\operatorname{Spec} R = \bigcup_i^n (X_i \cap \operatorname{Spec} R)$$

Then as each X_i is irreducible, it is a topological argument to see that the sets $(X_i \cap \operatorname{Spec} R)$ form an irreducible decomposition of $\operatorname{Spec} R$. Hence, they must each correspond have a generic point hat corresponds to a minimal prime ideal. Hence, if the X_i are distinct, R_i must contains at least two distinct minimal prime ideals. Then if $[\mathfrak{p}] \in X_i \cap X_j$, $\mathcal{O}_{X,[\mathfrak{p}]} = (R_i)_\mathfrak{p}$ contains two distinct minimal prime ideals. But integral domains could only contain a single minimal prime ideal (namely the one associated to the generic point $[(0)]$), and so this stalk cannot be integral. Thus, it must be that $X_i = X_j$ for all i, j, showing that X is irreducible and completing the proof.

Exercise 2.4.2

1. Show that every point $x \in X$ in a topological space must be contains in an irreducible component (you will use Zorn's Lemma).
2. Show that connected components (maximal connected subset) are closed. Show that the connected components of $\operatorname{Spec}\left(\prod_i^\infty \mathbb{F}_2\right)$ are not open.
3. Show that $D(f)$ is empty if and only if f is nilpotent. Show that if X is an integral scheme, then $D(f)$ is always nonempty and contains the generic point.
4. If $f \in R$ is nilpotent, then R_f is the zero-ring.
5. Let X be an irreducible scheme. Show that there is a unique generic point η such that $\overline{\eta} = X$ (note that lack of reducedness). More generally, show that if $Z \subseteq X$ is a (nonempty) irreducible subset, there is a unique generic point η such that $\overline{\eta} = Z$
6. Let X be a reduced scheme and $Y \hookrightarrow X$ a closed immersion where $\iota(Y) = X$. Then ι is an isomorphism.

2.4.2 Normality and Factoriality

Recall that a normal ring R has the property that it an integral domain that is is integrally closed in it's field of fractions, $\overline{R}^{\operatorname{Frac}(R)} = R$. Due to their importance in number theory, we shall take the time to develop the theory of Schemes for which all the rings of sections are normal. These shall correspond to schemes that have some nicer behaviour for their singularities. It can be thought of as a condition weaker than smoothness (smoothness shall be called regular in section REF:HERE) as normality shall still allow for singularities, however these singularities shall be more tame. In particular, the reader can intuitively have the following geometric picture in mind for subschemes of Normal Schemes:

1. In codimension 1, all singularities are resolved, meaning it is "smooth" in codim 1

2. In codimension ≥ 2, being normal means being nice enough so that rational functions defined on the compliment of the singularities can be extended to the singularity (i.e. Algebraic Hartogs's lemma applies)

Definition 2.4.27: Normal Scheme

Let X be a scheme. Then X is said to be *normal* if every stalk $\mathcal{O}_{X,\mathfrak{p}}$ is normal domains.

Proposition 2.4.28: Normal Scheme Stalk Local

Let X be a scheme. Then X is a normal scheme if and only if $\mathcal{O}_{X,\mathfrak{p}}$ is normal for each $\mathfrak{p} \in X$

Proof **:**
Recall (or reprove) that normality is preserved when localizing (hint: recall the proof that a UFD is normal)

Note that reducedness was stalk-local, and so it is immediate that normal schemes are reduced. As being integrally closed is well-behaved under localization (recall the proof that a UFD is normal), for an if R is normal then $\operatorname{Spec}(R)$ is a normal scheme. If each stalk is a UFD, then it is immediately a Normal scheme. Let's give a name to such a scheme

Definition 2.4.29: Factorial Scheme

Let X be a scheme. Then X is said to be *normal* if every stalk $\mathcal{O}_{X,\mathfrak{p}}$ is a UFD.

If R is a UFD, then $\operatorname{Spec} R$ is factorial (by the above reasoning), however the converse need be true! Namely, we may have factorial schemes where the stalks are all UFD's but the global sections do not form a UFD. A quintessential example will be the spectra of elliptic curves.

Example 2.4.30: Normal Schemes

1. The schemes $\mathbb{A}^n_k$, $\mathbb{P}^n_k$ and $\operatorname{Spec}\mathbb{Z}$ are normal. If R is normal, by definition $\operatorname{Spec} R$ is normal.
2. Starting on p.164 section 5.4.8, lots and lots of good examples

2.5 Associated Points

Associated points can be thought of as the points that capture the components structure of a scheme and how it comes together at intersection points. These points include points that come from schemes $k[x,y]/(y^2,xy)$ that "remember" that $V(xy)$ would be two components if it were not for the "fuzz factor" of y^2.

Example 2.5.1: Support Of Non-reduced Ring

Let $X = \text{Spec}\,(k[x,y]/(y^2,xy))$. Show that the support of any function X is either empty, the origin, or all of X. Note that the function y evaluated to 0 everywhere on X, but the support is nonzero at the origin!

This will be linked to the irreducible components of the above X, namely the x-axis and the origin. This gives the following definition tha shall be expanded upon:

Definition 2.5.2: Associated Points

Let X be a scheme. Then the associated points are precisely the generic points of irreducible components of the support of some element of X

Proposition 2.5.3: Affine Integral Schemes And Associated Points

Let R be an integral domain. Then the generic point is the only associated point of $\text{Spec}\,R$.

Proof **:**
Certainly the generic point is an associated point as it is associate to the support of the function 1 on the irreducible component $\text{Spec}\,R$ (i.e. the entire space itself). We must show the nonzero support of all functions (besides 0) is all of $\text{Spec}\,R$. Let $f \in R$, and let $f_{\mathfrak{p}} \in R_{\mathfrak{p}}$ be the associated localization at the point $[\mathfrak{p}]$. If $f_{\mathfrak{p}} = 0_{\mathfrak{p}}$, then there must be a $g \in R \setminus \mathfrak{p}$ such that $f/1 = fg/g = 0/g = 0$. But R is an integral domain, hence this is not possible.

Finish this here, p.167

Exercise 2.5.1

1. Find the structure for the following rings:
   a) $\text{Spec}\,\mathbb{C}[x]/(x^3 - x^2)$
   b) $\text{Spec}\,\mathbb{C}[x]/(x^2 - d)$
   c) $\text{Spec}\,\mathbb{C}[x]/(x^2 + 1)$
2. Let X be a quasicompact scheme. Show that the closure of each point contains a closed point (this is not true for non quasicompact schemes in general)
3. Let X be a quasicompact scheme. Suppose f vanishes at all point of X. Then there exists some N st $f^n = 0$. Show that this fails when X is not quasicompact (hint: take infinite disjoint unions of $\text{Spec}\,(R_n)$ where $R_n := k[x]/(x^n)$)

4. Let X be a quasicompact scheme. Then it suffices to cheek reducedness at closed points (show that any nonreduced point has a nonreduced closed point in its closure)

5. Show that reducedness is not in general an open condition by considering:

$$X = \text{Spec}\left(\frac{\mathbb{C}[x, y_1, y_2, ...,}{(y_1^2, y_2^2, y_3^3, ..., (x-1)y_1, (x-2)y_2, (x-3)y_3), ...}\right)$$

and $Y = \text{Spec}\,[x]$. Show the nonreduced points of X are the closed points corresponding to the positive integers. The compliment of this set is *not* Zariski open.

6. Let X be an integral scheme. Show that if $\emptyset \neq V \subseteq U$, then the map $\text{res}_{U,V} : \mathcal{O}_X(U) \to \mathcal{O}_X(V)$ is an inclusion. very important, see p.156 under exercise

7. Let X be an integral scheme, γ the generic point, and $\text{Spec}\,(R) \cong U \subseteq X$ is any nonempty affine open subset of X. Then $\mathcal{O}_X(U) \to \mathcal{O}_{X,\gamma} = K(A)$ is an inclusion. Conclude that gluing is easy: every f_i in U_i for an open cover for U glue together if they are same element in $\text{Frac}(X)$ (recall that $\text{Frac}(X)$ is defined to be $\text{Frac}(R)$ where R is given by $\mathcal{O}_X(\text{Spec}\,(R))$

8. Show that integrality is not stalk local via the example $\text{Spec}\,(A) \coprod \text{Spec}\,(B) = \text{Spec}\,(A \times B)$.

9. Show that reducedness is an affine local property.

10. Let X be a locally Noetherian Scheme. Then X is quasiseparated.

11. Let X be a Noetherian Scheme. Then it has a finite number of connected components, each a finite union of irreducible components.

12. Show that the ring of sections of a Noetherian Scheme need not be Noetherian.

13. Let X be a scheme of locally finite type over $\mathbb{Z}$ or k. Then X is locally Noetherian.

14. A scheme that is of finite type over any Noetherian ring is Neotherian.

15. Let X be a quasiprojective R-scheme. Then if R is Noetherian, X is a Noetherian scheme, and hence has a finite number of irreducible components.

16. Let X be a locally Noetherian scheme. Then the reduced locus is open. (Vakil p.156)

17. Let $U \subseteq X$ be an open subscheme of a projective R-scheme X. Show that U is locally of finite type R

18. Find a counter-example showing that normal schemes are not in general integral schemes (hint: think of the disjoint union of two normal scheme. For more hint, Vakil p. 162).

19. Let X be a Noetherian scheme. Then X is normal if and only if if is a finite disjoint union of integral normal schemes.

2.6 Dimension

The reader may want to review [8] for an in depth look at the algebraic counter-part. Recall that the Krull dimension of a [commutative] ring R is the supremum of the proper chain of prime ideas with indexing starting at 0 . Geometrically, for $\text{Spec}\,R$, and topological spaces more generally (including schemes), its dimension is the supremum of proper chains of closed irreducible subsets with indexing starting at 0 . Naturally, $\dim \operatorname{Spec} R = \mathrm{k}\dim R$. Also naturally,

$$\dim \operatorname{Spec} R = \dim \operatorname{Spec} R/\sqrt{(0)}$$

If the space is is not irreducible, then the dimension per component can vary (for example $\mathbb{A}^1 \cup \mathbb{A}^2 \subseteq \mathbb{A}^3$ interpreted as the x-axis union the yz-plane), and hence we usually work with irreducible sets. If it is necessary, we shall say that a space has pure dimension if all of it's irreducible components have the same dimension. Dimension respects subsets: if $X \subseteq Y$ then $\dim X \leq \dim Y$. If a space is 1 dimensional, it is called a curve. If it is 2 -dimensional, a surface, and ≥ 3 is sometimes generally referred to as an n-fold.

Lemma 2.6.1: Dimension of Open Subset

Let $U \subseteq X$ be an open subset. Then there is a bijection between irreducible closed subsets of U and the irreducible closed subsets of X meeting U.

***Proof* :**
exercise

Proposition 2.6.2: Dimension of Scheme

Let X be a scheme. Then X is of dimension n if and only if it admits an open cover of affine open subsets of dimension at most n, where equality must be achieved by at least one affine open subset.

***Proof* :**
If X is n-dimensional, certainly each open subset in an affine cover is dimension $\leq n$. For the converse, we may take the contra-positive by supposing $\dim X \neq n$. It certainly can't be less, for at least one $\dim U_i = n$, which then implies $n = \dim U_i = \dim X \cap U_i \geq \dim X < n$, but $n < n$ is a contradiction, so suppose $\dim X > n$. Then there exists a chain of irreducible closed subsets:

$$X_0 \subsetneq X_1 \subsetneq \cdots \subsetneq X_n \subsetneq X_{n+1}$$

Then given an affine cover, it must be that $X_0 \cap U_i \neq \emptyset$ for at least one open set in the cover. Then by lemma 2.5.1, there is a bijective correspondence between the irreducible closed subsets of U_i and the irreducible closed subsets of X meeting U_i. As $X_0, \ldots, X_{n+1}$ meet U_i and the correspondence is bijective:

$$X_0 \cap U_i \subsetneq X_1 \cap U_i \subsetneq \cdots \subsetneq X_n \cap U_i \subsetneq X_{n+1} \cap U_i$$

However this is a contradiction as $\dim U_i \leq n$, meaning the last $\subsetneq$ can't be proper. Then this establishes the if and only if, completing the proof.

Lemma 2.6.3: Integral Extensions and Dimension

Let $f : \operatorname{Spec} R \to \operatorname{Spec} S$ correspond to an integral extension. Then

$$\dim \operatorname{Spec} R = \dim \operatorname{Spec} S$$

Proof **:**
use going-up theorem and going-down theorem.

2.6.1 A word on Noetherian Scheme It is not the case that a Noetherian scheme needs to be finite-dimensional: Nagata came up with a counter-example (it is outlined in Vakil in chapter 11 in the first section).
Noetherian Local rings will have to be finite dimensional, and so points of locally Noetherian schemes must have finite codimension.

We are often concerned with spaces that are a few dimensions lower than the ambient space X. As schemes can the union of irreducible components, we can have a codimension object that have lower dimensional components. To avoid this, we shall limit codimensions to irreducible subschemes:

Definition 2.6.4: Codimension

Let X be a scheme and $Y \subseteq X$ an irreducible subscheme. Then:

$$\operatorname{codim}_X Y = \sup\{ \text{ proper chains of irreducible closed subsets starting at } \bar{Y}\}$$

where indexing starts at 0

Naturally, for $\operatorname{Spec} R/\mathfrak{p} \subseteq \operatorname{Spec} R$, then the codimension of $\operatorname{Spec} R/\mathfrak{p}$ corresponds to the supremum of lengths of decreasing proper chains of prime ideals starting at $\mathfrak{p}$ with indexing starting at 0 : this is called the *height* of $\mathfrak{p}$

$$\mathfrak{p}_0 \subsetneq \mathfrak{p}_1 \subsetneq \cdots \subsetneq \mathfrak{p}_h = \mathfrak{p}$$

that when R is a here, most likely Cohen–Macaulay ring, but maybe in general, $\operatorname{codim}_{\operatorname{Spec} R} \operatorname{Spec} R/\mathfrak{p} = \operatorname{kdim} R_{\mathfrak{p}}$. For example, the generic point η as a subset of $\operatorname{Spec} \mathbb{Z}$ has codimension 0, while any other point has codimension 1. If Y is codimension 0 in X, it must be an irreducible component.

Definition 2.6.5: Hypersurface

Let $Y \subseteq X$ be an irreducible closed subscheme. Then Y is called a hypersurface if $\operatorname{codim}_X Y = 1$.

Lemma 2.6.6: Sum of Dimension and Codimension - Scheme

Let X be a scheme and $Y \subseteq X$ an irreducible subscheme. Then:

$$\operatorname{codim}_X Y + \dim Y \leq \dim X$$

> ***Proof*** **:**
> this is a simple topological argument

The equality holds for varieties (see ref:HERE), but does not hold in general; consider the following scheme (image from Vakil)

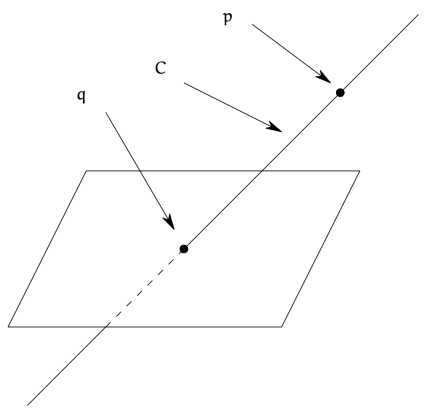


Figure 2.7: Codimension Pathology

The scheme is dimension 2, the point p is dimension 0, but codimension 1 . The point q is codimension 2 with no codimension 1 subset in-between. Note too that this scheme has open subsets of *different dimension*! This *does not* occur for open subsets of n-manifolds.

Working the Krull dimension can be difficult. It is thus useful to remember the following equating the transcendental dimension and krull dimension. First, recall the following definition:

Definition 2.6.7: Transcendental Dimension

Let A be a k-algebra. Then the transcendental dimension of A is the transcendental degree of $\operatorname{Frac}(A)$ over k, where $A \cong k\left[u_1, \ldots, u_n\right]$ for appropriate (not necessarily transcendental) generators.

Theorem 2.6.8: Transcendental and Krull Dimension Equality

Let $\operatorname{Spec} A$ be an affine k-scheme where A is a finitely generated integral domain over k. Then:

$$\dim \operatorname{Spec} A = \operatorname{trdeg} \operatorname{Frac}(A)/k$$

Thus, if X is an irreducible k-variety, then

$$\dim X = \operatorname{trdeg} k(X)/k$$

Proof **:**
For the first part, see [8]. The second part follows when interpreting variety in the classical sense, and follows for the general sense by theorem ref:HERE (the correspondence between Var and a subcategory Sch)).

Using this result, we can prove the equality of lemma 2.5.6 using transcendental degree. We shall also requires a few more concepts, one important one establishes which rings "strongly" connect the dimension using prime ideals and transcendental elements:

Definition 2.6.9: Catenary Rings

Let R be a commutative ring. Then R is said to be catenary if for every nested chain $\mathfrak{q} \subseteq \mathfrak{p} \subseteq R$ of prime ideals, the maximal chains prime ideals between $\mathfrak{q}$ and $\mathfrak{p}$ all have the same length.

Many rings are catenary rings: localization of finitely generated $\mathbb{Z}$-algebras, complete Noetherian local rings, Dedekind domains, and so forth (see exercise 2.5.1). Importantly, localization of finitely generated k-algebras are catenary (which you need to prove for the following)

Theorem 2.6.10: Sum of Dimension and Codimension - locally finite k-Scheme

Let X be a locally finite k-scheme, $Y \subseteq X$ an irreducible subset, and η the generic point of Y. Then:

$$\operatorname{codim}_X Y + \dim Y = \dim X$$

equivalently:

$$\dim Y + \dim \mathcal{O}_{X,\eta} = \dim X$$

Proof **:**
exercise (you may want to reference [8] for some useful algebraic results, Or see Vakil p. 311 for a guided exercise).

Krull's Height Theorem and Hartog's lemma

In this section, we shall show how on some nice schemes, the number of equations gives an upper bound on the codimension. This shall give us some nice algebraic computational tools to determine, or at least give information, on what type of objects equations cut out.

Hartog's Lemma

(Cool for bringing in complex analysis into algebraic geometry)
move this to where it's appropriate

Exercise 2.6.1

1. some of these questions should eventually be moved to their appropriate section
2. Show that a Noetherian scheme of dimension 0 has finitely many points
3. Let $f : X \to Y$ be an integral morphism. Then the dimension of fibers are 0 .
4. If $\nu : \tilde{X} \to X$ is the normalization of a scheme, then $\dim \tilde{X} = \dim X$
5. If K/k is an algebraic field extension and X a k-scheme, $X_k = X \times_k K$ the base change, then $\dim X = \dim X_K$, in particular, $\dim X = n$ if and only if $\dim X_K = n$.
6. Generalize the above result for arbitrary field extensions K/k.
7. Show the following rings are catenary: localization of finitely generated $\mathbb{Z}$-algebras, complete Noetherian local rings, Dedekind domains.

2.7 Regular and Smooth schemes

(I'm considering Vakil material optional over Hartshorne rn, so I'll cover this in detail later)

Definition 2.7.1: Tangent Space

Let X be a scheme and $x \in X$ a point Then the tangent space at x is given by

$$\mathfrak{m}_x/\mathfrak{m}_x^2$$

where $\mathfrak{m}_x \subseteq R_x R$ is given by an affine local neighborhood of $x, x \in \operatorname{Spec}(R) \subseteq X$.

Definition 2.7.2: Regular Local Ring

Let $(R, \mathfrak{m})$ be a local ring. Then R is regular if:

$$\dim R = \dim_{\operatorname{Frac}(R)} \mathfrak{m}/\mathfrak{m}^2$$

Definition 2.7.3: Regular Scheme

Let X be a scheme. Then X is regular if $\mathcal{O}_{X,\mathfrak{p}}$ is a regular local ring for all points $[\mathfrak{p}] \in X$.

Example 2.7.4: Regular Scheme

1. A warm-up example would be $\mathbb{A}^2_k$. For the closed points, first show that $\kappa(\mathfrak{m}) = k$. Then show that $\mathfrak{m}/\mathfrak{m}^2$ is indeed a 2 -dimensional vector space for any closed point. Finally, $k[x, y]_\mathfrak{m}$ has a maximal chain $0 \subsetneq (x - a) \subsetneq (x - a, y - b)$ (you can invoke strong result about localization of Noetherian rings to show it is maximal length, or do this more manually). Then we see that the two dimension math.

   As we are working over schemes, we have higher-dimensional points! Consider $\left[\left(x^2 - y\right)\right] = [\mathfrak{p}] \in \mathbb{A}^2_k$. Then show its local ring has dimension 1 and that $\mathfrak{m}_\mathfrak{p}/\mathfrak{m}_\mathfrak{p}^2$ is a dimension $1\kappa(\mathfrak{p})$ vector space.

An important observation is that $\left[\left(x^3-y^2\right)\right] \in \mathbb{A}_k^2$ is also a regular point, even though it seems non-regular (check this)! This is an important distinction between the generic points and the irreducible closed subscheme associated to it! You can think of this as being "generically regular" along the point, motivating the name generic.

2. Take $X = \operatorname{Spec} k[x,y]/\left(x^3-y^2\right)$. Then this is not regular: show that $\operatorname{kdim} \mathcal{O}_{X,(0,0)} = 1$ but $\dim_k \mathfrak{m}/\mathfrak{m}^2 = 2$.

3. If X is normal scheme, then we know that it's stalks are integrally closed, and all codimension 1 points are regular. However, this does not imply that X is regular! Take for example Spec $k[x,y,z]/\left(x^2+y^2-z^2\right)$ which is not regular at $(0,0,0)$

   Note that if X was 1-dimensional (and say connected, Noetherian to avoid edge cases), then this does imply that it is regular because we have that the codimension 1 points are simply the points of the scheme, and the generic point will be regular as the reader ought to check. In fact, a 1-dimensional normal scheme is a regular scheme (this was even shown purely algebraically in [8])).

3

Schemes II: Morphisms and Constructions

Scheme morphisms shall be inherently more *algebraic* than *smooth* in nature: they should be in some ways more stringent than holomorphic maps as only "polynomial-like" functions are allowed, but they also will allow singularities. In this chapter, we shall properly define morphisms of schemes, classify many types of morphisms, and interpret many important types of morphisms. In the process, we be able to "recover" the right notion of Hausdroffness and compactness for scheme in the form of *separatedness* and *properness*. The condition of being proper will be closer to being projective (all proper schemes are covered by a projective scheme) showing that projective schemes are the natural "compact schemes" to consider. The chapter concludes by defining an abstract variety and showing that $\mathbf{Var}_k$ is a fully-faithful subcategory of of $\mathbf{Sch}_k$.

3.1 Elementary Concepts

Some of the discussion here shall overlap with the content in section 1.4; this was done as the self-completeness of the material in this section is a priority.

Let us start with comparing regular maps between varieties and smooth maps between manifolds to compare and contrast intuitions. In [4] it was shown that continuous maps between manifolds can be arbitrarily well approximated by a smooth map (as a consequence of Whitney's Approximation Theorem). Hence, smooth maps can be thought of as smoothing out continuous maps. This is not at all the case for the Zariski topology, for example, the "indicator function" $f : \mathbb{A}^1_k \to \mathbb{A}^1_k$ where $f(0) = 1$ and $f(a) = 0$ for all other points is continuous (notice the importance of the coarsens of the Zariski topology!), but it is not possible to get f arbitrarily uniformally close to a regular map.

Thus, scheme morphisms behave fundamentally differently from maps between manifolds, they are much more strict.

However, they shall share in important feature with scheme morphisms: A continuous map $f : M \to N$ between smooth manifolds is a smooth map if and only if $f^*(C^\infty(N)) \subseteq f^\infty(M)$, and in particular if f is a homeomorphism, it is a diffeomorphism if and only if f^* is an ring-isomorphism. In essence, f "preserves" and is "compatible" with all the relations given on N given by all the smooth maps. As the relations on schemes is generally more diverse than what can be found on smooth manifolds (recall again that the affine patches on schemes can be wildly different), this approach generalizes better [1].

With this in mind, we shall the following properties from scheme morphisms:

1. ring homomorphisms and affine scheme morphism are dual to each other (this should generalize that a map between k-varieties is regular if and only if the contra-variant map between global sections is a k-algebra homomorphism)
2. For general scheme morphisms, they must preserve the structure sheaf (which holds all the algebraic information). Namely given a continuous map $f : X \to Y$, if $s_i \in \mathcal{O}_Y(V)$ is a local section and $V \subseteq Y$ is an open set, then there should be a map $f^\#(V) : \mathcal{O}_Y(V) \to \mathcal{O}_X(f^{-1}(V))$ that maps s_i to another local section, that is $f^\#(V)$ is a ring homomorphism that directly tells us how the functions should behave in the pre-image.
3. The evaluations of regular functions in $\mathcal{O}_Y(V)$ are compatible with the evaluation of functions on $\mathcal{O}_X(f^{-1}(V))$.

Recall (definition 1.4.3) that a morphism of ringed spaces $\pi : X \to Y$ is a continuous map of topological spaces along with the map $\pi^\# : \mathcal{O}_Y \to \pi_*\mathcal{O}_X$, i.e. the pushforward map, or equivalently by adjointness the map $\pi^{-1} : \pi^*(\mathcal{O}_Y) \to \mathcal{O}_X$. As we saw in example 2.1.20, not all spectra maps are induced by ring homomorphisms, and there is no natural way to induce a ring homomorphism from a general spectra map. If we add a ringed space structure and get an affine scheme, then the global sections give a natural ring homomorphism. This also allows us to distinguish between spectra with different sheaves placed on them, or similarly have the same underlying topological but different sheaf morphisms:

Example 3.1.1: Multiple ringed space maps for Spectra Map

Let $X = Y = \operatorname{Spec} k$ with char $k = 0$ and $\operatorname{Gal}_{\mathbb{Q}}(k) \neq 0$. Let $f : \operatorname{Spec} k \to \operatorname{Spec} k$ be the identity. Then for each galois automorphism $\varphi \in \operatorname{Gal}_{\mathbb{Q}}(k)$, $f^\#_\varphi : k \to k$ is a distinct sheaf morphism.

Notice how this differs for the theory of smooth manifolds, where a single point does not have distinct differential structures (up to isomorphism) and distinct smooth maps between them.

As we want our definition to work well for geometric spaces like smooth manifolds, we may ask if a ringed space homomorphism between two affine schemes induces a unique ring map. However, this is not quite enough: not all ringed space maps between spectra are induced by ring homomorphisms, and there can still be multiple ringed space maps associated to a ring homomorphism.

The key realization is that ring homomorphism induce some more local structure that needs to be preserved! For the commutative algebra background for this intuition, see [8, chapter 20.3] (specifically

[1] Note that in the smooth case, the "if and only if" can make this the equivalent definition of smooth map, where since we only have one type "affine patch" we can then deduce the usual definition of a smooth map; proposition 3.1.9 shows the equivalent for schemes

section 20.3.3). By proposition 1.1.22 , the stalks of a sheaf can determine a sheaf morphism (with suitable compatibility on open subsets). By proposition 2.3 .14 , the stalk of schemes are all local rings. The geometrical interpretation is that $\mathcal{O}_{X,x}$ remembers the information of functions defined around x and whether they evaluate to 0 or not. The unique maximal ideal $\mathfrak{m}_x \subseteq \mathcal{O}_{X,x}$ contains all function defined around x where $f(x) = 0$, and we may only evaluate at the point x[2](hence why there can only be one maximal ideal, to evaluate at that point).

If $f : R \to S$ is a ring homomorphism and $\mathfrak{p} \subseteq S$, then we get a natural map $f_{\mathfrak{p}} : R_{f^{-1}(\mathfrak{p})} \to S_{\mathfrak{p}}$ given by $f_{\mathfrak{p}}(r/t) = f(r)/s$ where $s = f\left(f^{-1}(s)\right)$ with $s \notin \mathfrak{p}$. Let $\mathfrak{m}_R$ and $\mathfrak{m}_S$ be the unique maximal ideals of both rings, and notice that $f_{\mathfrak{p}}(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$. Equivalently, we have $\mathfrak{m}_R \subseteq f_{\mathfrak{p}}^{-1}(\mathfrak{m}_S)$, and as $\mathfrak{m}_R$ is maximal it must be that $\mathfrak{m}_R = f_{\mathfrak{p}}^{-1}(\mathfrak{m}_S)$[3]. This need not be true for general homomorphisms between local rings (consider $\mathbb{Z}_{(p)} \hookrightarrow \mathbb{Q}$), and so we give a name to such ring homomorphisms between local rings:

Definition 3.1.2: Local Ring Homomorphism

let $\varphi : R \to S$ be a ring homomorphism between local rings with maximal ideals $\mathfrak{m}_R$ and $\mathfrak{m}_S$ respectively Then φ is a local ring homomorphism if $\varphi(\mathfrak{m}_R) \subseteq \mathfrak{m}_S$.

In particular, elements in R that are zero in the quotient must remain being 0 in the image. Now, given a ringed space morphism $\varphi : \operatorname{Spec} S \to \operatorname{Spec} R$, does it in general preserve the local-ring structure on stalks? The answer is no:

Example 3.1.3: Ringed Space Morphism Not Locally Ringed

Take $R = \mathbb{Z}_{(p)}$ and $S = \mathbb{Q}$ so that $\operatorname{Spec}(R) = \{[(p)], [(0)]\}$ and $\operatorname{Spec}(S) = \{[(0)]\}$. The open sets of $\operatorname{Spec}(R)$ are

$$\emptyset \quad \{[(0)]\} \quad \operatorname{Spec}(R)$$

and the stalks of $\operatorname{Spec}(R)$ are $\mathbb{Z}_{(p)}$ at (p) and $\mathbb{Q}$ at (0) respectively. The reader can verify that there is only one possible ring homomorphism $\iota : R \to S$; it must be the embedding.
Now, ι induces a morphism $\iota^a : \{(0)\} \hookrightarrow \{(0), (p)\}$:

$$\iota^a : \operatorname{Spec}(S) \to \operatorname{Spec}(R) \quad \iota^a(0) = (0)$$

and $\iota^\# : \mathcal{O}_{\operatorname{Spec}(R)} \to (\iota^a)_* \mathcal{O}_{\operatorname{Spec}(S)}$. Now, to show the sheaf morphism is well-defined, we shall show that the map on stalks is well-defined, which suffices by proposition 1.1.22. Indeed, this map is determined by its stalk at 0 and p :

$$\begin{gathered} \iota_0^\# : \mathbb{Z}_{(p)} \to \mathbb{Q} \\ \iota_p^\# : \mathbb{Q} \to \mathbb{Q} \end{gathered}$$

Now, we can also define a map φ that maps $(0) \mapsto (p)$. The pre-image of every open set (notably

[2]of course, we can evaluate at the generic point to get back the function as well, but we currently interested in the non-trivial behavior

[3]Note this does not imply $\varphi(\mathfrak{m}_R) = \mathfrak{m}_S$, can you think of a counter-example

the pre-image of zero is now the empty-set). Then the we need the stalk maps:

$$\varphi_0^{\#} : \mathbb{Z}_{(p)} \to \mathbb{Q}$$
$$\varphi_p^{\#} : \mathbb{Q} \to \mathbb{Q}$$

which is well-defined by being an inclusion map. Hence we have another distinct ringed-space morphism. However, we know there is only one ring homomorphism, and hence we managed to produce an extra map.

As an exercise, show that $\operatorname{Spec} k(x) \to \operatorname{Spec} k[y]_{(y)}$ sending the unique point η to the closed points of the codomain has the same problem (exercise 3.1.1(1)).

In general, a ring map $R \to S$ will induce a ringed space map $\operatorname{Spec}(S) \to \operatorname{Spec}(R)$ which induces a ring map $R \to S$, however the first mapping is in general injective but not surjective, while the second mapping is surjective but not injective (example 3.2 being one of the canonical examples). We require that the maps between stalks be a local ring homomorphism, and as we want this definition to work well with other geometric objects like manifolds, we define them for ringed spaces:

3.1.1 Scheme Morphisms

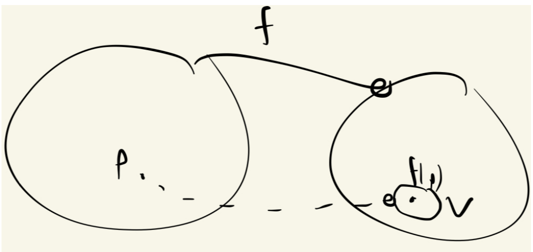
Definition 3.1.4: Locally Ringed Space Morphism

Let X, Y be locally ringed spaces. Then a morphism of locally ringed spaces $(\pi, \pi^{\#}) : X \to Y$ is a morphism of ringed spaces such that the induced map of stalks $\pi^{\sharp} : \mathcal{O}_{Y,q} \to \pi_* \mathcal{O}_{X,p}$ (where $\pi(p) = q$) are local ring homomorphisms.

The geometry behind this is better motivated when considering a continuous function between topological spaces $f : X \to Y$. Say $p \mapsto q = f(p)$ and that there is a function g on an open subset $V \subseteq Y$ that vanishes on $q : g(q) = 0$. Algebraically, the vanishing of g at q means that q has some algebraic information that is represented by g. We want that the pullback of g via f to vanish at p, namely:

$$g(f(p)) = 0 \quad \text{if and only if} \quad (g \circ f)(p) = 0$$

Visually, we can see this as:



In other words, we want scheme morphism to preserve algebraic relations. This is "obvious" when g, f are our usual functions since $g(f(p)) = (g \circ f)(p)$, however the above example shows how this is not necessarily true for ringed space morphism, precisely because the domain we are evaluating at differs for g and $g \circ f$. To give a concrete demonstrate, consider again:

$$f : \operatorname{Spec} \mathbb{Q} \to \operatorname{Spec} \mathbb{Z}_{(p)} \quad f([0]) = [p]$$

with $f^{\#} : \mathcal{O}_{\operatorname{Spec} \mathbb{Z}_{(p)}} \to f_* \mathcal{O}_{\operatorname{Spec} \mathbb{Q}}$, in particular for each open subset:

$$f^{\#}(V) : \mathcal{O}_{\operatorname{Spec} \mathbb{Z}_{(p)}}(V) \to \mathcal{O}_{\operatorname{Spec} \mathbb{Q}}\left(f^{-1}(V)\right)$$

As $\operatorname{Spec} \mathbb{Z}_{(p)}$ has two non-empty open subsets

$$\begin{gathered} f^{\#}(\{[0],[p]\}) : \mathbb{Z}_{(p)} \to \mathbb{Q} \\ f^{\#}(\{[0]\}) : \mathbb{Q} \to T \end{gathered}$$

where T is the trivial ring where $0 = 1$, since $f^{-1}([0]) = \emptyset$, and $\mathcal{O}_{\operatorname{Spec} \mathbb{Q}}(\emptyset)$ is the terminal object, which for Ring is the ring with a single element where $0 = 1$. The 2nd map is the obvious surjection, and the first map must be the identity map.
Now, take the function $p \in \mathcal{O}_{\operatorname{Spec} \mathbb{Z}_{(p)}}\left(\operatorname{Spec} \mathbb{Z}_{(p)}\right)$. Then:

$$p([p]) = 0 \in \operatorname{Frac}\left(\mathbb{Z}_{(p)}/(p)\right) = \mathbb{Q}$$

Then we may "pullback" this function via the map $f^{\#}$:

$$f^{\#}\left(\operatorname{Spec} \mathbb{Z}_{(p)}\right)(p) = p \in \mathbb{Q} = \mathcal{O}_{\operatorname{Spec} \mathbb{Q}}\left(f^{-1}\left(\operatorname{Spec} \mathbb{Z}_{(p)}\right)\right.$$

Then the function p, at the point $f^{-1}([p]) = [0]$ evaluates to:

$$p([0]) = p \in \mathbb{Q} \in \mathbb{Q} = \operatorname{Frac}(\mathbb{Q}/(0))$$

But notice that it no longer evaluates to 0!! Hence, if we consider general ringed space morphisms between spectra, evaluation and composition do not behave well together, we require the extra condition on stalks to make it hold!
Now, we know that ring homomorphisms induce local homomorphisms on stalks. Let us use this information to construct ring homomorphism between the local sections defined on distinguished open sets (from which we can readily define ring homomorphisms on local section over an arbitrary open subset). Let $f : R \to S$ be a ring homomorphism and $(f^a, f^{\#}) : \operatorname{Spec} S \to \operatorname{Spec} R$ be the induced spectrum map. We shall show that for each $r \in R$ we get a map:

$$\mathcal{O}_{\operatorname{Spec}(R)}(D(r)) \to \mathcal{O}_{\operatorname{Spec}(S)}\left(\left(f^a\right)^{-1}(D(r))\right) \stackrel{!}{=} \mathcal{O}_{\operatorname{Spec}(S)}(D(f(r)))$$

where $\stackrel{!}{=}$ is the desired equality to make this map well-defined and independent of choice of r. Indeed:

$$\begin{aligned}
(f^a)^{-1}(D(r)) &= \left\{[\mathfrak{p}] \in \operatorname{Spec}(S) \middle| [f^a([\mathfrak{p}])] \in D(r)\right\} \\
&= \left\{[\mathfrak{p}] \in \operatorname{Spec}(S) \middle| r([f^a([\mathfrak{p}])]) \neq 0\right\} \\
&= \left\{[\mathfrak{p}] \in \operatorname{Spec}(S) \middle| r \bmod f^{-1}(\mathfrak{p}) \neq 0 \in \kappa\left(f^{-1}(\mathfrak{p})\right)\right\} \\
&\overset{!}{=} \{[\mathfrak{p}] \in \operatorname{Spec}(S) | f(r) \bmod \mathfrak{p} \neq 0 \in \kappa(\mathfrak{p})\} \\
&= \{[\mathfrak{p}] \in \operatorname{Spec}(S) | f(r)([\mathfrak{p}]) \neq 0\} \\
&= D(f(r))
\end{aligned}$$

To explain the $\overset{!}{=}$ equality, let us focus on $\kappa\left(f^{-1}(\mathfrak{p})\right)$ and $\kappa(\mathfrak{p})$. First, notice that:

$$\begin{aligned}
\kappa\left(f^{-1}(\mathfrak{p})\right) &= \operatorname{Frac}\left(\mathcal{O}_{\text{Spec } R, f^{-1}(\mathfrak{p})}\right) = R_{f^{-1}(\mathfrak{p})}/\mathfrak{m}_{f^{-1}(\mathfrak{p})} \\
\kappa(\mathfrak{p}) &= \operatorname{Frac}\left(\mathcal{O}_{\text{Spec } S, \mathfrak{p}}\right) = S_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}
\end{aligned}$$

Next, the ring homomorphism f naturally induces the local ring homomorphism $f_{\mathfrak{p}} : R_{f^{-1}(\mathfrak{p})} \to S_{\mathfrak{p}}$ given by $\frac{r}{t} \mapsto \frac{f(r)}{f(t)} = \frac{f(r)}{s}$ is well-defined where $s = f\left(f^{-1}(s)\right)$ for appropriate $s \notin \mathfrak{p}$. Then if $\mathfrak{m}_{\mathfrak{p}} \subseteq S_{\mathfrak{p}}$ is the unique maximal ideal,

$$f^{-1}(\mathfrak{m}_{\mathfrak{p}}) = \left\{\frac{r}{t} \in R_{f^{-1}(\mathfrak{p})} \middle| \frac{f(r)}{s} \in \mathfrak{m}_{\mathfrak{p}}, s \notin \mathfrak{p}\right\}$$

Notice that the elements $r/1$ map into $\mathfrak{m}_s$, and so $\mathfrak{m}_{f^{-1}(\mathfrak{p})} \subseteq f^{-1}(\mathfrak{m}_{\mathfrak{p}})$. As the ideal is maximal, we have $\mathfrak{m}_{f^{-1}(\mathfrak{p})} = f^{-1}(\mathfrak{m}_{\mathfrak{p}})$. Hence, an element is zero in $R_{f^{-1}(\mathfrak{p})}/\mathfrak{m}_{f^{-1}(\mathfrak{p})}$ if and only if it is zero in $S_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}$. As this must be the case for each point (as $f_{\mathfrak{p}}$ is a local homomorphism for each stalk), this is exactly the condition we need so that $\overset{!}{=}$ holds! If starting with a map $\varphi : \operatorname{Spec} S \to \operatorname{Spec} R$, so that $\varphi^{\#} : \mathcal{O}_{\text{Spec } S} \to \varphi_* \mathcal{O}_{\text{Spec } R}$, then the codomain of the sheaf map on each distinguished open set has the following interesting interpretation that shows the map is natural

$$\begin{gathered}
\varphi^{\#}(D(f)) : S_f = \mathcal{O}_{\operatorname{Spec} S}(D(f)) \to \mathcal{O}_{\operatorname{Spec} R}\left(\varphi^{\#}(f)\right) = R_{\varphi^{\#}(f)} \cong R \otimes_S S_f \\
\varphi^{\#}(D(f))\left(\frac{s}{t}\right) = 1 \otimes_S \frac{s}{t}
\end{gathered}$$

Thus, over general schemes, we want the same structure to be preserved:

Definition 3.1.5: Morphism Of Schemes

Let X, Y be schemes. Then a morphism of locally ringed spaces $\pi : X \to Y$ is called a morphism of schemes. Schemes along with scheme morphisms form a category Sch.

We have shown how to induce the map between rings and locally ringed space on affine schemes. Let us make sure that these maps are in bijective correspondence:

Theorem 3.1.6: Ring Homomorphism and Locally Ringed Space

Let $\varphi : R \to S$ be a ring homomorphism. Then

1. φ induces a natural morphism of locally ringed space[a]

$$\left(\varphi, \varphi^\sharp\right) : \left(\operatorname{Spec}(S), \mathcal{O}_{\operatorname{Spec}(S)}\right) \to \left(\operatorname{Spec}(R), \mathcal{O}_{\operatorname{Spec}(R)}\right)$$

2. Any morphism of locally ringed space from $\operatorname{Spec}(S) \to \operatorname{Spec}(R)$ induces a ring homomorphism $\varphi : R \to S$

3. If the ring homomorphism φ induces a locally ringed space morphism φ^* which induces a ring homomorphism ψ, then $\varphi = \psi$.

[a]ringed space morphism that preserves local rings

Proof **:**

1. Let $\varphi : R \to S$ be a morphism which induces the continuous map $f : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ (where the notation f is used to not overcrowd notation later in the proof). For each $\mathfrak{p} \in \operatorname{Spec}(S)$, localize S at $\mathfrak{p}$ (i.e. $S \backslash \mathfrak{p}$) to obtain the local homomorphism of the local rings $\varphi_{\mathfrak{p}} : R_{\varphi^{-1}(\mathfrak{p})} \to S_{\mathfrak{p}}$. Now, as these stalks follow the condition of proposition 1.1 .22 the map $f^\sharp : \mathcal{O}_{\operatorname{Spec}(R)}(U) \to \mathcal{O}_{\operatorname{Spec}(R)}\left(f^{-1}(U)\right)$ is a ring homomorphism for all $U \subseteq \operatorname{Spec}(U)$. Hence $f^\sharp$ is a morphism of sheaves which induces local rings

2. Let $\left(f, f^\sharp\right) : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ be a morphism of locally ringed spaces with the usual structure sheaf. By proposition 2.3.5, the global section induce a ring homomorphism:

$$\varphi : \mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R)) \to \mathcal{O}_{\operatorname{Spec}(S)}(\operatorname{Spec}(S)) \quad \text{i.e.} \quad \varphi : R \to S$$

   Then by the above discussion for any prime $\mathfrak{p} \in \operatorname{Spec}(S)$, the induce map on the stalks $R_{f^{-1}(\mathfrak{p})} \to S_{\mathfrak{p}}$ is compatible with φ, namely $f(\mathfrak{p}) = \left(f^\#\right)^{-1}(\mathfrak{p})$ (i.e. the commutative diagram between $\varphi : R \to S$ and $\varphi_{\mathfrak{p}} : R_{f^\#_{f^{-1}(\mathfrak{p})}} \to S_{\mathfrak{p}}$) forces this equality). Namely, this implies that the map f mappings prime ideals uniquely induces a map φ between rings.:

$$\begin{tikzcd}
X & \operatorname{Spec} R \arrow[d, dash, dashed] & \operatorname{Spec} S \arrow[l, "f"'] \arrow[d, dash, dashed] & Y \\
\mathcal{O}_X(X) & R \arrow[r, "f^\#(X)=\varphi"] \arrow[d, dash, dashed] & S \arrow[d, dash, dashed] & f_*\mathcal{O}_Y(Y) \\
\mathcal{O}_{X,f^{-1}(\mathfrak{p})} & R_{f^{-1}(\mathfrak{p})} \arrow[r, "f^\#_{f^{-1}(\mathfrak{p})}"] & S_{\mathfrak{p}} & \mathcal{O}_{Y,\mathfrak{p}}
\end{tikzcd}$$

3. As $f^\sharp$ is a local homomorphism, $\varphi^{-1}(\mathfrak{p}) = f(\mathfrak{p})$, hence f concise with the map $\operatorname{Spec}(S) \to \operatorname{Spec}(R)$ induced by φ. Furthermore, $f^\sharp$ is also induced by φ, since the morphism $\left(f, f^\sharp\right)$ of locally ringed spaces comes from the homomorphism φ, as we sought to show.

s are the correct category to be the opposite category of [commutative] rings:

Corollary 3.1.7: Equivalence of Rings And Affine Schemes

The category of rings and the opposite category of s are equivalent.

Furthermore, the functor Spec and Γ (the global section functor) are adjoint.

This shows that schemes are the proper object to capture the information of rings.

Proof **:**
exercise.

This equivalence will allow us to completely control the morphisms of schemes by looking at ring homomorphisms, namely by reducing a morphism of schemes to a morphism of s which is then in correspondence with the morphism of affine schemes. First, we must insure that gluing morphisms is well-defined. The result can work in the slightly more general context of ringed spaces:

Lemma 3.1.8: Morphisms Defined on Open Sets

Let X, Y be two ringed spaces where $X = \cup_i U_i$ is an open cover for X. Then if there exists morphism of ringed space morphisms $\pi_i : U_i \to Y$ that agree on overlaps

$$f_i|_{U_I \cap U_j} = f_j|_{U_i \cap U_j}$$

Then there is a unique morphism of ringed space $f : X \to Y$ such that $f|_{U_i} = f_i$

Proof **:**
First, by the gluing lemma in topology the map $f : X \to Y$ is certainly a continuous map, we must show that we have a sheaf morphism $f^{\#} : \mathcal{O}_Y \to f_* \mathcal{O}_X$. For each $f_i : U_i \to Y$ we have a sheaf morphism $f_i^{\#} : \mathcal{O}_Y \to (f_i)_* \mathcal{O}_{U_i}$. If $f_i|_{U_i \cap U_j} = f_j|_{U_i \cap U_j}$ then:

$$f_i^{\#}(U_i \cap U_j) = f_j^{\#}(U_i \cap U_j)$$

Now for each open $V \subseteq Y$, We can define for each $V \cap f^{-1}(U_i)$

$$f^{\#}\left(V \cap f^{-1}(U_i)\right) : \mathcal{O}_Y\left(V \cap f^{-1}(U_i)\right) \to \mathcal{O}_X\left(f^{-1}\left(V \cap f^{-1}(U_i)\right)\right)$$
$$f^{\#}\left(V \cap f^{-1}(U_i)\right) = f_i^{\#}\left(V \cap f^{-1}(U_i)\right)$$

where $f_i^{\#}$ and $f_j^{\#}$ agree on $V \cap f^{-1}(U_i) \cap f^{-1}(U_j)$. As these maps cover V and agree on intersections, then for each $s \in \mathcal{O}_Y(V)$, there is a unique section $t \in \mathcal{O}_X\left(f^{-1}(V)\right)$ where $f^{\#}(V)(s) = t$, namely this section exists by the sheaf axioms and respects restrictions. As this is true for each open $V \subseteq Y$, then this is exactly the criterion for the sheaf morphism $f^{\#}$, completing the proof.

Proposition 3.1.9: Scheme Morphism Locally Affine

Let $\pi : X \to Y$ be a morphism of schemes. Then it is a morphism of locally ringed spaces that locally looks like a morphism of affine schemes, that is if $\operatorname{Spec} R \subseteq X$ and $\operatorname{Spec} S \subseteq Y$ where $\pi(\operatorname{Spec} R) \subseteq \operatorname{Spec}(S)$, then the induced morphism of ringed spaces is a morphism of affine schemes. Conversely, a morphisms of ringed spaces that locally looks like a morphism of affine scheme is a morphism of schemes.

Proof **:**
Let $\pi : X \to Y$ be a morphism of schemes, and let $Y = \bigcup_j \operatorname{Spec} S_j$ be an affine open cover. Let $y \in \pi(X)$, and consider $x \in \pi^{-1}(y)$. As $y \in \operatorname{Spec} S_j$ for some $j, x \in \pi^{-1}$ (Spec S_j) and the pre-image is open, and hence can be covered by affine open subschemes, on of which must contain the point x, say this is $\operatorname{Spec} R_i$. As the open subset of $\pi^{-1}(\operatorname{Spec} S_j)$ is open in X, we have that each point $x \in X$ is contain in an affine open subscheme $\operatorname{Spec} R_i$ such that $\pi(\operatorname{Spec} R_i) \subseteq \operatorname{Spec} S_j$ and the map is certainly a morphism of affine schemes.
Conversely, if a map $\pi : X \to Y$ between schemes is locally a morphism of schemes on s, then by lemma 3.1.6 we get a ringed space morphism, which is certainly a locally ringed space morphism and hence $\pi : X \to Y$ is a scheme morphism, completing the proof.

Example 3.1.10: Morphisms Of Schemes

1. Recall the map $\varphi : \mathbb{C}[y] \to \mathbb{C}[x]$ given by $y \mapsto x^2$, which on closed points can be interpreted as $a \mapsto a^2$. Let's look at the pullback of functions. Consider $3/(y-4)$ on $D([(y-4)]) = \mathbb{A}^1 \backslash \{4\}$. The pull-back is $3/\left(x^2-4\right)$ on $\mathbb{A}^1 \backslash \{-2, 2\}$.

2. Let $U \subseteq Y$ be an open subsets of a scheme Y. Then there is a natural morphism of ringed space
$$(U, \mathcal{O}_Y|_U) \to (Y, \mathcal{O}_Y)$$
More is true, the above map is an open embedding, namely the image of U under the above map is an isomorphism, where the image is also open in Y. Later, this will be shown to be an immersion in the geometrical sense, that is the stalks maps are injective (mirroring the tangent-space maps being injective).

3. Let us look at a non- morphisms. Consider:
$$\iota : \mathbb{A}_k^{n+1} \backslash \{0\} \to \mathbb{P}_k^n \quad (x_0, x_1, \ldots, x_n) \mapsto [x_0, x_1, \ldots, x_n]$$
It certainly continuous, so let's us define a sheaf morphism compatible with it. By lemma 3.1.6, it suffices to show it on an open cover and showing gluing works. Take the usual affine cover of $\mathbb{P}_k^n, H_0, \ldots, H_n$, and define:
$$\iota^{\#}(H_i) : \mathcal{O}_{\mathbb{P}_k^n}(H_i) \to \mathcal{O}_{\mathbb{A}_k^{n+1}}\left(f^{-1}(H_i)\right)$$
where
$$\mathcal{O}_{\mathbb{P}_k^n}(H_i) \cong k[x_0, \ldots, \hat{x}_i, \ldots, x_n]$$

where $(\hat{-})$ represents omitting the element, and $f^{-1}(H_i) = \left\{x \in \mathbb{A}_k^{n+1} \middle| x_i \neq 0\right\} \cong D(x_i) \subseteq \mathbb{A}_k^{n+1}$. Thus, we define a map:

$$\iota^{\#}(H_i) : k[x_0, \ldots, \hat{x}_i, \ldots, x_n] \to k[x_0, \ldots, x_n]_{x_i} \quad f \mapsto \frac{f}{1}$$

On $H_i \cap H_j$, we have that the homogeneous coordinates $x_i, x_j \neq 0$ and:

$$\iota^{\#}(H_i \cap H_j) : \mathcal{O}_{\mathbb{P}_k^n}(H_i \cap H_j) \to \mathcal{O}_{\mathbb{A}_k^{n+1}}\left(f^{-1}(H_i \cap H_j)\right)$$

where the domain is:

$$\mathcal{O}_{\mathbb{P}_k^n}(H_i \cap H_j) = k[x_0, \ldots, \hat{x}_i, \ldots, x_n]_{x_j} \cong k[x_0, \ldots, \hat{x}_i, \ldots, \hat{x}_j, \ldots, x_n]$$

and the codomain is:

$$\mathcal{O}_{\mathbb{A}_k^{n+1}}\left(f^{-1}(H_i \cap H_j)\right) = \mathcal{O}_{\mathbb{A}_k^{n+1}}(D(x_i) \cap D(x_j)) \cong k[x_0, \ldots, x_n]_{x_i, x_j}$$

And hence we have

$$k[x_0, \ldots, \hat{x}_i, \ldots, \hat{x}_j, \ldots, x_n] \to k[x_0, \ldots, x_n]_{x_i, x_j} \quad f \mapsto \frac{f}{1}$$

Certainly all the restriction maps commute, and the H_i cover $\mathbb{P}_k^n$, hence we have a well-defined sheaf morphism. Intuitively, we would expect that on every function on a hyper-surface H_i be identically a function on $D(x_i) \subseteq \mathbb{A}_k^n$, and this scheme morphisms shows exactly that intuition!

4. Let $S_\bullet$ be a finitely generated graded R-algebra. Describe the morphism

$$\operatorname{Proj} S_\bullet \to \operatorname{Spec}(R)$$

5. Let $R_\bullet \to S_\bullet$ be a surjective homogeneous homomorphism. Show this gives a clsoed embedding $\operatorname{Proj} R_\bullet \to \operatorname{Proj} S_\bullet$.

6. Take the map $\pi : \operatorname{Spec}(\mathbb{C}[x, y]) \to \operatorname{Spec}(\mathbb{C}[x])$ induced by the inclusion $\mathbb{C}[x] \hookrightarrow \mathbb{C}[x, y]$. This map is the projection map, any closed point $[(x - 3, y - a)]$ for all $a \in \mathbb{C}$ map to $[(x - 3)]$. What about non-closed points such as $\left[\left(y - x^2\right)\right]$? Check that this point maps to the generic point $[(0)]$!

7. Let $X_\mathfrak{p}$ be the local scheme at the point $\mathfrak{p}$ of a scheme $X = \mathbb{A}_k^1$. The global sections are given by $k[x]_{(x)}$, and there is an inclusion map $k[x] \hookrightarrow k[x]_{(x)}$ inducing a map $X_\mathfrak{p} \to \mathbb{A}_k^1$. Show that the induced morphism of sheaves is an isomorphism (that is, if $\iota : \operatorname{Spec}(\mathcal{O}_{X,\mathfrak{p}}) \to \mathbb{A}_k^1$ is the natural inclusion, the map $\iota^{\#}$ is an isomorphism). This shows that local schemes can be thought of as "almost" open sets.

If the codomain of a scheme morphism is an affine scheme, then we can completely determine the behaviour of the scheme morphisms by looking at the ring morphism between the global sections:

Proposition 3.1.11: Morphisms to Affine Schemes

Let $\mathrm{Hom}_{\mathbf{Sch}}(X, \mathrm{Spec}(R))$ be the collection of morphism of scheme from a scheme X to an $\mathrm{Spec}(R)$. Then it is in natural bijection to the ring homomorhpisms $\mathrm{Hom}_{\mathrm{Ring}}\ (R, \mathcal{O}_X(X))$:

$$\mathrm{Hom}_{\mathbf{Sch}}(X, \mathrm{Spec}(R)) \cong \mathrm{Hom}_{\mathbf{Ring}}\,(R, \mathcal{O}_X(X))$$

In particular, The map $\theta : X \to \mathrm{Spec}\,(\Gamma(X, \mathcal{O}_X))$ is an addiction. Furthermore, this is an isomorphism if and only if X is an affine scheme.

Note that the ring $\mathcal{O}_X(X)$ can be very complex, even in nice cases where X is a finite-type k-scheme, for example it need not be finitely generated.

Proof **:**
If $X = \mathrm{Spec}(S)$, then theorem 3.1 .4 gives the bijection. Next, map $\pi : X \to \mathrm{Spec}(R)$ certainly induces a ring homomorphism $\varphi : R \to \mathcal{O}_X(X)$, we shall show that a ring homomorphism $\varphi : R \to \mathcal{O}_X(X)$ induces a unique scheme map $\pi : X \to \mathrm{Spec}(R)$ and that this map is the inverse of the above map. By definition of an :

$$\pi : \bigcup_i \mathrm{Spec}\,(S_i) \to \mathrm{Spec}(R)$$

where we get a restriction map $\pi_i : \mathrm{Spec}\,(S_i) \to \mathrm{Spec}(R)$ along with sheaf morphisms, for which the global section map is $\varphi_i : R \to \mathcal{O}_X\,(\mathrm{Spec}\,(S_i)$. As $\mathrm{Spec}\, S_i \subseteq X$ is an open subset, there is a natural restriction map $\mathcal{O}_X(X) \to \mathcal{O}_X\,(\mathrm{Spec}\, S_i)$. We thus get a composition:

$$R \to \mathcal{O}_X(X) \to \mathcal{O}_X\,(\mathrm{Spec}\, S_i)$$

The map $R \to \mathcal{O}_X\,(\mathrm{Spec}\, S_i)$ is uniquely determined by the ring homomorphism $S_i \to R$, and as X is covered by the $\mathrm{Spec}\, S_i$, by lemma 3.1 .6 there is a unique way to glue the maps of $\mathcal{O}_X\,(\mathrm{Spec}\, S_i)$ to $\mathcal{O}_X(X)$, completing the proof.

The converse is not true:

Example 3.1.12: Maps From Affine To Projective
Take $\mathrm{Hom}_{\mathrm{Sch}}\,\left(\mathrm{Spec}\,\mathbb{Q}[x], \mathbb{P}^1_{\mathbb{Q}}\right)$ and $\mathrm{Hom}_{\mathbf{Ring}}(\mathbb{Q}, \mathbb{Q}[x])$ (recall that $\mathcal{O}_X\,\left(\mathbb{P}^1_{\mathbb{Q}}\right) \cong \mathbb{Q})$). Then

$$\mathrm{Hom}_{\mathrm{Ring}}\,(\mathbb{Q}, \mathbb{Q}[x]) = \{\mathrm{id}\}$$

namely since since the units of $\mathbb{Q}$ must map to the units of $\mathbb{Q}[x]$, which are all contains in $\mathbb{Q}$, and as it must be that $1 \mapsto 1$, we are forced to have the identity. However, there are multiple maps from Spec $\mathbb{Q}[x]$ to $\mathbb{P}^1_k$; simply map Spec $\mathbb{Q}[x]$ into the open subset $U_0 \cong k[y] \subseteq \mathbb{P}^1_k$ by taking $x \mapsto p(y)$ for some polynomial $p(y) \in k[y]$. More generally, we can let choose a field k[a]

The intuition is that $\mathcal{O}_X(X)$ in general contains less information than X as the global sections on a general scheme need to restrict to local sections on affine open subsets, and hence there may be in general less of them.

Show that the canonical morphism $X \to \mathrm{Spec}\,(\mathcal{O}_X(X))$ is an isomorphism if and only if X is a affine.

[a]Note that an [injective] ring homomorphism $k \to k$ need not be surjective, consider $F(x) \to F(x)$ given by $x \mapsto x^2$, hence the argument requires a bit more work

Corollary 3.1.13: Characterizing Affine Schemes

Let X be a scheme. Then X is an affine scheme if and only if there is a finite subset of elements $f_1, \ldots, f_n \in \mathcal{O}_X(X)$ such that X_f is affine and $(f_1, \ldots, f_n) = \mathcal{O}_X(X)$

***Proof* :**
If X is affine, take $f_1 = 1$. Conversely, assume each X_{f_i} is affine and $(f_1, ..., f_n) = \mathcal{O}_X(X)$ so that for some $a_i \in \mathcal{O}_X(X)$, $\sum_i a_i f_i = 1$. Let $R = \mathcal{O}_X(X)$. I claim that $X \cong \operatorname{Spec} R$. Define $\varphi : X \to \operatorname{Spec} R$ where x maps to the prime ideal in R that vanishes at x. It is not too hard to show this map is a homeomorphism. For the scheme isomorphism, we have by definition the isomorphisms $X|_{D(f_i)} \cong \operatorname{Spec} R|_{D(f_i)}$, hence by proposition 3.1.9 we have our desired isomorphism, completing the proof.

3.1.2 S-Schemes

Recall that all abelian rings can be seen as $\mathbb{Z}$-algebras, and that $\mathbb{Z}$ is initial in the category of commutative rings. From this perspective, all ring homomorphism $\varphi : R \to S$ are in bijective correspondence with morphisms that make the following diagram commute:

$$\begin{tikzcd} R \arrow[rr] & & S \\ & \mathbb{Z} \arrow[ul] \arrow[ur] & \end{tikzcd}$$

Then we may generalize the notion of abelian rings to R-algebras by replacing $\mathbb{Z}$ with a ring R. We may further augment the morphisms to R-algebra homomorphisms as we are working with commutative rings. This is the motivation to the following generalization; we can generalize the notion of R-schemes given in definition 2.2.14:

Definition 3.1.14: S-Scheme

The category consisting of all schemes X that have scheme morphisms $X \to S$ and whose morphisms are:

$$\begin{tikzcd} X \arrow[rr] \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

is denoted $\mathbf{Sch}_S$, and it's object are called *S-scheme*. The morphism of schemes $X \to S$ is called the *structure morphism*.

To see what extra structure is added, let us start the case of $S = \operatorname{Spec}(R)$. Then each open subset of X has the structure of an R-algebra, and these generalize definition 2.2.14. If $S = \operatorname{Spec}(R)$, then we would often say that the S-scheme (i.e. $\operatorname{Spec}(R)$-scheme) is an R-scheme

Working over S-schemes shall restrict to much "simpler" cases than over general schemes. For example, if studying complex analysis and complex varieties, we would expect Spec($\mathbb{C}$) to have trivial automorphisms, and that Spec $\left(\mathbb{C}[x]/\left(x^2+1\right)\right)$

Example 3.1.15: S-scheme

1. When working with polynomials, we are often interested in k-homomorphisms $k[x] \to k[x]$ where k is fixed (or else our morphisms would include field automorphisms, which may be undesirable). We can consider this condition as the following diagram commuting:

$$\begin{tikzcd} k[x] \arrow[rr] & & k[x] \\ & k \arrow[ul, "\mathrm{id}"] \arrow[ur, "\mathrm{id}"'] & \end{tikzcd}$$

   This would correspond to k-scheme maps:

$$\begin{tikzcd} \operatorname{Spec}(k[x]) \arrow[dr] & & \operatorname{Spec}(k[x]) \arrow[ll] \arrow[dl] \\ & \operatorname{Spec}(k) & \end{tikzcd}$$

   Note the importance of these maps commuting as scheme morphisms and not just topological maps.

2. Projective schemes over a field k are k-projective schemes.

3. Let L/K be a field extension. Then Spec K, Spec L are K-schemes, Notice that Spec $\to$ Spec K has as a sheaf morphism the injective map $K \to L$, that is it is the field extension L/K. We shall later see that we can find an appropriate scheme that will act as a covering space for K, and we shall generalize the Galois group to the fundamental group scheme which shall remember multiplicity information of roots of polynomials.

Lemma 3.1.16: Stalks of k-schemes

Let X be a k-scheme and $x \in X$ so that $x = [\mathfrak{p}]$ for some prime $\mathfrak{p} \subseteq R_i = k\left[u_i\right]_i$. Then

$$\mathcal{O}_{X,\mathfrak{p}} \cong \left(k\left[u_i\right]_i\right)_{\mathfrak{p}}$$

In particular, we have that the quotient by $\mathfrak{m}_x$ is a k-algebra:

$$\kappa(x) = k(x)$$

Proof **:**
As X is a k-scheme, each $\mathcal{O}_X(U)$ is a k-algebra, and the colimit of k-algebra's is a k-algebra (they are closed under products and equalizers, see [8]). The rest comes from proposition 2.3.14.

Definition 3.1.17: Locally Finite Type And Finite Type

Let X, Y be S-schemes. Then a morphism $f : X \to Y$ is locally of finite type if there exists an affine open cover $V_j = \text{Spec}\,(S_j)$ of Y where $f^{-1}(V_i)$ can be covered by affine open subsets $U_{ij} = \text{Spec}\,(A_{ij})$ where A_{ij} is a finitely generated S_i-algebra, that is

$$f^{\#}(V_i) : S_j = \mathcal{O}_Y(V_j) \to \mathcal{O}_X\left(f^{-1}(V_i)\right) \xrightarrow{\text{res}} \mathcal{O}_Y(U_{ij}) = R_{ij}$$

induces a finitely generated S_j-algebra on each R_{ij}. If furthermore each $f^{-1}(V_j)$ can be covered by finite number of affine subsets, f is said to be of finite type.
An S-scheme X is said to be of locally finite type (resp. finite type) over S if $f : X \to S$ is of locally finite type (resp. finite type).

Proposition 3.1.18: Final Object Of Scheme

In **Sch**, the scheme $\text{Spec}\,\mathbb{Z}$ is the final object. This makes the category of schemes isomorphic to the category of $\mathbb{Z}$-schemes.

If k is a field, $\text{Spec}\,k$ is the final object in the category of k-schemes.

Proof **:**
Recall that $\mathbb{Z}$ is initial in Ring, and thus it is easy to show it is initial in Comm-Ring, then as Schemes (resp. k-schemes) are the co-categories to rings (resp. k-algebras), the result is immediate.

Finally, with k-schemes defined, we can see that there is a natural way in which we can embed (quasi-projective) varieties into the category $\mathbf{Sch}_{\bar{k}}$. In section 3.4 . we shall understand better what the image of this functor is:

Theorem 3.1.19: Varieties and Schemes: Fully Faithful Functor

Let $\bar{k}$ be an algebraically closed field. Then there exists a fully faithful functor $\mathbf{Var}_{\bar{k}} \to \mathbf{Sch}_{\bar{k}}$ where Var is the usual category of varieties as defined in [8].

Recall that a fully faithful means the hom-sets are in bijection, and it implies injectivity on objects (but not necessarily surjective, for example Ab $\to$ Grp is a fully faithful functor). We shall also stick with Hartshorne's notation for the functor, which is the letter t, it shall come up a few times later in the book.

Proof **:**
For any (quasi-projective) variety, we shall first find the "points" that shall be the points of the scheme by working with the irreducible closed subsets. We shall then find the structure sheaf to put on it by finding all the regular functions, which shall be enough to define the functor.

First off, for any topological space X, and let $t(X)$ be the set of (nonempty) irreducible closed subsets of X. If Y is a closed subset of X, then $t(Y) \subseteq t(X), t(Y_1 \cup Y_2) = t(Y_1) \cup t(Y_2)$, and $t\left(\bigcap Y_i\right) = \bigcap t(Y_i)$. Hence, we can define a topology $t(X)$ by taking as closed sets the subsets of the form $t(Y)$, where Y is a closed subset of X. If $f : X_1 \to X_2$ is a continuous map, then we obtain a map $t(f) : t(X_1) \to t(X_2)$ by sending an irreducible closed subset to the closure of its image. Thus t is a functor on topological spaces. Furthermore, define a continuous map $\alpha : X \to t(X)$ by $\alpha(P) = \{P\}^-$. Note that α induces a bijection between the set of open subsets of X and the set of open subsets of $t(X)$.

Next, let k be an algebraically closed field, V be a variety over k, and let $\mathcal{O}_V$ be its sheaf of regular functions. I claim that $(t(V), \alpha_*(\mathcal{O}_V))$ is a scheme over k. Since any variety can be covered by open affine sub-varieties, it will be sufficient to show that if V is affine, then $(t(V), \alpha_*(\mathcal{O}_V))$ is a scheme. Hence, let V be an affine variety with affine coordinate ring A. Define a morphism of locally ringed spaces

$$\beta : (V, \mathcal{O}_V) \to X = \operatorname{Spec} A$$

as follows: for each point $P \in V$, let $\beta(P) = \mathfrak{m}_P$, the ideal of A consisting of all regular functions which vanish at P. Then β is a bijection of V onto the set of closed points of X. It is easy to see that β is a homeomorphism onto its image. Now for any open set $U \subseteq X$, we will define a homomorphism of rings $\mathcal{O}_X(U) \to \beta_*(\mathcal{O}_V)(U) = \mathcal{O}_V(\beta^{-1}U)$. Given a section $s \in \mathcal{O}_X(U)$, and given a point $P \in \beta^{-1}(U)$, we define $s(P)$ by taking the image of s in the stalk $\mathcal{O}_{X,\beta(P)}$, which is isomorphic to the local ring $A_{\mathfrak{m}_P}$, and then passing to the quotient ring $A_{\mathfrak{m}_P}/\mathfrak{m}_P$ which is isomorphic to the field k. Thus s gives a function from $\beta^{-1}(U)$ to k. It is easy to see that this is a regular function, and that this map gives an isomorphism $\mathcal{O}_X(U) \cong \mathcal{O}_V(\beta^{-1}U)$. Finally, since the prime ideals of A are in 1-1 correspondence with the irreducible closed subsets of V, these remarks show that $(X, \mathcal{O}_X)$ is isomorphic to $(t(V), \alpha_*(\mathcal{O}_V))$, so the latter is indeed an affine scheme, completing the proof.

3.1.3 Z-valued Points

Let's say we have the scheme Spec $\mathbb{R}[x,y]/(x^2+y^2-1)$ whose closed points is the real circle[4]. Then if we are looking for the rational-valued points on the scheme, we are looking for the function:

$$\mathrm{ev}_{a,b} : \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)} \to \mathbb{Q}$$

mapping $x \to a$ and $y \to b$ with the property (given by the quotient) that $a^2+b^2=1$. Another way of looking at this situation is that we have a collection of functions $\mathrm{ev} : \mathbb{R}[x,y] \to \mathbb{Q}$ and we are taking the functions where $a^2+b^2-1=0$, which is exactly the functions that factor through $\mathrm{ev}_{a,b} : \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)} \to \mathbb{Q}$. Examples of such functions are $\mathrm{ev}_{1,0}$, $\mathrm{ev}_{\frac{3}{5},\frac{4}{5}}$ for $\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2 - 1 = 0$. This ring homomorphism is part of the global section morphism, namely given the scheme morphism $\mathrm{ev} :$ Spec $\mathbb{Q} \to$ Spec $\frac{\mathbb{R}[x,y]}{(x^2+y^2-1)} = C$, we have the sheaf morphism on global sections:

$$\mathrm{ev}^\#(C) : \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)} \to \mathbb{Q}$$

Now, if we change the ring to $\mathbb{Z}$, or more generally some (local or global) number ring, we shall get different sets of "legal" evaluation functions. For example, if $R = \mathbb{Z}$, then the only evaluation maps

[4]Note that it non-closed points make this bigger in a meaningful way, recall example 2.14

are the 4 maps $\mathrm{ev}_{\pm 1,0}$ and $\mathrm{ev}_{0,\pm 1}$. Hence, we may think of the collection of ring homomorphisms:

$$\mathrm{ev}_{a,b} : \frac{\mathbb{R}[x,y]}{(x^2+y^2-1)} \to \mathbb{Z}$$

as representing the solution over $\mathbb{Z}$ of $x^2+y^2=1$. Dualizing this to affine schemes, this is equivalent to asking for a scheme morphism:

$$\mathrm{Spec}\,(\mathrm{ev}) : \mathrm{Spec}\,(\mathbb{Z}) \to \mathrm{Spec}\,\frac{\mathbb{R}[x,y]}{(x^2+y^2-1)}$$

which geometrically can be thought out as picking the $\mathbb{Z}$-valued points. More generally, we may imagine we want to have the values come from a scheme, not just an affine scheme, i.e. a ring R (for example, we are evaluating points on a torus with values coming from a curve). This looses the interpretation of evaluation on a global level (ex. a subscheme of projective space), however this is exactly what the sheaf was made to recover: evaluation can be done locally. Hence, we more generally want to have any scheme morphism $S \to X$ to be considered the points of X that take on values on S. We make this precise in the following definition:

Definition 3.1.20: Z-valued Points

Let X be a scheme. Then the *Z-valued points* of X are the morphisms $Z \to X$, and is denoted $X(Z)$. If $Z = \mathrm{Spec}\,(R)$, then this is often denoted $X(R)$.

The reader should be aware that the morphism $Z \to X$ is called the "Z-valued point", and not any particular point in Z or the image of Z. This terminology can be thought of from the following idea: a morphisms of schemes $X \to Y$ is not determined by the underlying map of points (the continuous map).

Proposition 3.1.21: Z-valued Points and Scheme Morphisms

Let $X \to Y$ be a scheme morphism. Show that it is determined by all the maps of Z-valued points, namely given all Z-valued point maps to X and Y, we get a unique scheme morphism map

Proof **:**
Here is the high-level: first show that any scheme morphism $X \to Y$ induced a map $X(Z) \to Y(Z)$. Then, take $Z = X$. Then we have a version of Yoneda's lemma.

This proof may seem trivial, but the idea behind it will become important when looking at cohomological ideas.

Example 3.1.22: Z-valued Points

1.

2. One of the most common example's is $\bar{k}$-valued points on given on schemes (not yet necessarily over any field). The $\bar{k}$-valued points are the collection of maps $\varphi : \mathrm{Spec}(\bar{k}) \to X$. Consider $\varphi([0]) = y \in X$. Taking the stalk-map, where $\varphi_y : \bar{k}O_{X,y} \to \bar{k}$. Taking the residue field, we get: the naturally extended map:

$$\varphi_y : \bar{k}(y) \to \bar{k}$$

This map forces y to be a closed point: If y is not closed, $\bar{k}(y)$ contains transcendental elements (see ref:HERE), say $x \in \bar{k}(y)$, but then there is no morphism from $\bar{k}(y)$ to $\bar{k}$ as for $\varphi(x) \in \bar{k}$ we have $p\left(\varphi_y(x)\right) = 0$, but then $\varphi_y(p(x)) = 0$ for $p(x) \in \bar{k}(y)$ which is impossible as field homomorphisms are injective (naturally $\varphi(0) = 0$ as well).
Hence it is always the case that there exists a natural map $\bar{k}(y) \to \bar{k}$ from $\mathrm{Spec}(\bar{k})$ to the closed points of a scheme X. If X is locally finite type over $\bar{k}$, each stalk is a finitely generated $\bar{k}$-algebra and, hence by Zariski's Lemma $\bar{k}(y) = \bar{k}[y]$, and as $\bar{k}$ is algebraically closed, $\bar{k}[y] = \bar{k}$. In fact, we have a one-to-one correspondence:

$$\{\mathrm{Spec}(\bar{k}) \to X\} \xrightarrow{\sim} X_{\max}$$

where $X_{\max}$ is the set of closed points of a scheme X of finite type over $\bar{k}$. Show this is even a functor.
Note that this is another counter-example of the "co" of proposition 3.1.8

3. Consider $\mathbb{C}(t)$-valued points over $\mathbb{C}$-schemes. Any such point would fit into the following commutative diagram:

$$\begin{tikzcd} \mathrm{Spec}\,\mathbb{C}(t) \arrow[rr] \arrow[dr] & & X \arrow[dl] \\ & \mathrm{Spec}\,\mathbb{C} & \end{tikzcd}$$

4. Next, take $D = k[\epsilon]/(\epsilon^2)$, the dual numbers. Let us classify all scheme morphisms $\mathrm{Spec}\,(D) \to X$ of finite type over k (a scheme morphism is of finite type if every affine open subset $\mathrm{Spec}\,(S) \subseteq X$ has an affine pre-image, $\mathrm{Spec}\,(R) = \varphi^{-1}(\mathrm{Spec}\,(S)$ whose ring R is a finitely generated S-algebra). As D has a unique maimxla ideal (ϵ), $\mathrm{Spec}\,(D)$ has a unique maximal closed point, let's label it $\bar{o}$. Then the homomorphism $D \to k$ with kernel (ϵ) is associated to the canonical embedding $\iota : \mathrm{Spec}\,(k) \hookrightarrow \mathrm{Spec}\,(D)$ with $\bar{o} = \iota(o)$ (hence the bar notation). Then any morphism $\varphi : \mathrm{Spec}\,(D) \to X$ is in correspondence to the composite-morphism $\varphi \circ \iota : \mathrm{Spec}\,(k) \to X$, which itself is in correspondence to the closed points of X. If $x \in X$ is a closed points, then for appropriate φ we have $x = \varphi(\bar{o})$.

   Now, for a choice of $x \in X$, pick an affine open neighborhood $x \in \mathrm{Spec}\,(R) \subseteq X$, and take $\mathrm{Hom}_x(\mathrm{Spec}\,(D), X) \cong \mathrm{Hom}_x(\mathrm{Spec}\,(D), \mathrm{Spec}\,(R))$ where Hom_x is the collection of scheme morphisms st $\varphi(\bar{o}) = x$. Then all these morphisms are determined by a local k-homomorphism $f : R \to D$ where $f(\mathfrak{m}_x) \subseteq (\epsilon)$ by locality. Then as R, as a vector-space, is equal to

$$R = k + \mathfrak{m}_x$$

   we have that the homomorphism is determined by what it does to $\mathfrak{m}_x$, namely it defines some linear transformation $\mathfrak{m}_x \to (\epsilon) \cong k$. As $\epsilon^2 = 0$, $f(\mathfrak{m}_x^2) = 0$, and so f is a *linear function* on $\mathfrak{m}_x/\mathfrak{m}_x^2$. Convesely, any function f to extended to be 0 on $\mathfrak{m}_x^2$ determines a homomorphism $R \to D$ taking $\mathfrak{m}_x$ into (ϵ).

   Thus, if X is a k-scheme and $x \in X$, the set

$$\mathrm{Hom}_x(\mathrm{Spec}\,(D), X) \cong \mathfrak{m}_x/\mathfrak{m}_x^2 = \Theta_{X,x}$$

Furthermore, if $f : X \to X'$ is another morphism, this induces a differential

$$\mathrm{d}_x : \Theta_{X,x} \to \Theta_{X',f(x)}$$

3.1.4 Category of Scheme

(I moved this discsussion under S-schemes and Z-valued points because it seemed important to mention those here)

I want to essentially write this:

1. affine schemes are complete and cocomplete because Ring is cocomplete and complete respectively
2. schemes are not closed under infinite limits: $\prod_{i=1}^{\infty} \mathbb{P}^1_k$ is not a scheme. However it is closed under finite limits. There is a proof (found on Wikipedia click here) that all finite limits exist in a category with pullbacks (i.e. fibered products) and terminal objects. The fibered product which shall be covered soon.
3. Schemes are not closed under even finite colimits (the example of Spec $(K) \rightrightarrows \mathbb{A}^1_K$ gluing the generic point is the usual counter-example, it comes up later in example 3.2.5)
4. Though pushout don't exist in general, if both maps are open immersions, it exists (open immersion here means $f : X \to Y$ is open and $f^{\#}$ is a sheaf isomorphism; in Emmanuel's notes)
5. schemes have cofiltered limits (sometimes called inverse limits; example is $\mathbb{Z}_p$ as a limit)
6. schemes have filtered colimits (sometimes called projective limits; example is $\mathbb{P}^{\infty}_k$ as a colimit)

Exercise 3.1.1

1. Show that Spec $k(x) \to$ Spec $k[y]_{(y)}$ sending the unique point η to the closed points of the codomain is a ringed-space morphism that is not a locally ringed space morphism.
2. Let $\mathfrak{p} \in X$ be a point in a scheme. Show there is a canonical map Spec $(\mathcal{O}_{X,\mathfrak{p}}) \to X$
3. Using the above, define a canonical morphism Spec$(\kappa(\mathfrak{p})) \to X$.
4.

3.1.5 Rational Maps

When working with [quasiprojective] varieties, we had the weaker notion of a rational function that was defined almost everywhere. A rational function was more flexible, allowing for some important classification results, the definition of Kodaira dimension, study singularities, more invariances by looking at the birational automorphisms, and some "local to global" principles. The following brings these concepts over to schemes

Definition 3.1.23: Rational Map

Let X, Y be schemes. Then a *rational map* $\pi : X \dashrightarrow Y$, is an equivalence relation:

$$(\alpha U \to Y) \sim (\beta : V \to Y) \qquad \text{if and only if} \qquad Z \subseteq U \cap V \text{ s.t. } \alpha|_Z = \beta|_Z$$

where U, V, Z are all dense open subsets[a].

If X, Y are S-schemes, then rational maps of S-schemes are defined similarly.

If the image of one of the representatives is dense, then the rational map is said to be *dominant* or *dominating*.

[a] When Y is separated, we will let $Z = U \cap V$

Note that a rational map is is an equivalence relation, not necessarily defined on a single domain.

Definition 3.1.24: Birational Map

A rational map $\pi : X \dashrightarrow Y$ is *birational* if it is dominant, and it has a dominant rational inverse. Two schemes X, Y are *birationally equivalent* if there is a birational map $\pi : X \dashrightarrow Y$.

Example 3.1.25: Rational Map

1. A scheme morphisms always gives a rational map, similarily to how all regular functions are special rational functions
2. The projection $\mathbb{P}^n_R \dashrightarrow \mathbb{P}^{n-1}_R$ projecting down a dimension is a rational map (recall the same reasoning for rational functions in [8, chapter 22.3]

Lemma 3.1.26: Dominating Rational Map on Irreducible Schemes

Let $\pi : X \dashrightarrow Y$ be a rational map between irreducible schemes. Then π is dominating iff it sends the generic point of X to the generic point of Y.

***Proof* :**
recall the properties of generic points in irreducible schemes.

We may naturally want to have a result that is similar to the correspondence of rational functions and field extensions as given in [8]. The full correspondence requires irreducible varieties (integral finite type k-schemes), however we may at least always induce a map of function fields:

Proposition 3.1.27: Rational Maps of Integral Schemes induces Function Field Map

The collection of integrals schemes with dominant maps induces morphisms of function fields in the opposite direction.

Proof **:**
First show that the form a category (importantly, composition works). Then think of how to create this correspondence.

Corollary 3.1.28: Birational Map on Irreducible Schemes

Let $\pi : X \dashrightarrow Y$ be a birational map between irreducible schemes. Then the induced map of function fields is an isomorphism.

For the converse failing

Example 3.1.29: Function Field Not Inducing Rational Map
Take Spec $(k[x])$ and Spec $(k(X)$ which have the same function field $k(x)$. Then certainly there is no corresponding rational map

$$\operatorname{Spec} k[x] \dashrightarrow \operatorname{Spec} k(x)$$

of k-schemes, as such a morphism would correspond to a morphism from $U \subseteq \operatorname{Spec} k[x]$ (ex. $U = \operatorname{Spec} k[x, 1/f(x)]$) to $\operatorname{Spec} k(x)$, but not map of rings $k(X) \to k[x, 1/f(x)]$ satisfies the right properties for any $f(x)$.

Proposition 3.1.30: Reduced Schemes and Birational Maps

Let X, Y be reduced schemes. Then X and nY are birational if and only if there exists a dense open subscheme $U \subseteq X$ and $V \subseteq Y$ such that

$$U \cong V$$

Proof **:**
Generalize the proof from [8] (If stuck, look at Vakil p. 190)

Vakil looks here at rational maps of irreducible varieties that look like translating the results over from varieties, so I will omit for now

I like his translation of the Pythagorean problem into a scheme problem

non-rational complex curve, cool presentation

3.1.6 More on Projective Schemes

We shall now look at closed subschemes of projective schemes

Definition 3.1.31: Projective Hypersuface

A *hypersurface* of $\mathbb{P}^n_k$ is a closed subscheme given by a single homogeneous polynomial f. The *degree* of the hypersurface is the degree of the homogeneous polynomial.

A hypersurface of degree 1 is called a *hyperplane*. A hypersurface of degree $2, 3, 4, 5...$ is called a *quartic*, *cubic*, *quartic*, *quintic*, and so froth.

If $n = 2$ ($\mathbb{P}^2_k$) a hypersurface is called a *curve*. A hypersurface of degree $1, 2, 3$ is called a *line*, *conic curve* (or *conic*), and *surface* respectively.

Recall that $\mathbb{P}^n_k$ has only constant functions defined globally, and hence any such cut-out would be done from regular functions defined on open subsets.

Example 3.1.32: Closed Projective Subschemes

1. *Twisted Cubic*: Take $V(xy - zw, x^2 - wy, y^2 - xz) \subseteq \mathbb{P}^3_k$. This shape is called the *twisted cubic*. This is a *curve*, and it is isomorphic to $\mathbb{P}^1_k$ by the induced map:
$$k[w, x, y, z] \to k[s, t] \qquad (w, x, y, z) \mapsto (s^3, s^2t, st^2, t^3)$$
To see it is an isomorphism, take the standard open sets and show it gives closed embeddings, for example if $w \neq 0$, $s \neq 0$ then:
$$k\left[\frac{x}{w}, \frac{y}{w}, \frac{z}{w}\right] \to k\left[\frac{t}{s}\right] \qquad \frac{x}{w} \mapsto \frac{t}{s}, \ \frac{y}{w} \mapsto \frac{t^2}{s^2}, \ \frac{z}{w} \mapsto \frac{t^3}{s^3}$$

2. *Surjective graded ring map*: Show that $R_\bullet \twoheadrightarrow S_\bullet$ is a surjective graded ring homomorphism, then it induces a closed embedding $\operatorname{Proj} S_\bullet \to \operatorname{Proj} R_\bullet$.

3. Let $X \to \operatorname{Proj} S_\bullet$ be a closed embedding in a projective R-scheme (so that $S_\bullet$ is a finitely generated graded R-algebra). Show that X is projective as $\operatorname{Proj}(S_\bullet/I)$ (mirror the proof for the affine case, see Shafarevich).

4. Any injective linear transformation $V \to W$ induces closed embedding $\mathbb{P}V \to \mathbb{P}W$. The closed subscheme defined here is called a *linear space*.

5. Take $S_\bullet$ to be a finitely generated graded ring generated by elements of degree 1. Note that S_1 is a finitely generated S_0-module, and the irrelevant ideal S_+ is generated by degree 1 elements. Then $\operatorname{Proj} S_\bullet$ is isomorphic to a closed subscheme of $\mathbb{P}^n_R$. To see this, take the free module R^{n+1} generated by $t_0, ..., t_n$. Then take:
$$\operatorname{Sym}^\bullet(R^{n+1}) = R[t_0, .., t_n] \twoheadrightarrow S_\bullet \qquad t_i \mapsto x_i$$
Then this implies:
$$S_\bullet = R[t_0, t_1, ..., t_n]/I$$
Thus, we always get that such rings can be interpreted as closed projective subschemes! Conceptually, this is the generalization of the affine case where if S is a finitely generated R-algebra, w may choose n generators of S and describe $\operatorname{Spec}(S)$ as a closed subscheme of $\mathbb{A}^n_R$; we are choosing coordinates (now in the projective case).

6. The following shows a way to parameterize certain projective varieties. Take $V(wz-xy) \subseteq \mathbb{P}^3_k$ (with projective coordinates w,x,y,z) Then for any two elements $\alpha, \beta \in k$ that are not both zero, notice that as $[c,d] \in \mathbb{P}^1_k$ varies, $[\alpha c, \beta c, \alpha d, \beta d]$ is a line on the quadratic surface. Show there is one other parameterization of this surface via lines, an show it must be the case that these are the only 2 (hint: one parameterization contains $V(w,x)$ and the other $V(w,y)$). The following gives a visualization of the parameterization that contains $V(w,x)$:

A common important closed projective subscheme is those given by the different gradings $S_{d\bullet}$:

Definition 3.1.33: Veronese Embedding

Let $S_\bullet = k[a_0, ..., a_n]$. Let $N = \dim_k S_{d\bullet} = \binom{n+d}{d}$. Then $\operatorname{Proj} S_{d\bullet} \subseteq \mathbb{P}^{N-1}$ is called the *d-uple Veronese Embedding.*

Most of the projective schemes are asociated to graded rings generated by elements of degree 1. Having different degrees, or weighting, can

Definition 3.1.34: Weighted Projective Space

Let $S_\bullet = k[a_1, ..., a_n]$ where each a_i has degree d_i. Then $\operatorname{Proj} k[a_1, ..., a_n]$ is called the weighted projective space and is denoted $\mathbb{P}(d_1, ..., d_n)$

Example 3.1.35: Weighted Projective Space

1. Show that $\mathbb{P}(m,n) \cong \mathbb{P}$
2. Show that $\mathbb{P}(1,1,2) \cong \operatorname{Proj} k[u,v,w,z]/(uw - v^2)$

Exercise 3.1.2

1. Put Exercise 8.1.M from Vakil
2. Show that the composition of two locally closed embeddings is locally closed

3.1.7 Group Schemes

I'm putting this here because I looked into algebraic groups and so I want a place to put it down

Exercise 3.1.3

1. If $\varphi : A \to B$ is a ring homomorphism, then the corresponding morphism $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is dominant if and only if $\ker \varphi$ is contained in the nilradical of A.

3.2 Fibered Product and Fibers

With scheme morphisms defined, we may bring in one more tool from commutative algebra to analyze schemes, namely we can translate the tensor product to the fibered product (or pullback). As a high-level idea, the fibered product can be thought of as a product of schemes with gluing, while the fibered coproduct is the disjoint union plus gluing, and will be a scheme if it satisfies the conditions of proposition 2.3.19, but it will not exist in general, see example 3.10 .

Theorem 3.2.1: Fibered Product Of Schemes

Let X, Y be schemes over S. Then then the fibered product of X and Y over S exists, that is the limit of the diagram exists:

$$\begin{tikzcd} X \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

The fibered product is denoted $X \times_S Y$

More commonly, we would say that the fibered product satisfies the universal property that if $Z \to X$ and $Z \to Y$ commutes in the diagram, then there exists a unique morphism $Z \to X \times_S Y$ such that

$$\begin{tikzcd} & Z \arrow[d] \arrow[ddl, bend right] \arrow[ddr, bend left] & \\ & X \times_S Y \arrow[dl] \arrow[dr] & \\ X \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

commutes. If S is the final object, or more concisely it is empty or "doesn't matter", then we have the usual product. If S is not specified, the product shall be defined as $X \times Y := X \times_{\mathrm{Spec}\,(\mathbb{Z})} Y$

Proof **:**
We shall construct the products of s, and then glue them together.

Let $X = \mathrm{Spec}\,(A)$, $Y = \mathrm{Spec}\,(B)$, and $S = \mathrm{Spec}\,(R)$ be affine. Then A, B are R-algebras, and the product of X and Y over S is:

$$\mathrm{Spec}\,(A \otimes_R B)$$

This comes form the universal property of tensor products along with the property of adjoints. Indeed, if $Z \to \mathrm{Spec}\,(A \otimes_R B)$ is a scheme morphism, then this gives a ring homomorphism $A \otimes_R B$ into $\mathcal{O}_Z(Z)$. By the universal property of tensor products, a morphism of $A \otimes_R B$ into is the same as a morphism from A and B to that ring. Then by proposition 3.1.11, this gives a morphism of Z into $\mathrm{Spec}\,(A \otimes_R B)$, and this morphism will satisfy the universal property of fibered products given by the universal property of tensor products.

Let us now construct the more general case via gluing. Let X, Y be schemes. To do this, we shall reduce to a covering of $X \times_S Y$. For any open affine $U \subseteq X$, if the product existed then $\pi_1^{-1}(U) \cong U \times_S Y$ (check this[a]). Now if $\{X_i\}$ is an open covering for x, then I claim that if $X_i \times_S X$ exists, so does $X \times_S Y$. Indeed, for each $U_{ij} = \pi_1^{-1}(X_i) \subseteq X_i \times_S Y$, by the uniqueness of the product there are unique isomorphisms: $\varphi_{ij} : U_{ij} \to U_{ji}$ for each i, j for compatible projections. Furthermore, these isomorphisms are compatible with the gluing condition (as you should check if you are not convinced!). As a remind, the condition was:

$$\varphi_{ik}|_{X_{ij} \cap X_{ik}} = \varphi_{jk} \circ \varphi_{ij}|_{X_{ij} \cap X_{ik}}$$

Thus, we get that we may glue them together by lemma 3.1.8 we have that this glued scheme is isomorphic to $X \times_S Y$, and through this isomorphism we can check that the universal property is satisfied.

By interchanging X and Y, we can now conclude from the proof on affine schemes that if X and Y are any schemes, and S is an affine, then $X \times_S Y$ exists. To show that S can be an arbitrary scheme, let $\{S_i\}$ be an open affine cover of S. Let $\alpha : X \to S$ and $\beta : Y \to S$ be two scheme morphisms, and let $X_i = \alpha^{-1}(S_i)$, $Y_i = \beta^{-1}(S_i)$. Then $X_i \times_{S_i} Y_i$ is well-defined.

Now, what's key is that this scheme is the same as the scheme $X_i \times_S Y$. To see this, we must invoke the uniqueness of the universal property, and indeed if $f : Z \to X_i$ and $g : Z \to Y$ are scheme morphisms over S, by the properties of the commutative diagram we must have that the image of v must land inside Y_i. But by universality they must be equal. Thus $X_i \times_S Y$ is well-defined for each i, and so by the above argument it glues to create $X \times_S Y$, as we sought to show.

[a] If $f : Z \to U$, compose with the inclusion map and invoke the appropriate universal properties

Example 3.2.2: Fibered Products

1. Show that $\mathbb{A}_k^n \times_k \mathbb{A}_k^m \cong \mathbb{A}_k^{n+m}$, which is exactly what we would hope from a product. Notably, prove $k[x] \otimes_k k[y] \cong k[x, y]$.
2. Similarly, if k is a characteristic 0 field show that $\mathbb{Z}[x] \otimes_{\mathbb{Z}} k[y] \cong k[x, y]$. Hence $\mathbb{A}_{\mathbb{Z}}^1 \times \mathbb{A}_k^1 \cong \mathbb{A}_k^2$.
3. Let $X = \operatorname{Spec} \mathbb{C}[x, y]/(x^2 - y)$ and $Y = \mathbb{C}[z, w]/(z^3 - w^4)$. Then:

$$X \times_{\mathbb{C}} Y \cong \operatorname{Spec} \frac{\mathbb{C}[x, y, z, w]}{(x^2 - y, z^3 - w^3)}$$

   The reader should stare at this until they understand why it is called a fibered product.

4. There is new behaviour when considering schemes being tensored over non-algebraically closed fields. Take $\operatorname{Spec} \mathbb{C} \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C}$. As these are affine, this is $\operatorname{Spec} (\mathbb{C} \otimes \mathbb{R}\mathbb{C})$. As $\mathbb{C} \otimes \mathbb{R}\mathbb{C} \cong \mathbb{C} \times \mathbb{C}$ (exercise),

$$\operatorname{Spec} \mathbb{C} \times_{\operatorname{Spec} \mathbb{R}} \operatorname{Spec} \mathbb{C} \cong \operatorname{Spec}(\mathbb{C}) \amalg \operatorname{Spec}(\mathbb{C})$$

   Notice here how the choice of sheaf completely changed the answer, notably $\operatorname{Spec} \mathbb{C}$ and $\operatorname{Spec} \mathbb{R}$ are schemes over a single point, hence the fibered product diagram when only considering points is an identity map on points:

$$
\begin{tikzcd}
\bullet \arrow[dr] & & \bullet \arrow[dl] \\
& \bullet &
\end{tikzcd}
$$

and it is easy to see that in **Top** this results is a single point. However, the resulting scheme is *two* points. Thus, the fibered product here somehow "unravelled" the fact that $\operatorname{Spec}\mathbb{R}$ should be thought of as "gluing" conjugates together (think of the intuition for $\operatorname{Spec}\mathbb{R}[x]$ as a $\operatorname{Spec}\mathbb{C}[x]/\mathrm{Gal}_{\mathbb{R}}(\mathbb{C})$, the real line then don't glue to another points, but their residue field is $\mathbb{R}$ instead of $\mathbb{C}$, and hence it codimension 2 with respect to $\mathbb{C}$).

More generally show that if L/K is a finite galois extension, then

$$L \otimes_K L \cong \prod_{|\mathrm{Aut}(L/K)|} K$$

is isomorphic to $L^{|G|}$($|G|$ copies of L), hint: this is the linear independence of characters proof in geometric form. Hence, over fields fibered products capture some idea properties of galois groups.

5. Recall that the intersection of rings is a ring. This is a special case of the fibered product in the category of commutative rings, namely that if $S \to R$ and $S' \to R$ exists, then:

$$
\begin{tikzcd}
& S \cap S' \arrow[dl] \arrow[dr] & \\
S \arrow[dr] & & S' \arrow[dl] \\
& R &
\end{tikzcd}
$$

Let us see how this translates to schemes and affine schemes. Recall that the product and coproduct of affine schemes is an affine scheme:

$$\operatorname{Spec} S \times \operatorname{Spec} S' = \operatorname{Spec} S \otimes_{\mathbb{Z}} S'$$
$$\operatorname{Spec} S \amalg \operatorname{Spec} S' = \operatorname{Spec} S \times S'$$

If $U, U' \subseteq \operatorname{Spec} R$ are open affine, $U = \operatorname{Spec} S$ and $U' = \operatorname{Spec} S'$, Then it is not the case that:

$$\operatorname{Spec} S \cap \operatorname{Spec} S' = \operatorname{Spec} S \cap S'$$

namely because adjoint relation between CRing and Af-Sch is contravariant. Consider for example $R = \mathbb{C}[x], D(t), D(t-1) \subseteq \operatorname{Spec} R$. Then $D(t) \cap D(t-1) = D(t(t-1))$ and

$$\operatorname{Spec} D(t(t-1)) = \operatorname{Spec}\mathbb{C}\left[t, \frac{1}{t(t-1)}\right]$$

while $\mathbb{C}\left[t, \frac{1}{t}\right] \cap \mathbb{C}\left[t, \frac{1}{t-1}\right] = \mathbb{C}[t]$, hence:

$$\operatorname{Spec} \mathbb{C}[t]_{t(t-1)} = D(t) \cap D(t-1) = \operatorname{Spec} \mathbb{C}[t]_t \cap \operatorname{Spec} \mathbb{C}[t]_{t-1} \neq \operatorname{Spec} \mathbb{C}[t] = \mathbb{A}^1_{\mathbb{C}}$$

We may not take $S \cup S'$, as this is not a ring, and as these contain units $\langle S, S' \rangle = R$. Instead, we get:

$$\operatorname{Spec} S \cap \operatorname{Spec} S' = \operatorname{Spec} S \otimes_R S'$$

As the tensor product is a very versatile operation, the fibered product is as well. Let us demonstrate by translating some important results:

Proposition 3.2.3: Changing Basis Strategies

The following are common simple fibered products:

1. Let $\iota : U \to Z$ be an open embedding and $\varphi : Y \to Z$ be any scheme morphism. Then $U \times_Z Y$ is
$$\left(\varphi^{-1}(U), \mathcal{O}_Y|_{\varphi^{-1}(U)}\right)$$
2. Let X be an R scheme, then $S \times_R X$ for $\varphi : R \to S$ is an S-scheme
3. Let $Y \subseteq X$ be a closed scheme. Then the fibered product with a closed subscheme is a closed subscheme of the fibered product. In particular closed immersion are preserved under change of basis.

***Proof* :**
In each case 2, 3 and 4, you can reduce to the affine case (just like for the proof of the fibered product)

1. trace the argument to show this is the only option. A special consequence of this is that if $U, V \to Z$ are open embeddings, this is just their intersection.
2. They key result is that $S[t] \cong S \otimes_R R[t]$.
3. the key result is that if $\varphi : S \to R$ is a ring morphism with $I \subseteq S$ and $I^e \subseteq R$ the extension, then:
$$R/I^e \cong R \otimes_S S/I$$
Which can be shown using the right-exactness of $- \otimes_S R$ on $I \to S \to S/I$.
4. The key result is that if $\varphi : S \to R$ is a morphism $D \subseteq S$ a multiplicative subset, Then:
$$R\left[\varphi(D)^{-1}\right] \cong R \otimes_S \left(S\left[D^{-1}\right]\right)$$

This motivates an important definition generalizing affine and projective schemes (idk if this should be here)

Definition 3.2.4: Affine Schemes over a Scheme

Let X be a scheme. Then:
$$\mathbb{A}^n_X = X \times_{\mathbb{Z}} \mathbb{Z}[x_1, \dots, x_n]$$

Example 3.2.5: Pushout Need Not Exist

Consider two copies of $\mathbb{A}^1_{\mathbb{C}}$. Consider:

$$\begin{tikzcd} & \operatorname{Spec}\mathbb{C} \arrow[dl] \arrow[dr] & \\ \mathbb{A}^1_{\mathbb{C}} & & \mathbb{A}^1_{\mathbb{C}} \end{tikzcd}$$

where $[(0)] \mapsto [(0)]$ in both cases. This is like the double-origin example, except every point is a double origin. This shall be shown to not be a scheme as it is "too non-separated".
We cannot fix this by limiting to the category of affine schemes. Consider for example:

$$\begin{tikzcd} & \operatorname{Spec} k[t]_t \arrow[dl] \arrow[dr] & \\ \operatorname{Spec} k[t] & & \operatorname{Spec} k[t] \end{tikzcd}$$

where $\operatorname{Spec} k[t]_t = \mathbb{A}^1_k \backslash \{0\}$, and the maps are the maps given by $t \mapsto t$ and $t \mapsto 1/t$. Then we saw in example 2.15 that the colimit of this diagram is projective space, $\mathbb{P}^1_k$. Hence affine schemes are not in general closed under pushout. Note that the corresponding fibered product does exist in CRing, namely it will have to be:

$$\begin{tikzcd} & k \arrow[dl, dashed] \arrow[dr, dashed] & \\ k[t] \arrow[dr] & & k[t] \arrow[dl] \\ & k[t]_t & \end{tikzcd}$$

However, applying Spec to this will not return the pushout, namely the pair of adjoint functors (Spec, Γ) does not preserve pushouts.

Note however that colimits do exist in the category of locally ringed spaces. See this link (which link??) for more information.

3.2.1 Fibers and family of Schemes

This gives a huge class of new object's that we may define. Let us start with defining fibers:

Definition 3.2.6: Fibers

Let $f : X \to Y$ be a morphism of scheme. Let $y \in Y$, and $k(y)$ be the residue field of y. Let $\mathrm{Spec}\,(k(y)) \to Y$ be the natural morphism[a]. Then the *fiber* of the morphism f over the point y is the scheme

$$X_y = X \times_Y \mathrm{Spec}\,(\kappa(y))$$

In the case where $X = \mathrm{Spec}\,(R), Y = \mathrm{Spec}\,(S)$ are affine, then for a point $\mathfrak{p} \in Y$:

$$f^{-1}(\mathfrak{p}) = \mathrm{Spec}\, R \times_S \mathrm{Spec}\, \kappa(\mathfrak{p}) = \mathrm{Spec}\,(R \otimes_S S_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}})$$

[a] exercise ref:HERE

Proposition 3.2.7: Fiber Property

Let $f : X \to Y$ be a morphism of schemes, $y \in Y$, and X_y the fiber of f over y. Then X_y is homeomorphic to $f^{-1}(y) \subseteq X$.

If this is unclear, we should expect the fibers of the projection $\mathbb{A}^2_{\mathbb{C}} \to \mathbb{A}^1_{\mathbb{C}}$ to be lines, that is $\mathbb{A}^1_{\mathbb{C}}$, and indeed for some $a \in \mathbb{A}^1_{\mathbb{C}}$:

$$\begin{aligned}
\mathbb{A}^2_{\mathbb{C}} \times_{\mathbb{A}^1_{\mathbb{C}}} \mathrm{Spec}(\kappa(a)) &= \mathrm{Spec}\left[(\mathbb{C}[x,y]) \otimes_{\mathbb{C}[x]} \left(\frac{\mathbb{C}[x]_{(x-a)}}{\mathfrak{m}_{(x-a)}}\right)\right] \\
&\cong \mathrm{Spec}\left[\mathbb{C}[x,y] \otimes_{\mathbb{C}[x]} \mathrm{Frac}\left(\frac{\mathbb{C}[x]}{(x-a)}\right)\right] \\
&= \mathrm{Spec}\left[\frac{\mathbb{C}[x,y]}{(x-a)} \otimes_{\mathbb{C}[x]} \mathrm{Frac}\left(\frac{\mathbb{C}[x]}{(x-a)}\right)\right] \\
&\cong \mathrm{Spec}\left[\mathbb{C}[y] \otimes_{\mathbb{C}[x]} \mathbb{C}\right] \\
&\cong \mathrm{Spec}\, \mathbb{C}[y] \\
&= \mathbb{A}^1_{\mathbb{C}}
\end{aligned}$$

Proof **:**
Let $X'_y = f^{-1}(y) = \{x \in X \,|\, f(x) = y\}$ and consider the following commutative diagram:

$$\begin{tikzcd}
 & Z \arrow[d, dashed] \arrow[ddl, bend right, "g"'] \arrow[ddr, bend left, "h"] & \\
 & f^{-1}(y) \arrow[dl, "\mathrm{id}"'] \arrow[dr, "f"] & \\
X \arrow[dr, "f"'] & & \{y\} \arrow[dl, "\mathrm{id}"] \\
 & Y &
\end{tikzcd}$$

As $f(g(Z)) = h(Z) = y, f(g(Z)) = y$, so $f^{-1}(y) = g(Z)$. Hence, there is a natural map from Z to

$f^{-1}(y)$ that makes the diagram commute. Now letting $\{y\} = \operatorname{Spec} \kappa(y)$, $f^{-1}(y)$ is homeomorphic to X_y by uniqueness of the universal object. As the subspace topology of $f^{-1}(y)$ and the Zariski topology match, $f^{-1}(y)$ is scheme-isomorphic to X_y, completing the proof.

In some ways, fibers of schemes are more "regular" than fibers of sets, as the extra information of schemes will provide some more possible pre-images:

Example 3.2.8: Fibers

1. Take the familiar projection of the parabola onto the x axis induced by the ring map $\mathbb{Q}[x] \to \mathbb{Q}[y]$ mapping $x \mapsto y^2$. We can think of this as the map $\mathbb{Q}[x] \to \mathbb{Q}[y^2]$ where the codomain is isomorphic to:
$$\mathbb{Q}[y^2] \cong \frac{\mathbb{Q}[x,y]}{x - y^2}$$
This is sometimes called the parameterization of the map. Then the pre-image of 1 is:
$$\begin{aligned}\operatorname{Spec} \mathbb{Q}[x,y]/\left(y^2 - x\right) \otimes_{\mathbb{Q}[x]} \mathbb{Q}[x]/(x-1) &\cong \operatorname{Spec} \mathbb{Q}[x,y]/\left(y^2 - x, x - 1\right) \\ &\cong \operatorname{Spec}\left(\mathbb{Q}[y]/\left(y^2 - 1\right)\right) \\ &\cong \operatorname{Spec}(\mathbb{Q}[y]/(y-1)(y+1)) \\ &\overset{\text{C.R.T}}{\cong} \operatorname{Spec}(\mathbb{Q}[y]/(y-1) \times \mathbb{Q}[y]/(y+1)) \\ &\cong \operatorname{Spec}(\mathbb{Q}[y]/(y-1)) \amalg \operatorname{Spec}(\mathbb{Q}[y]/(y+1))\end{aligned}$$
That is, it is the points ± 1. The pre-image of 0 is:
$$\operatorname{Spec} \mathbb{Q}[x,y]/(y^2 - x, x) \cong \operatorname{Spec} \mathbb{Q}[y]/(y^2)$$
Notice that the pre-image remembers multiplicity information!
As the spectrum of a polynomial ring over rationals is the quotient by the galois action on the polynomial ring over $\bar{k}$, the pre-image of -1 exists! It is:
$$\operatorname{Spec} \mathbb{Q}[x,y]/(y^2 - x, x + 1) \cong \operatorname{Spec} \mathbb{Q}[y]/(y^2 + 1) \cong \operatorname{Spec} \mathbb{Q}[i] = \operatorname{Spec} \mathbb{Q}(i)$$
This can be thought of as "size 2" point over the base field. The pre-image of the generic point is a reduced point, and can be thought of as "size 2" over the residue field:
$$\operatorname{Spec} \mathbb{Q}[x,y]/(y^2 - x) \otimes_{\mathbb{Q}[x]} \mathbb{Q}(x) \cong \operatorname{Spec} \mathbb{Q}[y] \otimes_{\mathbb{Q}[y]} \mathbb{Q}(y^2)$$
Try working through the pre-image of other points that are not commonly associated to $\mathbb{Q}$. Notice that the pre-image is the two points, or you get nonreduced behaviour that is "of degree 2" or you get a field extension of degree 2.

2. Find the fibers of $\operatorname{Spec} \mathbb{Z}[i] \to \operatorname{Spec} \mathbb{Z}$ over different points.

3. As $X \to \operatorname{Spec}(\mathbb{Z})$ is always well-defined and unique, there is always fibers over $\operatorname{Spec}(\mathbb{Z})$ and hence they are usually given a name: The fiber over (0) is often labeled $X_{\mathbb{Q}}$ (over $\mathbb{Q}$), and the fibers for all the primes are X_p (over $\mathbb{F}_p$). A common terminology is that X_p *arises by reduction* mod p of the scheme X.

On p. 260 of Vakil he gives more on general fibers, generic fibers, and generically finite morphisms

The reader may recall that we used tensor products to extend R-modules to S-modules and called it a *base change*. We can now motivate this terminology. If we have collection S-scheme, i.e. scheme morphisms $X \to S$, then we may want to change the "base" S to another scheme S'. This would be done by a *base change map* $S \to S'$. The resulting new S'-scheme would be $X' = X \times_S S'$. More generally

Definition 3.2.9: Scheme-theoretic Pullback

Let $Y \to Z$ and $X \to Z$ be scheme morphisms. Then the map $X \times_Z Y \to X$ is called the *scheme-theoretic pullback* of Y (or just pullback if it is clear we are working with schemes).

Example 3.2.10: Parameterization of Curves using Schemes

Take

$$X = \operatorname{Spec} \frac{k[x,y,t]}{(ty - x^2)}$$
$$Y = \operatorname{Spec} k[t]$$

Let $f : X \to Y$ be given by the natural ring homomorphism $k[t] \to \frac{k[x,y,t]}{(ty-x^2)}$. Note both these schemes are integral schemes of finite type over k.

Now, identify the closed points of Y with k. For $a \neq 0 \in k$, we get the fibers correspond to the parabola's $ay = x^2$ or $y = x^2/a$. All of these are irreducible, reduced curves. At $a = 0$, we get $x^2 = 0$, which is a reduced scheme.

More generally, S-schemes can be geometrically interpreted as schemes paramaterized by the points of S whose maps are fiber-bundle morphisms. Hence, the scheme $\operatorname{Spec} k[t]$ *paramaterized* the curve.

A quick note must be done about general products and why it may be preferable to take the fibered product over a field:

Example 3.2.11: Strange Fibered Products

1. Show that $\operatorname{Spec}(\mathbb{Z}/m\mathbb{Z}) \times_{\operatorname{Spec}(\mathbb{Z})} \operatorname{Spec}(\mathbb{Z}/n\mathbb{Z}) = \emptyset$.
2. Another product of taking the "absolute" fibered product is that if $X = \operatorname{Spec} \mathbb{Z}[x]$ and $Y = \operatorname{Spec} \mathbb{Z}[y]$ then:

$$\dim X \times Y = \dim(X) + \dim(Y) - \dim(\operatorname{Spec} Z) = \dim X + \dim Y - 1$$

If we restrict to k-schemes for a field k, these two pathologies do not occur.

As these examples show, the fibered product of two schemes need not be the schemes of these two fibered products. In a concrete sense, this is because we must be more specific with what types of points we are dealing with. If we specify the points, we in fact recover the usual "sets"[5].

[5] The topological product can also be thought of as being over the right valued points

Proposition 3.2.12: Fibered Product With Z-valued Points

Let X, Y be schemes and take Z-valued points on each scheme (i.e. morphisms in $\mathrm{Hom}(Z, X)$ and $\mathrm{Hom}(Z, Y)$. Then show that the fibered product will be:

$$\mathrm{Hom}(W, X \otimes_W Y) = \mathrm{Hom}(Z, X) \times_{\mathrm{Hom}(Z,W)} \mathrm{Hom}(Z, Y)$$

***Proof* :**
exercise.

Put an appropriate definition here for the following

Hence, given a surjective map $Y \to X$, Y can be regarded as a collection of schemes paramaterized by Y. Conversely, we may want to study a scheme X_0, and see if there are some deformations we may want to do. Then a common way of defining a *family of algebraic deformation* is by a scheme morphisms $f : X \to Y$ where one of the fibers of f is isomorphic to X_0.

Some more here on representable Functors

Exercise 3.2.1

1. Let $\varphi : X \to Y$ be a morphism of S-scheme of finite type. Show that if $S \to S'$ is any base extension, $\varphi' : X' \to Y'$ is an S'-scheme morphism preserves finite type.
2. Let $\varphi : X \to Y$ be a scheme morphism of integral schemes. Show the fibers needn't be irreducible or reduced.
3. Let $\mathfrak{p} \in X$ be a point in a scheme. Show there is a canonical map
$$\mathrm{Spec}\,(\mathcal{O}_{X,\mathfrak{p}}) \to X$$
4. Using the above, define a canonical morphism $\mathrm{Spec}\,(\kappa(\mathfrak{p})) \to X$.
5. Perhaps put here some base-change exercises relating the tensor product exercises you had in [8] and now give them their geometric counter-part.
6. Show if $U, V \subseteq X$ are open subschemes then $U \times_X V \cong U \cap V$.
7. Let $\varphi : B \to A$ be a ring morphism, $S \subseteq B$ a multiplicative subset, so $\varphi(S)$ is a multiplicative subset of A. Show that $\varphi^(S)^{-1}A \cong A \otimes_B (S^{-1}B)$. What does this imply for the relation between localization and the fibered product?
8. Let X be a k-scheme. Show that if L/k is a field extension, $X \otimes_{\mathrm{Spec}\,(k)} \mathrm{Spec}\,(L)$ is an L-scheme. Show that this in fact gives a functor (define the appropriate maps, and verify).

3.3 Finiteness Conditions on Morphism

Many of these concepts are the generalizations of the types of schemes covered so far. All of these concepts can be expressed in terms of morphisms.

As schemes are rather general notions, so to scheme morphisms can be very general. The following notions are guiding principles presented by Vakil on what a good type of morphism should have. This terminology will be very useful in labeling many of the following lemmas an propositions. Let's say that C is a collection of morphisms between schemes. Then:

1. it is *local on target* if

   (a) $\pi : X \to Y$ is in the class, then for any open subset $V \subseteq Y$, the restricted homomorphism $\pi^{-1}(V) \to V$ is in the class

   (b) $\pi : X \to Y$ is a morphism, $\cup_i V_i = Y$ is an open cover where each $\pi^{-1}(V_i) \to V_i$ is in this class, then π is in this class

   Usually we consider affine opens, but it need not be the case.

2. is *local on source* if there is

   (a) $\pi : X \to Y$ is in the class, then for any open subset $U \subseteq X$, the restricted homomorphism $U \to \pi(U)$ is in the class

   (b) $\pi : X \to Y$ is a morphism, $\cup_i U_i = X$ is an open cover where each $U_i \to \pi(U_i)$ is in this class, then π is in this class

3. it is closed under composition

4. it is closed under fibered products. As a special case, is closed under change of basis from an S-scheme to a T-scheme

Finiteness Conditions

Definition 3.3.1: Quasicompact Morphism

Let $\pi : X \to Y$ be a morphism of scheme. Then π is *quasicompact* if every open affine subset $U \subseteq Y$ has quasicompact pre-image, that is $\pi^{-1}(U)$ is quasicompact.

Naturally, Quasicompactness is is closed under composition, and is local on target.

Example 3.3.2: Quasicompact Morphism

We shall in this section that other types of morphism shall be quasicompact (finite, embeddings, proper, etc). At the moment, we shall give concrete examples:

1. Any scheme morphism of affine schemes is quasicompact, this relies on the use of the distinguished open base: Let $f : \operatorname{Spec} S \to \operatorname{Spec} R$ and $U \subseteq \operatorname{Spec} R$ be quasi-compact. Then $U = \bigcup_i^n D(g_i)$ for appropriate distinguished open sets. Then since $f^{-1}(D(g_i)) = D(f^{\#}(g_i))$, we have that
$$f^{-1}(U) = \bigcup_i^n D(f^{\#}(g_i))$$
   showing the pre-image is quasi-compact.

2. Let $X = \coprod_i^\infty \operatorname{Spec} k$ and $Y = \operatorname{Spec} k$. Then the natural projection $\pi : X \to Y$ is certainly not quasicompact, the pre-image of $\operatorname{Spec} k$ not being quasicompact.

3. Let X be Noetherian schemes and $f : X \to Y$ a scheme morphism. Then f is quasicompact. This follows from the following topological property: If X is a Noetherian topological space, every subset is Noetherian.

   Hence, any non-quasicompact map *must* have X as a non-Noetherian scheme, and so in practice we shall not often see non-quasicompact schemes.

4. Let $X = \operatorname{Spec} k[x]$ and $\tilde{X}$ be the blow up at every point. Then the natural map $\varphi : \tilde{X} \to X$ is not quasicompact.

For future reference, the following definition is put down (see ref:HERE for its application):

Definition 3.3.3: Quasiseparated Morphism

Let $\pi : X \to Y$ be a morphism of scheme. Then π is *quasiseparated* if every open affine subset $U \subseteq Y$ has quasicompact pre-image.

Example 3.3.4: Quasiseparated Morphism

We shall see examples of such morphisms later.

The next important type of morphism that will always have affine open pre-images:

Definition 3.3.5: Affine Morphism

Let $\pi : X \to Y$ be a morphism of schemes. Then π is *affine* if for every affine open subset $U \subseteq Y$, $\pi^{-1}(U)$ is an (interpreted as an open subscheme of X)

Affine morphisms can be thought of as being the closest types of morphisms between schemes that "behave" like morphisms of affine schemes, namely since the pre-image of every affine is affine we can just cover in affine sets and study those without needing to "think" about the gluing structure. Proposition 3.3.9 shall show we only need to check a morphism is affine on a single affine cover, which is not immediately obvious as it is not *a priori* clear that we cannot "gerrymander" a partition to force affine pre-images on the open subsets of the cover.

3.3.1 Advanced Intuition This is for the reader already familiar with all this material. Let $f : X \to Y$ is a morphism. Then f is an affine morphisms if and only if there exists a quasi-coherent sheaf of $\mathcal{O}_Y$-algebras $\mathcal{A}$ such that $X \cong \operatorname{Spec}_Y(\mathcal{A})$ where $\operatorname{Spec}_Y(\mathcal{A})$ is the relative spectrum with respect to S (this is a generalization of X as an S-algebra)

We shall give examples of affine morphisms after we prove some nice results, one important such example shall be that when the domain and codomain are affine, the morphism must be affine. Let us start by showing that if the domain/codomain are not affine, the morphism may not be affine:

Example 3.3.6: Non Affine Morphisms

The prototypical example of a non affine morphism is $\mathbb{P}^1_k \to \operatorname{Spec} k$ (i.e. the structure morphism). Then the pre-image is *not* an (as shown in proposition 2.3.5). Example of a non- where the codomain is not affine is the embedding $\mathbb{A}^n_k \to \mathbb{P}^n_k$ for $\mathbb{A}^n_k$ embeds in a choice of U_i ($n = 1$ it is enough to just do $n = 1$ to see why).

Certainly affine morphisms are stable under composition, and are stable under base change. This condition would be nice to be affine-local on target. It takes some build-up:

Lemma 3.3.7: Affine Morphisms and Qcqs

Let $\pi : X \to Y$ be an affine morphism. Then it is quasicompact and quasiseparated.

***Proof* :**

The key is that are quasi-separated (see proposition 2.4.10)

Lemma 3.3.8: Qcqs Lemma

Let X be a quasicompact and quasiseparated scheme and $s \in \mathcal{O}_X(X)$. Then

$$\mathcal{O}_X(X)_s \cong \mathcal{O}_X(X_s)$$

This theorem can be thought of as the generalization of the result where $X = \operatorname{Spec}(R)$, we we already know that:

$$\mathcal{O}_{\operatorname{Spec}(R)}\left(\operatorname{Spec}(R)\right)_s \cong, \mathcal{O}_{\operatorname{Spec}(R)}(\operatorname{Spec}(R)_s) = \mathcal{O}_{\operatorname{Spec}(R)}(D(s))$$

***Proof* :**

As X is quasicompact, cover it with finitely many affine open sets $U_i = \operatorname{Spec} R_i$. take $U_{ij} = U_i \cap U_j$. Then the following is exact

$$0 \to \mathcal{O}_X(X) \to \prod_i R_i \to \prod_{i,j} \mathcal{O}_X(U_{ij})$$

As X is quasiseparated, each U_{ij} is covered by finitely many affine open sets $U_{ijk} = \operatorname{Spec}(R_{ij})$ so that the following is exact:

$$0 \to \mathcal{O}_X(X) \to \prod_i R_i \to \prod_{(i,j),k} R_{ijk}$$

Now, localization (being an exact functor) gives:

$$0 \to \mathcal{O}_X(X)_s \to \left(\prod_i R_i\right)_s \to \left(\prod_{(i,j),k} R_{ijk}\right)_s$$

which is still exact. As localization commutes with finite products (importance on finiteness), we get for appropriates values:

$$0 \to \mathcal{O}_X(X)_s \to \prod_i (R_i)_{s_i} \to \prod_{i,j,k} (R_{ijk})_{s_{ijk}}$$

which again is still exact, where the s_i and s_{ijk} is is the restriction to the rings R_i and R_{ijk}. Now, the scheme X_s can be covered by the affine open sets $\operatorname{Spec}(R_i)_{s_i}$ and $\operatorname{Spec}(R_i)_{s_i} \cap \operatorname{Spec}(R_{ijk})_{s_{ijk}}$ giving almost the same exact sequence:

$$0 \to \mathcal{O}_X(X_s) \to \prod_i (R_i)_{s_i} \to \prod_{i,j,k} (R_{ijk})_{s_{ijk}}$$

Hence, the kernel of the map

$$\mathcal{O}_X(X)_s \to \mathcal{O}_X(X_s)$$

is the same, and we get an isomorphism.

Proposition 3.3.9: Affine Morphisms Are Affine-local On Target

Let $\pi : X \to Y$ be a scheme morphism. Then π is an affine morphism if and only if there is a affine open cover $\cup_i U_i = Y$ such that $\pi^{-1}(U_i)$ is affine.

***Proof* :**
We shall want to apply the affine communication lemma. For that, we must show the conditions are satisfied. First, suppose $\pi : X \to Y$ is affine over $\operatorname{Spec}(S)$ so that $\pi^{-1}(\operatorname{Spec}(S)) = \operatorname{Spec}(R)$. Then notice that for any $s \in S$:

$$\pi^{-1}(\operatorname{Spec}(S_s)) = \operatorname{Spec}(R)_{\pi^\# s}$$

showing the first condition. For the next, suppose that $\pi : X \to \operatorname{Spec}(S)$ and $(s_1, ..., s_n)$ with $X_{\pi^\# s_i} = \operatorname{Spec}(R_i)$. We'll show that X is affine as well. To that end, let $R = \mathcal{O}_X(X)$. Then $X \to \operatorname{Spec}(S)$ factors through $\alpha : X \to \operatorname{Spec}(R)$ (given by the isomorphism $R \to \mathcal{O}_X(X)$), giving us:

$$\begin{tikzcd} \bigcup_i X_{\pi^\# s_i} \arrow[rr, "\alpha"] \arrow[dr, "\pi"'] & & \operatorname{Spec}(R) \arrow[dl, "\beta"] \\ & \bigcup_i D(s_i) = \operatorname{Spec}(S) & \end{tikzcd}$$

Then, as we are working with scheme morphisms, the pre-image of a distinguished open set is a distinguished open set and so

$$\beta^{-1}(D(s_i)) = D(\beta^\#(s_i)) \cong \operatorname{Spec} R_{\beta^\# s_i}$$

and $\pi^{-1}(D(s_i)) = \operatorname{Spec}(R_i)$. Now, as X is quasicompact and quasiseparated, by the Qcqs lemma we have the isomorphism:

$$A_{\beta^\# s_i} = \mathcal{O}_X(X)_{\beta^\# s_i} = \mathcal{O}_X(X_{\beta^\# s_i}) = R_i$$

Hence, α induces an isomorphism $\operatorname{Spec}(R_i) \cong \operatorname{Spec}(R_{\beta^\# s_i})$, and so α is an isomorphism over each $\operatorname{Spec}(R_{\beta^\# s_i})$, which covers $\operatorname{Spec}(R)$, and hence α is an isomorphism, and so $X \cong \operatorname{Spec}(R)$, completing the proof.

From this, we can conclude that if the domain/codomain are affine, the morphisms must be affine

Corollary 3.3.10: Morphism Between Affine is Affine

Let $f : \operatorname{Spec} R \to \operatorname{Spec} S$ be a morphism of schemes. Then f is an affine morphism

Proof :
By proposition 3.3.9, it suffices to prove it on an affine open cover of the codomain. Then naturally, take an open cover by distinguished open sets, $D(g_i)$ whose pre-image $D(f^{\#}(g_i))$ which is an .

Corollary 3.3.11: Hypersurfaces and Affine Morphissm

Let $X \subseteq \operatorname{Spec}(R)$ be a closed subset which is locally cut out by one equation (Z is covered by open subsets which are each given by the quotient of one equation). Then the complement of Z, Y, is affine

Note that in the global case, $Y = \operatorname{Spec}(R_f)$, but Y need not be of this form for the local case.

Proof :
exercise.

Example 3.3.12: Affine Morphisms

1. By corollary 3.3.10, if the domain and codomain are affine, the morphism must be affine. Example 3.3.6 shows that if the domain/codomain are not affine, the morphism is not affine. For a concrete example of an affine morphism, consider L/K.
2. Take $\mathbb{A}^2_k \to \mathbb{P}^1_k$ given by $(a, b) \mapsto [a : b]$ which was shown to be a scheme morphism in example 3.1.10. This is an affine morphism: take the usual open cover U_0, U_1. Then:
$$f^{-1}(U_0) \cong \operatorname{Spec} k[x, y]_x \qquad \text{and} \qquad f^{-1}(U_1) \cong \operatorname{Spec} k[x, y]_y$$
hence by proposition 3.3.9 it is affine.
3. More generally let $\iota : X \to Y$ be either an open or closed embedding. Show that ι is an affine morphism.

The next two types of morphisms are affine morphisms where the pre-image has some extra properties. The next one can be thought of as the generalization of finite integral extensions (or more simply a finite extension or finite ring homomorphism):

Definition 3.3.13: Finite Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *finite* if for every affine open set $\operatorname{Spec}(S) \subseteq Y$, $\pi^{-1}(\operatorname{Spec}(S))$ is the spectrum of a S-algebra that is finitely generated as a S-module, i.e. it is a finite S-algebra.

Note how being a finitely generated S-module is a rather stronger condition than being a finitely generated S-algebra: it forces each element to be integral and "globally bounded" (since R is a finitely generated S-module, there can't be elements of arbitrary order). The reader should see that the composition of two finite morphisms is finite (see exercise ref:HERE)

Lemma 3.3.14: Algebra Build-up For Local On Target

Let $R \to S$ be a ring map, and $(f_1, ..., f_n) = R$ for appropriate $f_i \in R$. Then:

1. If $R_{f_i} \to S_{f_i}$ is integral, $R \to S$ is integral
2. If $R_{f_i} \to S_{f_i}$ is finite, $R \to S$ is finite

***Proof* :**

1. Take $s \in S$, and consider the ideal
$$(p \in R[x] \,|\, p(s) = 0) \subseteq R[x]$$
Then take the subset $J \subseteq R$ of the leading coefficients of the elements in I. This is in fact an ideal, as the reader could verify or see [8, section 20.5]. By clearing denominators, we get that:
$$f_i^{n_i} \in J$$
Hence, as shown in [8, section 9.6], $1 \in J$
2. The above shows that $R \to S$ is integral, for the finiteness note that we may choose a maximal N_i as the degree's of the generators are globally bounded.

Proposition 3.3.15: Finite Morphism Affine Local On Target

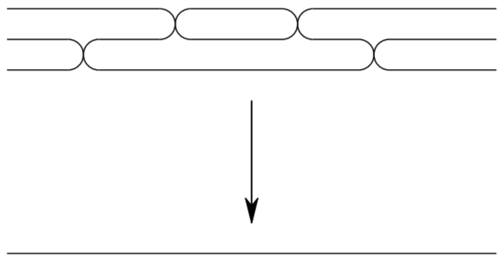
Let $f : X \to Y$ be a scheme morphism. Then it is finite if there is an affine cover $\bigcup_i \text{Spec}\,(S_i) = Y$ such that $f^{-1}(\text{Spec}\,(S_i))$ is the spectrum of a finite S_i-algebra.

***Proof* :**

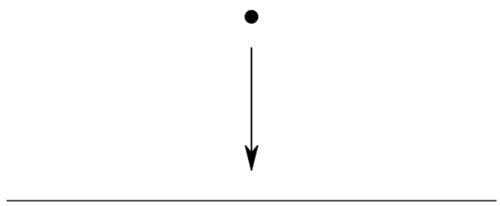
If f is finite, the result is *a fortiori* true, for the converse take $\text{Spec}\, S_i \subseteq Y$, cover it in distinguished open sets, and apply corollary 3.1.13 and lemma 3.3.14.

Example 3.3.16: Finite Morphism

1. If L/K is a finite extension, $\text{Spec}\,(K) \to \text{Spec}\,(L)$ is a finite morphism.
2. Take the map $\text{Spec}\, k[t] \to \text{Spec}\, k[u]$ given by $u \mapsto p(t)$ for some degree n polynomial $p(t) \in k[t]$. Then this map is finite: notice that $k[t]$ is generated as a $k[u]$-module by $1, t, ..., t^{n-1}$. This can be visualized as:



3. Let $I \subseteq R$ be an ideal and take the natural schemes and map $\operatorname{Spec}(R/I) \to \operatorname{Spec}(R)$. Then this is a finite morphism, as R/I is generated as an R-module by $1 \in R/I$. This can be visualized as:



4. More generally, all closed embeddings are finite morphisms.
5. Let $f \in R[x]$ be a polynomial whose leading coefficient is *not* a unit. Show that

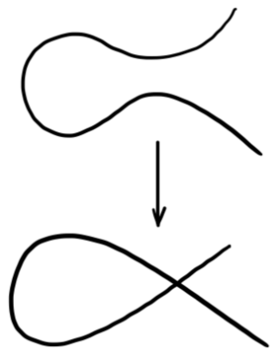
$$\operatorname{Spec} \frac{R[x]}{(f)} \to \operatorname{Spec} R$$

is *not* a finite morphism. Show that it *is* a finite morphism if the leading coefficient is a unit, making this an iff.

6. Take $\operatorname{Spec} k[t] \to \operatorname{Spec} \frac{k[x,y]}{(y^2-x^2-x^3)}$ corresponding to the ring map given by:

$$x \mapsto t^2 - 1 \qquad y \to t^3 - t$$

Namely, this shows this curve is rational. This is a finite morphism, since $k[t]$ is generated as a $\frac{k[x,y]}{(y^2-x^2-x^3)}$-module by 1 and t. This can be visualized as:



As the reader may have noticed, there is only one point that is 2-to-1. The reader should check that this will induce an isomorphism between $D(t^2 - 1)$ and $D(x)$, i.e. if the node is removed there is an isomorphism. This shall come up many times as an example of normalizing a curve.

7. Let $X \to \operatorname{Spec} k$ be a scheme morphism. Show that if the morphism if finite, X is the finite union of points with the discrete topology, each points having as residue field a finite extension of k. This can be visualized as:

Notice that all the fibers were finite. This is no happenstance of the chosen examples:

Proposition 3.3.17: Finite Morphism is Finite-to-one

Let $\pi : X \to Y$ be a finite morphism. Then it's fibers are finite

***Proof* :**
exercise. (Vakil p.214 for hint if you're stuck)

The converse is not the case

Example 3.3.18: Finite Fibers But Not Finite Morphisms

The open embedding $\mathbb{A}_k^2 \setminus \{(0,0)\} \hookrightarrow \mathbb{A}_k^2$ has finite fibered (they are one-to-one), but it is not affine (as $\mathbb{A}_k^2 \setminus \{(0,0)\}$ is not affine).

It also possible to have finite fibers and affine, but not finite: Take $\mathbb{A}_{\mathbb{C}}^1 \setminus \{0\} \hookrightarrow \mathbb{A}_{\mathbb{C}}^1$ (intuitively, localizing is "changing perspectives" to the infinitesimal).

It looks important to add 7.3.I to get a grasp on finite morphisms to affine spectrums relation to projective scheme

Definition 3.3.19: Integral Morphism

Let $\pi : X \to Y$ be a scheme morphism. The nit is an *integral morphism* if π is affine, and for every affine open subset $\text{Spec}\,(S) \subseteq Y$, where $\pi^{-1}(\text{Spec}\,(S)) = \text{Spec}\,(R)$, the induced map $S \to R$ is an integral extension.

As integral extensions must be finitely generated over the source ring, an finite morphism is a integral morphism, however an integral morphism need not be finite (think $\text{Spec}\,\overline{\mathbb{Q}} \to \text{Spec}\,\mathbb{Q}$). It is further easy to verify that the composition of integral maps is integral.

Proposition 3.3.20: Integral Morphisms Affine-local On Target

Let $\pi : X \to Y$ be an scheme morphism. Then π is an integral morphism if and only if there is a affine open cover $\cup_i U_i = Y$ such that we get integral extension for each pre-image.

Proof **:**
exercise.

Proposition 3.3.21: Integral Morphism Are Closed Maps

Let $\pi : X \to Y$ be a Integral morphism. Then it is a closed map: it maps closed subsets to closed subsets

As a consequence, finite morphisms are closed

Proof **:**
idea: reduce first to ring map., and consider the induced map. Now take $I \subseteq R$ and $J = (\pi^{\#})^{-1}(I)$ and consider the integral extension $S/J \subseteq A/I$. Apply the Lying Over theorem.

Definition 3.3.22: Locally Finite Type Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *locally of finite type* if for every affine open subset $\text{Spec}\,(S) \subseteq Y$, and every affine open subset $\text{Spec}\,(R) \subseteq \pi^{-1}(\text{Spec}\,(R))$, the induced morphism $S \to R$ makes R a finitely generated S-algebra. Equivalently, $\pi^{-1}(\text{Spec}\,(S))$ can be covered by affine open subsets $\text{Spec}\,(R_i)$ where each R_i is a finitely generated S-algebra.

Definition 3.3.23: Locally Finite Type Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *of finite type* if it is locally of finite type and quasicompact.

Proposition 3.3.24: Locally Finite Type Morphism Affine-local On Target

Let $\pi : X \to Y$ be an scheme morphism. Then π is locally of finite type (resp. of finite type) if there is a cover of Y by affine open sets $\operatorname{Spec}(S_i)$ such that $\pi^{-1}(\operatorname{Spec}(S_i)$ is locally finite type over S_i.

***Proof* :**
exercise.

Example 3.3.25: Finite Type Morphisms

1. All Schemes that are finite type over k are example of finite type scheme morphisms $X \to \operatorname{Spec} k$
2. The map $\mathbb{P}^n_R \to \operatorname{Spec} R$ is of finite type ($\mathbb{P}^n_R$ is covered by $n+1$ open sets)

Proposition 3.3.26: Integral and Finite Type iff Finite

1. finite morphisms are of finite type
2. A morphism is finite if it is integral and of finite type

***Proof* :**
exercise.

Definition 3.3.27: Quasifinite Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *quasifinite* if it is of finite type, and for each $\mathfrak{q} \in Y$, $\pi^{-1}(\mathfrak{q})$ is a finite set.

By proposition proposition 3.3.17 and proposition 3.3.26(1), finite morphisms are quasifinite. The converse is not true (think of $\mathbb{A}^2_k \setminus \{(0,0)\} \to \mathbb{A}^2_k$. The finite type condition is necessary, there exists morphisms with finite fibers that are not quasifinite

Example 3.3.28: Finite Fibers, But Not Quasifinite
The morphism $\operatorname{Spec}(\mathbb{C}(t)) \to \operatorname{Spec}(\mathbb{C})$ has finite fibers, but is not quasifinite. Similarly to $\operatorname{Spec}(\overline{\mathbb{Q}}) \to \operatorname{Spec}(\mathbb{Q})$.

Quasifinite morphisms shall return when discussing Zariski Main Theorem (see ref:HERE).

Definition 3.3.29: Locally Finite Presentation Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *locally of finite presentation* if for every affine open subset $\text{Spec}\,(S) \subseteq Y$, and every affine open subset $\text{Spec}\,(R) \subseteq \pi^{-1}(\text{Spec}\,(R))$, the induced morphism $S \to R$ makes R a finitely presented S-algebra. Equivalently, $\pi^{-1}(\text{Spec}\,(S))$ can be covered by affine open subsets $\text{Spec}\,(R_i)$ where each R_i is a finitely presented S-algebra.

There is one exception for the definition of finitely presented schemes that must be underlined in the following definition:

Definition 3.3.30: Finite Type Morphism

Let $\pi : X \to Y$ be a scheme morphism. Then π is said to be *finitely presented* if it is locally of finite type and quasicompact *and quasiseparated.*

Vakil promises exercise 9.3.H shows why this would naturally arise

Exercise 3.3.1

1. Let $\pi : X \to Y$ be an open embedding. Then if Y is locally Noetherian, X is locally Noetherian. If Y is Noetherian, X is Noetherian.
2. Let $\pi : X \to Y$ be an open embedding. Show that if Y is quasicompact, X need not be (there is an affine counter-example)
3. Show that the composition of two finite morphism is a finite morphism.
4. Show that being an open embedding is not local on the source.
5. Let B/A be an integral extension. Then show that any ring homomorphism $B \to C$ induces an integral extensions $C \otimes_B A/C$. Conclude that integrality is preserved by base change (see scalar extensions in [8] if you are unsure what it means to change base).
6. Show that every open embedding is locally of finite type. Hence, every quasicompact open embedding is of finite type.
7. Every open embedding into a locally Noetherian scheme is of finite type
8. The composition of two morphisms locally of finite type is locally of finite type.
9. Let $\pi : X \to Y$ be locally of finite type and Y locally Noetherian. Then X is locally Noetherian. IF $\pi : X \to Y$ is of finite type and Y is Noetherian, then X is Noetherian,
10. Show that quasifinite morphisms to $\text{Spec}\,(k)$ are finite)
11. Let $k = \overline{\mathbb{F}_p}$. Show that the morphism $F : \mathbb{A}^n_k \to \mathbb{A}^n_k$ of k-schemes corresponding to the bijection $x_i \mapsto x_i^p$ is a bijection, and no power of F is the identity (as sets)
12. (Generalizing the above) Let X be an $\mathbb{F}_p$-scheme. Define an endomorphism $F : X \to X$ such that for each affine open subset $\text{Spec}\,(R) \subseteq R$¡ F corresponds to the map $R \to R$ given by $f \mapsto f^p$. The morphism F is called the *absolute Frobenius morphism.*

3.4 Separated and Proper Schemes

The last two major types of schemes we must cover are *separated schemes* that recover the notion of Hausdroffness for schemes, and *proper schemes* which recover a notion of compactness for schemes. As is the theme in this section, these two properties can be defined purely in terms of scheme morphisms.

From an algebraic-geometric perspective, these conditions allow for better behaviour of closed sets, whether the intersection of two closed sets is closed or whether the image of closed sets is closed is governed by separatedness and properness respectively. Both separatedness and properness can be defined in terms of morphisms, and this is in fact the more practical approach. As closed sets are locally closed set of affine scheme (i.e. vanishing sets of global sections of $\operatorname{Spec} R_i$), the preservation of closedness of these sets under various kinds of mappings is of key interest in scheme theory.

3.4.1 Separated Morphisms

Recall that if $f : X \to Y$ is a continuous map and $K \subseteq X$ is compact, then $f(K)$ is compact. In euclidean topology, this implies closed and bounded maps to closed and bounded, importantly closed maps to closed. As affine schemes are quasi-compact (recall we use the term quasi-compact instead of compact due to the differing behaviour without Hausdorffness), if $K \subseteq X$ is closed it is quasi-compact, and hence its image is quasi-compact if $f : X \to Y$ is a continuous map (say, a regular map between affine schemes). However, such a map *need not be closed*, namely it must not map closed sets to closed sets (recall example ref:HERE). In particular a quasicompact set need not be closed (see [11, section 3.2]).

Another example that is shows the failure of separatedness is the following: Take X to be the line with double origin (recall example 2.2.38). Then the two natural inclusions $\iota_1, \iota_2 : \mathbb{A}^1_k \to X$ do not intersect on a closet set!

To recover the good properties of Hausdroffness, the notion of *separatedness* is defined. The idea is that we want to avoid cases where there can be "double points" or even more points that are intuitively arbitrarily close to each other.

In the topological setting, we saw X is Hausdorff if and only if $\Delta \subseteq X \times X$ is closed. In the case of scheme every "diagonal" (when working with fibered products) is a locally closed embedding:

Proposition 3.4.1: Diagonal Locally Closed Embedding

Let $f : X \to S$ be a morphism of schemes so that X is an S-scheme. Then the diagonal morphism $\delta : X \to X \times_S X$ is a locally closed embedding.

***Proof*:**

We will describe a union of open subsets of $X \times_S X$ covering the image of X, such that the image of X is a closed embedding in this union.

Say S is covered with affine open sets V_i and X is covered with affine open sets U_{ij}, with $f : U_{ij} \to V_i$. Take $W = \bigcup_{i,j} U_{ij} \times_{V_i} U_{ij}$. This is an affine open subscheme of the product $X \times_S X$ (recall this is how the product was constructed). Then the diagonal is covered by these affine open subsets $U_{ij} \times_{V_i} U_{ij}$. Any point $p \in X$ lies in some U_{ij}; then $\delta(p) \in U_{ij} \times_{V_i} U_{ij}$. Note that $\delta^{-1}(U_{ij} \times_{V_i} U_{ij}) = U_{ij}$ since certainly $U_{ij} \subset \delta^{-1}(U_{ij} \times_{V_i} U_{ij})$, and as $\pi_1 \circ \delta = id_X$ (as π_1 is the first projection), $\delta^{-1}(U_{ij} \times_{V_i} U_{ij}) \subset U_{ij}$. Finally, to check that $U_{ij} \to U_{ij} \times_{V_i} U_{ij}$ is a closed embedding, say $V_i = \operatorname{Spec} B$ and $U_{ij} = \operatorname{Spec} A$. Then this corresponds to the natural ring map $A \otimes_B A \to A$ $(a_1 \otimes a_2 \mapsto a_1 a_2)$, which is certainly surjective, completing the proof.

The key is that these open subsets need not cover $X \times_S X$, hence this does not imply the embedding is closed.

In our case, a subset is closed if it is locally closed in an affine covering, this means that the diagonal can be expressed (locally) as vanishing sets. Furthermore, the product is not well-defined for schemes, as it must always be fibered over some other scheme. This leads to the following:

Definition 3.4.2: Separated Scheme

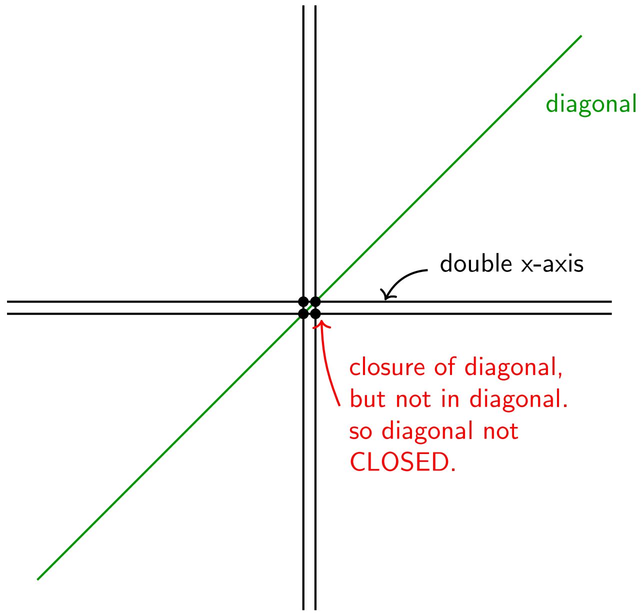
Let X be a scheme. Then X is *separated* if the map $\delta : X \to X \times_{\mathbb{Z}} X$ is a closed embedding.

More generally, a map $X \to S$ between schemes is *separated* if the map $\delta : X \to X \times_S X$ is a closed embedding. Denote the image of δ as $\Delta(X)$ or Δ_X.

Note that the first condition is equivalent to saying that the morphism $X \to \operatorname{Spec}(\mathbb{Z})$ is separated. The fact that separatedness is given by the purely topological condition of $\Delta(X)$ being closed shows its closeness to the original topological condition of Hausdorffness.

Example 3.4.3: Separated Morphisms

1. Take the line with two origins over k, label it X. Then when considering $X \times_{\operatorname{Spec} k} X$, we can think to it as in image bellow. The double origin will correspond to 4 points, but the diagonal will only have 2 points. Taking the closure of the diagonal will include all 4 points (show that any open set containing either $(0_1, 0_2)$ or $(0_2, 0_1)$ must intersect $\Delta(X)$, and hence is in the closure; see [11, section 1.5]), showing that it is *not* separated.



Note that $X \to X \times_k X$ is still a locally closed embedding. To show that it is not closed, we can take an open cover that covers *all* of $X \times_k X$ and show that for the affine open W_i containing $P_1 = (0_1, 0_2)$, $\Delta(X) \cap W_i$ is not closed. Indeed, As $P_1 \notin \Delta(X)$, $P_1 \notin \Delta(X) \cap W_i$. It is easy to see that P_1 is in the closure of this set, $P_1 \in \overline{\Delta(X) \cap W_i}$, we see that $\Delta(X) \cap W_i$ is *not* closed.

Note that if we take $X \to X$ to be the identity map, then this *is* a separated morphism, since:

$$X \cong X \times_X X$$

showing the relativeness of separatedness when picking different base schemes.

2. Let $X = \operatorname{Spec}(A) \to \operatorname{Spec}(B) = Y$ be any morphism of schemes between affine schemes. Then this is *always* a separated morphism. Unwind the definitions: when converting into rings we get the map:

$$A \otimes_B A \to A$$

which is surjective, and hence we get:

$$A \cong A \otimes_B A/\mathfrak{a}$$

that is, we are taking the quotient by an ideal, which we know will give us a closed subscheme of $X \times_Y X$ isomorphic to X.

3. More generally, if $X \to Y$ is an affine morphism, it is separated (follows from proposition 3.4.1).

4. The following scheme is a famous example of non-separatedness appearing naturally. Consider $\mathbb{C}^* \curvearrowright \mathbb{A}^2_{\mathbb{C}}$ via $t \cdot (x, y) = (tx, t^{-1}x)$; this is usually sated as "the algebraic torus acts on the affine plane with weights $(1, -1)$". Then taking $(\mathbb{A}^2_{\mathbb{C}} \setminus \{0\})/\mathbb{C}^*$ where the origin is removed as the action is non-free, the resulting space is a non-separated scheme.

In many cases, non-separated scheme appear naturally in moduli theory.

As a historical footnote, the term scheme used to mean separated scheme back before the 1960s, and the term pre-scheme was used for the modern term scheme.

The next concept will not be too useful to us for awhile, and shall come back up when we see algebraic stacks and algebraic spaces, but is useful to define:

Definition 3.4.4: Quasi-separated

Let X be a scheme. Then X is quasi-separated if the image of $X \to X \times X$ is quasi-compact. More generally, a map $X \to S$ between schemes is quasi-separated if the image map of $X \to X \times {}_S X$ is quasi-compact.

Proposition 3.4.5: Separated Implies Quasi-separated

Let X be a separated scheme. Then then X is quasi-separated

Proof **:**
Show that any closed embedding is quasi-compact.

Separatedness gives us nicer behaviour on closed sets. The following results outline some important examples:

Proposition 3.4.6: Intersection of Separated affine schemes

Let $X \to S$ be a separated scheme morphism where S is affine. Then if $U, V \subseteq X$ are open affine

$$U \cap V$$

is open affine.

Proof **:**
Here is the high-level

1. First show that $U \times_S V \subseteq X \times_S X$ is open and affine (it is the tensor-product of the corresponding algebras)
2. $U \cap V$ is the preimage image of $U \times_S V$ of the diagonal map, which is *closed* by assumption, and hence,
3. For any closed embedding, the inverse image of an open affine is open-affine. Hence as f is separated, $U \cap V$ is affine, completing the proof.

For a counter-example, taking affine *plane* with double origin we get that taking two affine lines each containing a different origin will intersect to produce the plane without the origin, which in example 2.2.20 is not affine.

Proposition 3.4.7: Graphs Are Closed

Let X be a separated S-schemes and Y any S-scheme. Then the graph of the S-morphism $f : X \to Y$:

$$\Gamma_f = \operatorname{im}\left((\mathrm{id}_Y, f)Y \to Y \times_S X\right)$$

is closed

Proof **:**

1. Show the graph is constructed via the pullback of the diagonal $\Delta_X \subseteq X \times_S X$ along the map $id_Y \times g : Y \times_S X \to X \times_S X$.
2. pullbacks preserve closed embeddings (see exercise ref:HERE

For the case where it is not separated as usual let X be the line with two origins (not separated). Let Y be the affine line $\mathbb{A}^1$. Consider the morphism $g : Y \setminus \{0\} \to X$ which is the natural inclusion into the "un-doubled" part of X. What is the graph of this morphism? It's a subset of $(Y \setminus \{0\}) \times_k X$.

The key is that this map cannot be extended to the origin of Y in a unique way. Indeed, $0 \in Y$ could map to either $0_1 \in X$ or $0_2 \in X$. This ambiguity means the graph of any potential extension is *not* closed; the point $(0, 0_1)$ and $(0, 0_2)$ are "limit points" of the graph but there is no natural choice of which one is in the graph. The graph is not well-behaved at the boundary. This intuition will in fact be a central way of interpreting separatedness, as shall be explored in the next section.

Proposition 3.4.8: Morphism Determined By Dense Open Subset

Let X be a repeated S-scheme, Z any scheme, and $U \subseteq Z$ a dense open subset. Then if two morphisms $f, g : Z \to X$ agree on U, $f|_U = g|_U$, then they agree everywhere:

$$f|_U = g|_U \qquad \Rightarrow \qquad f = g$$

***Proof* :**
Consider the set of points in Z where f and g agree. The key is this can be described as the preimage of the diagonal of X, namely, take $(g, h) : Z \to X \times_S X$, and:

$$E = (g, h)^{-1}(\Delta_X)$$

Then, we have a closed subset E that contains U (as every point in E are points f, g agree on), and as U is open and dense, it must be that $E = Z$, completing the proof.

Valuation Criterion for Separatedness

We shall next characterize separated schemes and morphisms in using valuation rings. This will be analogous to the fact that in a Hausdorff space, the limit of a sequence has at most 1 point. From this result, think of a continuous map $(0, 1) \to X$. If we want to extend this to a map $[0, 1] \to X$, if X is Hausdroff, there is only one possible extension. If X wasn't, say it had two origins, then there can be more than one map that can extend (the new 0 point will have two possible places to map to).

Let us now extend this idea to a map $X \to Y$, and we have an embedding $(0, 1) \to X$ an identity map $(0, 1) \to [0, 1]$ and an embedding $(0, 1] \to Y$ where we are only adding one point. Then there should only be one lift. Note that this lift is not dependent on X and Y. Let's say Y was the double-origin scheme and X was the double y-axis scheme. Then the embedding $(0, 1)$ can be extended to two way into Y, but that will correspond to a unique lift.

With this intuition in mind, let us now look at what is the equivalent of the open interval for schemes. This would be a Discrete valuation ring (DVR) R. It has only two points. We can think of as $\mathrm{Spec}\,(\mathrm{Frac}(R))$ as the open interval and $\mathrm{Spec}\,(R)$ as adding the closed 0 (notice that the new point in $\mathrm{Spec}\,(R)$ is also a closed point, while the other point is "open" and is the generic point that can be thought of was closed to everywhere else. Then we have the following situation

$$\begin{tikzcd}
\mathrm{Spec}\,(\mathrm{Frac}(R)) \arrow[r] \arrow[d] & Y \arrow[d] \\
\mathrm{Spec}\,(R) \arrow[r] \arrow[ur, dashed] & X
\end{tikzcd}$$

With the hope that having at most one lift corresponding to the map being separable. This is *sometimes* the case

Go over Harshorne's proofs

Theorem 3.4.9: Valuation Criterion for Separatedness

Let $f : X \to Y$ be a morphism of schemes, and assume that X is Noetherian. Then f is separated if and only if for any field K, and for any valuation ring R with quotient field K, $T = \operatorname{Spec} R, U = \operatorname{Spec} K$, and the morphism $i : U \to T$ is the morphism induced by the inclusion $R \subseteq K$, given a morphism of T to Y, and given a morphism of U to X, the following diagram commutes:

$$\begin{tikzcd} U \arrow[r] \arrow[d, "\iota"] & X \arrow[d, "f"] \\ T \arrow[r] \arrow[ur, dashed] & Y \end{tikzcd}$$

and there is at most one morphism of T to X making the whole diagram commutative.

***Proof* :**
see Hartshorne theorem 4.3

Rational maps to Separated Schemes

here

Exercise 3.4.1

1. add exercise 4.5 of Hartshorne: important for Weil divisors to show characterization of irred. closed subscheme of (separated noethean integral) reg. in codim 1 scheme
2. Show that $X \cong \Delta(X)$. Show that in general, $\Delta(X)$ is a locally closed embedding.
3. Show that a scheme X over S is separated if and only if it is separated over $\mathbb{Z}$.

3.4.2 Proper Morphism

Proper morphisms capture the idea of compactness. Compactness allows us to introduce some finiteness condition to our problem. In our case, this shall translate to the map from a proper scheme to another being closed over any basis. We shall see that almost all schemes we have encountered so far that are proper shall be able to be embedded into projective space, and projective space is in some way the prototypical proper scheme, and maps to projective space are the prototypical proper morphisms.

Recall that scheme morphisms need not be closed: their image need not be the vanishing set. It was shown in [8] that when working with projective varieties, their image is always closed with the intuition being that there are no "gaps" that can be created (the canonical example being $V(xy - 1)$ projected onto the x-axis). For schemes, we shall require something a bit stronger as schemes geometry is "inherent" rather than obtained from an ambient space; the consequence of the geometry being "inherent" is that we can have closed maps where when we change basis and then it is longer closed (hence, the closedness was somehow in the choice of basis of the scheme). To fix this, the maps have to always be closed under any basis:

Definition 3.4.10: Universally Closed

Let $f : X \to Y$ be a continuous map. Then f is universally closed if for every map $Z \to Y$, the induced map $Z \times_Y X \to Y$ is closed.

The idea is that we want the image to always be closed, even if we extend the map by a "non-proper" space.

Example 3.4.11: Non Proper Map

The map $\mathbb{R} \to 0$ is a closed map, but $\mathbb{R} \times \mathbb{R} \to 0 \times \mathbb{R}$ is no longer a closed map, as it take $V(xy - 1)$ to a non-closed set. To link this to the usual notion of proper: A continuous map $f : X \to Y$ on locally compact Hausdroff spaces with a countable bases is universally closed if and only if it is proper (i.e. the pre-image of compact subsets are compact)

Another useful equivalent is $f : X \to Y$ is proper if f is closed and the fibers are compact.

On schemes, we shall also want the maps to be locally finite and be separate so that proper maps behave very similarly to embedding in projective space $\mathbb{P}_k$[6]

Definition 3.4.12: Proper Morphism

A morphism $f : X \to Y$ is called *proper* if it is separated, finite type, and universally closed. A scheme X is said to be proper if the morphism $X \to \operatorname{Spec}(\mathbb{Z})$ is proper, and a k-scheme is proper if the maps $X \to \operatorname{Spec}(k)$ is proper. An R-scheme is said to be *proper over* R if the map $X \to \operatorname{Spec}(R)$ is proper.

Rarely, a proper k-scheme is called a complete k-scheme. This terminology is a holdover from the Italian school of algebraic geometry and is meant to reflect how they "completed" a variety by adding a points at infinity. However, the word "complete" is already an overloaded word in algebraic geometry (ex. completion of a field with respect to to a valuation). The term proper from topology was used as the appropriate new term.

Example 3.4.13: Proper Morphisms

1. The map $\mathbb{A}^1_{\mathbb{C}} \to \operatorname{Spec}(\mathbb{C})$ is separated, finite type, and closed. It is not proper, as:

$$\mathbb{A}^2_{\mathbb{C}} \cong \mathbb{A}^1_{\mathbb{C}} \times_{\mathbb{C}} \mathbb{A}^1_{\mathbb{C}} \to \mathbb{A}^1_{\mathbb{C}}$$

2. All closed embeddings are proper

[6]We shall define projective morphisms later. The two concepts are similar, but a proper scheme need not have an ample bundle

Proposition 3.4.14: Finite Morphisms Are Proper

Let $f : X \to Y$ be a finite morphism. Then f is proper

***Proof* :**
Here is the high-level: A finite morphism is certainly separated and finite type, so let us show it is universally closed. First, let's say $X \to Y$ is finite and we have a morphism $Z \to Y$. Then the fibered product $X \times_Y Z$ along with the map $X \times_Z Y \to Z$ is finite. Hence, it suffices to show that finite morphism maps are closed. By covering Y in affine open subsets, we reduce to the affine case. Hence, the problem comes down to having $\varphi : R \to S$ be a ring homomorphism where S is a finite R-module, then $\operatorname{Spec} S \to \operatorname{Spec} R$ is closed. Next, if $I = \ker \varphi$, then $R \to R/I \subseteq S$ is a composition of closed maps, and so we can assume $R \subseteq S$.
Now, a closed set of S is given by an ideal $J \subseteq S$. Hence we get:

$$\frac{R}{R \cap \varphi^{-1}(J)} \subseteq \frac{S}{J}$$

As $\operatorname{Spec} \frac{R}{R\cap\varphi^{-1}(J)} \subseteq \operatorname{Spec} R$, we have now reduced to showing the map $\operatorname{Spec} S \to \operatorname{Spec} R$ is surjective. But this is given by the Going-up Theorem (see [8]), which completes the proof.

Note that quasi-finite morphisms need not be proper: $\mathbb{A}^1_k \backslash \{0\} \to \mathbb{A}^1_k$ is one of the popular counterexamples.
Another very important family of proper morphism are projective morphism. Let us first define general projective space:

Definition 3.4.15: Projective Space Over Scheme

Let Y be a scheme. Then the projective space over Y is the scheme:

$$\mathbb{P}^n_Y := Y \times_{\operatorname{Spec} \mathbb{Z}} \mathbb{P}^n_{\mathbb{Z}}$$

Definition 3.4.16: Projective Morphism

Let $f : X \to Y$ be a scheme morphism. Then f is said to be projective if it factors into the following:

$$\begin{tikzcd} X \arrow[r] \arrow[rr, bend left, "f"] & \mathbb{P}^n_Y \arrow[r] & Y \end{tikzcd}$$

A morphism is said to be quasiprojective if it factors into an open immersion $X \to X'$ followed by a projective morphism $X' \to Y$.

We shall show that projective morphisms are proper, and along with finite morphisms shall form a large class of proper morphisms (though not all proper morphisms fall under these two categories, see example ref:HERE).

Proposition 3.4.17: Properties of Proper Morphisms

Consider only Noetherian schemes. Then:

1. A closed embedding to X is proper
2. A composition of proper maps is proper
3. Proper morphism are stable under base-extension
4. Product of proper morphisms are proper
5. If $f : X \to Y$ and $g : Y \to Z$ are morphisms, and g is separated, and $g \circ f$ is proper, then f is proper.
6. Properness is local on the base

Proof **:**
exercise

Proposition 3.4.18: Global Sections of Some Proper Schemes

Let X be a connected, reduced, proper k-scheme. Then $\mathcal{O}_X(X) = k$.

Proof **:**
exercise.

We conclude this section on proper morphism and schemes by giving one interpretation of why projective schemes are "natural" and how they relate to proper schemes:

Theorem 3.4.19: Chow's Lemma

Let X be a proper morphism over a Noetherian scheme S; an S-scheme. Then there exists a scheme X' and a morphism $\varphi : X' \to X$ such that

1. X' is a projective over S.
2. there exists an open dense subset $U \subseteq X$ such that $\varphi^{-1}(U) \to U$ is an isomorhpism

Proof **:**
exercise

Exercise 3.4.2

1. Show that two copies of $\operatorname{Spec}(k[x_1, x_2, ...])$ minus the origin, glued at the origin, is not quasi-separated.

Valuation Criterion for Properness

I heard that the proof for projective being proper using the valuation criterion is more complicated than it has to be! The properness criterion should in a sense be its own thing, again useful for moduli spaces.

To show that projective morphisms are proper, we require a local description of properness similar to the one given for separatedness. To motivate it, let us consider the case in euclidean topology. Consider a function $f : \mathbb{R}_{>0} \to \mathbb{P}^1$. Then such a function may or may not have a limit as $x \to 0$, namely if it isolates arbitrarily fast. As the codomain is $\mathbb{P}^1$ we shall say it does have a limit of f shoots off to infinity. Now, for rational functions, it is impossible to oscillate arbitrarily quickly. Hence, it always has a limit as we tend to 0 .
The fact that the limit always exists is a consequence of $\mathbb{P}^1$ being compact. Consider now a diagram:

$$\begin{tikzcd}
\mathbb{R}_{>0} \arrow[r, "f"] \arrow[d, "\iota"'] & X \arrow[d, "\pi"] \\
\mathbb{R}_{\geq 0} \arrow[r, "\text{extension}"'] \arrow[ur, dashed] & \mathbb{R}
\end{tikzcd}$$

Then if there is always a lift, represented by the dotted line, we have a proper space. Visually, we can see this as:

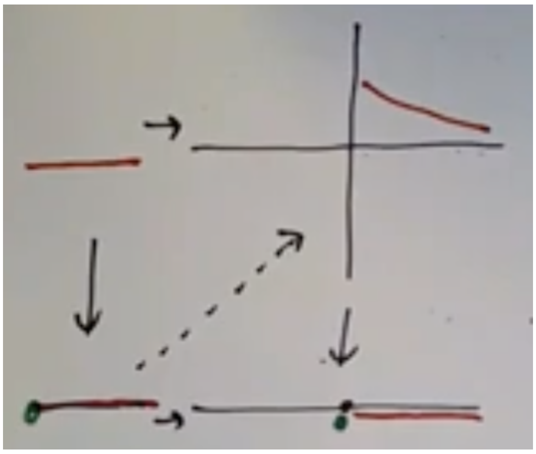


Figure 3.1: Visualizing point-wise condition

To change this into the language of scheme theory, we would replace $\mathbb{R}_{>0}$ with $\operatorname{Spec}\operatorname{Frac}(R)$ and $\mathbb{R}_{\geq 0}$ with Spec R for a DVR (or just a valuation ring) R. Then the top-right and bottom-right should be X and Y, and we want the map to be proper. Just like for separatedness, this works with a couple of extra relatively weak conditions:

Theorem 3.4.20: Valuation Criterion For Properness

Let $f: X \to Y$ be a scheme morphism of finite type and X is Noetherian. Then f is proper if and only if for every valuation ring R and every morphism $U \to X$ and $T \to Y$ there exists a unique lifting $T \to X$ such that following diagram commutes:

$$\begin{tikzcd} U \arrow[r] \arrow[d, "\iota"'] & X \arrow[d, "f"] \\ T \arrow[r] \arrow[ur, dotted] & Y \end{tikzcd}$$

***Proof* :**
First assume that f is proper. Then by definition f is separated, so the morphism $T \to X$ is unique once we know it exists. For the existence, we consider the base extension $T \to Y$, and let $X_T = X \times_Y T$. We get a map $U \to X_T$ from the given maps $U \to X$ and $U \to T$.

$$\begin{tikzcd} U \arrow[r] & X_T \arrow[r] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ & T \arrow[r] & Y \end{tikzcd}$$

Let $\xi_1 \in X_T$ be the image of the unique point t_1 of U. Let $Z = \{\xi_1\}^-$. Then Z is a closed subset of X_T. Since f is proper, it is universally closed, so the morphism $f': X_T \to T$ must be closed, so $f'(Z)$ is a closed subset of T. But $f'(\xi_1) = t_1$, which is the generic point of T, so in fact $f'(Z) = T$. Hence there is a point $\xi_0 \in Z$ with $f'(\xi_0) = t_0$. So we get a local homomorphism of local rings $R \to \mathcal{O}_{\xi_0,Z}$ corresponding to the morphism f'.
Now the function field of Z is $k(\xi_1)$, which is contained in K, by construction of ξ_1, and R is maximal for the relation of domination between local subrings of K. Hence R is isomorphic to $\mathcal{O}_{\xi_0,Z}$, and in particular R dominates it. Hence by lemma ref:HERE (4.4 in Hartshorne) we obtain a morphism of T to X_T sending t_0, t_1 to ξ_0, ξ_1. Composing with the map $X_T \to X$ gives the desired morphism of T to X.
Conversely, suppose the condition of the theorem holds. To show f is proper, we have only to show that it is universally closed, since it is of finite type by hypothesis, and it is separated by theorem 3.3.26. So let $Y' \to Y$ be any morphism, and let $f': X' \to Y'$ be the morphism obtained from f by base extension. Let Z be a closed subset of X', and give it the reduced induced structure.

$$\begin{tikzcd} Z \subseteq X' \arrow[r] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r] & Y \end{tikzcd}$$

We need to show that $f'(Z)$ is closed in Y'. Since f is of finite type, so is f' and so is the restriction of f' to Z (Ex. 3.13). In particular, the morphism $f': Z \to Y'$ is quasi-compact, so by (4.5) we have only to show that $f(Z)$ is stable under specialization. So let $z_1 \in Z$ be a point, let $y_1 = f'(z_1)$, and let $y_1 \rightsquigarrow y_0$ be a specialization. Let $\mathcal{O}$ be the local ring of y_0 on $\{y_1\}^-$with its reduced induced structure. Then the quotient field of $\mathcal{O}$ is $k(y_1)$, which is a subfield of $k(z_1)$. Let $K = k(z_1)$, and let R be a valuation ring of K which dominates $\mathcal{O}$.

From this data, by lemma ref:HERE (4.4 in Hartshorne) we obtain morphisms $U \to Z$ and $T \to Y'$ forming a commutative diagram

$$\begin{tikzcd} U \arrow[r] \arrow[d, "\iota'"] & Z \arrow[d] \\ T \arrow[r] & Y' \end{tikzcd}$$

Composing with the morphisms $Z \to X' \to X$ and $Y' \to Y$, we get morphisms $U \to X$ and $T \to Y$ to which we can apply the condition of the theorem. So there is a morphism of $T \to X$ making the diagram commute. Since X' is a fibred product, it lifts to give a morphism $T \to X'$. And since Z is closed, and the generic point of T goes to $z_1 \in Z$, this morphism factors to give a morphism $T \to Z$. Now let z_0 be the image of t_0. Then $f'(z_0) = y_0$, so $y_0 \in f'(Z)$. This completes the proof.

Proposition 3.4.21: Projective Morphism Are Proper

1. Let $f : X \to Y$ be a projective morphism between Noetherian Schemes. Then f is proper.
2. Let $f : X \to Y$ be a quasi-projective morphism between Noetherian Schemes. Then f is of finite type and separated.

***Proof* :**
It suffices to show that $X = \mathbb{P}^n_{\mathbb{Z}}$ is proper over $\operatorname{Spec}\mathbb{Z}$. Recall that X is a union of open affine subsets $V_i = D(x_i)$, and that V_i is isomorphic to $\operatorname{Spec}\mathbb{Z}[x_0/x_i, \ldots, x_n/x_i]$. Thus X is of finite type. To show that X is proper, we will use the valuation criterion for proneness and imitate the proof of that a regular map on a projective variety minus a point can uniquely be extended to that point. So suppose given a valuation ring R and morphisms $U \to X, T \to \operatorname{Spec}\mathbb{Z}$ as shown:

$$\begin{tikzcd} U \arrow[r] \arrow[d] & X \arrow[d] \\ T \arrow[r] \arrow[ur, dashed] & \operatorname{Spec}\mathbb{Z} \end{tikzcd}$$

Let $\xi_1 \in X$ be the image of the unique point of U. Using induction on n, we may assume that ξ_1 is not contained in any of the hyperplanes $X - V_i$, which are each isomorphic to $\mathbb{P}^{n-1}$. In other words, we may assume that $\xi_1 \in \bigcap V_i$, and hence all of the functions x_i/x_j are invertible elements of the local ring $\mathcal{O}_{\xi_1}$.
We have an inclusion $k(\xi_1) \subseteq K$ given by the morphism $U \to X$. Let $f_{ij} \in K$ be the image of x_i/x_j. Then the f_{ij} are nonzero elements of K, and $f_{ik} = f_{ij} \cdot f_{jk}$ for all i, j, k. Let $v : K \to G$ be the valuation associated to the valuation ring R. Let $g_i = v(f_{i0})$ for $i = 0, \ldots, n$. Choose k such that g_k is minimal among the set $\{g_0, \ldots, g_n\}$, for the ordering of G. Then for each i we have

$$v(f_{ik}) = g_i - g_k \geq 0$$

hence $f_{ik} \in R$ for $i = 0, \ldots, n$. Then we can define a homomorphism

$$\varphi : \mathbb{Z}\left[x_0/x_k, \ldots, x_n/x_k\right] \to R$$

by sending x_i/x_k to f_{ik}. It is compatible with the given field inclusion $k(\xi_1) \subseteq K$. This homomorphism φ gives a morphism $T \to V_k$, and hence a morphism of T to X which is the one required. The uniqueness of this morphism follows from the construction and the way the V_i patch together.

3.5 Abstract Varieties

Let us use the tools we've developed to our usual notion of variety. We shall see that defining varieties abstractly (other common adverbs being inherently, integrally, intrinsically) shall give us many of the same benefits as defining schemes as spectrum. Most results about quasi-projectile varieties from [8] readily translate to the abstract setting with a few exceptions that we shall go over.

Recall that the category of varieties embed into the category of schemes (theorem 3.1.19). We have expanded upon enough properties to characterize the image of this embedding:

Definition 3.5.1: k-Variety

Let X be a k-scheme. Then if X is reduced, irreducible, separated, and of finite type over k, then X is said to be a *variety over* k or a k*-variety*. If X is affine/(quasi)projective, then it is an affine/(quasi)projective variety over k respectively.

It is important to note that in [8], a variety was an affine algebraic set that was irreducible, which motivates adding irreducible in the definition. However some authors drop the condition and "variety" would be "algebraic set". Furthermore, some authors require the field $k = \overline{k}$ be algebraically closed as part of the definition to more closely reflect the geometric results of the Nullstellensatz. Note too that we have define k-varieties and not varieties in general.

Definition 3.5.2: Complete Variety

Let X be a k-variety. Then it is complete if it is proper

The name complete for a proper variety comes from the fact that we are in some way thinking of a complete variety as an extension into projective Space. It can be shown (see ref:HERE[7]) that if C is complete, then there exists a morphism $C \to \mathbb{P}^n$ to a projective scheme that is isomorphic to an open dense subset. There do exist complete non-projective varieties, but they are hard to come by, partly due to how badly the singularity must be[8].

[7] see Borcherds scheme video abstract and projective varieties

[8] again see the same Borcherds video for an overview. The example he gave also allows for producing algebraic spaces that are not schemes

Proposition 3.5.3: Affine and Projective Varieties

1. $\mathbb{A}^n_k/I$ is an affine k-variety if and only if $I \subseteq k[x_1, ..., x_n]$ is a radical ideal
2. Let $I \subseteq k[x_1, ..., x_n]$ be a radical graded ideal. Then $\operatorname{Proj} k[x_0, ..., x - n]/I$ is a projective k-variety [a]

[a] I need not be radical, ex. $I = (x_0^2, x_0x_1, ..., x_0x_n)$

Proof **:**
exercise.

In theorem 3.1.19, we shall show that a certain subset of schemes will fully and faithfully map to the category of $\overline{k}$-varieties.

(Put somewhere that the fibered product of finite type k-schemes over a finite type k-scheme is a finite type k-scheme, and so varieties behave well under fibered product)

4

Schemes III: Sheaves of Modules – Projectives, Divisors, Differentials

So far, we have only worked with the structure sheaf $\mathcal{O}_X$ over a scheme X; we can put other sheaves that shall give use many important insights, similarity to how different [vector, principal G-] bundles on manifolds provide a lot of useful information. All of these sheaves shall have an $\mathcal{O}_X$-action on them, similarly to sections of bundles have an $C^\infty(M)$-action (or $G \subseteq \mathrm{GL}_r(K)$-action) on them, making them the generalization of modules to ringed spaces. The most important of these sheaves shall be ones that are locally isomorphic to sheaves corresponding to R_i-modules M_i. Such sheaves shall be called quasi-coherent sheaves, and if they are finitely generated they shall be called coherent shaves. The term “coherent” comes from Serre who saw these sheaves as ”cohering” (i.e. sticking together). The additional ”quasi-” for non-finitely generated modules comes from loosing some results when omitting finite generation (for example, they are not in general preserved under pushforward).

Quasi-coherent sheaves can be interpreted as generalization of bundles, in particular, *locally free sheaves* (over X) with sheaf morphisms shall be categorically equivalent to the vector bundles (over X) with bundle morphisms. Locally free sheaves are not closed under kernels and cokernels, and hence the same holds for bundles (see [7, chapter 1] on the detail of how to form sub/quotient bundles). The category of quasi-coherent sheaves can be constructed by adding the kernels and cokernels of locally free sheaves (similarly to how we can form projective/injective resolutions of non-free modules), and so one way of thinking about quasi-coherent sheaves is that they are the natural smallest extension of the category of vector bundles to form an abelian category; all of this shall be demonstrated in section 4.1.2.

By working with quasi-coherent sheaves, we shall be able to define ideal sheaves, image schemes, twisted sheaves on projective space (which leads to recovering $S_\bullet$ from Proj $S_\bullet$), sheaves remembering information at just some points point with multiplicity (the natural generalization skyscraper sheaves from example 1.1.10), and differentials (from which we recover forms and tangent bundles), among many more new interesting sheaves on schemes that will give lots more information on schemes.

4.1 Definition and Basic Properties

Let us start by defining the module-equivalent for ringed spaces:

Definition 4.1.1: $\mathscr{O}_X$-Module

Let $\mathscr{O}_X \in \mathbf{Sh}_{\mathbf{Ring}}(X)$. Then an $\mathscr{O}_X$-*module* on X is a sheaf $\mathscr{F} \in \mathbf{Sh}_{\mathbf{Ab}}(X)$ that has the following data:

1. For each open $U \subseteq X$ $\mathscr{F}(U)$ is an $\mathscr{O}_X(U)$-module
2. if $U \subseteq V$, then the following diagram commutes:

$$\begin{tikzcd} \mathscr{O}_X(V) \times \mathscr{F}(V) \arrow[r, "\curvearrowright"] \arrow[d, "\mathrm{res}_{V,U} \times \mathrm{res}_{V,U}"'] & \mathscr{F}(V) \arrow[d, "\mathrm{res}_{V,U}"] \\ \mathscr{O}_X(U) \times \mathscr{F}(U) \arrow[r, "\curvearrowright"] & \mathscr{F}(U) \end{tikzcd}$$

If $\mathscr{F}, \mathscr{G}$ are $\mathscr{O}_X$-modules, then a *morphism of $\mathscr{O}_X$-modules* $\varphi : \mathscr{F} \to \mathscr{G}$ is a sheaf-morphism such that for each open $U \subseteq X$, $\mathscr{F}(U) \to \mathscr{F}(X)$ is a $\mathscr{O}_X(U)$-module homomorphism. Their collection is often denoted $\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{F}, \mathscr{G})$ or sometimes just $\mathrm{Hom}_X(\mathscr{F}, \mathscr{G})$ if it is unambiguous.

These should really be thought of as fibre bundles or even vector bundles on X. We define the pushforward and pullback of $\mathscr{O}_X$-modules on ringed spaces for future cohomological purposes:

Definition 4.1.2: $\mathscr{O}_X$-module Pushforward and Pullback

let $(f, f^\#) : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a ringed space morphism, and $\mathscr{F}$ is a $\mathscr{O}_X$-module on X. Then $f_*\mathscr{F}$ is a $f_*\mathscr{O}_X$-module on Y. Via the map $f^\# : \mathscr{O}_Y \to f_*\mathscr{O}_X$, we see that this has a natural $\mathscr{O}_Y$-module structure, and it is called the direct image of $\mathscr{F}$ via f. We can do a similar chain of reasoning for $f^{-1}\mathscr{O}_Y$ creating the inverse image of $\mathscr{F}$ which is adjoint to the direct image.

Example 4.1.3: General $\mathscr{O}_X$-Modules Constructions

Fixed a locally ringed space $(X, \mathscr{O}_X)$

1. Take $\mathscr{F} = \mathscr{O}_X$ with action on each section being defined by multiplication. Then then this $\mathscr{O}_X$-module as any ringed space $\mathscr{O}_X$ is a $\mathscr{O}_X$-module over itself. As each restriction map is a ring-homomorphism, it is naturally $\mathscr{O}_X(U)$-module homomorphism, and hence the action is preserved.
2. Just like how every abelian group is a $\mathbb{Z}$-module, every sheaf of abelian groups $\mathcal{G}$ is a $\underline{\mathbb{Z}}$-module (where $\underline{\mathbb{Z}}$ is the constant sheaf associated to $\mathbb{Z}$), showing how this is a natural generalization.
3. If $\varphi : \mathscr{F} \to \mathscr{G}$ is a $\mathscr{O}_X$-module homomorphism, then the kernel, image, and cokernel are all

$\mathcal{O}_X$-modules (generalizing the result form the case of sheaf morphisms). It is also closed under products and coproducts, making $\mathcal{O}_X$-modules an abelian category.

4. For cohomological purposes, we can always define the sheaf-hom $\mathcal{O}_X$ module given by

$$U \mapsto \operatorname{Hom}_{\mathcal{O}_x|_U}\left(\mathscr{F}|_U, \mathscr{G}|_U\right)$$

where $\mathcal{O}_X|_U$ is given the natural restriction to a $\mathcal{O}_X|_U$-module on U.

5. The tensor product $\mathscr{F} \otimes_{\mathcal{O}_X} \mathscr{G}$ is the sheafification of the presheaf given by $U \mapsto \mathscr{F}(U) \otimes_{\mathcal{O}_X(U)} \mathscr{G}(U)$. The sheafification is necessary and shall in fact cause very non-trivial behaviour, namely it shall not be the case that:

$$\mathscr{F} \otimes \mathscr{G}(U) \cong \mathscr{F}(U) \otimes \mathscr{G}(U)$$

and in fact it can go spectacularly wrong, see example 4.12

4.1.1 Quasi-Coherent and Coherent Sheaves

A natural question then is whether the category R-Mod and the category of $\mathcal{O}_X$-Mod are equivalent when $X = \operatorname{Spec} R$. As it turns out, there are in fact too many $\mathcal{O}_X$-modules:

Example 4.1.4: Too Many $\mathcal{O}_X$-modules Over Affine Schemes

Let's say there was this correspondence so that for an R-module M, $\widetilde{M}$ is the correspondence sheaf. For this correspondence to work, we would need that $\widetilde{M}_\mathfrak{p} = M_\mathfrak{p}$. Let us show we can find a sheaf that does not have this property.

Let $X = \operatorname{Spec} \mathbb{Z}_{(2)}$ which consists of two open sets: $\{(2), \eta\}$ and $\{\eta\}$. We have the corresponding $\mathcal{O}_X(X) = \mathbb{Z}_{(2)}$ and $\mathcal{O}_X(\{\eta\}) = \mathbb{Q}$. To define the $\mathcal{O}_X$-module on X, we shall need to define a pair of modules, an $\mathbb{Z}_{(2)}$-module M and a $\mathbb{Q}$-module N, along with a restriction module homomorphism:

$$M \to N$$

Here is a combo that works:

$$M = \mathbb{Z}_{(2)} \quad N = 0 \qquad M \twoheadrightarrow N$$

However, this cannot come from a module. Notice that:

$$M_{(0)} = M\left[\frac{1}{2}\right]$$

But the above implies that $\widetilde{M}_{(0)} = 0$.

The existence shows that the potential ways of "localizing" is more than the algebraic structure of modules via localization provides. To fix this, we require that our $\mathcal{O}_X$-module must locally look like sheaf modules. Let us start with defining the "affine scheme", namely the sheaf modules over $X = \operatorname{Spec} R$. Then given any R-module M, we shall want an $\mathcal{O}_X$-modules $\widetilde{M}$ that locally looks like modules with a structure sheaf. We saw that every ring R has is naturally a scheme $(\operatorname{Spec} R, \mathcal{O}_{\operatorname{Spec} R})$. In the same vein, every R-module is naturally a $\mathcal{O}_{\operatorname{Spec} R\text{-module}}$

Definition 4.1.5: Sheaf Associated to M

Let R be a ring and M and R-module. Then the *sheaf associated to M* on $\operatorname{Spec} R$ is a sheaf given by the basis:

$$\widetilde{M}(D(f)) = M_f \qquad \text{(an } R_f\text{-module)}$$

that is, the localization of module M at the ring element f, and define the natural restriction maps. Denote this $\mathcal{O}_{\operatorname{Spec} R}$-module $\widetilde{M}$.

The judicious reader may check that this is indeed affine schemes sheaf on Spec R. As we have now introduced another sheaf structure on X, we will start being conscious of our representation of the local and global sections and write $\Gamma(X, \widetilde{M})$. The usual $\mathcal{O}_X(X)$ notation for a structure sheaf can be written as $\Gamma(X, \mathcal{O}_X)$.

Definition 4.1.6: Quasi-coherent Sheaf

Let $(X, \mathcal{O}_X)$ be a scheme. Then a $\mathcal{O}_X$-module $\mathscr{F}$ is called *quasi-coherent* if X can be covered by open affine subsets $U_i = \operatorname{Spec} R_i$ such that for each i there exists an R_i-module M_i such that:

$$\mathscr{F}|_{U_i} \cong \widetilde{M_i}$$

We shall also have the finitely generated equivalent of the above, however we must be a bit careful as the kernel of between two finitely generated R-modules need not be finitely generated, for example take the ring homomorphism between these modules over themselves

$$k[x_1, x_2, \dots] \to k[x_1, x_2, \dots] \quad x_i \to 0$$

for all $i \in \mathbb{N}$. Then the kernel is $(x_1, x_2, \dots)$ which is not finitely generated. The reader could check that the image and cokernel are always finitely generated, hence this is specifically a kernel problem. We simply force the required condition

Definition 4.1.7: Coherent Module

Let R be an R-module. Then if R is finitely generated and all kernel of maps $\operatorname{Hom}_R(R^n, M)$ are finitely generated for all $n \in \mathbb{N}$, then M is called *coherent*

A ring is called *coherent* if it is a coherent R-module over itself.

Then if a sheaf if covered by coherent modules, it is coherent

Definition 4.1.8: Coherent Sheaf

Let $(X, \mathcal{O}_X)$ be a scheme. Then a $\mathcal{O}_X$-module $\mathscr{F}$ is called quasi-coherent if X can be covered by open affine subsets $U_i = \operatorname{Spec} R_i$ such that for each i there exists a coherent R_i-module M_i such that:

$$\mathscr{F}|_{U_i} \cong \widetilde{M_i}$$

For the reader interested in complex geometry, coherent and quasi-coherent sheaves can be defined on ringed spaces, and this is done in particular done by complex geometers (and in fact the terminology originates in complex geometry).

4.1.1 Distinction from Hartshorne

In Hartshorne, coherent sheaves are defined as sheaves over finitely generated modules, but then Hartshorne says they are a bit pathological and so he always imposes the Noetherian condition, which for us is a sufficient condition for a module to be coherent.

We will immediately show that quasi-coherent sheaves are independent of choice of covering:

Proposition 4.1.9: Quasi-coherence is Affine-local

Let X be a scheme. Then an $\mathcal{O}_X$-module $\mathscr{F}$ is quasi-coherent if and only if for every open affine subset $U = \operatorname{Spec} R \subseteq X$, there is an R-module M such that

$$\mathscr{F}|_U \cong_R \widetilde{M}$$

If X is coherent, then we can replace quasi-coherent with coherent and M is a coherent R-module

Proof **:**
As $X = \bigcup_i U_i$ can be covered in affine open sets and each of these are covered by distinguished open sets, we may reduce to proving this in the case of $X = \operatorname{Spec} R$. To that end, let $M = \Gamma(X, \mathscr{F})$. We need to show there is an isomorphism $\varphi : \widetilde{M} \to \mathscr{F}$. As $\mathscr{F}$ is quasi-coherent, for each $g_i \in X$ we have:

$$\mathscr{F}|_{D(g_i)} \cong \widetilde{M_i}$$

for some R_{g_i}-module M_i. But then

$$\mathscr{F}(D(g_i)) \cong M_{g_i}$$

and so $M_i = M_{g_i}$, and so raising these maps we get that φ is an isomorphism.

By simply addition the conditions of X being Noetherian and M finitely generated we get the corresponding strengthening.

Hence, we have the equivalence of the correspondence between rings and affine schemes for [quasi] coherent sheaves:

Corollary 4.1.10: Equivalence of Categories: Module Version

If $X = \operatorname{Spec} R$, then the functor $M \mapsto \widetilde{M}$ is an equivalence of categories between R-modules and quasi-coherent $\mathcal{O}_X$-modules with inverse function $\mathscr{F} \mapsto \Gamma(\operatorname{Spec} R, \mathscr{F})$. If R is Noetherian, then the equivalence is between finitely generated R-modules and coherent $\mathcal{O}_X$-modules.

Proof **:**
Follows immediately

The following should also give a strong intuition on locally free $\mathcal{O}_X$-modules;

Proposition 4.1.11: Locally Free If and only If Stalk Free

Let $\mathscr{F}$ be an quasi-coherent sheaf over X. Then $\mathscr{F}$ is locally free iff each stalk $\mathscr{F}_x$ is free.

Proof **:**
Exercise: key is to remember localization preserves exactness, and that if all local modules are 0 then the global module is 0. Conclude the right result after by restricting at distinguished open sets, i.e. localizing at f.

Example 4.1.12: Quasi-Coherent Sheaves

1. Certainly, the usual structure sheaf $\mathcal{O}_X$ on X is quasi-coherent (even coherent). At each point [$\mathfrak{p}$, we have the $R_\mathfrak{p}$-module $R_\mathfrak{p} \cong \mathcal{O}_{X,\mathfrak{p}}$, that is every point has the germ as it's stalk. We can think of all of these stalks as being one-dimension vector-spaces (as they are all $R_\mathfrak{p}$-modules over themselves).

2. Let $X = \operatorname{Spec} k[x, y]$ and Take the $k[x, y]$-module $k[x, y]/(f)$ for some irreducible f. Now take a point $[\mathfrak{p}] \in X$. If $f \notin \mathfrak{p}$, then
$$\left(\frac{k[x, y]}{(f)}\right)_\mathfrak{p} \cong 0$$
On the other hand, if $f \in \mathfrak{p}$, we essentially get an 1-dimensional module. Visually, there is a line-bundle on Spec $k[x, y]/(f) \subseteq \operatorname{Spec} k[x, y]$ which is zero everywhere else.
For example, if $f = (x)$, and $x \in \mathfrak{p}$ then $(k[y])_\mathfrak{p}$ is just the germ at that point.
More generally, let $Y \subseteq X$ be a closed subscheme defined by an ideal $\mathfrak{a} \subseteq R$. Then given the inclusion map $\iota : Y \to X, \iota_*\mathcal{O}_Y$ is a quasi-coherent $\mathcal{O}_X$-module (even coherent); show it is isomorphic to $\widetilde{R/\mathfrak{a}}$.

3. Take $X = \operatorname{Spec} \mathbb{Z}$, let $M = \mathbb{Z}/2\mathbb{Z}$ be the usual $\mathbb{Z}$-module, and take $\widetilde{M}$. Then this sheaf has a single non-trivial stalk at 2 for:
$$M_{(2)} \cong \mathbb{Z}/2\mathbb{Z}$$
For any other prime p (and 0):
$$M_{(p)} \cong 0$$
Hence, we can think of this module just being a fibre bundle with only one non-trivial fiber. For another example, take the $\mathbb{Z}$-module $M = \mathbb{Z}/12\mathbb{Z}$. Then we have:
$$M_{(2)} \cong \mathbb{Z}/4\mathbb{Z} \quad M_{(3)} \cong \mathbb{Z}/3\mathbb{Z}$$

and zero everywhere else. We can picture this as a stalk at 2 with "twice" the size of the vector space (note that $\mathbb{Z}/2\mathbb{Z}$ is a vector-space over itself), and the stalk at 3 has just a vector-space.

4. Let $U \subseteq X$ be an open subscheme with inclusion map $\iota : U \to X$. Inspired by exercise 1.4.1, define the sheaf $\iota!\left(\mathcal{O}_U\right)$ by extending $\mathcal{O}_U$ by zero outside of 0 . Then this is an $\mathcal{O}_X$-module, but not in general quasi-coherent! To see this, let X be any integral scheme so that each $\mathcal{O}_X(W)$ is an integral domain. Take $V = \operatorname{Spec} R$ to be an open affine subset of X not contained in U but $U \cap V \neq \emptyset$. Then $\iota_!\left(\mathcal{O}_U\right)|_V$ contains no global sections of V, but is not the zero sheaf. But then it cannot be of the form $\widetilde{M}$ for any R-module M.

5. It is important the we took the pushforward in the above example. Let Y be a closed subscheme of X (not necessarily from an ideal). Then $\mathcal{O}_X|_Y$ is not in general quasi-coherent on Y, and usually not even a $\mathcal{O}_Y$-module. Let $X = \operatorname{Spec} k[x,y]$ and Y be the closed subscheme given by $\left(y^2\right)$, so that $Y = \operatorname{Spec} k[x,y]/\left(y^2\right)$. Then in $\mathcal{O}_Y(U)$ where $U \subseteq Y$, we have that the function y is nilpotent: $y^2 = 0$. However, on $\mathcal{O}_X|_Y(U)$ there is no nilpotence! In particular, when restricting, we are restricting to Y as a set! Hence, we see that $\mathcal{O}_X|_Y$ is not even a $\mathcal{O}_Y$-module

6. Some of the simplest quasi-coherent sheaves are free sheaves where $\mathscr{F} \cong \mathcal{O}_X^{\oplus I}$, and locally free sheaves, where there is an open cover such that $\mathscr{F}|_{U_i} \cong \mathcal{O}_X^{\oplus I}$. In fact, we can think of quasi-coherent sheaves as the smallest sub-category of $\mathcal{O}_X$-Mod that is an abelian category and includes locally-free sheaves (show that the category of locally-free sheaves along with $\mathcal{O}_X$-homomorphisms is not abelian, take the trivial line bundle on $\mathbb{A}^1_{\mathbb{C}}$ with coordinate t and multiply it by t). The way we take the closure is simply by adding the kernels and cokernels of of locally-free sheaves, showing that they can be thought of as the "building blocks" of quasi-coherent sheaves.

7. Let us give the scheme equivalent of the Hedgehog (or Hairy Sphere[a]) Theorem. Take $X = \operatorname{Spec} \mathbb{R}[x,y,z]/\left(x^2+y^2+z^2-1\right)$. All the closed points of this spectrum correspond to the sphere, and there are many non-closed points. Labeling this ring as R, take the R-module M to be the submodule of $R \oplus R \oplus R$ of elements (X,Y,Z) where $xX + yY + zZ = 0$ (note that x, y, z here represent the indeterminates of the ring). This module represent all vectors that are orthogonal to the points of the sphere. Then this sheaf will on each open set look like $\mathcal{O}_X \oplus \mathcal{O}_X$ (similar to the sphere example), but it cannot look like it globally (or else its change of basis to $\mathbb{C}$ and then analytification would contradict the hairy ball theorem).

8. As mentioned, quasi-coherent sheaves are the (proper) closure of locally free sheaves, which means there must be non-locally free sheaves. We shall see that for the curve $y^2 - x^3 - x = 0$ in $\mathbb{A}^2_{\mathbb{C}}$ (more generally, for elliptic curve), every nonempty set has a nontrivial invertible sheaves, see section 6.1.1.

9. (If you know some number theory) Let K be a number field and $\mathcal{O}_K$ the ring of integers. Then the fractional ideals form an invertible sheaf. Show that two invertible sheaves that

differ by a principal ideal yield the same invertible sheaf.
Let us do a concrete example: Take $R = \mathbb{Z}[\sqrt{-5}]$ which the reader should recall is not a UFD as

$$2 \cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$$

It was shown in number theory that $J = (2, 1+\sqrt{-5})$ is not principal. We shall use this to find a non-trivial line bundle. First, if $I \subseteq R$ is principal, it is easy to see that $I \cong_R R$, and this is not the case if I is not principal[b]. Next, $D(2), D(3)$ over R. Over both of these, J becomes principal:

$$D(2) : (2, 1+\sqrt{-5}) = R[1/2] \quad \text{as 2 is now a unit}$$

$$D(3) : (2, 1+\sqrt{-5}) = (1+\sqrt{-5}) \quad \text{as} \quad 2 = \frac{(1+\sqrt{-5})(1-\sqrt{-5})}{3}$$

hence, taking J as an R-module, we see that it is locally free, but is is not globally free.

10. Take $\mathbb{A}^1_k$ and let $\mathscr{F}$ be the sky-scrapper sheaf supported at the generic point with (abelian) group $k(t)$. Show this is a quasi-coherent sheaf. Show that this sheaf is not locally free. More generally, let X be an integral Noetherian scheme. Put the constant sheaf $\kappa(X)$ on it. Then this is a quasi-coherent $\mathcal{O}_X$-module, but not in general finitely generated on open subsets (and hence not coherent, unless X is restricted to a point).

11. Let $X = \mathbb{P}^n_k$. Then there is a natural line bundle on X where each point $p \in X$ is associated to the 1-dimensional sub-space in $\mathbb{A}^{n+1}_k$ of the points that make up the equivalence class. We shall later see that all line bundles on $\mathbb{P}^n_k$ will be isomorphic to this line bundle, an n-times tensor product of this line bundle or a n-times tensor product of the dual of the bundle (which can be thought of as the inverse).

12. On affine space $\mathbb{A}^n_k$ locally free $\mathcal{O}_X$-modules must be trivial. It is highly nontrivial to prove (the Quillen-Suslin Theorem, formally known as Serre's Theorem[c]), but it is enlighetning: Every projective module over a polynomial ring is free. This translates to every vector bundle over affine space is globally trivial. We shall see that we want non-trivial quasi-coherent sheaves on $\mathbb{A}^n_k$.

13. Let $X = \operatorname{Spec} R$ be an and let M be a projective R-module. Recall projective modules over local rings are free. Then the germ $\widetilde{M}_{\mathfrak{p}}$ is a free $R_{\mathfrak{p}}$-module (see [8]). From this perspective, projective modules are one of the closest modules to how we usually view non-trivial vector-bundles

14. let $X = \operatorname{Spec} R$ again and let M be a flat R-module. Then taking $\widetilde{M}$, the localization at each point remains flat (see [8]). This can geometrically be interpreted as the fibers of M over X varying smoothly!

[a]Also known more humorously as the Hairy Ball Theorem
[b]Notice that this means we can take $\widetilde{I}$, something we can't do with schemes as ideals are not rings. We shall return to this in ref:HERE
[c]The title got moved as there are enough important things named Serre's Theorem

Let us establish some basic properties of quasi-coherent sheaves. The following proposition is a direct translation of the results from structure sheaves and affine schemes and is boxed here for reference:

Proposition 4.1.13: Properties of Sheaf Module

Let M be an R-module, $X = \operatorname{Spec} R$, $\widetilde{M}$ the associated sheaf of M, and $f : R \to S$ a ring homomorphism with $f^a : \operatorname{Spec}(S) \to \operatorname{Spec}(R)$ Then:

1. The stalks of $\widetilde{M}$ at $\mathfrak{p}$ is isomorphic to $M_{\mathfrak{p}}$
2. $\Gamma(\operatorname{Spec} R, \widetilde{M}) \cong M$
3. $M \to \widetilde{M}$ is an exact fully faithful functor from R-**Mod** to $\mathcal{O}_{\operatorname{Spec} R}$-**Mod**.
4. If N is another R-module
$$(\widetilde{M \otimes_R N}) \cong \widetilde{M} \otimes_{\mathcal{O}_{\operatorname{Spec} R}} \widetilde{N}$$
5. If $\{M_i\}_{i \in I}$ is a collection of R modules:
$$\widetilde{\left(\bigoplus_{i \in I} M_i\right)} \cong \bigoplus_{i \in I} \widetilde{M_i}$$
6. If N is an S-module,
$$f_*(\widetilde{N}) = \widetilde{{}_R N}$$
where ${}_R N$ is where we consider N with the natural R-module structure given by f. Furthermore, $f_*(-)$ is an exact functor (note: X must be affine)
7. If $\widetilde{N}$ is the associated sheaf of N and $\widetilde{M}$ is some sheaf module, then
$$f^*(\widetilde{M}) = \widetilde{M \otimes_R S}$$

It is important for exactness that $f_*(-)$ is pushing a quasi-coherent sheaf over an . This fails for non-affine schemes, as we shall see with some of the simplest exact with projective schemes. In chapter 5, we shall find right-derived functors for this map.

***Proof* :**

1. Direct generalization of proposition 2.3 .14
2. Direct generalization of corollary 3.1.5
3. The fully faithful part can be deduced by generalizing corollary 3.1.5. For exactness, recall from [8] that localization is exact.

4. Recall from [8] that $(M \otimes_R N)_f \cong M_f \otimes_{R_f} N_f$.
5. exercise
6. exercise
7. exercise

Let us next show that being 0 on an open subset is almost globally zero, and that it is also relatively easy to find global sections form local section on (distinguished) open subsets:

Lemma 4.1.14: Quasi-coherence and Extending From $D(f)$

Let $X = \operatorname{Spec} R$ be an , $f \in R$, $D(f) \subseteq X$, and $\mathscr{F}$ a quasi-coherent sheaf on X. Then:

1. If $s \in \Gamma(X, \mathscr{F})$ is a global section of $\mathscr{F}$ whose restriction to $D(f)$ is zero, then there exists some $n > 0$ such that $f^n s = 0$
2. If $t \in \mathscr{F}(D(f))$, then or some $n > 0$, $f^n t \in \mathscr{F}(X)$.

The idea behind (2) is that we can eliminate the denominator to have a globally defined function; think $1/x \in \mathcal{O}_X(\mathbb{A}^1_k \setminus \{0\})$. For part (1), think of this as reflecting the same property as complex holomorphic functions, but having to take into consideration nilpotent elements. This idea works for "vector bundles" on affine schemes.

***Proof* :**

1. Suppose $s \in \Gamma(X, \mathscr{F})$ such that $s|_{D(f)} = 0$. Cover X in finitely many $D(g_i)$ (as X is quasicompact), and consider $s|_{g_i} = s_i \in M_i = \mathscr{F}(D(g_i))$. Then as $D(f) \cap D(g_i) = D(fg_i)$,
$$\mathscr{F}|_{D(fg_i)} = \widetilde{M_{if}}$$
Then the homomorphism restriction map must imply $s_i = 0 \in (M_i)_f$, which by definition of localization implies $f^{n_i} s = 0$. Doing this for each M_i and taking the $n = \max_i n_i$, by gluing we get that $f^n s = 0$.

2. Let $t \in \mathscr{F}(D(g))$, and like above consider $t_i \in \mathscr{F}(D(fg_i)) = (M_i)_f$. Then by definition, for some $n > 0$ there is some element $t_i \in M_i = \mathscr{F}(D(g_i))$ that restricts to $f^{n_i} t$ on $D(fg_i)$. Again, take $n = \max_i n_i$ to make it independent of choice.

   With n chosen to be large enough, on $D(g_i) \cap D(g_j) = D(fg_ig_j)$, the two sections t_i, t_j restrict to $f^n t$. Hence, by part a there exists an $m > 0$ such that
$$f^{m_{ij}}(t_i - t_j) = 0 \qquad \text{on} \qquad D(g_ig_j)$$
   Naturally take $m = \max_{i,j} m_{ij}$. Now the local sections $f^m t_i \in \mathscr{F}(D(g_i))$ glue to give a global section $s \in \mathscr{F}(X)$ where $s|_{D(f)} = f^{n+m} t$.

Next, recall that the global section functor need not be exact. In the case where $X = \operatorname{Spec} R$, and we are working with quasi-coherent sheaves, it is exact:

Proposition 4.1.15: Exactness of Global Section with sheaves of Modules

Let $X = \operatorname{Spec} R$ be an and $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ an exact sequence of $\mathcal{O}_X$-modules, where at least $\mathscr{F}'$ is quasi-coherent. Then:

$$0 \to \Gamma(\operatorname{Spec} R, \mathscr{F}') \to \Gamma(\operatorname{Spec} R, \mathscr{F}) \to \Gamma(\operatorname{Spec} R, \mathscr{F}'') \to 0$$

is exact

***Proof* :**
By proposition 1.2.16, Γ is left-exact, hence it suffices to show it is right-exact when working with $\mathcal{O}_X$-modules and at least $\mathscr{F}'$ is quasi-coherent, namely we must show the map is surjective.

Let $s \in \Gamma(X, \mathscr{F}'')$. As $\mathscr{F} \to \mathscr{F}''$ is surjective, by exercise 1.2.1[1] there is an open neighborhood $D(f)$ of x such that $s|_{D(f)} \in \mathscr{F}''(D(f))$ lifts to a section $t_x \in \mathscr{F}(D(f))$. I claim now that there exists an appropriate $n > 0$ such that $f^n s \in \Gamma(\mathscr{F}'', X)$ lifts to a global section $t \in \mathscr{F}(X)$.

As X is an , it is quasicompact, and so cover it by a finite principal open sets $D(g_i)$, and for g_i $s|_{D(g_i)}$ lifts to a section $t_i \in \mathscr{F}(D(g_i))$. Then, on $D(f) \cap D(g_i) = D(fg_i)$, there is $t_x, t_i \in \mathscr{F}(D(fg_i))$ which both lift $s_{D(fg_i)}$.

$$t_x - t_i \in \mathscr{F}'(D(fg_i))$$

As $\mathscr{F}'$ is quasi-coherent, by lemma 4.1.14[1], there exists some $n_i > 0$ such that $f^{n_i}(t_x - t_i)$ extends to a section:

$$u_i \in \mathscr{F}'(D(g_i))$$

Pick $n = \max_i n_i$. Now, taking the image of u_i (as the function is injective), consider $t_i' = f^n t_i + u_i \in \mathscr{F}(D(g_i))$. Then t_i' is a lifting of $f^n s|_{D(g_i)}$, and

$$t_i'|_{D(g_i)} = f^n t_x|_{D(g_i)}$$

Then on $D(g_i g_i)$, t_i', t_j' both lift $f^n s$, and so $t_i' - t_j' \in \mathscr{F}'(D(g_i g_j))$. As $t_i'|_{D(fg_ig_j)} = t_j'|_{D(fg_ig_j)}$, by lemma 4.1.14[2] there exists an m_{ij} such that $f^{m_{ij}}(t_i' - t_j') = 0$ on $D(g_i g_j)$. As usual, take $m = \max_{i,j} m_{ij}$. Then the section $f^m t_i' \in \mathscr{F}(D(g_i))$ glue to a global section $t'' \in \mathscr{F}(X)$, which is a lift of $f^{n+m} s$.

Now, to show each s has a nonempty pre-image. As X is quasicompact, cover X by $D(f_1), ..., D(f_r)$. Then we know that each $s|_{D(f_i)} \in \mathscr{F}''(D(f_i))$ lifts to a section in $\mathscr{F}(D(f_i))$. Then by what was just shown there exists a global section $t_i \in \mathscr{F}(X)$ such that t_i is a lift of $f_i^n s$ where n is the maximum value of all lifts.

Now, As $D(f_i)$ cover X, $(f_1^n, ..., f_r^n) = R$, so let

$$1 = \sum_i a_i f_i^n$$

Take $t = \sum_i a_i t_i \in \mathscr{F}(X)$. Then the image t is

$$\sum_i a_i f_i^n s = s$$

showing each s has a nonempty preimage, as we sought to show.

The reader could think of the above result as something special about affine schemes. In chapter 5 we shall see that it is exactly the non-s that will have a non-exact global-section function (over quasi-coherent sheaves), see theorem 5.1.17

Proposition 4.1.16: All Objects Are Quasi-Coherent

Let X be a scheme. Then

1. the kernel, cokernel, and image of any morphism of quasi-coherent sheaves are quasi-coherent.
2. Any extension are quasi-coherent.

If X is Noetherian, the same is true for coherent. If $f : X \to Y$ is a morphism of schemes then:

1. If $\mathscr{G}$ is a quasi-coherent $\mathcal{O}_Y$-module, then $f^*\mathscr{G}$ is a quasi-coherent $\mathcal{O}_X$-module
2. If X, Y and Noetherian, $\mathscr{G}$ is coherent,then $f^*\mathscr{G}$ is coherent
3. If either X is Noetherian of f is quasicompact and separated, then if $\mathscr{F}$ is quasi-coherent, $f_*\mathscr{F}$ is quasi-coherent.

sheaves.

Proof **:**
As shown in chapter 1, these properties can be shown locally, so it suffices to assume X is affine. Then as $M \mapsto \widetilde{M}$ is exact and fully faithful, kernels, cokernels, and images of quasi-coherent sheaves are preserved. We must still show that the extension of two quasi-coherent sheaves is quasi-coherent, so consider an exact sequence of $\mathcal{O}_X$-modules where $\mathscr{F}', \mathscr{F}''$ is quasi-coherent:

$$0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$$

Then by proposition 4.1.13, the global sections are exact as $\Gamma(X, -)$ is an exact functor:

$$0 \to \Gamma\left(X, \mathscr{F}'\right) \to \Gamma(X, \mathscr{F}) \to \Gamma\left(X, \mathscr{F}'\right) \to 0$$

relabel for simplicity to:

$$0 \to M' \to M \to M'' \to 0$$

Now apply the $\widetilde{(-)}$ exact functor. Then there is a natural commutative diagram:

$$\begin{tikzcd}
0 \arrow[r] & M' \arrow[r] \arrow[d] & M \arrow[r] \arrow[d] & M'' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathscr{F}' \arrow[r] & \mathscr{F} \arrow[r] & \mathscr{F}'' \arrow[r] & 0
\end{tikzcd}$$

As $\mathscr{F}', \mathscr{F}''$ are quasi-coherent, Γ and $\widetilde{(-)}$ returns itself by proposition 4.1.11, we have that $\mathscr{F}' \to M'$ and $\mathscr{F}'' \to M''$ are isomorphism. By the 5-lemma, $M \to \mathscr{F}$ is an isomorphism.

Naturally, if X is Noetherian, all sheaves are coherent.
Now let $f : X \to Y$ be a scheme morphism. As these properties are local we may assume X, Y are affine. As Γ and $\widetilde{(-)}$ are equivalence of categories and inverses of each other, and
$f^*(\widetilde{M}) \cong \left(\widetilde{M \otimes_R S}\right)$, a quasi-coherent $\mathcal{O}_Y$-module $\mathscr{G}$ get's pushed out to a quasi-coherent $\mathcal{O}_X$-module $f^*\mathscr{G}$. If Y is Noetherian, everything becomes coherent.
the push-forward requires more conditions (see example 4.5). If X is Noetherian of f is quasi-compact, X can be covered by finitely many affine open subsets, and as X is Noetherian of f is separated, the intersection of two affine open subsets is either affine or the union of affine open subsets. Label the affine open U_i and the affine local subsets whose union is the intersections $U_i \cap U_j$ as U_{ijk}. Now for any $s \in \mathscr{F}\left(f^{-1}(V)\right), V \subseteq Y$ open, there is a unique collection of $s_i \in \mathscr{F}\left(f^{-1}(V) \cap U_i\right)$ whose restriction to $f^{-1}(V) \cap U_{ijk}$ must all be equal. Thus, we get the following exact sequence:

$$0 \to f_*\mathscr{F} \to \bigoplus_i f_*\left(\mathscr{F}|_{U_i}\right) \to \bigoplus_i f_*\left(\mathscr{F}|_{U_{ijk}}\right)$$

where f_* is a bit overloaded for the second and third position, As it should be the function induced by $f : U_i \to Y$ and $f : U_{ijk} \to Y$. Now, as U_i are affine, by proposition 4.1.11[4], $f_*\left(\mathscr{F}|_{U_i}\right)$ and $f_*\left(\mathscr{F}|_{U_{ijk}}\right)$ are quasi-coherent. As the kernel of a $\mathcal{O}_X$-module morphism of a quasi-coherent sheaf is quasi-coherent, $f_*(\mathscr{F})$ is quasi-coherent, completing the proof.

If X were not Noetherian or f not quasi-compact and separated, the pushforward need not be quasi-coherent:

Example 4.1.17: Pushforward Not Quasi-Coherent
Let $X = \coprod_{i \in \mathbb{N}} \operatorname{Spec} \mathbb{Z}, Y = \operatorname{Spec} \mathbb{Z}$, and $f : X \to Y$ be the identity for each disjoint element of $\operatorname{Spec} \mathbb{Z}$. Take $\mathscr{F} = \mathcal{O}_X$.
Now, for any quasi-coherent sheaf $\mathscr{G}$ on Y, as we have a change of basis for each open set we must have:

$$\mathscr{G}(Y) \otimes_{\mathcal{O}_Y(Y)} \mathcal{O}_X(U) \xrightarrow{\sim} \mathscr{G}(U)$$

We'll show this is not the case when $\mathscr{G} = f_*\mathcal{O}_X$ and $U = D(2) \subseteq \operatorname{Spec} \mathbb{Z}$. In this case we get:

$$\left(\prod_{i \in \mathbb{N}} \mathbb{Z}\right) \otimes_{\mathbb{Z}} \mathbb{Z}\left[\frac{1}{2}\right] \to \prod_{i \in \mathbb{N}} \mathbb{Z}\left[\frac{1}{2}\right]$$

Next, for $t \in \left(\prod_{i \in \mathbb{N}} \mathbb{Z}\right) \otimes_{\mathbb{Z}} \mathbb{Z}\left[\frac{1}{2}\right]$, we have hat:

$$t = \left(\frac{a_i}{2^n}\right)_{i \in \mathbb{N}}$$

for some fixed n. The image ought to be the identity. However, the codomain contains elements that are not of this form, namely there are:

$$s = \frac{b_i}{2^{n_i}} \in \prod_{i\in\mathbb{N}} \mathbb{Z}\left[\frac{1}{2}\right]$$

where n_i grows arbitrarily large and cannot be an image the image for any t !

If X were not Noetherian or f not quasi-compact and separated, the pushforward need not be quasi-coherent:

If X, Y are Noetherian, it is not necessarily the case that f_* of a coherent is a coherent sheaf!

Example 4.1.18: Pushforward of Noetherian Not Coherent

Take $f : \operatorname{Spec} k[x] \to \operatorname{Spec} k$ given by $k \to k[x]$. Then $f_*\left(\mathcal{O}_{\operatorname{Spec} k[x]}\right) = k[x]$ is not a finitely generated k-module.
There are however many sufficient properties that can be added to f to make the image coherent, for example finite, projective, or proper. (see ref:HERE)

4.1.2 Locally Free Sheaves and Vector Bundles

An important class of examples (and historically the first studied class of examples) were inspird by vector bundles. Recall that a vector-bundle can be thought of as a local-trivialization, that is for every point $x \in M$ there is a open neighborhood $x \in U \subseteq M$ such that there is a map $\pi^{-1}(U) \cong_{\mathbf{Top}} U \times \mathbb{R}^k$ respecting projecting and isomorphic on fibers. In other words, it is locally free. If there is a global trivialization, than it is simply free. As a reminder, here is the definition of a vector bundle, for simplicity defined over a quasi-projective variety

Definition 4.1.19: Vector Bundle

Let X be a quasi-projective variety over k. Then a *vector bundle* E of rank r is a surjective morphism $\pi : E \to X$ along with an open covering $\{U_i\}$ and isomorphism:

$$\varphi_i : \pi^{-1}(U_i) \to U_i \times \mathbb{A}^r_k$$

such that for every intersection $U_i \cap U_j$:

$$\varphi_j \circ \varphi_i^{-1}|_{U_i\cap U_j} = (\mathrm{id}, \varphi_{ij}) \qquad \text{for} \qquad \varphi_{ij} \in \mathrm{GL}_r(k)$$

A vector bundle has a natural topology impose by the pre-images and gluging. A *section* is a continuous map $s : X \to E$ such that $\pi \circ s = \mathrm{id}$. On $U \subseteq X$, we can define a local section $s : U \to E$ such that $\pi|_U \circ s = \mathrm{id}_U$.

Note that $\varphi_{jk} \circ \varphi_{ij} = \varphi_{ik}$ and $\varphi_{ii} = \mathrm{id}$. This is equivalent to the usual local trivialization definition of vector bundle as the reader can verify. This construction is given by gluing the disjoint union of $U_i \times \mathbb{A}^r_k$. From this, we can define a morphism of vector bundles $E \to E'$ to be a morphism of varieties such that the natural commutative diagram projecting to X holds and the morphism restricts to a linear morphism on each fiber. We can define the direct sum and tensor product of vector bundles fiber-wise as well.

Next, we define the algebraic counter-part:

Definition 4.1.20: Free and Locally-Free $\mathcal{O}_X$-module

Let X be a ringed space and $\mathscr{F}$ a $\mathcal{O}_X$-module. Then $\mathscr{F}$ is free if

$$\mathscr{F} \cong \mathcal{O}_X^n$$

$\mathscr{F}$ is locally free if X can be covered in open sets U such that $\mathscr{F}|_U$ is a free $\mathcal{O}_X|_U$-module. The rank of each $U \subseteq \mathscr{F}$ is the number of copies of $\mathcal{O}_X$ (the "dimension", whether finite or infinite).

If X is connected then the rank must be globally the same.

Consider an open cover $U_i \subseteq X$ that are locally trivializable. Then each $U_i \cap U_j \hookrightarrow U_i$ and $U_i \cap U_j \hookrightarrow U_j$ are two different local trivializations of rank r. Therefore, there is an invertible $r \times r$ matrices A_{ij} of functions given an isomorphism:

$$(\mathcal{O}(U_i \cap U_j))^r \to (\mathcal{O}(U_i \cap U_j))^r$$

Naturally, $A_{ii} = \mathrm{id}$ and

$$A_{jk}A_{ij} = A_{ik}$$

Notice these satisfy the cocycle condition. Hence, we can define a sheaf using these.

Lemma 4.1.21: Coherent Sheaf Locally Free Criterion

Let $\mathcal{E}$ be a coherent sheaf on X. Then

1. $\mathcal{E}$ is locally free if and only if each fiber $\mathcal{E}_p$ is free.
2. A subsheaf of a locally free sheaf is locally free
3. If the rank is 1, a nonzero map between locally free sheaves is injective

***Proof* :**
exercise.

Naturally, the quotient of a locally free sheaf need not be locally free (which is why it is not an abelian category); the problem is the torsion of the module. However, there is a natural way of fixing this:

Lemma 4.1.22: Natural Quotient of Locally Free Sheaf

Let $\mathcal{E}$ be a locally free sheaf of rank r and $\mathcal{E}' \subseteq \mathcal{E}$ a locally free sheaf of rank $r' \leq r$. Then there exists a subsheaf $\overline{\mathcal{E}}' \subseteq \mathcal{E}$ of rank r' containing $\mathcal{E}'$ such that $\mathcal{E}/\overline{\mathcal{E}}'$ is a locally free sheaf of rank $r - r'$. if $\mathcal{E}'$ is a maximal subsheaf of rank r', then the quotient is free.

***Proof* :**
Let T be the torsion submodule of $M = \mathcal{E}/\mathcal{E}'$. The fiber of M at a point of C is a finitely generated module over a principal ideal domains. If $T = 0$, then the fibers of M are torsion-free and therefore free. Then, using lemma 4.1.21(1), M is free.

Now define $\bar{\mathcal{E}}'$ to be the kernel of the natural composition

$$\mathcal{E} \to M \to M/T$$

The new quotient of $\mathcal{E}$ by $\bar{\mathcal{E}}'$ is M/T which has no torsion, hence it is locally free, hence By definition of $\bar{\mathcal{E}}'$, it is clear that $\mathcal{E} \subset \bar{\mathcal{E}}'$. Moreover, from the commutative diagram

$$\begin{tikzcd} 0 \arrow[r] & \mathcal{E}' \arrow[r] \arrow[d] & \mathcal{E} \arrow[r] \arrow[d] & M \arrow[r] \arrow[d] & 0 \\ 0 \arrow[r] & \bar{\mathcal{E}}' \arrow[r] & \mathcal{E} \arrow[r] & M/T \arrow[r] & 0 \end{tikzcd}$$

thus by the snake's lemma, $\bar{\mathcal{E}}'/\mathcal{E}' \cong T$. Hence, $\bar{\mathcal{E}}'$ and $\mathcal{E}'$ have the same rank, completing the proof.

Theorem 4.1.23: Vector-bundle/Locally Free Sheaf Correspondence

The category of Vector bundles with bundle morphism is equivalent to the category of locally free sheaves with sheaf morphisms

The base space can be more general than what we do in the proof, however generalities add complications not intrinsic to the problem and so is left to the reader to sort out the extra details if they so wish.

***Proof* :**

Throughout, fix X to be a complete k-variety where $k = \overline{k}$ is algebraically and let $\mathcal{O}_X$ be the structure sheaf. Let us start with a vector bundle and construct a locally free sheaf. Let E be a vector bundle. Define a sheaf $\mathcal{E}$ by letting

$$\mathcal{E}(U) = \{\text{local sections on E}\}$$

This certainly respects the sheaf axioms of restriction and gluing. Define an $\mathcal{O}_X$-action on $\mathcal{E}$ by taking:

$$(f \cdot s)_p = f(p) \cdot s_p$$

and addition is clear, turning it into a $\mathcal{O}_X$-module. We need to show that $\mathcal{E}$ is locally free. Take an locally-trivializable open covering of $E \to X$ so that

$$\pi^{-1}(U_i) \cong U_i \times \mathbb{A}^r_k$$

Define the canonical local "coordinate" sections:

$$x_i : U_i \to U_i \times \mathbb{A}^r_k$$
$$p \mapsto (p, (0, ..., 1, ..., 0)$$

Then every section $s \in \mathcal{E}(U_i)$ can be written as:

$$s = f_1 x_1 + \cdots + f_r x_r$$

for some functions $f_i \in \mathcal{O}_X$. As each U_i is locally trivial it follows that there exists a unique s for each r-tuple $(f_1, ..., f_r)$, hence the map:

$$\mathcal{E}(U_i) \to (\mathcal{O}(U_i))^r s \mapsto (f_1, ..., f_r)$$

is bijective, and is certainly a $\mathcal{O}_X(U_i)$-module morphism, hence an isomorphism.

Conversely, the direction requires a bit more work due to non-closed points. Let us first show what we are aiming for by demonstrating the result for a curve. Suppose that $\mathcal{E}$ is a locally free sheaf over a curve C. Define a vector bundle by taking

$$E = \{(p,t) \mid p \in X,\, p \text{ closed point}\}\, t \in \mathcal{E}_p/\mathfrak{m}_p\mathcal{E}_p \tag{4.1}$$

Then the map $\pi : E \to X$ projecting onto the first component shall have the required properties. This same construction does not generalize as we miss the non-closed points. We shall thus create a more general construction, and show how this generalizes the above. First, a $\mathcal{O}_X$-algebra $\mathcal{A}$ is defined in the same way a $\mathcal{O}_X$-module were each $\mathcal{A}(U)$ is a $\mathcal{O}_X(U)$-algebra. Then we have $\mathrm{Spec}\,(\mathcal{A}(U))$. We shall glue these together to form a new scheme $\mathrm{Spec}_X(\mathcal{A})$ call the *relative scheme* $\mathcal{A}$ with respect to X. Namely, for any $V \subseteq U$, we have a ring homomorphism $\mathcal{A}(U) \to \mathcal{A}(V)$ giving spectrum maps $\mathrm{Spec}\,(\mathcal{A}(V)) \to \mathrm{Spec}\,(\mathcal{A}(U))$ that give natural gluing conditions. This forms a scheme, denoted $\mathrm{Spec}_X(\mathcal{A})$ with the natural projection:

$$\pi : \mathrm{Spec}_X(\mathcal{A}) \to X$$

where if $U \subseteq X$, $\pi^{-1}(U) \cong \mathrm{Spec}\,(\mathcal{A}(U))$. In particular, we can characterize the set $\mathrm{Spec}_X(\mathcal{A})$ as:

$$\mathrm{Spec}_X(\mathcal{A}) = \{(p,\mathfrak{q}) \mid p \in X,\ \mathfrak{q} \in \mathrm{Spec}\,(\mathcal{A}_p)\}$$

In this way, we defined a natural "bundle" on X. This construction is even universal (see exercise ref:HERE). For us, it is important that the fibers are:

$$\mathrm{Spec}_X(\mathcal{A})_x \cong \mathrm{Spec}\,(\mathcal{A}_x \otimes \mathcal{O}_X, \kappa(x))$$

where $\mathcal{A}_x$ is the stalk at x and $\kappa(x)$ is the residue field at x. What shall be important to us is that:

$$\mathrm{Spec}_X(\mathrm{Sym}(\mathcal{E}^\vee))$$

shall be our desired vector bundle. To see this, the first thing is that Sym is the usual symmetric algebra; if $\mathscr{F}$ is a $\mathcal{O}_X$-algebra then:

$$\mathrm{Sym}\mathscr{F} = \mathcal{O}_X \oplus (\mathscr{F}) \oplus (\mathscr{F} \otimes \mathscr{F}/\sim) \oplus \cdots$$

where $\sim$ is the usual symmetric tensor relations. Then by some elementary ring theory we see that if $\mathscr{F}|_U$ has generators $\{s_1, ..., s_n\}$ then:

$$\mathrm{Sym}(\mathscr{F})|_U \cong \mathcal{O}_X(U)[t_1, ..., t_n]$$

for some indeterminates t_i. Next, we take the dual $\mathcal{E}^\vee = \mathcal{H}om(\mathcal{E}, \mathcal{O}_X)$ simply to reverse the directions of the arrows in the right way (you the intuition from differential geometry [4]). But as wee see in the above, the localization give us n-dimension affine space over the appropriate ring. Looking at this spectrum as set, we get:

$$\{(p,v) \mid p \in X,\ \mathcal{E}_p \otimes \kappa(p)\}$$

which we can compare with equation 4.1 to see this is indeed the right generalization!

We may now return to our discussion. By letting $E = \mathrm{Spec}_X(\mathrm{Sym}(\mathcal{E}^\vee))$, we have the surjection $E \to X$ onto the first component. Let us show that we get a vector-bundle structure. Choose an open cover U_i which is locally trivializable so that $\mathcal{E}(U_i) \cong (\mathcal{O}_X(U_i))^r$. As $\overline{k}$ is algebraically closed, we have that each fiber is isomorphic to $\mathbb{A}^r_k$.

Now, on $U_i \cap U_j$, we get a transition function on $\mathcal{E}(U_i \cap U_j)$ given by the $r \times r$ matrix A_{ij}. Then it is clear that these respect the cocycle condition and $A_{ii} = I$. Now, fiber-wise, we see that reducing mod $\mathfrak{m}_p$ where $\mathfrak{m}_p$ is the maximal ideal $\mathcal{E}_p$ gives isomorphisms, and tensoring preserves isomorphisms. Certainly the cocycle condition is preserved, and hence we have that E is a vector-space.

Finally, for the maps, let us start with a map of locally free sheaves. Then the vector-bundle map can be defined by fiber-wise by reducing and tensoring. Conversely, if $\varphi : E \to E'$ is a vector bundle map and a section $s : X \to E$, composing gives a section $\varphi(s) : X \to E'$. Certainly these maps respect restrictions, giving the desired result.

Thus,we have:

$$\left\{ \begin{array}{c} \text{Vector Bundles } V \\ \text{on } X \end{array} \right\} \underset{\mathscr{F} \mapsto \operatorname{Spec} \operatorname{Sym}(\mathscr{F}^\vee)}{\overset{V \mapsto \mathscr{F}}{\rightleftarrows}} \left\{ \begin{array}{c} \text{locally free} \\ \text{sheaves on } X \end{array} \right\}$$

A subset of locally free sheaves to point out are those that are rank 1. A locally free $\mathcal{O}_X$-module of rank 1 (over a locally ringed space) is called an invertible sheaf. If the reader knows some K theory (see [NateKTheory]) to quickly draw the connection recall that we can take the monoid of $\mathcal{O}_X$-module along with the tensor product. The prototypical example in algebra is the monoid of k-modules. Then if $\mathscr{F} \otimes \mathscr{F} \vee \cong \mathcal{O}_X$ (or for R-modules $M \otimes_R M^\vee \cong R$), then these would be the invertible elements in this monoid. Then every locally free sheaf of rank 1 is invertible. If X is a locally ringed space (not just a ringed space) then this becomes and iff. Hence, it is common to call them a line bundle as well. Due to the fact that they make a group, they shall appear a lot, and shall play a central role in section 4.3 .

Definition 4.1.24: Invertible Sheaf or Line Bundle

Let X be a ringed space and $\mathscr{F}$ a $\mathcal{O}_X$-module. Then $\mathscr{F}$ is called an invertible sheaf or line bundle if it is locally free of rank 1.

4.1.3 Closed subschemes and Ideal Sheaves

Let us return to closed subschemes, which were defined to be closed subsets $X \subseteq Y$ where $f^\# : \mathcal{O}_Y \to f_*\mathcal{O}_X$ is a surjective map. As the map is surjective, the kernel at each open set must give an ideal. Given a closed embedding $\pi : X \to Y$, there is an ideal sheaf that we can put on Y that is given by the kernel of the map:

$$\mathcal{O}_Y \to \pi_*\mathcal{O}_X$$

Definition 4.1.25: Ideal Sheaf

Let $\mathcal{O}_X$ be a structure sheaf for a scheme X. Then an ideal sheaf is a $\mathcal{O}_X$-submodule of $\mathcal{O}_X$.

Lemma 4.1.26: Kernel and Ideal Sheaves

Let Y be a closed subscheme of X and $\iota : Y \to X$ the inclusion map. Then the kernel of $\iota^{\#} : \mathcal{O}_X \to \iota_* \mathcal{O}_Y$ is an ideal sheaf, and is usually denoted $\mathcal{I}_Y$.

Proof **:**
exercise

The reader should find no difficulty in seeing this is a coherent sheaf over $\mathcal{O}_Y$. We get the following short exact sequence of $\mathcal{O}_Y$-modules:

$$0 \to \mathcal{K} \to \mathcal{O}_Y \to \pi_* \mathcal{O}(X) \to 0$$

For each affine open subset with a closed embedding $\mathrm{Spec}(R) \to Y$, let $I(R)$ be the corresponding ideal (they define a base for the sheaf). We may ask if we can study the closed embedding condition locally using $\mathcal{K}$ and the ideals $I(R)$. The first thing we may want is an equivalent of the fact that every ideal sheaf should correspond to a closed subscheme (i.e. make lemma 4.1.16 and iff). This is not the case, though mainly since we have not limited ourselves to quasi-coherent sheaves. The following necessary property for ideal sheaves given by kernels shall be shown to not apply to all ideal sheaves:

Proposition 4.1.27: Sheaf Ideal For Closed Embedding

Let $X \to Y$ be a scheme morphism. Then $X \to Y$ is a closed embedding if and only if for each affine open subset $\mathrm{Spec}\, S \subseteq Y$ and corresponding inclusion $\mathrm{Spec}\, S \to Y$ giving the ideals $I(S) \subseteq S$, the restriction morphism $S \to S_f$ induces an isomorphism:

$$I(S)_f \cong I(S_f)$$

for all $f \in S$. In other words the closed embeddings $\mathrm{Spec}\; S/I \to \mathrm{Spec}\, S$ glue to get a closed embedding $X \to Y$.

Proof **:**
The $(\Rightarrow)$ is the easier direction. For the $(\Leftarrow)$ direction, take an ideal $I(S) \subseteq S$ corresponding to an affine open subset $\mathrm{Spec}(S) \subseteq X$. Suppose that for each closed embedding $\mathrm{Spec}\;(S) \to Y$ and $f \in S$, the restriction $S \to S_f$ induces an isomorphism $I(S)_f \cong (S_f)$. Using this, show you can glue together closed embeddings $\mathrm{Spec}(S/I) \to \mathrm{Spec}(S)$ in a well-defined manner to get a closed embedding $X \to Y$.

Example 4.1.28: Not All Ideal Sheaves Correspond To Closed Embeddings

Take $X = \mathrm{Spec}\, k[x]_{(x)}$. This scheme has two points: the closed point and the generic point, say η. Then take:

$$\mathcal{K}(X) = 0 \subseteq \mathcal{O}_X(X) = k[x]_{(x)}$$
$$\mathcal{K}(\eta) = k(x) = \mathcal{O}_X(\eta)$$

Then using the above proposition, we see that this sheaf cannot define a closed embedding.

We recover a correspondence if we limit to quasi-coherent sheaves:

Proposition 4.1.29: Quasi-coherent Ideal Sheaves

Let X be a scheme. Then for any closed subscheme $Y \subseteq X$, the corresponding ideal sheaf $\mathcal{I}_Y$ is a quasi-coherent sheaf of ideals on X. If X is Noetherian, $\mathcal{I}_Y$ is coherent.
Conversely, any quasi-coherent sheaf of ideals on X corresponds uniquely to a closed subscheme of X.

Proof **:**
This follows by the Qcqs lemma (lemma 3.3.5) to get the required condition for proposition 4.1.17

Corollary 4.1.30: Ideal Sheaves On Affine Schemes

Let $\operatorname{Spec} R$ be an . Then there is a one-to-one correspondence between the ideals $\mathfrak{a} \subseteq R$ and the closed subscheme $Y \subseteq X$ given by

$$\mathfrak{a} \mapsto \operatorname{Spec} R/\mathfrak{a}$$

With further control of closed subschemes, we can define the following interesting analogy to vanishing sets:

Definition 4.1.31: Vanishing Scheme

Let X be a scheme and $s \in \mathcal{O}_X(X)$. Then a vanishing scheme corresponding to $s, V(s)$, is the scheme corresponding to gluing $\operatorname{Spec}(R/(s_R))$ where $s_R = s|_{\operatorname{Spec}(R)}$. More generally, we can take the ideal $(S) \subseteq \mathcal{O}_X(X)$ generated by the elements S.

The reader should use proposition 4.1.17 to insure this is well-defined. The advantage of working with vanishing schemes over vanishing sets is that it remembers extra information, such as the multiplicity of intersection.

Proposition 4.1.32: Properties of Vanishing Schemes

Let X be a scheme. Then:

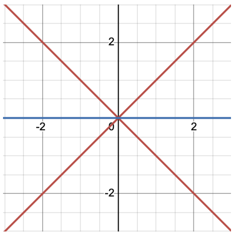
1. $\cup_i^n V(S_i) = V(\cap_i^n S_i)$
2. $\cap_{i \in \mathcal{I}} V(S_i) = V\left(\left\langle S_i \middle| i \in \mathcal{I}\right\rangle\right)$

Proof **:**
Note that the underlying sets of the schemes will just be finite unions and intersections.
The harder point to verify is that the union becomes intersection. In the case of vanishing sets, the proof takes advantage of the fact that $I \cap J \subseteq IJ \subseteq (I \cap J)^2$. Note that it must be the intersection,

as we would need that that $V(S) \cap V(S) = V(S)$ and not $V\left(S^2\right)$.

Example 4.1.33: Vanishing Scheme Intersection

1. Take $V\left(y - x^2\right), V(y) \subseteq \mathbb{A}_k^2$. Then with vanishing sets, this would just be the point $(0,0)$. However, this point now has multiplicity 2.
2. Take $V\left(y^2 - x^2\right), V(y) \subseteq \mathbb{A}_k^2$. The visual of these two are:



Use this to show that if X, Y, Z are closed subscheme of a scheme W, then

$$(X \cap Z) \cup (Y \cap Z) \neq (X \cup Y) \cap Z$$

Hence, not all properties of unions and intersection can carry over!

Closed subscheme are in analogy to closed sets, which can be made very precise with the following generalization of the vanishing set

Definition 4.1.34: Vanishing Scheme

Let X be a scheme and $s \in \mathcal{O}_X(X)$. Then a *vanishing scheme* corresponding to s, $V(s)$, is the scheme corresponding to gluing $\operatorname{Spec}(R/(s_R))$ where $s_R = s|_{\operatorname{Spec}(R)}$. More generally, we can take the ideal $(S) \subseteq \mathcal{O}_X(X)$ generated by the elements S.

The advantage of working with vanishing schemes over vanishing sets is that it remembers extra information, such as the multiplicity of intersection.

Proposition 4.1.35: Properties of Vanishing Schemes

Let X be a scheme. Then:

1. $\cup_i^n V(S_i) = V\left(\cap_i^n S_i\right)$
2. $\cap_{i \in \mathcal{I}} V(S_i) = V\left(\langle S_i \mid i \in \mathcal{I}\rangle\right)$

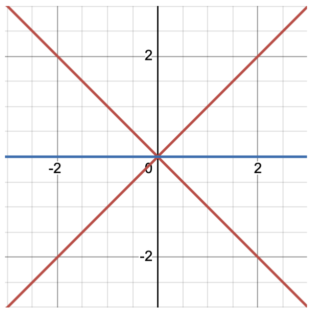
Proof **:**
Note that the underlying sets of the schemes will just be finite unions and intersections.

The harder point to verify is that the union becomes intersection. In the case of vanishing sets, the proof takes advantage of the fact that $I \cap J \subseteq IJ \subseteq (I \cap J)^2$. Note that it must be the intersection, as we would need that that $V(S) \cap V(S) = V(S)$ and not $V(S^2)$.

Example 4.1.36: Vanishing Scheme Intersection

1. Take $V(y - x^2), V(y) \subseteq \mathbb{A}^2_k$. Then with vanishing sets, this would just be the point $(0, 0)$. However, this point now has multiplicity 2.

2. Take $V(y^2 - x^2), V(y) \subseteq \mathbb{A}^2_k$. The visual of these two are:



Use this to show that if X, Y, Z are closed subscheme of a scheme W, then

$$(X \cap Z) \cup (Y \cap Z) \neq (X \cup Y) \cap Z$$

Hence, not all properties of unions and intersection can carry over!

Locally Closed Embedding

A lot on when it is nicer, on locally closed embeddings, etc. Vakil p.238-240

Recall that the line without the origin is not a closed subscheme of $\mathbb{A}^2_k$. This is This is however still a pretty reasonable subset, and almost a closed subscheme: it is a closed subscheme of an open subscheme of $\mathbb{A}^2_k$ (namely the subscheme given by $k[x, 0]$). This motivates the following:

Definition 4.1.37: Locally Closed Embedding

Let $\pi : X \to Y$ be a scheme morphism. Then it is a *locally closed embedding* if π can be factored into:

$$X \xrightarrow{\alpha} Z \xrightarrow{\beta} Y$$

where α is a closed embedding, and b is an open embedding.

The above is such an example: more formally the morphism $\operatorname{Spec} k[t, t^{-1}] \to \operatorname{Spec} k[x, y]$ is a locally closed embedding. Note that we could have also taken an open subset of a closed subset (in exercise ref:HERE you make sure these are equivalent)

Proposition 4.1.38: Locally Closed embedding is Locally Finite

Let $\pi : X \to Y$ be a locally closed embedding. Then it is locally finite.

Proof **:**
exercise.

Give a characterization of intersection of locally closed embeddings as you've moved the fibred product to earlier

4.1.4 Image Scheme

In general, the image of a scheme is badly behaved, for example check that $(x, y) \mapsto (x, xy)$ does not give a scheme as an image.

Definition 4.1.39: Image Scheme

Let $\iota : Z \to Y$ be a closed subscheme, which gives the short exact sequence:

$$0 \to \mathcal{K}_{Z/Y} \to \mathcal{O}_Y \to \iota_* \mathcal{O}_Z \to 0$$

Then the *image* of $\pi : X \to Y$ *lies in* Z if the composition

$$\mathcal{K}_{Z/Y} \to \mathcal{O}_Y \to \pi_* \mathcal{O}_X$$

is zero. Then the *scheme-theoretic image* of π (a closed subscheme of Y) is the smallest closed subscheme containing the image

Intuitively, locally functions vanishing on Z pullback to zero functions on X. If the image of π line sin some subschemes Z_j, it lies in their intersection, and hence it is the scheme-theoretic image. Note that the scheme theoretic image may not necessarily be an open subscheme:

Example 4.1.40: Scheme-theoretic Image

1. Take
$$\pi : \operatorname{Spec}(k[\epsilon]/(\epsilon^2)) \to \operatorname{Spec}(k)$$
given by $x \mapsto \epsilon$. Then the scheme-theoretic image of π ks
$$\operatorname{Spec}(k[x]/(x^2))$$
as the polynomials pulling back to 0 are precisely the multiples of x^2.

2. Now take
$$\pi : \operatorname{Spec}(k[\epsilon]/(\epsilon^2)) \to \mathbb{A}^1_k$$

given by $x \mapsto 0$. Then the scheme-theoretic image is

$$\operatorname{Spec}\left(k[x]/(x)\right)$$

which is reduced.

3. Take

$$\operatorname{Spec} k[t, t^{-1}] = \mathbb{A}^1_k \setminus \{0\} \to \mathbb{A}^1_k = \operatorname{Spec}\left(k[u]\right)$$

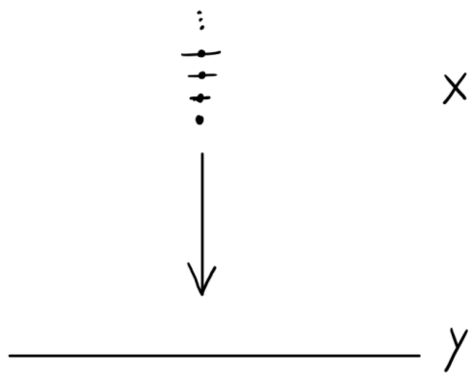
given by $u \mapsto t$. Then any function $g(u)$ which pulls back to 0 as a function of t must be the zero function. Hence,the scheme-theoretic image is

$$\operatorname{Spec}\left(k[u]\right)$$

4. Consider the following more pathological example:

$$\pi : \coprod \operatorname{Spec}\left(\frac{k[\epsilon_n]}{(\epsilon_n)^n}\right) \to \operatorname{Spec} k[x]$$

which well denote $\pi : X \to Y$, and it is given by $x \mapsto \epsilon_n$ on the nth component. Now, if $g(x)$ on Y pulls back to 0 on X, then its Taylor expansion must be 0. But then the scheme-theoretic image is $V(0)$ on Y, that is it is Y, even though the scheme-theoretic image is the origin (the closed point 0). Visually:



4.2 Quasi-Coherent Sheaves on Projective Schemes

Let us now explore quasi-coherent shaves on projective schemes. The main result of this section is to recover the graded ring of a projective scheme as a special kind of sheaf. Let us start by defining the natural way we can have a sheaf over a projective scheme:

Definition 4.2.1: Sheaf Associated to M [on projective scheme]

Let S be a graded ring and M a graded S-module. Then the sheaf associated to M on Proj S, denoted M is given by:

$$\widetilde{M}\Big|_{D_+(f)} = \left(\widetilde{M\left[f^{-1}\right]_0}\right)$$

for homogeneous $f \neq 0$.

Like in projective schemes, if S is the usual polynomial ring with grading by degree, then open sets $D(x_i) \subseteq \operatorname{Proj} S$ are isomorphic to Spec $(S_{x_i})_0$, and so the module $\widetilde{M}(D(x_i))$ becomes a module over an affine scheme.

Proposition 4.2.2: Sheaf on Projective Scheme is Quasi-Coherent

Let S be a graded ring, M a graded S-module, and $X = \operatorname{Proj} S$. Then:

1. $\widetilde{M}_{\mathfrak{p}} = M_{\mathfrak{p}}$
2. $\widetilde{M}$ is a quasi-coherent $\mathcal{O}_X$-module, and if S is Noetherian and M finitely generated, it is coherent.

Proof **:**
a repeat of similar result earlier

Let us start by showing how we can define some natural sheaves on $\operatorname{Proj} S$ whose direct sum of global sections will recover S :

Definition 4.2.3: Twisting Sheaf

Let $S_\bullet = \bigoplus_n S_n$ be a graded ring and $X = \operatorname{Proj} S$. Then for every $n \in \mathbb{Z}$, define the sheaf:

$$\mathcal{O}_X(n) := \widetilde{S(n)}$$

where $S(n)$ is the graded ring $S(n)_i = S_{n+i}$. The sheaf $\mathcal{O}_X(1)$ will be called the twisting sheaf.

Note that $S(0) = S$, so $\mathcal{O}_X(0) = \mathcal{O}_X$ which implies $\Gamma(\operatorname{Proj} S, S(0)) = \Gamma(\operatorname{Proj} S, \mathcal{O}_{\operatorname{Proj} S})$ represents the constant global sections. Let us next consider $\widetilde{S(n)}$. Let us work on our usual $S = R[x_0, \dots, x_n]$ (proposition 4.2.5 shall generalize this discussion). On $D(x_i)$ we will then have our usual localization, except we will not have the degree 0 elements but degree m elements:

$$\widetilde{S(m)}\left(D_+(x_i)\right) = \text{ degree } m \text{ of } R\left[x_0, \dots, x_n, x_i^{-1}\right]$$

Note that this is no longer a ring, but a $R[x_0/x_i, \dots, x_n/x_i]$-module (it is closed under the action as the elements are degree 0). As an abelian group, it is generated by:

$$x_0^m, x_0x_1^{m-1}, x_0x_2^{m-1}, x_3^{m-6}x_2^3x_8^3, \dots, x_n^m$$

Is this module any different from $\widetilde{S}\left(D_+\left(x_i\right)\right)$ as modules? In fact they are not! Note that on $D_+\left(x_i\right)$, multiplying by x_i is an isomorphism from the degree m component of $R\left[x_0, \ldots, x_n x_i^{-1}\right]$ to its degree $m+1$ component! Thus, $\widetilde{S(n)}$ is locally isomorphic to $\widetilde{S}$, and hence it is locally free of rank 1 , that is a *line bundle* (or invertible sheaf).
These sheaves are not be isomorphic, because the gluing of these sheaves is subtly different and shall motivate why it is called the twisted sheaves. Let us first distinguish the $\widetilde{S(m)}$ by showing they have different global sections. Going about it a bit informally, let's consider $\widetilde{S(m)}$ and cover it with $D\left(x_i\right)$ so that $\Gamma\left(\widetilde{S(m)}, D\left(x_i\right)\right)=\left(R\left[x_0, \ldots, x_n x_i^{-1}\right]\right)_m$. Then whenever gluing two $D\left(x_i\right), D\left(x_i\right)$, their intersection shall have polynomials in degree n that have no x_i, x_j term. The gluing is very similar to what we've done before: we can think of the transition from $D\left(x_i\right)$ to $D\left(x_j\right)$ on $D\left(x_i\right) \cap D\left(x_j\right)$ would map $x_k \mapsto x_k$ and

$$\frac{x_k}{x_i} \mapsto \frac{x_k}{x_j}$$

For a concrete example, if $X=\operatorname{Proj} R[x, y]$, then homogeneous polynomial of degree 1 are of the form $F(x, y)=a x+b y$, which on the two distinguished open sets $D_+(x)$, $D_+(y)$ are represented as can be (where $t=x / y$ and $s=y / x$):

$$f(t)=a t+b \qquad g(s)=a+b s$$

and have transition function

$$f(t)=t \cdot t g(1 / t)$$

Contrast this with the regular gluing which doesn't have the t factor. More generally, on $S(n)$, we would have:

$$f(t)=t^n g(1 / t)$$

Then that means that the global section is:

$$\Gamma(\widetilde{S(n)}, \operatorname{Proj} S)=S_n$$

and so we can use dimension counting to show they are distinct for each classical projective space!

$\dim \Gamma(\widetilde{S(n)}, \operatorname{Proj} S)$	$\mathbb{P}_R^0$	$\mathbb{P}_R^1$	$\mathbb{P}_R^2$	$\mathbb{P}_R^3$	$\mathbb{P}_R^4$
$S(0)$	1	1	1	1	1
$S(1)$	1	2	3	4	5
$S(2)$	1	3	6	10	15

After introducing some properties of twisted sheaves, we shall show the $\widetilde{S(-n)}$ (given by the multiplicity of poles, or similarly by the dual space of $S(n)$) are not isomorphic as well
We should take a moment to justify the term "twisting". First off, only $\widetilde{S(0)}$ is globally free, hence there is something going on for each $\widetilde{S(n)}$. To see that it is a twisting, let us look more closely at the transition for $\widetilde{S(1)}$ on $\mathbb{P}_R^1$. On $D(x_0) \cap D(x_1)$, we have only degree 1 terms.Then we see from the map that

$$\frac{x_0}{x_1} \mapsto \frac{x_0 x_0}{x_1 x_0}=\frac{x_1}{x_0}$$

as we see, we would have to "flip twice", to get back to the original. In higher twists, the power would be greater, showing the number of twists increases.
Let us now introduce the grading to any sheaf defined on projective space:

Definition 4.2.4: n-twisted sheaf of $\mathscr{F}$

For any $\mathcal{O}_X$-module $\mathscr{F}$, let:

$$\mathscr{F}(n) := \mathscr{F} \otimes_{\mathcal{O}_X} \mathcal{O}_X(n)$$

we call this sheaf the n-twisted sheaf of $\mathscr{F}$.

Proposition 4.2.5: Properties of Twisted Sheaves

Let S be a graded ring and $X = \operatorname{Proj} S$. Let S be generated by S_1 as an S_0-algebra. Then

1. The sheaf $\mathcal{O}_X(n)$ is an invertible sheaf and $\mathcal{O}_X(n) \not\cong_{\mathcal{O}_X} \mathcal{O}_X(m)$ for $n \neq m$
2. For any graded S-module M, $\widetilde{M}(n) = \widetilde{M(n)}$
3. $\mathcal{O}_X(n) \otimes \mathcal{O}_X(m) \cong \mathcal{O}_X(n+m)$
4. Let T be a graded ring generated by T_1 as a T_0-algebra. Let $\varphi : S \to T$ be a homogeneous homomorphism, and take $U \subseteq \operatorname{Proj} T$. Given $f : U \to X$:

$$f^*\left(\mathcal{O}_X(n)\right) \cong \mathcal{O}_Y(n)|_U \quad \text{and} \quad f_*\left(\mathcal{O}_Y(n)|_U\right) \cong \left(f_*\mathcal{O}_U\right)(n)$$

***Proof* :**

1. We must show that $\mathcal{O}_X(n)$ is an invertible sheaf, i.e. locally free of rank 1 . We shall use the fact that S is generated by S_1 as an S_0-algebra, meaning it is finitely generated and hence will be "locally S_n ". Take $\mathcal{O}_X(n)|_{D(f)} \cong \widetilde{(S_n)_f}$ on $\operatorname{Spec}(S_f)_0$. Now, notice that $(S_n)_f$ is isomorphic to $(S_0)_f$ via the map $s \mapsto f^n s$. As S is generated by S_1 as an S_0-algebra, we see that each open subsets are isomorphic to $\widetilde{S_n}$, and hence it is an invertible sheaf (i.e. a line bundle).
For the second claim, formalize the global section calculations from earlier.
2. immediate
3. Show that for any two graded modules $\widetilde{M \otimes_S N} \cong \widetilde{M} \otimes_{\mathcal{O}_X} \widetilde{N}$ (not too difficult generalization of an earlier result), and that $\left((M \otimes_S N)_f\right)_0 = (M_f)_0 \otimes_{(S_f)_0} (N_f)_0$.
4. Generalize form the affine case.

Using the above properties, show that the negative degree elements are not isomorphic and fill out the table:

$\dim \Gamma(\widetilde{S(n)}, \operatorname{Proj} S)$	$\mathbb{P}^0_R$	$\mathbb{P}^1_R$	$\mathbb{P}^2_R$	$\mathbb{P}^3_R$	$\mathbb{P}^4_R$
$S(-2)$	1	0	0	0	0
$S(-1)$	1	0	0	0	0
$S(0)$	1	1	1	1	1
$S(1)$	1	2	3	4	5
$S(2)$	1	3	6	10	15

We shall show that $S(-d) = 0$ in theorem theorem5.2.7. Let us now show there is a natural graded ring associated to each projective scheme:

Definition 4.2.6: Graded s-module Associated to $\mathscr{F}$

Let S be a graded ring, $X = \operatorname{Proj} S$, and $\mathscr{F}$ a sheaf of $\mathcal{O}_X$-modules. Then the graded S-module associated to $\mathscr{F}$ as:

$$\Gamma_*(\mathscr{F}) = \bigoplus_{n \in \mathbb{Z}} \Gamma(X, \mathscr{F}(n))$$

where the product is given by the properties of proposition 4.2.5.

We can now naturally recover S in the case of it being a polynomial ring:

Proposition 4.2.7: Recovering Graded Ring

Let R be a ring, $S = R[x_0, \ldots, x_n]$, and $X = \operatorname{Proj} S$. Then:

$$\Gamma_*(\mathcal{O}_X) \cong S$$

Proof **:**

We cover X with the open sets $D_+(x_i)$. Then to give a section $t \in \Gamma(X, \mathcal{O}_X(n))$ is the same as giving sections $t_i \in \mathcal{O}_X(n)(D_+(x_i))$ for each i, which agree on the intersections $D_+(x_i x_j)$. Now t_i is just a homogeneous element of degree n in the localization S_{x_i}, and its restriction to $D_+(x_i x_j)$ is just the image of that element in S_{x_i, x_j}. Summing over all n, we see that $\Gamma_*(\mathcal{O}_X)$ can be identified with the set of $(r+1)$-tuples $(t_0, \ldots, t_r)$ where for each $i, t_i \in S_{x_i}$, and for each i, j, the images of t_i and t_j in S_{x_i, x_j} are the same.
Now the x_i are not zero divisors in S, so the localization maps $S \to S_{x_i}$ and $S_{x_i} \to S_{x_i, x_j}$ are all injective, and these rings are all subrings of $S' = S_{x_0 \cdots x_r}$. Hence $\Gamma_*(\mathcal{O}_X)$ is the intersection $\bigcap S_{x_i}$ taken inside S'. Now any homogeneous element of S' can be written uniquely as a product $x_0^{i_0} \cdots x_r^{i_r} f(x_0, \ldots, x_r)$, where the $i_j \in \mathbb{Z}$, and f is a homogeneous polynomial not divisible by any x_i. This element will be in S_{x_i} if and only if $i_j \geq 0$ for $j \neq i$. It follows that the intersection of all the S_{x_i} (in fact the intersection of any two of them) is exactly S.

Note that if S is not a polynomial ring, this result need not hold, though it is be related (see Hartshorne exercise II.5.14). A case where it fails is if we have that $M_0 = k$ (or some finite dimensional vector space) and $M_n = 0$ for all other integers; then $\tilde{M} = 0$. Hence, this shows that can't start with an M, do $\widetilde{M}$ and recover M by Γ_*. Unfortunately, the map $M \mapsto \Gamma_*(\widetilde{M})$ is neither injective nor surjective.

However, if you start with a coherent sheaf $\mathscr{F}$, apply Γ_* to get $\Gamma_*(\mathscr{F})$, and then do $\widetilde{\Gamma_*(\mathscr{F})}$, then:

$$\widetilde{\Gamma_*(\mathscr{F})} \cong \mathscr{F}$$

To prove this, we first require a technical lemma:

Lemma 4.2.8: Extending Sections for Line Bundles

Let X be a scheme, $\mathcal{L}$ an invertible sheaf on $X, f \in \Gamma(X, \mathcal{L}), X_f$ be the open set of points $x \in X$ where $f_x \notin \mathfrak{m}_x \mathcal{L}_x$, and let $\mathscr{F}$ be a quasi-coherent sheaf on X. Now:

1. Suppose that X is quasi-compact, and $s \in \Gamma(X, \mathscr{F})$ is a global section of $\mathscr{F}$ whose restriction to X_f is 0 . Then for some $n > 0$, we have $f^n s = 0$, where $f^n s$ is considered as a global section of $\mathscr{F} \otimes \mathcal{L}^{\otimes n}$.
2. Suppose furthermore that X has a finite covering by open affine subsets U_i, such that $\mathcal{L}|_{U_i}$ is free for each i, and such that $U_i \cap U_j$ is quasi-compact for each i, j (it is quasi-separated). Given a section $t \in \Gamma(X_f, \mathscr{F})$, then for some $n > 0$, the section $f^n t \in \Gamma(X_f, \mathscr{F} \otimes \mathcal{L}^{\otimes n})$ extends to a global section of $\mathscr{F} \otimes \mathcal{L}^{\otimes n}$.

***Proof* :**
This lemma is a direct generalization of lemma 4.1.12, with an extra twist due to the presence of the invertible sheaf $\mathcal{L}$.

1. First, cover X with a finite number (possible since X is quasi-compact) of open affines $U = \operatorname{Spec} A$ such that $\mathcal{L}|_U$ is free. Let $\psi : \mathcal{L}|_U \cong \mathcal{O}_U$ be an isomorphism expressing the freeness of $\mathcal{L}|_U$. Since $\mathscr{F}$ is quasi-coherent, by proposition 4.1 .13 there is an A-module M with $\mathscr{F}|_U \cong \widetilde{M}$. Our section $s \in \Gamma(X, \mathscr{F})$ restricts to give an element $s \in M$. On the other hand, our section $f \in \Gamma(X, \mathcal{L})$ restricts to give a section of $\mathcal{L}|_U$, which in turn gives rise to an element $g = \psi(f) \in A$. Clearly $X_f \cap U = D(g)$. Now $s|_{X_f}$ is zero, so $g^n s = 0$ in M for some $n > 0$, just as in the proof of lemma4.1.12. Using the isomorphism

$$\mathrm{id} \times \psi^{\otimes n} : \mathscr{F} \otimes \mathcal{L}^n|_U \cong \mathscr{F}|_U,$$

we conclude that $f^n s \in \Gamma(U, \mathscr{F} \otimes \mathcal{L}^n)$ is zero. This statement is intrinsic (i.e., independent of ψ). So now we do this for each open set of the covering, pick one n large enough to work for all the sets of the covering, and we find $f^n s = 0$ on X.
2. Proceed as in the proof of lemma 4.1.12, keeping track of the twist due to $\mathcal{L}$ as above. The hypothesis $U_i \cap U_j$ quasi-compact is used to be able to apply part (1) there.

Proposition 4.2.9: Partial Equivalence For Graded Rings

Let S be a graded ring, which is finitely generated by S_1 as an S_0-algebra. Let $X = \operatorname{Proj} S$ and let $\mathscr{F}$ be a quasi-coherent sheaf on X. Then there is a natural isomorphism

$$\beta : \widetilde{\Gamma_*(\mathscr{F})} \to \mathscr{F}$$

In particular, all quasi-coherent sheaves on projective schemes come from a graded module. Note the important of S being generated by S_1 and not any other S_n; if it were then the grading of the two rings would be different and we won't recover the result.

Proof **:**

First, define the morphism β for any $\mathcal{O}_X$-module $\mathscr{F}$. Let $f \in S_1$. Since $\widetilde{\Gamma_*(\mathscr{F})}$ is quasi-coherent in any case, to define β, it is enough to give the image of a section of $\widetilde{\Gamma_*(\mathscr{F})}$ over $D_+(f)$. Such a section is represented by a fraction m/f^d, where $m \in \Gamma(X, \mathscr{F}(d))$, for some $d \geq 0$. We can think of f^{-d} as a section of $\mathcal{O}_X(-d)$, defined over $D_+(f)$. Taking their tensor product, we obtain $m \otimes f^{-d}$ as a section of $\mathscr{F}$ over $D_+(f)$. This defines β.
Now let $\mathscr{F}$ be quasi-coherent. To show that β is an isomorphism we have to identify the module $\Gamma_*(\mathscr{F})_{(f)}$ with the sections of $\mathscr{F}$ over $D_+(f)$. We apply lemma 4.2 .8 . considering f as a global section of the invertible sheaf $\mathcal{L} = \mathcal{O}(1)$. Since we have assumed that S is finitely generated by S_1 as an S_0-algebra, we can find finitely many elements $f_0, \ldots, f_r \in S_1$ such that X is covered by the open affine subsets $D_+(f_i)$. The intersections $D_+(f_i) \cap D_+(f_j)$ are also affine, and $\mathcal{L}|_{D_+(f_i)}$ is free for each i, so the hypotheses of lemma 4.2.8 are satisfied. The conclusion of lemma 4.2.8 tells us that $\mathscr{F}(D_+(f)) \cong \Gamma_*(\mathscr{F})_{(f)}$, which is just what we wanted.

We shall later show this that all resolution of quasi-coherent sheaves by locally free sheaves must have finite resolutions, see ref:HERE. For now, this shows we can translate our characterization of closed subschemes with ideal sheaves to projective schemes, and that

Corollary 4.2.10: Characterizing Projective sub-Schemes

Let R be a ring. Then:

1. If Y is a closed subscheme of $\mathbb{P}^r_R$, then there is a homogeneous ideal $I \subseteq S = R[x_0, \ldots, x_r]$ such that Y is the closed subscheme determined by I.
2. A scheme Y over $\operatorname{Spec} R$ is projective if and only if it is isomorphic to $\operatorname{Proj} S$ for some graded ring S, where $S_0 = R$, and S is finitely generated by S_1 as an S_0-algebra.

Proof **:**

1. Let $\mathcal{I}_Y$ be the ideal sheaf of Y on $X = \mathbb{P}^r_R$. Now $\mathcal{I}_Y$ is a subsheaf of $\mathcal{O}_X$; the twisting functor is exact; the global section functor Γ is left exact; hence $\Gamma_*(\mathcal{I}_Y)$ is a submodule of $\Gamma_*(\mathcal{O}_X)$. But by proposition 4.2.9, $\Gamma_*(\mathcal{O}_X) = S$. Hence $\Gamma_*(\mathcal{I}_Y)$ is a homogeneous ideal of S, which we will call I. Now I determines a closed subscheme of X, whose sheaf of ideals will be $\widetilde{I}$. Since $\mathcal{I}_Y$ is quasi-coherent, we have $\mathcal{I}_Y \cong \widetilde{I}$ by proposition 4.2.9, and hence Y is the subscheme determined by I. In fact, $\Gamma_*(\mathcal{I}_Y)$ is the largest ideal in S defining Y.

2. Recall that by definition Y is projective over $\operatorname{Spec} R$ if it is isomorphic to a closed subscheme of $\mathbb{P}^r_R$ for some r. By part (1), any such Y is isomorphic to $\operatorname{Proj} S/I$, and we can take I to be contained in $S_+ = \bigoplus_{d>0} S_d$, so that $(S/I)_0 = R$. Conversely, any such graded ring S is a quotient of a polynomial ring, so $\operatorname{Proj} S$ is projective.

Corollary 4.2.11: Characterizing Hypersurfaces

Let R be a UFD. Then any hypersurface $X \subseteq \mathbb{P}^n_R$ corresponds to a principal ideal

***Proof* :**
The key result is that in a UFD, every height-1 prime ideal (i.e. minimal ideal) is principal. In fact, R is a UFD if and only if every height 1 prime ideal is principal. If R is a UFD, let $\mathfrak{p}$ be a height 1 prime ideal. Choose $x \in \mathfrak{p}$. As R is a UFD, $x = up_1 \cdots p_k$ for some choice of unit u. As $\mathfrak{p}$ is prime, at least one $p_i \in \mathfrak{p}$. Then $(p_i) \subseteq \mathfrak{p}$, and as $\mathfrak{p}$ has height 1 $(p_i) = \mathfrak{p}$. Conversely, assume $(p_i) = \mathfrak{p}$ for all height-1 prime ideals. Let r be an irreducible element. We want to show (r) is prime. As Let $\mathcal{P}_r$ be the collection of all prime ideals containing r. As R is Noetherian, there is a minimal element; say $\mathfrak{p}$. As $\mathfrak{p}$ is a minimal prime ideal, it has height-1 (exercise 4.2.1(1)). Then by assumption, $\mathfrak{p}$ is principal so $\mathfrak{p} = (p)$ for some $p \in R$. Then $(r) \subseteq \mathfrak{p} = (p)$, hence $r = sp$ for some $s \in R$. But as r is irreducible, either s or p must be a unit, but p cannot be (as (p) is prime) so s is a unit. As R is an integral domain, $(r) = (p)$ are associates, and (r) is thus prime.

It is now a matter of putting everything together. As R is a UFD, $R[x_1, ..., x_n]$ is a UFD, and the graded structure respects unique decomposition (see [8]). Then by corollary 4.2.10, an irreducible codimension 1 subscheme $X \subseteq \mathbb{P}^n_R$ corresponds to a height-1 prime ideal $\mathfrak{p}$, which by the above must be principal; let $p \in R$ so that $(p) = \mathfrak{p}$. Then $V((p)) = X$, completing the proof.

Definition 4.2.12: Degree of Projective Hypersurface

Let R be a UFD and $X \subseteq \mathbb{P}^n_R$ be a hypersurface. By corollary 4.2.11, let $f \in R[x_0, ..., x_n]$ be the function corresponding to X. Then

$$\deg X := \deg f$$

We conclude this section by finally showing why tensoring quasi-coherent sheaves is not closed:

Example 4.2.13: Lack of Closure For Quasi-coherent Sheaves
Take $\mathscr{F} = \mathcal{O}(-1)$ and $\mathscr{G} = \mathcal{O}(1)$. Then let us compare:

$$(\mathscr{F} \otimes \mathscr{G})(U) \quad \text{and} \quad (\mathscr{F}(U) \otimes \mathscr{G}(U))$$

The left hand side is $\tilde{R}$ and hence is 1 dimensional, while the right hand side is 0 dimensional (tensoring a 0 dimensional space with a 2 dimensional space).

Another example I came across is very interesting:

Example 4.2.14: Free Sheaves Need Not Be Projective Objects
Really cool! here

Exercise 4.2.1

1. (Krull Principal Ideal Theorem) Let R be a ring and $\emptyset \neq (x) \subsetneq R$ a nontrivial proper ideal. Show that if $\mathfrak{p}$ is a minimal prime ideal containing (x), $\mathfrak{p}$ has height-1. (hint: you will localize, and at some point use Nakayama's lemma)

4.3 Divisors

Divisors are a way of measuring how far away the geometry of your space is constructed by "functions" (i.e. elements of the global section of the structure sheaf). There are two ways of approaching this problem. For the first, recall that if V is a variety and $f \in \mathcal{O}_V$, then $V(f)$ is a hypersurface. Recall further that a hypersurface is defined to be a codimension 1 subvariety (more generally a codimension 1 subscheme). Then we may ask which hypersurfaces are given as the zero's of global sections. The first approach shall yield *Weil Divisors*. For the second,

while the second approach shall yield *Cartier Divisors*. In most nice cases (think Integral Noetherian Schemes), these two shall give the same notion of divisor. If you have read [8, chapter 22.7], then in this section we shall generalize the notion of divisors to general schemes and show hey they relate to a *Picard Scheme*. We shall also show how divisors naturally correspond to line bundles which shall let us use divisors to gain further important results about schemes[1].

The intuition for divisors works well when starting with examples from complex analysis (in particular, see [3, chapter 3 and 6]). In this case, the two notions of divisors shall coincide and so we shall be finding the divisors for Riemann surface X. Starting with $X = \mathbb{C}$, consider the free-abelian group generated by the points of X, so that each element of this group is of the form:

$$\sum_i n_i p_i \qquad n_i \in \mathbb{Z},\ p_i \in X$$

We may ask if there is a meromorphic function associated to this function. As we are interested in algebraic constructions, we limit ourselves to function with a finite number of poles and zero, that is all *rational functions*. Then We get that for each $\sum_i n_i p_i$, we can associated to it the meromorphic function;

$$(z - p_i)^{n_i}$$

Let us call the set of meromorphic functions the *principle divisors*, and the finite formal $\mathbb{Z}$-linear combinations the *divisors*. Then the quotient of these two groups is zero:

$$\frac{\text{Divisors}}{\text{Principal Divisors}} = 0$$

Let us now consider X to be the Riemann surface. We shall once again take the divisor group to be the free group. This time, the principal divisor group (the group of meromorphic functions) has a fundamental limitations placed on it: recall that the sum of the zero's and pole of a meromorphic function is zero[2]. In this case, the sum of each divisor is some integer $n \in \mathbb{Z}$. Using this, you can show that:

$$\frac{\text{Divisors}}{\text{Principal Divisors}} \cong \mathbb{Z}$$

[1]For example, this section is foundational in the establishment of the *Riemann-Roch Theorem* which is one of the fundamental results in roughly classifying curves

[2]Recall that by adding the point at infinite, we shall add the "compliment" of each zero/pole; convince yourself that z^2 has two zeros and two poles

Finally, consider an elliptic curve given by $\mathbb{C}/L$ where L is a lattice. Richard Borcherds, finish off!

4.3.1 Weil Divisors

Weil divisors generalize the above discussion on (compact) Riemann surfaces. We shall requires that there is as notion of "degree of zero" at a point. There are many ways of doing this, and for clarity of presentation of the material we shall limit to scheme that have a valuation rings for each stalk to have a well-defined degree. For this, we shall require the schemes to be nice enough for this (we shall present schemes where this is not so obvious in example ref:HERE). Naturally, if we are working with curves, it suffices to work with nonsingular schemes as each stalk is a DVR. To emulate this in higher dimensions, we want the principal components to be associated to functions which vanish, but in higher dimensions such components would be codimension 1. Thus we shall want our schemes to be at least regular in the codimension components, that is we are going to be working over some relatively nice schemes:

Definition 4.3.1: Regular Codimension 1 Scheme

Let X be a scheme. Then X is said to be regular of codimension 1 if every local ring $\mathcal{O}_{X,\mathfrak{p}}$ of X of dimension 1 is regular

Example 4.3.2: Regular Codimension 1 Scheme

1. Naturally, all regular curves (and regular schemes) satisfy this criterion, as the codimension 1 local rings are the local rings of points, which are all regular by definition
2. if X is a Noetherian regular scheme, any local ring of dimension 1 is an integrally closed domain (a normal domain), and hence by [8] it is a DVR.

As all local rings at each prime of codimension 1 is regular, there exists a natural notion of valuation, which is the key tool we want to use. If they were not regular, we can instead take the length of $R/(f)$; we still have the main important property which is that the order of fg is the order of f plus the order of g by doing the following short exact sequence:

$$0 \to R/g \xrightarrow{\times f} R/fg \to R/f \to 0$$

which is key as it gives a homomorphism between K^* and the Weil divisor groupie shall defined shortly. It is a bit easier to work with valuations and it gives us the core information we need, hence we still with them. Note further that if X was not integral, we would also insure to take f to be a nonzero divisor.

For the sake of narrative clarity, we shall be adding some more conditions on our schemes: Noetherian is to keep things finite, integral is to have a generic point for the entire scheme to work with, and separated is to have injective restriction maps to not have any issue picking functions:

Definition 4.3.3: Weil Divisor

Let X be a Noetherian integral separated scheme, regular in codimension 1. Then a prime divisor is a closed integral subscheme. A Weil divisor is an element of the free abelian group generated by the prime divisors. This group is denoted:

$$\mathrm{WDiv}\, X$$

If $D = \sum_i n_i Y_i$ and all $n_i \geq 0$, we say that D is effective.

Let $Y \subseteq X$ be as in he above definition. Then Y has a generic point, say η, and hence it has a stalk $\mathcal{O}_{Y,\eta}$ which has dimension 1 by choice of X and Y and hence is a DVR. Label the valuation ν_Y. As X is separated, Y is uniquely determined by its valuation (see exercise 3.3.2.)

Now, let $k(X)$ denote the function field of X (remember we are assuming X is integral). Pick $f \in (k(X))^*$, a nonzero rational function. Then $\nu_Y(f)$ is an integer. If it is positive, it represents the order of vanishing of f on Y, and if it is negative, it represents the order of the pole along Y (we shall take the order of the pole to be the absolute value of the valuation). As we are dealing with the codimension 1 case, by differential geometry we would expect that almost all function do not vanish along Y, and indeed:

Lemma 4.3.4: Almost All Function Non Vanishing

Let X be a Noetherian integral separated scheme, regular in codimension 1. Let $f \in (k(X))^*$. Then $\nu_Y(f) = 0$ for all except finitely many prime divisors Y

***Proof* :**
As rational function must be regular on some open subset, let's say f is regular on $U = \mathrm{Spec}\, R \subseteq X$. Then $C = X \backslash U$ is a proper closed subset of X. As X is Noetherian,

$$C = \bigcup_i^n Z_i$$

for irreducible closed subsets Z_i which each must have codimension ≥ 1. Then any irreducible component contained in C, i.e. any prime divisors, would have to be one of the Z_i 's, and so C must contain at most finitely many prime divisors (the rest meet U). Thus, if we show here are only finitely many primes divides Y of U for which $\nu_Y(f) \neq 0$, we are done.

As f is regular on $U, \nu_Y(f) \geq 0$ and $\nu_Y(f) > 0$ if and only if Y is contained in a closed subset of U defined by the principal ideal (f). As $f \neq 0$, this closed subset is proper, and so it can only contains finitely many irreducible subsets of codimension 1 on U, but then we have that there are only finitely many prime divisors Y of U for which $\nu_Y(f) \neq 0$, as we sought to show.

Thus, our sum is well-defined:

Definition 4.3.5: Principal Divisor

Let $f \in (\kappa(X))^*$. Then the principal divisor associated to f is:

$$\mathrm{Div}(f) = \sum_i \nu_Y(f)Y \in \mathrm{Div}\, X$$

This group is sometimes denoted PDiv X.

Note that $\mathrm{Div}(f/g) = \mathrm{Div}\, f - \mathrm{Div}\, g$ by the properties of discrete valuations, and hence this is even a group homomorphism from the multiplicative group $(\kappa(X))^*$ to the additive group Div X !

Definition 4.3.6: Divisor Class Group

Let X be a Noetherian integral separated scheme, regular in codimension 1. Then the divisor class group of X is:

$$\mathrm{Cl}X = \frac{\mathrm{WDiv}\, X}{\mathrm{PDiv}\, X}$$

We shall give examples of the divisor class group for a few schemes, however let us first generalize a very important result from commutative algebra that will allows us to find many schemes with trivial divisor class groups. First, a technical lemma:

Lemma 4.3.7: Normal Domain: Intersection Height One Localizations

Let R be a normal domain (an integrally closed domain) Then

$$R = \bigcap_{\mathrm{ht}\mathfrak{p}=1} R_{\mathfrak{p}}$$

Proof **:**
By [8], we already have $R = \bigcap_{\mathfrak{p}} R_{\mathfrak{p}}$. If R is normal, then $R_{\mathfrak{p}}$ is normal, and since $\mathfrak{p}$ is of height 1 , then $R_{\mathfrak{p}}$ is a local normal domain of dimension 1, making it a DVR by [8]. Then it is not too hard to see that the height 1 prime ideals correspond to DVR's containing R, so we have:

$$R = \bigcap_{R \subseteq R_{\mathfrak{F}}} R_{\mathfrak{P}}$$

Then this reduces to a standard argument by contradiction: if $x \in \mathrm{Frac}(R) \backslash R$, as R is normal we must have x not integral over R. As x is not integral over $R, 1/x$ must be contained in some maximal ideal (consider the map $R[s,t] \to R[x, 1/x]$ mapping $s \mapsto x$ and $t \mapsto 1/x$; find the maximal idea via the image of the map); call this ideal $\mathfrak{m}$. Then we have $R_{\mathfrak{m}}$, and it certainly contains R, but then $x \in R$, hence our original assumption is void, and we get the equality.

Proposition 4.3.8: Characterizing UFD Geometrically

Let R be a Noetherian domain. Then R is a UFD if and only if $\operatorname{Spec} R$ is normal and Cl Spec $R = 0$.

Proof **:**
As R is a UFD, it is normal (common algebraic exercise, or see [8]), hence $\operatorname{Spec}(R)$ is normal. It was shown in the proof of corollary corollary 4.2.11 that R is a UFD iff every height-1 prime ideal is principal. From this, we'll show that when every height-1 prime is principal, $\mathrm{ClSpec} R = 0$. If every prime of height 1 is principal, then every prime divisor Y is mapped to by a p, and so $\operatorname{Cl}\operatorname{Spec} R = 0$.

Conversely, given R Is normal and $\operatorname{Cl}\operatorname{Spec} R = 0$, it suffices to show that every height-1 prime $\mathfrak{p}$ is principal. First, for choose a prime divisor Y. As ClSpec $R = 0$, we know that there is some $p \in (\kappa(X))^*$ that maps to Y; it corresponds to some prime $\mathfrak{p}$. We must show that $p \in R^*$, and that $(p) = \mathfrak{p}$. As $\mathfrak{p}$ is the prime associated to the prime divisor Y and p map to it,

$$\nu_Y(p) = 1$$

Hence $p \in R_{\mathfrak{p}}$ (as it is ≥ 0) and p generates $\mathfrak{p}R_{\mathfrak{p}}$ (as $\nu_Y(p) = 1$). Next, if $\mathfrak{p}'$ is any other height 1 prime ideal, it correspond to a prime divisor Y', and as $p \in \kappa(X)^*$ we have

$$\nu_{Y'}(p) = 0$$

and so $p \in R_{\mathfrak{p}'}$. Then by lemma 4.3 .6 we have $p \in R$. Finally, let us show that p generates $\mathfrak{p}$. Let $g \in \mathfrak{p}$. Then:

$$\nu_Y(g) \geq 1 \quad \text{and} \quad \nu_{Y'}(g) \geq 0$$

Then for all prime divisors Y' (including Y):

$$\nu_{Y'}(g/p) \geq 0$$

Thus, $g/p \in R_{\mathfrak{p}'}$ for all height 1 prime ideals, implying by lemma 4.3.6 that $g/p \in R$, that is $g \in pR = (p)$. Hence $\mathfrak{p} \subseteq (p)$, and so $\mathfrak{p} = (p)$, as we sought to show.

Example 4.3.9: Divisor Class Groups

1. Affine space has trivial class group, $\mathrm{Cl}\mathbb{A}_k^n = 0$, as it is affine and $k[x_1, \ldots, x_n]$ is a UFD.
2. Let R be a Dedekind domain and consider Spec R. Then it is not a UFD, hence it must have a non-trivial divisor class group. The height 1 primes are precisely the primes of R, which as we know generate an abelian group known as the ideal class group. But this is just the divisor class group under a different name, and similarly with the principal divisors corresponding to elements of $\mathrm{Frac}(R)$. Hence, the ideal class group corresponds with the divisor class group!

It is natural to look for the divisor class group for projective space. The reader familiar with some differential geometry would hope that we recover the group of integers, and this is indeed the case. Recall that the degree of a hypersurface in $\mathbb{P}^n_k$ is the degree of the homogeneous polynomials defining it

Proposition 4.3.10: Characterizing Divisors of Projective Space

Let $X = \mathbb{P}^n_k$. For any divisor $D = \sum_i n_i Y_i$, define $\deg D = \sum_i n_i \deg Y_i$, and let H be the hyperplane given by x_0. Then:

1. for any $f \in \kappa(X)^*, \mathrm{Div}\, f = 0$
2. if D is a divisor of degree d, then $[D] = [dH]$ in $\mathrm{Cl}\mathbb{P}^n_k$
3. The degree functions give an isomorphism $\deg : \mathrm{Cl}\mathbb{P}^n_k \to \mathbb{Z}$

***Proof* :**

1. Let $S = k[x_0, \ldots, x_n]$ be the homogeneous coordinate ring of X and $f \in S$ a homogeneous element of order d. Then f factors into irreducibles
$$f = f_1^{n_1} \cdots f_r^{n_r}$$
Each f_i defines a hypersurface Y_i of degree d_i (i.e. an irreducible codimension 1 subscheme), and so
$$\mathrm{div}\, f = \sum n_y Y_i$$
hence $\deg f = d$. Note that f is not a rational function, it is a global section of the line bundle $\mathcal{O}(D)$, but $\mathrm{div}\, f$ is still well-defined by corollary 4.2.11. Now if $F = f/g$ is a rational function on $\mathbb{P}^n_k$ so that the degree of f, g are equal, then by the homomorphism property we have $\mathrm{div}\, F = \mathrm{div}\, f - \mathrm{div}\, g$, and so $\deg F = 0$

2. let $D = \sum_i n_i Y_i$ be any divisor of degree d. Then D can be written as the difference of effective divisors with degrees $d_1, d_2 : D_1 - D_2$ where $d = d_1 - d_2$. As the irreducible hypersurfaces in $\mathbb{P}^n_k$ correspond to homogeneous prime ideals of height 1 in S and these are principal (corollary 4.2.11), let $f_{1,j}$ correspond to $Y_{1,j}$ and $f_{2,j}$ correspond to $Y_{2,j}$, then let $f_i = \prod_{j=1}^{n_i} f_{i,j}$ for $i = 1, 2$. Then $D_1 = \mathrm{Div}\, f_1$ and $D_2 = \mathrm{Div}\, f_2$ so that $D = \mathrm{div}\, f_1/f_2$. Then:
$$D - dH = \mathrm{div}\, \frac{f_1}{f_2 x_0^d}$$
Hence the two are off by a principal divisor, meaning $[D] = [dH]$.

3. To complete the proof, if $[dH] \neq [d'H]$ then $d \neq d'$. Since $\deg H = 1$, we get an isomorphism:
$$\deg : \mathrm{Cl}\, X \to \mathbb{Z}$$
completing the proof.

Proposition 4.3.11: Divisor Class Group and Closed Subsets

Let X be a Noetherian integral separated scheme, regular in codimension 1. Let $Z \subseteq X$ be a proper closed subset and $U = X \setminus Z$. Then:

1. There is a surjective homomorphism $\operatorname{Cl} X \to \operatorname{Cl} U$ given by:
$$\sum n_i Y_i \mapsto \sum n_i (Y_i \cap U)$$
2. If $\operatorname{codim}(Z, X) \geq 2$, then $\operatorname{Cl} X \to \operatorname{Cl} U$ is an isomorphism
3. If Z is an irreducible subset of codimension 1, then there is an exact sequence:
$$\mathbb{Z} \to \operatorname{Cl} X \to \operatorname{Cl} U \to 0$$
where the first map is given by $1 \mapsto 1 \cdot Z$

Proof **:**

1. If Y is a prime divisor, then $Y \cap U$ is either empty or a prime-divisor. Then if $f \in \kappa(X)^*$ and $\operatorname{div} f = \sum n_i Y_i$, then we can consider f as as rational function on U by:
$$\operatorname{div}_U f = \sum n_i (Y_i \cap U)$$
Then this is certainly a homomorphism, and is surjective as the prime divisors of U are the restriction of its closure in X.
2. As $\operatorname{div} X$ is generated on the codimension 1 subsets, and removing a codimension 2 subset does not effect the Dimensionality of the other subsets, we have an isomorphism.
3. The kernel of $\operatorname{Cl} X \to \operatorname{Cl} U$ are all divisors whose support is contained in Z. Then as Z is irreducible, the kernel is the subgroup of $\operatorname{Cl} X$ generated by $1 \cdot Z$, completing the proof.

Example 4.3.12: More Divisor Class Groups

let $Y \subseteq \mathbb{P}^2_k$ be a degree d irreducible curve. Then combining the above observations we get:
$$\operatorname{Cl}(\mathbb{P}^2_k \setminus Y) = \mathbb{Z}/d\mathbb{Z}$$
Intuitively, By Bézout's theorem we have that any line L intersects a curve of degree d a total of d, counting multiplicity. Using this, can you piece together what the torsion in this group represents?

For a concrete example, let $\operatorname{Spec} R = \operatorname{Spec} k[x, y, z]/(xy - z^2)$ be the affine cone of $xy = 1$. Let's show that $\operatorname{Cl} \operatorname{Spec} R = \mathbb{Z}/2\mathbb{Z}$, which will be generated by the "ruling of the cone" $y = z = 0$. Let Y represent this subscheme. Visually:

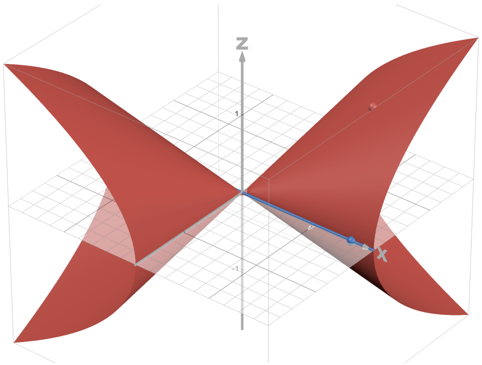


Figure 4.1: A ruling on the quadratic cone

As Y is irreducible and 1-codimension in $\operatorname{Spec} R$, Y is a prime divisor. Hence we can consider:

$$\mathbb{Z} \to \operatorname{Cl} X \to \operatorname{Cl}(\operatorname{Spec} R \setminus Y) \to 0$$

Now, on $\operatorname{Spec} R$, notice that Y can be cut out by the function y: $Y = V(y)$. It may then seem that $\operatorname{div} y = 1 \cdot Y$, however we in fact have

$$\operatorname{div} y = 2 \cdot Y$$

To see this, notice that $y = 0$ implies $z^2 = 0$. Then as z generates the maximal ideal at the generic point of Y, we see that $\nu_Y(y) = 2$. Now, certainly:

$$\operatorname{Spec} R \setminus Y \cong \operatorname{Spec} R_y \qquad R_y = \frac{k[x, y, z, y^{-1}]}{(xy - z^2)}$$

Then as $x = y^{-1}z^2$, we can eliminate x and reduce to:

$$R_y = k[y, z, y^{-1}]$$

This is now UFD, an thus:

$$\operatorname{Cl}(\operatorname{Spec} R \setminus Y) = 0$$

And so $\operatorname{Cl} X$ must be generated by Y, where $2 \cdot Y = 0$. Then:

$$\operatorname{Cl} X = \mathbb{Z}/2\mathbb{Z} \qquad \text{or} \qquad \operatorname{Cl} X = 0$$

If we show Y is not principal, then we have the former. As R is normal (classical exercise to show $k[x_1, ..., x_n, z]/(z^2 - f)$ is integrally closed where $\operatorname{char} k \neq 2$ and f is a square-free nonconstant polynomial), it is enough to show that the prime ideal of Y is not principal, that is $\mathfrak{p} = (y, z)$ is not principal. We reduce to the 3-dimensional vector space $\mathfrak{m}/\mathfrak{m}^2$ where $\mathfrak{m} = (x, y, z)$. Then as $\mathfrak{p} \subseteq \mathfrak{m}$, it is easy to see that $\mathfrak{p}$ in $\mathfrak{m}/\mathfrak{m}^2$ must be generated by the basis elements $\overline{y}$ and $\overline{z}$. But then $\mathfrak{p}$ cannot be principal, and hence we can conclude that

$$\operatorname{Cl} X = \mathbb{Z}/2\mathbb{Z}$$

Proposition 4.3.13: Class Group Stably Isomorphic

Let X be a Noetherian integral separated scheme, regular in codimension 1. Then $X \times \mathbb{A}^1_{\mathbb{Z}}$ also satisfies all these conditions, and:

$$\operatorname{Cl} X \cong \operatorname{Cl}\left(X \times \mathbb{A}^1_{\mathbb{Z}}\right)$$

Proof **:**
Certainly $X \times \mathbb{A}^1$ is Noetherian, integral, and separated (verify this if this is not clear). To see that it is regular in codimension one, first notice that there are two kinds of points of codimension one on $X \times \mathbb{A}^1$.

1. A point x whose image in X is a point y of codimension one. In this case x is the generic point of $\pi^{-1}(y)$, where $\pi : X \times \mathbb{A}^1 \to X$ is the projection. Its local ring is $\mathcal{O}_x \cong \mathcal{O}_y[t]_{\mathfrak{m}}$, which is clearly a discrete valuation ring, since $\mathcal{O}_y$ is. The corresponding prime divisor $\overline{\{x\}}$ is just $\overline{\pi^{-1}(\{y\})}$.
2. A point $x \in X \times \mathbb{A}^1$ of codimension one, whose image in X is the generic point of X. In this case $\mathcal{O}_x$ is a localization of $K[t]$ at some maximal ideal, where K is the function field of X. It is a discrete valuation ring because $K[t]$ is a principal ideal domain.

Thus $X \times \mathbb{A}^1$ also regular in codimension 1. For the rest of this proof, let's call these points *type 1* and *type 2*.

Now, define a map

$$\operatorname{Cl} X \to \operatorname{Cl}\left(X \times \mathbb{A}^1\right) \qquad \sum n_i Y_i \mapsto \pi^* D = \sum n_i \pi^{-1}(Y_i)$$

If $f \in \kappa(X)^*$, then $\pi^*((f))$ is the divisor of f considered as an element of $\kappa(X)(t)$, the function field of $X \times \mathbb{A}^1$ (from now on, let $K = \kappa(X)$). Thus we have a homomorphism $\pi^* : \operatorname{Cl} X \to \operatorname{Cl}\left(X \times \mathbb{A}^1\right)$.

To show π^* is injective, suppose $D \in \operatorname{div} X$, and $\pi^* D = (f)$ for some $f \in K(t)$. Since $\pi^* D$ involves only prime divisors of type 1, f must be in K, otherwise we could write $f = g/h$, with $g, h \in K[t]$, relatively prime. If g, h are both not in K, then $\operatorname{div} f$ will involve some prime divisor of type 2 on $X \times \mathbb{A}^1$. Now if $f \in K$, it is clear that $D = \operatorname{div} f$, so π^* is injective.

To show that π^* is surjective, it will be sufficient to show that any prime divisor of type 2 on $X \times \mathbb{A}^1$ is linearly equivalent to a linear combination of prime divisors of type 1. So let $Z \subseteq X \times \mathbb{A}^1$ be a prime divisor of type 2. Localizing at the generic point of X, we get a prime divisor in $\operatorname{Spec} K[t]$, which corresponds to a prime ideal $\mathfrak{p} \subseteq K[t]$. This is principal, so let f be a generator. Then $f \in K(t)$, and the divisor of f consists of Z plus perhaps something purely of type 1. It cannot involve any other prime divisors of type 2. Thus Z is linearly equivalent to a divisor purely of type 1, completing the proof.

4 more examples in Hartshorne

4.3.2 Cartier Divisors

We shall now generalize this to scheme. The problem with working over a general scheme is that we don't have a function field, and we can contain zero-divisors. We must thus be a bit more careful with our definition. The key intuition we want to preserve is how far away a scheme is from being "principal". To that end, we shall define a divisor as elements that are "locally principal". In most cases, we focus only on divisors that have no poles. For full generality, we shall give the general definition that includes poles and build up to the most common use case.

Definition 4.3.14: Maximal Quotient Sheaf

Let X be a scheme. Then for each affine open $\operatorname{Spec} R = U \subseteq X$, take all non zero-divisors $S \subseteq \Gamma(U, \mathcal{O}_X)$ and take $S^{-1}\Gamma(U, \mathcal{O}_X) = K(U)$, the *maximal quotient field* of $\Gamma(U, \mathcal{O}_X)$. Then this forms a presheaf, whose sheafification is called the *total quotient ring* of $\mathcal{O}_X$. This sheaf will be denoted $\mathscr{K}_X$,

Let $\mathscr{K}_X^*$ denote the abelian group sheaf of invertible elements for each open subset, and let $\mathcal{O}_X^*$ denote the abelian group sheaf of all invertible elements.

If X is integral, then this recovers the usual rational functions we'd define on each open subsets.

Definition 4.3.15: Cartier Divisor

Let $(X, \mathcal{O}_X)$ be a scheme with maximal quotient sheaf $\mathscr{K}$. Then a *Cartier divisor* is an element of the global section of $\mathscr{K}^*/\mathcal{O}_X^*$:

$$\Gamma(X, \mathscr{K}^*/\mathcal{O}_X^*)$$

The *principal Cartier divisors* are given by the image of The natural map

$$\Gamma(X, \mathscr{K}^*) \to \Gamma(X, \mathscr{K}^*/\mathcal{O}_X^*)$$

An equivalent way of describing Cartier divisors that is very useful is by taking an open cover $\{U_i\}$ of X and then choosing elements $f_i \in \Gamma(U_i, \mathscr{K}^*)$ such that for each overlap $U_i \cap U_j$

$$\frac{f_i}{f_j} \in \Gamma(U_i \cap U_j, \mathcal{O}_X^*)$$

or equivalently, $f_i = f_j$ up to a multiple of a section in $\Gamma(U_i \cap U_j, \mathcal{O}_Z^*)$. The idea is that we can loosen the definition of a function to only need to be equal on overlap by a unit, as units do no contribute to the "zero set".

Proposition 4.3.16: Cartier Divisors To Weil Divisors

Then there exists a homomorphism:

$$\operatorname{CDiv}(X) \to \operatorname{WDiv}(X)$$

Proof **:**
Let $f \in \mathrm{CDiv}\,(X) = \Gamma(X, \mathscr{K}^*/\mathcal{O}_X^*)$. By the above discussion, f can be given by a collection not $f_i \in K_i = \Gamma(U_i, \mathscr{K}^*)$. Then we can map each f_i to their corresponding Weil Divisor $\operatorname{div} f$ in U_i. Since K_i and K_j have the same zero's on the intersection, we get that we can glue these to get a Weil divisor D on X. Then it is routine to check this is a homomorphism, completing the proof.

Proposition 4.3.17: Weil and Cartier Equivalence Under Right Conditions

Let X be an integral separated Noetherian schemes, where all the local rings are UFD's. Then

$$\mathrm{WDiv}\,(X) \cong \mathrm{CDiv}\,(X)$$

and, under this same isomorphism, the principal Weil divisors correspond to the principal Cartier divisors

Proof **:**
As X is a UFD at each stalk, it must be normal, and hence as shown in example 4.3.2 it is regular in codimension 1. As X is integral $\mathscr{K}$ is just the constant sheaf that corresponds to the function field $\kappa(X)$ of X, which we shall simply denote as K in this proof.

Let us start by showing a Cartier divisor gives a Weil divisor. By our remark, we can choose an open cover $\{U_i\}$ of X an element $f_i \in \Gamma(U_i, \mathscr{K}^*)$ that agree on intersections up to a multiple of $\mathcal{O}_X^*$. Now, for each prime divisor Y, take it's coefficient to be $\nu_Y(f_i)$ if $Y \cap U_i$. If $Y \cap U_j$ for some other j, then notice that f_i/f_j is invertible on $U_i \cap U_j$, and so must have trivial valuation, $\nu_Y(f_i/f_j)$, hence $\nu_Y(f_i) = \nu_Y(f_j)$. Thus, we can get:

$$D = \sum_i \nu_Y(f_i) Y$$

where the sum is finite as X is Noetherian.

Conversely, let D be a Weil divisor. Take any point $x \in X$. Then D induces a Weil divisor D_x on $\operatorname{Spec} \mathcal{O}_{X,x}$ ($\sum_i n_i Y$ becomes $\sum_i n_y Y \cap \operatorname{Spec} \mathcal{O}_{X,x}$). As $\mathcal{O}_{X,x}$ is a UFD, we have that this divisor be principal, and so $D_x = \operatorname{div} f_x$ for some $f_x \in K$ (as we are working with the constant sheaf). Now, $\operatorname{div} f_x$ and D on X both restrict to $\operatorname{div} f_x$ on $\operatorname{Spec} \mathcal{O}_{X,x}$, thus they can only differ by prime divisors that do not pass through x. By the Noetherian property and that they are codimension 1, hence can only be finitely many nonzero such divisors, so there is an open neighborhood U_x of x such that D and $\operatorname{div} f_x$ have the same restriction to U_x. Finally, we may cover X with such U_x. I claim that the functions f_x now give the Cartier divisor. Indeed, if f, f' gave the same Weil divisor on an open set U, then $f/f' \in \Gamma(U, \mathcal{O}_X^*)$ as X is normal (recall from the proof of proposition 4.3.8), and so these are indeed well defined Cartier divisors.

Now, it should be clear that these two construction are inverses of each other and certainly map principal to principal, giving us our bijective correspondence as we sought to show.

If the conditions are not met, then we may get a different groups, as we shall soon show. Note that regular local ring are UFD's (see [8]) and so this equivalence applies to regular integral separated Noetherian schemes. If X is a normal scheme, then it is not necessarily locally factorial. Instead, the correspondence goes thusly:

Definition 4.3.18: Locally Principal Weil Divisor

let X be a normal scheme. Then D is *locally principal* if X can be covered by open sets U such that $D|_U$ is principal for each U.

In this case, Cartier divisors correspond to locally principal Weil Divisors.

Definition 4.3.19: Cartier Divisor Class Group

Let X be a scheme. Then the *Cartier divisor class group* is the Cartier divisors divided by the principal Cartier divisors, and is denoted

$$\operatorname{CaCl} X$$

Not to be confused with calcium-chloride which almost has the same symbol: CaCl_2.

Example 4.3.20: Cartier Divisors

1. Go over Hartshorne p.142
2. This should ring bells of the definition of the ideal group; recall that the ideal group of $\mathbb{Z}$ was isomorphic to $\mathbb{Q}^*/\mathbb{Z}^* \cong \mathbb{Q}^*_{\geq 0}$. If we take $\operatorname{Spec}\mathbb{Z}[\sqrt{-5}]$, Then it's total field is $\mathbb{Q}(\sqrt{-5})$ and it's group of units is ± 1, hence elements of $\Gamma(\operatorname{Spec}\mathbb{Z}[\sqrt{-5}], \mathbb{Q}(\sqrt{-5})_{>0})$ correspond to Cartier divisors. To see this in action, consider $(2, 1+\sqrt{-5}) \subseteq \mathbb{Z}[\sqrt{-5}]$ which we know is not principal. Taking the affine cover $D(2)$ and $D(1+\sqrt{-5})$ (as these correspond to coprime ideals), we can choose:

$$\frac{2}{1+\sqrt{-5}} \in \Gamma(D(1+\sqrt{-5}), \mathscr{K}^*) \qquad \text{and} \qquad \frac{1+\sqrt{-5}}{2} \in \Gamma(D(2), \mathscr{K}^*)$$

and we see that on the intersection:

$$\frac{2}{1+\sqrt{-5}} = \frac{1+\sqrt{-5}}{2}$$

up to a multiplication of -1 (which is in $\mathcal{O}^*_{\operatorname{Spec}\mathbb{Z}[\sqrt{-5}]}$), and hence this gives a Cartier divisor.

If we impose the same conditions as we did for Weil divisors, we shall see that these two groups are equivalent. In fact, we can weaken the codimension 1 regularity condition a bit:

4.3.3 Picard Group

Throughout the discussion of the divisors on projective schemes, it was clear that there is a strong connection to the line bundles $\mathcal{O}(d)$. In this section, we shall make this precise. This shall also be familiar to number theorists who recall that fractional ideals are finitely generated R-module J such $rJ \subseteq R$.

Proposition 4.3.21: Line Bundles K-theory Properties

let $\mathcal{L}, \mathcal{M}$ be invertible sheaves (line bundles) on a ringed space X. Then:

1. $\mathcal{L} \otimes \mathcal{M}$ is invertible
2. There exists an invertible sheaf $\mathcal{L}^{-1}$ such that $\mathcal{L} \otimes \mathcal{L}^{-1} \cong \mathcal{O}_X$

The way to think about the tensor product between line bundles is that they happen stalk-wise:

$$(\mathcal{L} \otimes \mathcal{M})_p \cong \mathcal{L}_P \otimes_p \mathcal{M}_p$$

which can be thought of as "multiplying" the values:

$$(s \otimes t)(p) = s(p) \otimes t(p)$$

As line bundles are one-dimensional, this shall not add dimensions (think of multiplying scalar vs wedging differential forms), hence why the tensor product stays the same dimension.

***Proof* :**

1. As both are locally free, on the appropriate restriction we get:
$$\mathcal{O}_X \otimes \mathcal{O}_X \cong \mathcal{O}_X$$
2. Take $\mathcal{L}^{-1} = \mathcal{H}om(\mathcal{L}, \mathcal{O}_X)$. Then it is an elementary algebra exercise to verify that:
$$\mathcal{L}^{-1} \otimes \mathcal{L} \cong \mathcal{H}om(\mathcal{L}, \mathcal{L}) \cong \mathcal{O}_X$$

Definition 4.3.22: Picard Group

Let X be a ringed space. Then the *picard group* is the group of invertible sheaves under $\otimes_{\mathcal{O}_X}$. The picard group is denoted

$$\operatorname{Pic} X$$

We shall see in ref:HERE that $\operatorname{Pic} X = H^1(X, \mathcal{O}_X^*)$.

Definition 4.3.23: Line Bundles Associated To Cartier Divisor

Let D be a Cartier divisor on a scheme X, represent by $\{U_i, f_i\}$. Then the *sheaf associated to* D as a sub-sheaf $\mathcal{L}(D) \subseteq \mathcal{K}$ for each U_i, $\mathcal{L}(D)(U_i)$ is a sub $\mathcal{O}_X$-module of $\mathcal{K}(U)$[a] generated by f_i^{-1} on U_i.

[a]Not that X need not be integral, hence we need not have a global rational function field K

Note that this is well-defined since on intersections $U_i \cap U_j$, f_i/f_j is invertible, and hence f_i^{-1} and f_j^{-1} will generate the same $\mathcal{O}_X$-module. To return to the usual $\operatorname{Spec} \mathbb{Z}[\sqrt{-5}]$, we see that $(2, 1+\sqrt{-5})$ is a $\mathcal{O}_X$-modules that is generated by $(\frac{2}{1+\sqrt{-5}})$ and $(\frac{1+\sqrt{-5}}{2})$ on their respective open sets, and hence we have a sheaf associated to the divisor mentioned earlier! It is often simpler to picture $\mathcal{L}(D)$ as follows:

$$\mathcal{L}(D)(U) = \{f \in \kappa(X)^* \mid \operatorname{div}(f) + D \geq 0 \text{ in } U\} \cup \{0\}$$

Intuitively, we see that $\mathcal{L}(D)$ at an open set U contains all functions that have have poles that are bounded by D, and zeros' that are prescribed by D (it has at most as many poles as D, and at least as many zeros as D); in fact this was historically the use of divisors on Riemann surfaces.

Proposition 4.3.24: Correspondence Between Line Bundles and Cartier Divisors

Let X be a scheme. Then $\mathcal{L}(D)$ is an invertible sheaf, and the map the map $D \mapsto \mathcal{L}(D)$ is an injective homomorphism that maps that respects principal ideals.

1. $\mathcal{L}(D_1 - D_2) = \mathcal{L}(D_1) \otimes \mathcal{L}(D_2)^{-1}$
2. $D_1 \sim D_2$ if and only if $\mathcal{L}(D_1) \cong \mathcal{L}(D_2)$ as invertible sheaves

As a consequence, we have an injective homomorphism:

$$\operatorname{CaCl} X \to \operatorname{Pic} X$$

Note how this pairs well with the intuition that the tensor product multiplies the values of the functions: it shall "gain" the zero's poles prescribed by each divisor.

***Proof* :**
Let's show the map $D \mapsto \mathcal{L}(D)$ is injective and maps to an invertible sheaf. First showing a Cartier divisor, let $\{U_i\}$ be an open covering and take $f_i \in \Gamma(U_i, \mathcal{K}^*)$ that respects the conditions of Cartier divisor. Then we can define:

$$\mathcal{O}_{U_i} \to \mathcal{L}(D)|_{U_i} \qquad 1 \mapsto f_i^{-1}$$

As $\mathcal{L}(D)|_{U_i}$ is generated by f_i^{-1} this is certainly an isomorphism. Then $\mathcal{L}(D)$ is indeed an invertible sheaf. Then for any invertible sheaf $\mathcal{L}(D)$, we can recover the Cartier divisor D (and its embedding in $\mathcal{K}$) by taking f_i on U_i to be the inverse of the local generators of $\mathcal{L}(D)$ (which is unique up to element of $\mathcal{O}_X^*$). Then this will by construction give us Cartier divisors, showing injectivity

Let us next show that the group structure is preserved. Let D_1 and D_2 be locally defined by collections f_i and g_i respectively. Then $\mathcal{L}(D - 1 - D_2)$ is locally generated by $f_i^{-1} g_i$, and so by some module theory we see that:

$$\mathcal{L}(D_1 - D_2) = \mathcal{L}(D_1) \cdot \mathcal{L}(D_2)^{-1}$$

Then with some elementary algebra we see that $\mathcal{L}(D_1) \cdot \mathcal{L}(D_2)^{-1} \cong \mathcal{L}(D_1) \otimes \mathcal{L}(D_2)^{-1}$

Finally, to show that taking the quotient by principals induces isomorphism on the sheaves of divisors (and vice-versa), by our above work it suffices to show that $D = D_1 - D_2$ is principal if and only if $\mathcal{L}(D) \cong \mathcal{O}_X$. If D is principal, then it is defined by $f \in \Gamma(X, \mathcal{K}^*)$, hence $\mathcal{L}(D)$ is globally generated by f^{-1}, so that $1 \mapsto f^{-1}$ gives the isomorphism:

$$\mathcal{O}_X \cong \mathcal{L}(D)$$

Conversely, given such an isomorphism, certainly the image of 1 gives an element of $\Gamma(X, \mathcal{K}^*)$, and so we have that D is principal, completing the proof.

Note the map need not be surjective, as there may be invertible sheaves on X which are not isomorphic to any invertible sheaf of $\mathcal{K}$ (Hartshorne gives a reference to check). If X is projective over a field or if X is integral, then this is an isomorphism.

Proposition 4.3.25: Isomorphism of Cartier Class Group and Picard Group

Let X be an integral scheme. Then the homomorphism $\operatorname{CaCl} X \to \operatorname{Pic} X$ given in proposition 4.3.24 is an isomorphism.

***Proof*:**
What's left to show is surjectivity. Let $\mathcal{L}$ be an invertible sheaf on $\mathcal{O}_X$. We must show it is isomorphic to a sub-sheaf of $\mathcal{K}$. As X is integral, we have a function field K for X. Consider:

$$\mathcal{L} \otimes_{\mathcal{O}_X} \mathcal{K}$$

On any open U where $\mathcal{L} \cong \mathcal{O}_X$, w have

$$\mathcal{L} \otimes \mathcal{K} \cong \mathcal{K}$$

so this is a constant sheaf on U. As X is irreducible, any sheaf whose restriction to each open set of a covering of$X$ is constant *must* be a constant sheaf (by the injectivity of the restriction maps). Hence $\mathcal{L} \otimes \mathcal{K}_*$ is isomorphic to the constant sheaf $\mathcal{K}$, and so:

$$\mathcal{L} \to \mathcal{L} \otimes \mathcal{K} \cong \mathcal{K}$$

gives $\mathcal{L}$ as a sub-sheaf of $\mathcal{K}$, as we sought to show.

Thus, under the right conditions on X:

Corollary 4.3.26: Divisor Class Group and Picard Group

Let X be a Noetherian, integral, separated, locally factorial scheme. Then

$$\operatorname{Cl} X \cong \operatorname{Pic} X$$

***Proof*:**
Follow from proposition 4.3.25 and proposition 4.3.17.

Corollary 4.3.27: Classifying Line Bundles On Projective Space

Let $X = \mathbb{P}^n_k$ for some field k. Then every invertible sheaf on X is isomorphic to $\mathcal{O}(d)$ for some $d \in \mathbb{Z}$

***Proof*:**
By proposition 4.3.10, $\operatorname{Cl} X \cong \mathbb{Z}$, and by corollary 4.3.26 $\operatorname{Pic} X \cong \mathbb{Z}$. As the generators of $\operatorname{Cl} X$ is a hyperplane, which corresponds to the invertible sheaf $\mathcal{O}(1)$, $\operatorname{Pic} X$ is a free group generated by $\mathcal{O}(1)$, and every invertible sheaf $\mathcal{L}$ is isomorphic to $\mathcal{O}(d)$ for some $d \in \mathbb{Z}$, as we sought to show.

This level of generality is what Hartshorne covers, however the Stacks project (as of writing this book) focus on *effect Cartier divisors*. Hence we shall take a moment to look at those:

Definition 4.3.28: Effective Cartier Divisor

Let X be a scheme and D a Cartier divisor. Then D is called *effective* if it can be represented by $\{(U_i, f_i)\}$ where all $f_i \in \Gamma(U_i, \mathcal{O}_{U_i})$.

The *associated subscheme of codimension* 1 is the closed subscheme defined by the sheaf of ideals $\mathscr{I}$ which is locally generated by f_i.

By the definition, we see almost immediately that there is an injective correspondence between effective Cartier divisors on X and *locally principal* closed subschemes (subschemes whose sheaf of ideals is locally generated by a single element). If X is an integral separated Noetherian locally factorial Scheme, then the Cartier divisors correspond to the Wald divisors, and it is not too hard to show that the effective Cartier divisors correspond to the effective Weil divisors.

Proposition 4.3.29: Effective Cartier Divisors and Line Bundles

Let D be an effective Cartier divisor on a scheme X and Y the associated locally principal closed subscheme. Then:

$$\mathscr{I}_Y \cong \mathcal{L}(-D)$$

***Proof* :**
As $\mathcal{L}(-D)$ is the sub sheaf of $\mathcal{K}$ generated by f_i, this is actually a sub-sheaf of $\mathcal{O}_X$, and it is by construction the ideal sheaf of $\mathscr{I}_Y$, completing the proof.

4.4 Maps to Projective Space

It has long been foreshadowed that many of the common schemes that are worked with embed into projective space. In this section, we shall give the right conditions for finding such a mapping. We shall see that a morphism of a scheme X to a projective space is determined by finding an invertible sheaf $\mathcal{L}$ on X and a set of global sections.

Note that maps from projective space are better determined using cohomological properties (see chapter 5); theorem 3.4.19 determined that all proper schemes can be covered by projective schemes, showing how they are natural in some sense of the word.

4.4.1 Build-up: Global Sections and Finite Generation

Definition 4.4.1: Twisting Sheaf On $\mathbb{P}^n_X$

Let X be a scheme. Then the twisting sheaf on $\mathcal{O}(1)$ on $\mathbb{P}^n_X$ is $g^*(\mathcal{O}(1))$ where:

$$g : \mathbb{P}^n_X \to \mathbb{P}^n_{\mathbb{Z}}$$

where $\mathbb{P}^n_X = X \times_{\mathbb{Z}} \mathbb{P}^n_{\mathbb{Z}}$ by definition 3.3.32.

Definition 4.4.2: Very Ample Sheaf

Let X be a Y-scheme, and $\mathcal{L}$ a line bundle on X. Then $\mathcal{L}$ is said to be very ample relative to Y if there is a locally closed embedding $\iota : X \to \mathbb{P}^n_Y$ for some n such that

$$\iota^*(\mathcal{O}(1)) \cong \mathcal{L}$$

If Y is Noetherian, then X is projective if and only if X is both proper and has a very ample sheaf relative to Y. Indeed, if X is projective over Y,
Recall proposition 4.1 .28 that showed that if if X a R-scheme and $f_0, \ldots, f_n$ are functions on X (global sections) with no common zeros. Then there exists a scheme morphism $[f_0 : \cdots f_n] : X \to \mathbb{P}^n_R$. It was commented that not all maps to projective space arise in this form, for example $\mathrm{id} : \mathbb{P}^n_k \to \mathbb{P}^n_k$. In this section, we remedy that. To start, we shall look for sheaves that are generated by their global section. The most important example will be when $S = k[x_0, \ldots, x_n]$ and $X = \operatorname{Proj} S$; then the global section of a coherent sheaf must be finitely generated.

Definition 4.4.3: Generated By Global Sections

Let X be a scheme and $\mathscr{F}$ an $\mathcal{O}_X$-module. Then $\mathscr{F}$ is generated by global sections if there is a family of global sections $\{s_i\}_{i \in I} \subseteq \Gamma(X, \mathscr{F})$ such that for each $x \in X$, the images of s_i in the stalk $\mathscr{F}_x$ generate $\mathscr{F}_x$ as an $\mathcal{O}_{X,x}$-module.

Naturally, $\mathscr{F}$ is generated by global sections iff $\mathscr{F}$ can be written as the quotient of a free-sheaf.

Example 4.4.4: Generated By Global Sections

1. Let $X = \operatorname{Spec} R$. Then any $\mathscr{F} = \widetilde{M}$ is generated by global sections, simply pick generators for M as an R-module
2. Let $X = \operatorname{Proj} S$ for a graded ring S genrated by S_1 as an S_0-algebra. Then the elements of S_1 can give global sections of $\mathcal{O}_X(1)$ that generate it.
3. The scheme $\mathcal{O}_X(-1)$ is not generated by global sections, as it has an empty global section.

Theorem 4.4.5: Global Section Generation of twisted Sheaf over Proj Noetherian

SchemeLet X be a projective scheme over a Noetherian ring R, let $\mathcal{O}(1)$ be a very ample invertible sheaf on X, and let $\mathscr{F}$ be a coherent $\mathcal{O}_X$-module. Then there is an integer n_0 such that for all $n \geq n_0$, the sheaf $\mathscr{F}(n)$ can be generated by a finite number of global sections.

Proof **:**
Let $i : X \to \mathbb{P}_R^r$ be a closed embedding of X into a projective space over R, such that $i^*(\mathcal{O}(1)) = \mathcal{O}_X(1)$. Then $i_*\mathscr{F}$ is coherent on $\mathbb{P}_R^r$ as closed embedding are finite morphisms and projective schemes are Noetherian, and $i_*(\mathscr{F}(n)) = (i_*\mathscr{F})(n)$ by proposition 4.2.5 and $\mathscr{F}(n)$ is generated by global sections if and only if $i_*(\mathscr{F}(n))$ is (in fact, their global sections are the same), so this can be reduced to the case $X = \mathbb{P}_R^r = \operatorname{Proj} R[x_0, \ldots, x_r]$.
Now cover X with the open sets $D_+(x_i), i = 0, \ldots, r$. Since $\mathscr{F}$ is coherent, for each i there is a finitely generated module M_i over $B_i = R[x_0/x_i, \ldots, x_n/x_i]$ such that $\mathscr{F}|_{D_+(x_i)} \cong \tilde{M}_i$. For each i, take a finite number of elements $s_{ij} \in M_i$ which generate this module. By lemma 4.1.12 there is an integer n such that $x_i^n s_{ij}$ extends to a global section t_{ij} of $\mathscr{F}(n)$. As usual, we take one n to work for all i, j. Now $\mathscr{F}(n)$ corresponds to a B_i-module M_i' on $D_+(x_i)$, and the map $x_i^n : \mathscr{F} \to \mathscr{F}(n)$ induces an isomorphism of M_i to M_i'. So the sections $x_i^n s_{ij}$ generate M_i', and hence the global sections $t_{ij} \in \Gamma(X, \mathscr{F}(n))$ generate the sheaf $\mathscr{F}(n)$ everywhere, as we sought to show.

Corollary 4.4.6: Coherent Sheaf is quotient of "free twisted sheaves"

Let X be projective R-scheme. Then any coherent sheaf $\mathscr{F}$ on X can be written as a quotient of a sheaf $\mathcal{E}$, where $\mathcal{E}$ is a finite direct sum of twisted structure sheaves $\mathcal{O}(n_i)$ for various integers n_i.

Proof **:**
Let $\mathscr{F}(n)$ be generated by a finite number of global sections. Then we have a surjection $\bigoplus_{i=1}^r \mathcal{O}_X \to \mathscr{F}(n) \to 0$. Tensoring with $\mathcal{O}_X(-n)$ we obtain a surjection $\bigoplus_{i=1}^r \mathcal{O}_X(-n) \to \mathscr{F} \to 0$ as required.

From this, we can now classify a large number of coherent sheaves of on projective space:

Theorem 4.4.7: Finite Generation of Coherent Sheaf Over Projective Scheme

Let k be a field, R a finitely generated k-algebra, $X = \mathbb{P}_R^n$, and $\mathscr{F}$ a coherent $\mathcal{O}_X$-module. Then $\Gamma(X, \mathscr{F})$ is a finitely generated R-module. If $R = k$, then $\Gamma(X, \mathscr{F})$ is a finite dimensional k-vector space.

This proof can be thought of the generalization of the proof that $\mathcal{O}_{\mathbb{P}_k^n}(\mathbb{P}_k^n) = k$. Note too that this is not true if the space is not projective, for example over $\mathbb{A}_k^n$ for any n the resulting space is not a finite dimensional k-vector space, as $\Gamma\left(\mathbb{A}_k^n, \mathcal{O}_{\mathbb{A}_k^n}\right) = k[x]$ which is infinite dimensional!

***Proof* :**
Let $X = \operatorname{Proj} S$, where S is a graded ring with $S_0 = R$ which is finitely generated by S_1 as an S_0-algebra, corollary 4.2 .10 . Let M be the graded S-module $\Gamma_*(\mathscr{F})$. Then by proposition 4.2.9, $\tilde{M} \cong \mathscr{F}$. On the other hand, by theorem 4.4.4 for n sufficiently large, $\mathscr{F}(n)$ is generated by a finite number of global sections in $\Gamma(X, \mathscr{F}(n))$. Let M' be the submodule of M generated by these sections. Then M' is a finitely generated S-module. Furthermore, the inclusion $M' \hookrightarrow M$ induces an inclusion of sheaves $\tilde{M}' \hookrightarrow \tilde{M} = \mathscr{F}$. Twisting by n we have an inclusion $\tilde{M}'(n) \hookrightarrow \mathscr{F}(n)$ which is actually an isomorphism, because $\mathscr{F}(n)$ is generated by global sections in M'. Twisting by $-n$ we find that $\tilde{M}' \cong \mathscr{F}$. Thus $\mathscr{F}$ is the sheaf associated to a finitely generated S-module, and so we have reduced to showing that if M is a finitely generated S-module, then $\Gamma(X, \tilde{M})$ is a finitely generated R-module.
Now, recall from [8] that there is a finite filtration

$$0 = M^0 \subseteq M^1 \subseteq \ldots \subseteq M^r = M$$

of M by graded submodules, where for each $i, M^i/M^{i-1} \cong (S/\mathfrak{p}_i)(n_i)$ for some homogeneous prime ideal $\mathfrak{p}_i \subseteq S$, and some integer n_i. This filtration gives a filtration of $\tilde{M}$, and the short exact sequences

$$0 \to \widetilde{M^{i-1}} \to \widetilde{M^i} \to \widetilde{M^i/M^{i-1}} \to 0$$

give rise to left-exact sequences

$$0 \to \Gamma\left(X, \widetilde{M^{i-1}}\right) \to \Gamma\left(X, \widetilde{M^i}\right) \to \Gamma\left(X, \widetilde{M^i/M^i - 1}\right)$$

Thus to show that $\Gamma(X, \tilde{M})$ is finitely generated over R, it will be sufficient to show that $\Gamma\left(X, (S/\mathfrak{p})^{\sim}(n)\right)$ is finitely generated, for each $\mathfrak{p}$ and n. Thus we have reduced to the following special case: Let S be a graded integral domain, finitely generated by S_1 as an S_0-algebra, where $S_0 = R$ is a finitely generated integral domain over k. Then $\Gamma\left(X, \mathcal{O}_X(n)\right)$ is a finitely generated R-module, for any $n \in \mathbf{Z}$.
Let $x_0, \ldots, x_r \in S_1$ be a set of generators of S_1 as an R-module. Since S is an integral domain, multiplication by x_0 gives an injection $S(n) \to S(n+1)$ for any n. Hence there is an injection $\Gamma\left(X, \mathcal{O}_X(n)\right) \to \Gamma\left(X, \mathcal{O}_X(n+1)\right)$ for any n. Thus it is sufficient to prove $\Gamma\left(X, \mathcal{O}_X(n)\right)$ finitely generated for all sufficiently large n, say $n \geq 0$.
Let $S' = \bigoplus_{n \geq 0} \Gamma\left(X, \mathcal{O}_X(n)\right)$. Then S' is a ring, containing S, and contained in the intersection $\bigcap S_{x_i}$ of the localizations of S at the elements $x_0, \ldots, x_r$.
Next, let's show that S' is integral over S. Let $s' \in S'$ be homogeneous of degree $d \geq 0$. Since $s' \in S_{x_i}$ for each i, we can find an integer n such that $x_i^n s' \in S$. Choose one n that works for all i. Since the x_i generate S_1, the monomials in the x_i of degree m generate S_m for any m. So by taking a larger n, we may assume that $ys' \in S$ for all $y \in S_n$. In fact, since s' has positive degree, we can say that for any $y \in S_{\geq n} = \bigoplus_{e > n} S_e, ys' \in S_{\geq n}$. Now it follows inductively, for any $q \geq 1$ that $y \cdot (s')^q \in S_{\geq n}$ for any $y \in S_{\geq n}$. Take for example $y = x_0^n$. Then for every $q \geq 1$ we have $(s')^q \in (1/x_0^n) S$. This is a finitely generated sub- S-module of the quotient field of S'. It follows by a well-known criterion for integral dependence (see [8]), that s' is integral over S. Thus S' is contained in the integral closure of S in its quotient field.
To complete the proof, since S is a finitely generated k-algebra, by commutative algebra we know that S' will be a finitely generated S-module. It follows that for every n, S_n is a finitely generated

S_0-module, as we sought to show.
In fact, this proof shows that $S_n = S'_n$ for all sufficiently large n, see exercises ref:HERE

Note that it was not technically needed that R is a finitely generated k-algebra, but just a "Nagata ring". See ref:HERE for a generalization. As almost all (Noetherian) rings are Nagata, this isn't much of a generalization.

Corollary 4.4.8: Pushforward of Coherent Sheaf by Projective Morphism

Let $f : X \to Y$ be a projective morphism of schemes of finite type over a field k. Let $\mathscr{F}$ be a coherent sheaf on X. Then $f_*\mathscr{F}$ is coherent on Y.

Recall this is not true if the map is not projective, see example 4.6

Proof **:**
The question is local on Y, so we may assume $Y = \operatorname{Spec} R$, where R is a finitely generated k-algebra. Then in any case, $f_*\mathscr{F}$ is quasi-coherent by proposition 4.1.14, so $f_*\mathscr{F} = \Gamma(Y, f_*\mathscr{F})^\sim = \Gamma(X, \mathscr{F})^\sim$. But $\Gamma(X, \mathscr{F})$ is a finitely generated R-module by the theorem, so $f_*\mathscr{F}$ is coherent.

This result shall have a cohomological proof which will allow us to generalize to proper morphisms, see ref:HERE.

4.4.2 Characterizing Morphisms to Projective Space

Let R be a fixed ring and $\mathbb{P}^n_R = \operatorname{Proj} R[x_1, ..., x_n]$ be our projective space over R. Then $\mathbb{P}^n_R$ has an invertible sheaf, denoted $\mathcal{O}(1)$. Let $x_1, ..., x_n$ be the homogeneous coordinates that generate the global section $\Gamma(\mathbb{P}^n_R, \mathcal{O}(1))$. Note that the images of $x_1, ..., x_n$ generate the stalks $\mathcal{O}(1)_p$ of the sheaf $\mathcal{O}(1)$ as a module over the local ring $\mathcal{O}_p$.

Now, let X be any R-scheme, and consider the R-morphism $\varphi : X \to \mathbb{P}^n_R$. Then it must be that $\varphi^*(\mathcal{O}(1)) = \mathcal{L}$ is an invertible sheaf on X, and the global sections $s_1, ..., s_n$ where $s_i = \varphi^*(x_i)$ generate $\mathcal{L}$.

The converse, that an invertible sheaf $\mathcal{L}$ with chosen global sections s_i determine φ, is also true:

Theorem 4.4.9: Determining Map To Projective Space

here

Proof **:**

4.5 Differentials

We next build-up to defining differential forms on a scheme. This shall give us a lot of information about the structure of a scheme and is one of the first computational tools you can use when studying a new algebraic object (similarly to how differential forms give a lot of information on manifolds)

Definition 4.5.1: A-derivation

Let A be a commutative ring, B an A-algebra, and M a B-module. Then an A-derivation of B into M is a map $d : B \to M$ such that

1. $d(b + b') = d(b) + d(b')$
2. $d(bb') = d(b)b' + bd(b')$
3. $d(a) = 0$ for all $a \in A$

Definition 4.5.2: Module of Relative Differential Forms

Let A be a commutative ring, B an A-algebra. Then a module of relative differential form is a B-module $\Omega_{B/A}$ with an A-derivation $d : B \to \Omega_{B/A}$ which satisfies the universal property of relative differential forms: if M is another B-module and $d' : B \to M$ an A-derivation, then there is a unique B-module homomorphism $f : \Omega_{B/A} \to M$ such that the following diagram commutes:

$$\begin{tikzcd} B \arrow[r, "d"] \arrow[dr, "d'"'] & \Omega_{B/A} \arrow[d, "f"] \\ & M \end{tikzcd}$$

Certainly $\Omega_{B/A}$ exists via the usual construction: take the free B-module F generated by the primitives $d(b)$ for $b \in B$ and quotient by the submodule generated by the relationships:

$$d(b + b') - d(b) - d(b') \quad \text{and} \quad d(bb') - d(b)b' - bd(b') \quad \text{and} \quad d(a)$$

Then the map $d : B \to \Omega_{B/A}$ is given by $b \mapsto d(b)$. Then it is an exercise in algebra to see that this satisfies the universal property and $\left(\Omega_{B/A}, d\right)$ is unique up to unique isomorphism. The following gives an alternative insightful construction:

Proposition 4.5.3: Natural Differential Structure

Let B be an A-algebra, $\delta : B \otimes_A B \to B$ the diagonal map given by $f(b \otimes b') = bb'$, and let $I = \ker \delta$. Then I/I^2 has the structure of a left B-module, and we can define a map

$$d : B \to I/I^2 \qquad d(b) = 1 \otimes b - b \otimes 1 \bmod I^2$$

In which case $(I/I^2, d)$ is a module of relative differential for B/A.

***Proof* :**
Hartshorne referenced Matsumura.

Example 4.5.4: Relative Differential

1. Take $A = \mathbb{R}$ and $B = \mathbb{R}[x]$. Then $I = \ker\delta$. Then as $\mathbb{R}[x]$ is an integral domain, it can be shown that any element is a linear combination of elements of the form:

$$f(x) \otimes g(x) - f(x)g(x) \otimes 1 \in I$$

Then it should be verified any element in I can be written as:

$$f(x)(1 \otimes g(x) - g(x) \otimes 1)$$

The reader should then verify that I/I^2 is generated by $f'(x)(x \otimes 1 - 1 \otimes x)$[a]. Let dx represent this element. Notice that this is the image of $d(x)$. Now, let's consider:

$$d(f(x) = d\left(\sum_i a_i x^i\right) = \sum_i a_i d(x^i)$$

Hence, let's compute $d(x^i)$. For $i = 2$, we get:

$$d(x^2) = d(x \cdot x) = xd(x) - xd(x) = 2xdx$$

Repeating this pattern, we can see that:

$$d(x^n) = nx^{n-1}dx$$

2. For example, taking $B = \mathbb{R}[x_1, ..., x_n]$, Then $\Omega_{B/A}$ is a free B-module generated by $dx_1, ..., dx_n$.

3. In proposition 4.5.9, we shall see that if E/F is a finitely generated separable extension (of characteristic 0), then

$$\Omega_{B/A} = \bigoplus_{b \in T} Bdb$$

where T is a transcendental A-basis for B.

4. What if we take $B = A[\epsilon]/(\epsilon^2)$?

5. Let $A = \mathbb{R}$ and $B = C^\infty(M)$ for some smooth n-manifold M. Then $C^\infty(M)$. Then $\Omega_{C^\infty(M)/\mathbb{R}}$ is a locally free $C^\infty(M)$-module (based off of the patches of M), consisting of the space of sections of the cotangent bundle T^*M.

[a] Hint: what is $(1 \otimes x^2 - x^2 \otimes 1)$. Repeat for higher powers

Proposition 4.5.5: Base Change For Differentials

Let A', B be A-algebras and $B' = B \otimes_A A'$. Then:

$$\Omega_{B'/A'} \cong \Omega_{B/A} \otimes_B B'$$

Furthermore, if S is a multiplicative subset of of S, then:

$$\Omega_{S^{-1}B/A} \cong S^{-1}\Omega_{B/A}$$

Proof **:**
Hartshorne referenced Matsumura.

Proposition 4.5.6: Differentials and Exact Sequence I

Let $A \to B \to C$ be a (not necessarily exact nor a complex) sequence of rings. Then there is an exact sequence of C-modules:

$$\Omega_{B/A} \otimes_B C \to \Omega_{C/A} \to \Omega_{C/B} \to 0$$

Proof **:**
Hartshorne referenced Matsumura.

Proposition 4.5.7: Differentials and Exact Sequence II

Let B be an A-algebra, I be an ideal of B, and $C = B/I$. Then there is a natural exact sequence of C-modules

$$I/I^2 \xrightarrow{\delta} \Omega_{B/A} \otimes_B C \to \Omega_{C/A} \to 0,$$

where for any $b \in I$, if $\bar{b}$ is its image in I/I^2, then $\delta\bar{b} = db \otimes 1$. In particular, I/I^2 has a natural structure of a C-module, and δ is a C-linear map, even though it is defined via the derivation d.

Proof **:**
Hartshorne referenced Matsumura.

Corollary 4.5.8: Finite Generation of Differential Forms

If B is a finitely generated A-module or a localization of a finitely generated A-module, then $\Omega_{B/A}$ is a finitely generated B-module

Proof **:**
As B is a finitely generated A-module, it is the quotient of polynomial rings. Hence, it follows from our direct computations from example 4.5.4 and, proposition 4.5.7, and the localization result

comes from proposition 4.5.5.

Proposition 4.5.9: Differentials and Field Extensions

Let K/k be a finitely generated field extension. Then:

$$\dim_k \Omega_{K/k} \geq \operatorname{trdeg} K/k$$

where equality holds iff K is separably generated over k. Then if K/k is a finite algebraic extension, $\Omega_{K/k} = 0$ if and only if K/k is separable.

Proof **:**
Hartshorne referenced Matsumura.

Proposition 4.5.10: Local Ring and Relative Differential

Let B be a local ring containing a field k isomorphic to $B/\mathfrak{m}$. Then given the map δ from proposition 4.5.7:

$$\delta : \mathfrak{m}/\mathfrak{m}^2 \xrightarrow{\cong} \Omega_{B/k} \otimes_B k$$

Proof **:**
By proposition 4.5.7 $\operatorname{coker}(\delta) = 0$, so δ is surjective. To show that δ is injective, we shall show the following dual vector space map is surjective:

$$\delta' : \operatorname{Hom}_k(\Omega_{B/k} \otimes k, k) \to \operatorname{Hom}_k(\mathfrak{m}/\mathfrak{m}^2, k)$$

The term on the left is isomorphic to $\operatorname{Hom}_B(\Omega_{B/k}, k)$, which by definition of the differentials, can be identified with the set $\operatorname{Der}_k(B, k)$ of k-derivations of B to k. If $d : B \to k$ is a derivation, then $\delta'(d)$ is obtained by restricting to $\mathfrak{m}$, and noting that $d(\mathfrak{m}^2) = 0$. Now, to show that δ' is surjective, let $h \in \operatorname{Hom}(\mathfrak{m}/\mathfrak{m}^2, k)$. For any $b \in B$, we can write $b = \lambda + c$, $\lambda \in k$, $c \in \mathfrak{m}$, in a unique way. Define $db = h(\bar{c})$, where $\bar{c} \in \mathfrak{m}/\mathfrak{m}^2$ is the image of c. Then one verifies immediately that d is a k-derivation of B to k, and that $\delta'(d) = h$. Thus δ' is surjective, as required.

We finish off this brief summary of the essential results about relative differentials by showing the relation between regular local rings and relative differentials. We require the following lemma:

Lemma 4.5.11: Characterizing Free Modules using Residue and Quotient Field

Let A be a Noetherian local integral domain, with residue field k and quotient field K. Then if M is a finitely generated A-module and $\dim_k M \otimes_A k = \dim_k M \otimes_A K = r$, then M is free of rank r.

Proof **:**
Since $\dim_k M \otimes k = r$, by Nakayama's lemma M can be generated by r elements. Hence, there is a surjective map $\varphi : A^r \to M$. Let R be its kernel. Then we obtain an exact sequence

$$0 \to R \otimes K \to K^r \to M \otimes K \to 0,$$

As $\dim_K M \otimes K = r$, we have $R \otimes K = 0$. But R is torsion-free, so $R = 0$, and M is isomorphic to A^r, completing the proof.

Theorem 4.5.12: Characterizing Regular Local Rings using Differential

Let B be the localization of a finitely generated k-algebra where $B/\mathfrak{m}$ is prefect (or it contains a field k isomorphic to $B/\mathfrak{m}$ where k is perfect). Then B is a regular local ring if and only if $\Omega_{B/k}$ is a free B-module of rank equal to $\dim B$.

Proof **:**
If $\Omega_{B/k}$ is free of rank $\dim B$, then by proposition 4.5.10

$$\dim_k \mathfrak{m}/\mathfrak{m}^2 = \dim B$$

which is an equivalent characterization of a regular local ring.

Conversely, suppose that B is a regular local ring of dimension r. Then $\dim_k \mathfrak{m}/\mathfrak{m}^2 = r$, so by proposition 4.5.10 we have $\dim_k \Omega_{B/k} \otimes k = r$. On the other hand, let K be the quotient field of B. Then by proposition 4.5.5, $\Omega_{B/k} \otimes_B K = \Omega_{K/k}$. As k is perfect, K is a separably generated extension field of k, and so $\dim_K \Omega_{K/k} = \operatorname{trdeg} K/k$ by 4.5.9. But we also have $\dim B = \operatorname{trdeg} K/k$. Finally, note that by corollary 4.5.8, $\Omega_{B/k}$ is a finitely generated B-module. But then by lemma 4.5.11 $\Omega_{B/k}$ is a free B module of rank r, as we sought to show.

4.5.1 Sheaves of Differentials

Let us now define sheaves of differentials on schemes. Let $f : X \to Y$ be a scheme morphism and $\Delta : X \to X \times_Y X$ the diagonal morphism. Then $X \cong \Delta(X)$ (see exercise ref:HERE, section 3.4.1). Namely, it is isomorphic to a closed subscheme of an open subset $W \subseteq X \times_Y X$.

Definition 4.5.13: Sheaf of Differentials

Let $\mathscr{I}$ be the sheaf of ideals of $\Delta(X)$ in W. Then the *sheaf of differentials* of X over Y is

$$\Omega_{X/Y} = \Delta^*(\mathscr{I}/\mathscr{I}^2)$$

Intuitively, this sheaf of modules ought to be a quasi-coherent $\mathcal{O}_X$-module. Indeed, it is first of all naturally a $\mathcal{O}_{\Delta(X)}$-module, and by the isomorphism $X \cong \Delta(X)$ we get a natural $\mathcal{O}_X$-module structure on it. By proposition 4.1.29 it is quasi-coherent, and by proposition 4.1.16 if Y is Noetherian and f is of finite-type it is coherent.

Reducing to the affine case, if $\operatorname{Spec} S = V \subseteq Y$ and $\operatorname{Spec} R = U \subseteq X$ such that $f(U) \subseteq V$, then $V \times_U V \subseteq X \times_Y X$ is an affine open subset isomorphic to $\operatorname{Spec} S \otimes_R S$, and $\Delta(X) \cap (V \times_U V)$ is a closed subscheme given by the kernel of the diagonal map $S \times_R S \to S$. Thus, $\mathscr{I}/\mathscr{I}^2$ is the usual I/I^2, and we get:

$$\Omega_{V/U} \cong \widetilde{\Omega_{S/R}}$$

Hence, $\Omega_{X/Y}$ could also be defined by giving an affine covering of X and Y which is glued together corresponding to the sheaves of modules $\widetilde{\Omega_{S/R}}$. Similarly, the maps $d : S \to \Omega_{S/R}$ glue to create a map $d : \mathcal{O}_X \to \Omega_{X/Y}$ of sheaves of abelian groups. This map is a derivation of local rings at each point. From this, we shall carry most of our results over to schemes.

(The other remark)

Proposition 4.5.14: Sheaf of Differentials: Base Change

Let $f : X \to Y$ be a morphism, let $g : Y' \to Y$ be another morphism, and let $f' : X' = X \times_Y Y' \to Y'$ be obtained by base extension. Then $\Omega_{X'/Y'} \cong g'^*(\Omega_{X/Y})$ where $g' : X' \to X$ is the first projection.

***Proof* :**
Follows from proposition 4.5.5

Proposition 4.5.15: Sheaf of Differentials: Short Exact Sequence I

Let $f : X \to Y$ and $g : Y \to Z$ be morphisms of schemes. Then there is an exact sequence of sheaves on X,

$$f^*\Omega_{Y/Z} \to \Omega_{X/Z} \to \Omega_{X/Y} \to 0.$$

***Proof* :**
Follows from proposition 4.5.6

Proposition 4.5.16: Sheaf of Differentials: Short Exact Sequence II

Let $f : X \to Y$ be a morphism, and let Z be a closed subscheme of X, with ideal sheaf $\mathscr{I}$. Then there is an exact sequence of sheaves on Z,

$$\mathscr{I}/\mathscr{I}^2 \xrightarrow{\delta} \Omega_{X/Y} \otimes \mathscr{O}_Z \to \Omega_{Z/Y} \to 0.$$

***Proof* :**
Follows from proposition 4.5.7

The next exact sequence will be specific to sheaves of differentials on projective space, and will be important for computations in the rest of the section:

Theorem 4.5.17: Sheaves of Differentials On Projective Space

Let R be a ring, $Y = \operatorname{Spec} R$, and $X = \mathbb{P}^n_R$. Then there exists an exact sequence of sheaves on X:

$$0 \to \Omega_{X/Y} \to \mathcal{O}_X(-1)^{\oplus n+1} \to \mathcal{O}_X \to 0$$

Recall that $\mathcal{O}(-1) \cong \operatorname{Hom}(\mathcal{O}(1), \mathcal{O})$, and so they are naturally interpreted as linear functionals. the final (non-trivial) map is naturally an evaluation map, giving us a great way of understanding the differentials $\mathcal{O}_{X/Y}$!

Proof **:**
Let $S = R[x_0, \ldots, x_n]$ be the homogeneous coordinate ring of X and E be the graded S-module $S(-1)^{n+1}$, with basis $e_0, \ldots, e_n$ in degree 1. Define a degree 0 homomorphism of graded S-modules $E \to S$ by sending $e_i \mapsto x_i$, and let M be the kernel. Then the exact sequence

$$0 \to M \to E \to S$$

of graded S-modules gives rise to an exact sequence of sheaves on X,

$$0 \to \tilde{M} \to \mathcal{O}_X(-1)^{n+1} \to \mathcal{O}_X \to 0.$$

Note that $E \to S$ is not surjective, but it is surjective in all degrees ≥ 1, and hence is surjective in all stalks, and so the corresponding map of sheaves is surjective!

Thus, if we show $\tilde{M} \cong \Omega_{X/Y}$, we're done. We shall show that we we get an isomorphism on patches that glue together. First note that if we localize at x_i, then $E_{x_i} \to S_{x_i}$ is a surjective homomorphism of free S_{x_i}-modules, so M_{x_i} is free of rank n, generated by $\{e_j - (x_j/x_i)e_i \mid j \neq i\}$. Then if U_i is the standard open set of X defined by x_i, $\tilde{M}|_{U_i}$ is a free $\mathcal{O}_{U_i}$-module generated by the sections $(1/x_i)e_j - (x_j/x_i^2)e_i$ for $j \neq i$. (Here we need the additional factor $1/x_i$ to get elements of degree 0 in the module M_{x_i}.)

To define the map $\varphi_i : \Omega_{X/Y}|_{U_i} \to \tilde{M}|_{U_i}$, recall that $U_i \cong \operatorname{Spec} R[x_0/x_i, \ldots, x_n/x_i]$, so $\Omega_{X/Y}|_{U_i}$ is a free $\mathcal{O}_{U_i}$-module generated by $d(x_0/x_i), \ldots, d(x_n/x_i)$. Thus, define φ_i by

$$\varphi_i(d(x_j/x_i)) = (1/x_i^2)(x_i e_j - x_j e_i).$$

Then φ_i is an isomorphism. Let us now show the isomorphisms φ_i glue together to give an isomorphism $\varphi : \Omega_{X/Y} \to \tilde{M}$ on all of X. But this is a simple calculation: on $U_i \cap U_j$, we have, for any k, $(x_k/x_i) = (x_k/x_j) \cdot (x_j/x_i)$. Hence in $\Omega|_{U_i \cap U_j}$ we have

$$d\left(\frac{x_k}{x_i}\right) - \frac{x_k}{x_j} d\left(\frac{x_j}{x_i}\right) = \frac{x_j}{x_i} d\left(\frac{x_k}{x_j}\right).$$

Now applying φ_i to the left-hand side and φ_j to the right-hand side, we get the same thing both ways, namely $(1/x_i x_j)(x_j e_k - x_k e_j)$. Thus the isomorphisms φ_i glue, which completes our proof.

Let us now focus down our attention to nonsingular varieties, as they are the most concrete example and most common place to apply these techniques. Let us first properly define the notion:

Definition 4.5.18: Abstract Nonsingular Variety

Let X be an abstract variety over an algebraically closed field k. Then X is *nonsingular* if all of its local rings are *regular*

4.5.1 Non Closed Points? Note that it we require the local rings at all points, not just closed points. It may thus seem that this condition is stronger, however as each local ring at a non-maximal prime ideal is the localization of a local ring at a maximal ideal, if localization preserves regularity the condition is equivalent, and this is indeed the case (see Matsumura p.199).

We can finally connect being nonsingular to the differential sheaves we've been working with:

Proposition 4.5.19: Nonsingular Varieties and Differential Sheaf

Let X be an irreducible (not necessarily reduced) separated scheme of finite type over an algebraically closed field k. Then $\mathcal{O}_{X/k}$ is locally free of rank $n = \dim X$ iff X is a nonsingular variety over k.

Proof **:**
If $x \in X$ is closed, then as $\mathcal{O}_{X/k}$ is locally free by proposition 4.1.11 $\mathcal{O}X, x$ is free of dimension n, has a residue field k, and its localization is a k-algebra of finite type. Let $B = \mathcal{O}_{X,x}$. Then the module $\mathcal{O}_{B/k}$ is equal to the stalks $(\mathcal{O}_{X/k})_x$ for the sheaf $\mathcal{O}_{X/k}$. Thus, by theorem 4.5.12 $(\mathcal{O}_{X/k})_x$ if free of rank n iff B is a regular local ring. Then remark 4.5.1 gives us the desired result.

From this result, we get a lovely new view of seeing why the set of nonsingular points are open and dense:

Corollary 4.5.20: Nonsingular Points Open and Dense using Differentials

Let X be a variety over k. Then there is an open dense subset of X which is nonsingular

Proof **:**
Let $n = \dim X$. Then the function field $k(X)$ has transcendence degree n over k and $k(X)/k$ is separable. Then by proposition 4.5.9, $\Omega_{K/k}$ is a $K(X)$-vector space of dimension n.

Now, notice that $\mathcal{O}_{K/k}$ is just the stalk of $\mathcal{O}_{X/k}$ at the generic point of X! Then by proposition 4.1.11 $\mathcal{O}_{X/k}$ is locally free in some neighbourhood of the generic point, that is some nonempty open dense set U. By proposition 4.5.19, the points of U are nonsingular, completing the proof.

An important result is to be able to cake closed subschemes whose points are nonsingular:

Theorem 4.5.21: Differentials: Nonsingular Closed Subschemes

Let X be a nonsingular variety over k and let $Y \subseteq X$ be an irreducible closed subscheme defined by a sheaf of ideals $\mathscr{I}$. Then Y is nonsingular if and only if

1. $\Omega_{Y,k}$ is locally free
2. the sequence of proposition 4.5.16 is exact on the left as well:

$$0 \to \mathscr{I}/\mathscr{I}^2 \to \Omega_{X,k} \otimes \mathcal{O}_Y \to \Omega_{Y,k} \to 0.$$

Given Y is nonsingular, then $\mathscr{I}$ is locally generated by $r = \operatorname{codim}_X Y$ elements, and $\mathscr{I}/\mathscr{I}^2$ is a locally free sheaf of rank r on Y.

Proof **:**
First suppose the two conditions hold so that $\Omega_{Y,k}$ is locally free, and the given sequence is also left exact. Then by proposition 4.5.19 it suffices to show that rank $\Omega_{Y,k} = \dim Y$. Let rank $\Omega_{Y,k} = q$. We know that $\Omega_{X,k}$ is locally free of rank n, so it follows from (2) that $\mathscr{I}/\mathscr{I}^2$ is locally free on Y of rank $n-q$. Hence by Nakayama's lemma, $\mathscr{I}$ can be locally generated by $n-q$ elements, and it follows that $\dim Y \geq n-(n-q) = q$ (see [8, chapter 21.5]). On the other hand, considering any closed point $y \in Y$, by proposition 4.5.10 we have $q = \dim(\mathfrak{m}/\mathfrak{m}^2)$, and so $q \geq \dim Y$ by (see [8, chapter 21.5.3]), but then by what we saw earlier $q = \dim Y$. From this, we conclude that that Y is nonsingular and that $n-q = \operatorname{codim}_X Y$ giving the desired result for the generators of $\mathscr{I}$.

Conversely, assume that Y is nonsingular. Then (1) is immediate as $\Omega_{Y,k}$ is locally free of rank $q = \dim Y$. By proposition 4.5.16 we have the exact sequence

$$\mathscr{I}/\mathscr{I}^2 \xrightarrow{\delta} \Omega_{X,k} \otimes \mathcal{O}_Y \xrightarrow{\varphi} \Omega_{Y,k} \to 0.$$

We need to show exactness on the left. Consider a closed point $y \in Y$. Then $\ker \varphi$ is locally free of rank $r = n-q$ at y, so it is possible to choose sections $x_1, \dots, x_r \in \mathscr{I}$ in an appropriate neighborhood of y such that $dx_1, \dots, dx_r$ generate $\ker \varphi$. Let $\mathscr{I}'$ be the ideal sheaf generated by $x_1, \dots, x_r$, and let Y' be the corresponding closed subscheme. Then by construction, the $dx_1, \dots, dx_r$ generate a free sub-sheaf of rank r of $\Omega_{X,k} \otimes \mathcal{O}_Y$ in a neighborhood of y. It follows that in the exact sequence of proposition 4.5.16 for Y',

$$\mathscr{I}'/\mathscr{I}'^2 \xrightarrow{\delta} \Omega_{X,k} \otimes \mathcal{O}_Y \to \Omega_{Y',k} \to 0,$$

Now, since its image is free of rank r and $\Omega_{Y',k}$ is locally free of rank $n-r$, δ injective. Then we demonstrates that Y' is irreducible and nonsingular of dimension $n-r$ (in a neighborhood of y). But $Y \subseteq Y'$, both are integral schemes of the same dimension, so we must have $Y = Y'$, $\mathscr{I} = \mathscr{I}'$, and this shows that

$$0 \to \mathscr{I}/\mathscr{I}^2 \xrightarrow{\delta} \Omega_{X,k} \otimes \mathcal{O}_Y$$

is injective, as we sought to show.

(Hartshorne here covers the very interesting *Bernini's Theorem* which is some Morse Theory on nonsingular projective varieties, but I will omit if for now)

We are now ready to bring in the notion of the tangent space of a nonsingular variety:

Definition 4.5.22: Tangent Sheaf and Canonical Sheaf

Let X be a nonsingular variety over k. Then the *tangent sheaf* of X is defined as:

$$\mathscr{T}_X = \mathcal{H}om_{\mathcal{O}_X}(\Omega_{X/k}, \mathcal{O}_X)$$

The *canonical sheaf* of X is the invertible sheaf:

$$\omega_X = \bigwedge^n \Omega_{X/k}$$

where $n = \dim X$.

Note that $\mathscr{T}_X$ is a locally free sheaf of rank $n = \dim X$. If is projective X, then the dimension of the global section of ω_X over X will give an important invariant:

Definition 4.5.23: Geometric Genus

Let X be projective and nonsingular. Then the *geometric genus* of X is:

$$p_g = \dim_k \Gamma(X, \omega_X)$$

4.5.2 Foreshadowing In [8, chapter 21.5.2], we introduced the Hilbert function and the *arithmetic genus*. This shall be the dual of the geometric genus in the case of curves, as we shall show in section 5.3. Note however these these two need not be equal in higher dimensions, see example ref:HERE.

When working over varieties, the Geometric genus is a rather stable result, it is in fact more stable than isomorphisms:

Theorem 4.5.24: Birational Invariance of Geometric Genus

Let X, X' be two nonsingular projective varieties. THen if X is birationally equivalent to X', then:

$$p_g(X) = p_g(X')$$

Note that the converse is not true: the geometric genus is in general not enough to determine a variety (this is only the case in dimension 1, see ref:HERE).

***Proof* :**
As X and X' are birationally equivalent , there exists a rational map $F : X \dashrightarrow X'$ that has a rational inverse. Let $V \subseteq X$ be the largest open set for which $f : V \to X'$ represents the rational map F. Then by proposition 4.5.15 we get $f^*\Omega_{X'/k} \to \Omega_{V/k}$. As X, X' are birational, these are locally free sheaves of the same rank, say $n = \dim X$, so there is an induced map on the exterior powers $f^*\omega_{X'} \to \omega_V$. This map in turn induces a map on the space of global sections

$$f^* : \Gamma(X', \omega_{X'}) \to \Gamma(V, \omega_V)$$

Now, since f is birational there is an open set $U \subseteq V$ such that $f(U)$ is open in X' and f induces an isomorphism from U to $f(U)$. Thus f induces:

$$\omega_V|_U \cong \omega_{X'}|_{f(U)}$$

Since a nonzero global section of an invertible sheaf cannot vanish on a dense open set, the map of vector spaces

$$f^* : \Gamma(X', \omega_{X'}) \to \Gamma(V, \omega_V)$$

is injective.

Next we will compare $\Gamma(V, \omega_V)$ with $\Gamma(X, \omega_X)$. First by the valuation criterion for properness (theorem 3.4.20), $X - V$ has codimension ≥ 2 in X. For, if $P \in X$ is a point of codimension 1, then $\mathcal{O}_{P,X}$ is a discrete valuation ring (as X is nonsingular). We already have a map of the generic point of X to X', and X' is projective, and hence proper over k, so there exists a unique morphism $\operatorname{Spec} \mathcal{O}_{P,X} \to X'$ compatible with the given birational map. This extends to a morphism of some neighborhood of P to X', so we must have $P \in V$ by definition of V.

Now, let us show there that the natural restriction map $\Gamma(X, \omega_X) \to \Gamma(V, \omega_V)$ is surjective, giving us bijectivity. This shall be an immediate consequence of lemma 4.3.7. It is enough to show that for any open affine subset $U \subseteq X$ such that $\omega_X|_U \cong \mathcal{O}_U$, that $\Gamma(U, \mathcal{O}_U) \to \Gamma(U \cap V, \mathcal{O}_{U \cap V})$ is bijective. Since X is nonsingular, it is normal, and since $U - U \cap V$ has codimension ≥ 2 in U, but this is now an immediate consequence of lemma 4.3.7.

Combining our results, we see that $p_g(X') \leq p_g(X)$. The other inequality is given by by symmetry, and thus conclude that

$$p_g(X) = p_g(X')$$

as we sought to show

We finish off this section by looking at the relation between tangent sheaves and their dual (normal sheaves):

Definition 4.5.25: Normal and Conormal Sheaf

Let $Y \subseteq X$ be a nonsingular subvariety of a variety X over k. Then $\mathscr{I}/\mathscr{I}^2$ is called the *conormal sheaf* of Y in X, and $\mathcal{N}_{Y/X} = \mathcal{H}om_{\mathcal{O}_Y}(\mathscr{I}/\mathscr{I}^2, \mathcal{O}_Y)$ is called the *normal sheaf* of Y in X.

Note that normal sheaf is locally free of rank $r = \operatorname{codim}_X(Y)$. The normal sheaf does indeed mimic the notion of a normal bundle on a manifold as we get the dual exact sequence:

$$0 \to \mathscr{T}_Y \to \mathscr{T}_X \otimes \mathcal{O}_Y \to \mathcal{N}_{Y/X} \to 0$$

Proposition 4.5.26: Subvarieties and Normal Sheaf

Let $Y \subseteq X$ be a nonsingular subvariety of codimension r from a nonsingular variety X over k. Then

$$\omega_Y \cong \omega_X \otimes \bigwedge^r \mathcal{N}_{Y/X}.$$

In case $r = 1$, considering Y as a divisor $\mathcal{L}$ be the associated invertible sheaf on X, Then

$$\omega_Y \cong \omega_X \otimes \mathcal{L} \otimes \mathcal{O}_Y$$

Proof **:**
First, take the highest exterior powers of the locally free sheaves in the exact sequence (by exercise ref:HERE)

$$0 \to \mathcal{I}/\mathcal{I}^2 \to \Omega_X \otimes \mathcal{O}_Y \to \Omega_Y \to 0$$

Thus we find that $\omega_X \otimes \mathcal{O}_Y \cong \omega_Y \otimes \bigwedge^r(\mathcal{I}/\mathcal{I}^2)$. Since formation of the highest exterior power commutes with taking the dual sheaf, we get

$$\omega_Y \cong \omega_X \otimes \bigwedge^r \mathcal{N}_{Y/X}$$

In the special case $r = 1$, $\mathcal{I}_Y \cong \mathcal{L}^{-1}$ by proposition 4.3.29. Thus, $\mathcal{I}/\mathcal{I}^2 \cong \mathcal{L}^{-1} \otimes \mathcal{O}_Y$, and $\mathcal{N}_{Y/X} \cong \mathcal{L} \otimes \mathcal{O}_Y$. So applying the previous result with $r = 1$, we obtain

$$\omega_Y \cong \omega_X \otimes \mathcal{L} \otimes \mathcal{O}_Y$$

completing the proof.

(Hartshorne gives many good examples here! One important one was showing that there exists non-rational varieties in all dimensions!

5

Cohomology of Schemes

Let X be a scheme with a sheaf $\mathscr{F}$. A natural question to ask is if we can quantify the obstruction to the gluing local sections to create global sections? This is precisely what *sheaf cohomology* shall study by taking the derived functors of the global-section functor $X \to \Gamma(X, \mathscr{F})$ for a sheaf $\mathscr{F}$. Important sheaves this will apply include (but are not limited to) the structure sheaf $\mathcal{O}_X$, the twisted sheaves $\mathcal{O}_X(n)$, the differential sheaves $\Omega_{B/A}$ and there wedge products, and the pushforward of closed sheaves (in particular to study a scheme X given information about closed subschemes $S \subseteq X$). Though this is theoretically nice, it is computationally difficult as it will require an injective resolution that uses acyclic sheaves that are generally difficult to compute. For computations, *Čech cohomology* is introduced as a more "geometric" approach in computing cohomology. From this we shall compute the cohomology of projective space $\mathbb{P}^n_R$ over twisted sheaves $\mathcal{O}_X(n)$.

We then bring over the generalization of deRham cohomology, and generalize Poincaré's duality to *Serre duality*. Through this, we can take $\mathscr{F}$ is a constant sheaf (ex. $\mathbb{R}$ or $\mathbb{C}$), and equate sheaf cohomology to the usual *singular cohomology* as studied in a first course in algebraic topology (see [2]).

The reader may want to review [8, Part VII] to get a refresher on the derived functor approach to (co)homology theory.

5.1 Definition and Basic Properties

Recall that if R is a ring, then every R-module is isomorphic to a submodule of an injective module (see [8, chatper 16.3]). From this, we shall be able to conclude that ringed space have enough injectives

Proposition 5.1.1: Ringed Space Has Enough Injectives

Let $(X, \mathcal{O}_X)$ be a ringed space. Then the category of sheaves of $\mathcal{O}_X$-modules, ${}_{\mathcal{O}_X}\mathbf{Mod}$ has enough injectives.

Proof **:**
Let $\mathscr{F}$ be a sheaf of $\mathcal{O}_X$-modules. Then for each $x \in X$, the stalk $\mathscr{F}_x$ is an $\mathcal{O}_{x,X}$-module. Then there is an injection

$$\mathscr{F}_x \to I_x$$

where I_x is an injective $\mathcal{O}_{x,X}$-module. We can do this for each point. Now, Let $j : \{x\} \to X$ denote the inclusion of the one-point space for each point, let $\{x\}$ have I_x as a sheaf, and define the sheaf:

$$\mathscr{I} = \prod_{x \in X} j_*(I_x)$$

where j_* is the usual pushforward. This shall be our injective object.

For any $\mathcal{O}_X$-module $\mathscr{G}$, consider

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{G}, \mathscr{I}) = \prod \mathrm{Hom}_{\mathcal{O}_X}(\mathscr{G}, j_*(I_x))$$

Then for each point $x \in X$, notice that:

$$\mathrm{Hom}_{\mathcal{O}_X}(\mathscr{G}, j_*(I_x)) \cong \mathrm{Hom}_{\mathcal{O}_{x,X}}(\mathscr{G}_x, I_x)$$

From this, we see that if we pick $\mathscr{G} = \mathscr{G}$, we get a natural injective $\mathcal{O}_X$-module map $\mathscr{F} \to \mathscr{I}$ given by $\mathscr{F}_x \to I_x$. Furthermore, $\mathrm{Hom}_{\mathcal{O}_X}(-, \mathscr{I})$ is exact as it is the product $\mathrm{Hom}_{\mathcal{O}_{x,X}}(-, I_x)$ which is exact as I_x is injective, and the stalk functor $\mathscr{G} \mapsto \mathscr{G}_x$ which is exact by 1.2.21. Hence, $\mathrm{Hom}(-, \mathscr{I})$ is exact, making $\mathscr{I}$ an injective $\mathcal{O}_X$-module, completing the proof.

Corollary 5.1.2: Sheaves of Abelian's Has Enough Injectives

Let X be a topological space. Then the category of sheaves of abelian groups on X has enough injectives

Proof **:**
Take $\mathcal{O}_X = \mathbb{Z}$ and apply proposition 5.1.1

Definition 5.1.3: Global Section Functor and Sheaf Cohomology

Let X be a topological space. Then the *global section functor*

$$\Gamma(X, -) : \mathrm{Ab}(X) \to \mathrm{Ab}$$

which is left- but not right-exact (by proposition 1.2.16). Then the cohomology functors $H^i(X, -)$, the right derived functors of $\Gamma(X, -)$, are the *cohomology groups* (with respect to a sheaf).

To be able to create resolutions, recall that it is enough to find acyclic objects ([8, chapter 27.4]). We shall show that the following types of objects shall be acyclic:

Definition 5.1.4: Flasque Sheaf

Let X be a topological space. Then a sheaf $\mathscr{F}$ on X is called *flasque* if for every inclusion $U \subseteq V$, $\mathrm{res} : \mathscr{F}(V) \to \mathscr{F}(U)$ is surjective.

Proposition 5.1.5: Injective Sheaf Modules Are Flasque

Let $(X, \mathcal{O}_X)$ be a ringed space. Then any injective $\mathcal{O}_X$-module is flasque

Proof **:**
For any open subset $V \subseteq X$, let $\mathcal{O}_V$ denote the sheaf $j_!(\mathcal{O}_X|_V)$, which is the restriction of $\mathcal{O}_X$ to V, extended by zero outside V. Now let $\mathscr{I}$ be an injective $\mathcal{O}_X$-module, and let $V \subseteq U$ be open sets. Then we have an inclusion $0 \to \mathcal{O}_U \to \mathcal{O}_V$ of sheaves of $\mathcal{O}_X$-modules. Since $\mathscr{I}$ is injective, we get a surjection $\mathrm{Hom}(\mathcal{O}_V, \mathscr{I}) \to \mathrm{Hom}(\mathcal{O}_U, \mathscr{I}) \to 0$. But $\mathrm{Hom}(\mathcal{O}_U, \mathscr{I}) = \mathscr{I}(U)$ and $\mathrm{Hom}(\mathcal{O}_V, \mathscr{I}) = \mathscr{I}(V)$, so $\mathscr{I}$ is flasque.

Corollary 5.1.6: Trivial Homology For Flask Sheaf

Let $\mathscr{F}$ be a flasque sheaf on X. Then

$$H^i(X, \mathscr{F}) = 0 \qquad i > 0$$

Proof **:**
As there are enough injective, embed $\mathscr{F}$ into $\mathscr{I}$, and take it's quotient $\mathscr{G}$. Most importantly, check that the quotient sheaf of two flasque sheaves is flasque, and so we have a short exact sequence of flasque sheaves:

$$0 \to \mathscr{F} \to \mathscr{I} \to \mathscr{G} \to 0$$

As $\mathscr{F}$ is flasque, we have the $\Gamma(X, -)$ will preserve this exact sequence:

$$0 \to \Gamma(X, \mathscr{F}) \to \Gamma(X, \mathscr{I}) \to \Gamma(X, \mathscr{G}) \to 0$$

As $\mathscr{I}$ is injective, $H^i(X, \mathscr{I}) = 0$ for $i > 0$, and so by taking the long exact sequence of cohomology:

$$H^1(X, \mathscr{F}) = 0, \qquad H^i(X, \mathscr{F}) \cong H^{i-1}(X, \mathscr{G}),\ i \geq 2$$

As $\mathscr{G}$ is also flasque, we get by induction that $H^i(X, \mathscr{F}) = 0$ for $i > 0$, completing the proof.

Hence, flasque sheaves are acyclic for $\Gamma(X, -)$, and so we may compute cohomology by taking Flasque resolutions! Now, let us link this to schemes. First, if $(X, \mathcal{O}_X)$ is a ringed space, by taking $R = \Gamma(X, \mathcal{O}_X)$, we always have a natural R-module structure on any $\mathcal{O}_X$-modules $\mathscr{F}$. Thus, the cohomology groups of $\mathscr{F}$ have a natural R-module structure on them, and the associated exact sequence are naturally R-modules. Then for R-schemes, the cohomology groups of any $\mathcal{O}_X$-module will have natural R-modules structure. We immediately get:

Corollary 5.1.7: Section and Hom Functor

Let $(X, \mathcal{O}_X)$ be a ringed space. Then the derived functors of $\Gamma(X, -)$ coincide with $H^i(X, -)$

Proof **:**
immediate from above.

Let us now prove our first sheaf cohomology result: let us try to bring in our results on dimension and Noetherian space. To prove this, let us first build-up the appropriate results for the cohomology of Noetherian space

Lemma 5.1.8: Noetherian Space and Flasque Sheaves

Let X be a Noetherian topological space. Then the colimit of flasque sheaves is flasque.

Proof **:**
Let $(\mathscr{F}_\alpha)$ be a directed system of flasque sheaves. Then for any inclusion $V \subseteq U$, for each α we have $\mathscr{F}_\alpha(U) \to \mathscr{F}_\alpha(V)$ is surjective. As $\xrightarrow{\lim}$ is an exact functor, we get that the map:

$$\varinjlim_\alpha \mathscr{F}_\alpha(U) \to \varinjlim_\alpha \mathscr{F}_\alpha(V)$$

is surjective. Then on a Noetherian topological space:

$$\underline{\lim}_{\longrightarrow} \mathscr{F}_\alpha(U) = \left(\underset{\longrightarrow}{\lim} \mathscr{F}_\alpha\right)(U)$$

Hence we have:

$$\left((\varinjlim_\alpha \mathscr{F}_\alpha \right) (U) \to \varinjlim_\alpha \mathscr{F}_\alpha()V \Big)$$

is surjective, hence $\lim_\to \mathscr{F}_\alpha$ is flasque, completing the proof.

Lemma 5.1.9: Colimit and Cohomology Commuting

Let X be a Noetherian topological space and $(\mathscr{F}_\alpha)$ a direct system of abelian sheaves. Then there is a natural isomorphism for each $i \geq 0$:

$$\varinjlim H^i\left(X, \mathscr{F}_\alpha\right) \cong H^i\left(X, \varinjlim \mathscr{F}_\alpha\right)$$

Proof **:**
For each α we have a natural map $\mathscr{F}_\alpha \to \varinjlim \mathscr{F}_\alpha$. This induces a map on cohomology, and then we take the direct limit of these maps. For $\vec{i} = 0$, this becomes

$$\Gamma\left(X, \varinjlim \mathscr{F}_i\right) \cong \varinjlim \Gamma\left(X, \mathscr{F}_i\right)$$

Which is readily true for Noetherian topological spaces. For the general case, consider the category $\operatorname{ind}_A(\mathbf{Ab}(X))$ consisting of all directed systems of objects of $\mathbf{Ab}(X)$, indexed by A. This is an abelian category. Furthermore, since $\varinjlim$ is an exact functor, we have a natural transformation of δ-functors

$$\varinjlim H^i(X, -) \to H^i(X, \varinjlim -)$$

from $\operatorname{ind}_A(\mathbf{Ab}(X))$ to $\mathbf{Ab}$. They agree for $i = 0$, so to prove they are the same, it will be sufficient to show they are both effaceable for $i > 0$. For in that case, they are both universal, and hence by uniqueness of universal object must be isomorphic.
So let $(\mathscr{F}_\alpha) \in \operatorname{ind}_A(\mathbf{Ab}(X))$. For each α, let $\mathscr{G}_\alpha$ be the sheaf of discontinuous sections of $\mathscr{F}_\alpha$. Then $\mathscr{G}_\alpha$ is flasque, and there is a natural inclusion $\mathscr{F}_\alpha \to \mathscr{G}_\alpha$. Furthermore, the construction of $\mathscr{G}_\alpha$ is functorial, so the $\mathscr{G}_\alpha$ also form a direct system, and we obtain a monomorphism $u : (\mathscr{F}_\alpha) \to (\mathscr{G}_\alpha)$ in the category $\operatorname{ind}_A(\mathbf{Ab}(X))$. Now the $\mathscr{G}_\alpha$ are all flasque, so $H^i\left(X, \mathscr{G}_\alpha\right) = 0$ for $i > 0$. Thus $\varinjlim H^i\left(X, \mathscr{G}_\alpha\right) = 0$, and the functor on the left-hand side is effaceable for $i > 0$. On the other hand, $\varinjlim \mathscr{G}_\alpha$ is also flasque. So $H^i\left(X, \varinjlim \mathscr{G}_\alpha\right) = 0$ for $i > 0$, and we see that the functor on the right-hand side is also effaceable, as we sought to show.

Corollary 5.1.10: Direct sum and cohomology commuting

Let X be a Noetherian topological space. Then there is a natural isomorphism for each $i \geq 0$:

$$\bigoplus_\alpha H^i\left(X, \mathscr{F}_\alpha\right) \cong H^i\left(X, \bigoplus_\alpha \mathscr{F}_\alpha\right)$$

Proof **:**
The direct is a special case of a colimit.

Lemma 5.1.11: Cohomology of Closed Subsets

Let $Y \subseteq X$ be a closed subset of $X, \mathscr{F}$ a sheaf of abelian groups, and $j : Y \to X$ the natural inclusion. Then:

$$H^i(Y, \mathscr{F}) = H^i(X, j_*\mathscr{F})$$

where $j_*\mathscr{F}$ is the extension of $\mathscr{F}$ by zero on Y (see exercise 1.3.1)

Proof **:**
If $\mathscr{I}*$ is a flasque resolution of $\mathscr{F}, j_*\mathscr{F}^*$ is a flasque resolution of $j_*\mathscr{F}$, and so for each i :

$$\Gamma\left(Y, \mathscr{I}^i\right) = \Gamma\left(X, j_*\mathscr{I}^*\right)$$

But then

$$H^i(Y, \mathscr{F}) = H^i(X, j_*\mathscr{F})$$

as we sought to show.

Due to this lemma, we shall often write $\mathscr{F}$ instead of $j_*\mathscr{F}$ without worrying about ambiguity. With these lemmas, we can prove our first main result:

Theorem 5.1.12: Noetherian Space and Dimension

Let X be a Noetherian topological space of dimension n. Then for all $i > n$, all sheaves of abelian groups $\mathscr{F}$ on X vanish:

$$H^i(X, \mathscr{F}) = 0 \qquad i > n$$

Proof **:**
Let us first set some notation: let $Y \subseteq X$ be a closed subset, $\mathscr{F}$ be any sheaf on X, and $\mathscr{F}_Y = j_*\left(\mathscr{F}|_Y\right)$ where $j : Y \to X$ is the inclusion. Similarly for open subset $U \subseteq X$ where $\mathscr{F}_U = i_*\left(\mathscr{F}|_U\right)$ with the inclusion $i : U \to X$. If $U = X \backslash Y$, we get the short exact sequence:

$$0 \to \mathscr{F}_U \to \mathscr{F} \to \mathscr{F}_Y \to 0$$

With this setup, we shall prove the result by induction on the dimension of X.
Step 1. Reduction to the case X irreducible. If X is reducible, let Y be one of its irreducible components, and let $U = X - Y$. Then for any $\mathscr{F}$ we have an exact sequence

$$0 \to \mathscr{F}_U \to \mathscr{F} \to \mathscr{F}_Y \to 0.$$

From the long exact sequence of cohomology, it will be sufficient to prove that $H^i(X, \mathscr{F}_Y) = 0$ and $H^i(X, \mathscr{F}_U) = 0$ for $i > n$. But Y is closed and irreducible, and $\mathscr{F}_U$ can be regarded as a sheaf on the closed subset $\bar{U}$, which has one fewer irreducible components than X. Thus using lemma 5.1.10 and induction on the number of irreducible components, we reduce to the case X

irreducible.
Step 2. Suppose X is irreducible of dimension 0 . Then the only open subsets of X are X and the empty set. For otherwise, X would have a proper irreducible closed subset, and $\dim X$ would be ≥ 1. Thus $\Gamma(X, \cdot)$ induces an equivalence of categories $\mathbf{Ab}(X) \rightarrow \mathbf{Ab}$. In particular, $\Gamma(X, \cdot)$ is an exact functor, so $H^{i}(X, \mathscr{F})=0$ for $i>0$, and for all $\mathscr{F}$.
Step 3. Now let X be irreducible of dimension n, and let $\mathscr{F} \in \mathbf{Ab}(X)$. Let $B=\bigcup_{U \subset X} \mathscr{F}(U)$, and let A be the set of all finite subsets of B. For each $\alpha \in A$, let $\mathscr{F}_{\alpha}$ be the subsheaf of $\mathscr{F}$ generated by the sections in α (over various open sets). Then A is a directed set, and $\mathscr{F}=\varinjlim \mathscr{F}_{\alpha}$. So by (2.9), it will be sufficient to prove vanishing of cohomology for each $\mathscr{F}_{\alpha}$. If $\alpha^{\prime}$ is a subset of α, then we have an exact sequence

$$0 \rightarrow \mathscr{F}_{\alpha^{\prime}} \rightarrow \mathscr{F}_{\alpha} \rightarrow \mathscr{G} \rightarrow 0$$

where $\mathscr{G}$ is a sheaf generated by $\#\left(\alpha-\alpha^{\prime}\right)$ sections over suitable open sets. Thus, using the long exact sequence of cohomology, and induction on $\#(\alpha)$, we reduce to the case that $\mathscr{F}$ is generated by a single section over some open set U. In that case $\mathscr{F}$ is a quotient of the sheaf $\underline{\mathbb{Z}}_{U}$ (where $\underline{\mathbb{Z}}$ denotes the constant sheaf $\underline{\mathbb{Z}}$ on X). Letting $\mathcal{K}$ be the kernel, we have an exact sequence

$$0 \rightarrow \mathcal{K} \rightarrow \underline{\mathbb{Z}}_{U} \rightarrow \mathscr{F} \rightarrow 0$$

Again using the long exact sequence of cohomology, it will be sufficient to prove vanishing for $\mathcal{K}$ and for $\underline{\mathbb{Z}}_{U}$.
Step 4. Let U be an open subset of X and let $\mathcal{K}$ be a subsheaf of $\underline{\mathbb{Z}}_{U}$. For each $x \in U$, the stalk $\mathcal{K}_{x}$ is a subgroup of $\underline{\mathbb{Z}}$. If $\mathcal{K}=0$, skip to Step 5 . If not, let d be the least positive integer which occurs in any of the groups $\mathcal{K}_{x}$. Then there is a nonempty open subset $V \subseteq U$ such that $\left.\mathcal{K}\right|_{V} \cong d \cdot \left.\underline{\mathbb{Z}}\right|_{V}$ as a subsheaf of $\left.\underline{\mathbb{Z}}\right|_{V}$. Thus $\mathcal{K}_{V} \cong \underline{\mathbb{Z}}_{V}$ and we have an exact sequence

$$0 \rightarrow \underline{\mathbb{Z}}_{V} \rightarrow \mathcal{K} \rightarrow \mathcal{K} / \underline{\mathbb{Z}}_{V} \rightarrow 0$$

Now the sheaf $\mathcal{K} / \underline{\mathbb{Z}}_{V}$ is supported on the closed subset $(U-V)^{-}$of X, which has dimension $<n$, since X is irreducible. So using lemma 5.1.10 and the induction hypothesis, we know $H^{i}\left(X, \mathcal{K} / \underline{Z}_{V}\right)=0$ for $i \geq n$. So by the long exact sequence of cohomology, we need only show vanishing for $\underline{\mathbb{Z}}_{V}$.
Step 5. To complete the proof, we need only show that for any open subset $U \subseteq X$, we have $H^{i}\left(X, \underline{Z}_{U}\right)=0$ for $i>n$. Let $Y=X-U$. Then we have an exact sequence

$$0 \rightarrow \underline{\mathbb{Z}}_{U} \rightarrow \underline{\mathbb{Z}} \rightarrow \underline{\mathbb{Z}}_{Y} \rightarrow 0$$

Now $\dim Y<\dim X$ since X is irreducible, so using lemma 5.1.10 and the induction hypothesis, we have $H^{i}\left(X, \underline{Z}_{Y}\right)=0$ for $i \geq n$. On the other hand, $\underline{\mathbb{Z}}$ is flasque, since it is a constant sheaf on an irreducible space (II, Ex. 1.16a). Hence $H^{i}(X, \underline{Z})=0$ for $i>0$ by (2.5). So from the long exact sequence of cohomology we have $H^{i}\left(X, \underline{\mathbb{Z}}_{U}\right)=0$ for $i>n$.

5.1.1 Application: Cohomology of Noetherian

In this section, we shall make rigorous the intuition that the cohomology of schemes measure some form of gluing by showing that if $X = \operatorname{Spec} R$, is a Noetherian , then for any quasi-coherent sheaf $\mathscr{F}$ over X,

$$H^i(X, \mathscr{F}) = 0 \quad \forall i > 0$$

The key insight is that if I is an injective R-module, $\tilde{I}$ on $\operatorname{Spec} R$ is flasque. This will motivate the idea that cohomology of schemes should be thought of how the gluing of schemes can be measured, which shall motivate introducing Čech cohomology in the next section and showing it is equivalent to the cohomology given by flasque resolutions, and shall be computationally much more manageable. First, let M be an R-module and $\mathfrak{a} \subseteq R$ be an ideal. Define the $\mathfrak{a}$-torsion submodule

$$\operatorname{Tor}_{\mathfrak{a}}(M) = \left\{ m \in M \middle| \mathfrak{a}^n m = 0, \text{ for some } n > 0 \right\}$$

5.1.1 Note This result is in fact true without the Noetherian condition, however it is harder to prove. The extra effort for building it up is not necessary for the main theory, for the curious see ref:HERE.

Lemma 5.1.13: $\mathfrak{a}$-Torsion Module Injective

Let R be a Noetherian ring, $\mathfrak{a} \subseteq R$ an ideal, and I an injective R-module. Then the submodule

$$J = \operatorname{Tor}_{\mathfrak{a}}(I)$$

is injective

***Proof* :**
To show that J is injective, it will be sufficient to prove Baer's Criterion, namely to show that for any ideal $\mathfrak{b} \subseteq A$, and for any homomorphism $\varphi : \mathfrak{b} \to J$, there exists a homomorphism $\psi : A \to J$ extending φ. Since A is noetherian, $\mathfrak{b}$ is finitely generated. On the other hand, every element of J is annihilated by some power of $\mathfrak{a}$, so there exists an $n > 0$ such that $\mathfrak{a}^n \varphi(\mathfrak{b}) = 0$, or equivalently, $\varphi(\mathfrak{a}^n \mathfrak{b}) = 0$. Now applying Artin-Rees lemma (see [8]) to the inclusion $\mathfrak{b} \subseteq A$, we find that there is an $n' \geq n$ such that $\mathfrak{a}^n \mathfrak{b} \supseteq \mathfrak{b} \cap \mathfrak{a}^{n'}$. Hence $\varphi\left(\mathfrak{b} \cap \mathfrak{a}^{n'}\right) = 0$, and so the map $\varphi : \mathfrak{b} \to J$ factors through $\mathfrak{b}/\left(\mathfrak{b} \cap \mathfrak{a}^{n'}\right)$. Now we consider the following diagram:

$$\begin{tikzcd}
A \arrow[r] & A/\mathfrak{a}^{n'} \arrow[dr, dashed, "\psi'"] & & \\
\mathfrak{b} \arrow[u, hook] \arrow[r] \arrow[rrr, bend right, "\varphi"'] & \mathfrak{b}/(\mathfrak{b} \cap \mathfrak{a}^{n'}) \arrow[u, hook] \arrow[r] & J \arrow[r] & I
\end{tikzcd}$$

Since I is injective, the composed map of $\mathfrak{b}/\left(\mathfrak{b} \cap \mathfrak{a}^{n'}\right)$ to I extends to a map $\psi' : A/\mathfrak{a}^{n'} \to I$. But the image of ψ' is annihilated by $\mathfrak{a}^{n'}$, so it is contained in J. Composing with the natural map $A \to A/\mathfrak{a}^{n'}$, we obtain the required map $\psi : A \to J$ extending φ, as we sought to show.

Lemma 5.1.14: Flasque on I

Let I be an injective R-module over a Noetherian ring R. Then for any $f \in R$, the natural map $I \to I_f$ is surjective

Proof **:**
Let $f \in R$ and $\varphi : I \to I_f$ be the natural map. I claim this map is surjective. Pick $x \in I_f$. By definition of localization and of φ :

$$\frac{\varphi(y)}{f^n} = x$$

We shall find an element that maps to the left hand side. To find this map, for each $i > 0$, let $\mathfrak{b}_i$ be the annihilator of f^i. Then $\mathfrak{b}_1 \subseteq \mathfrak{b}_2 \subseteq \cdots$, and as R is Noetherian there exists an m such that $\mathfrak{b}_m = \mathfrak{b}_{m+1} = \cdots$. Now, define from the ideal to the injective R-module $\psi : (f^{n+m}) \to I$ where $f^{n+m} \mapsto f^m y$. As I is injective, by Baer's criterion this map extends to $\Psi : A \to I$. Label $\psi(1)$ as z. Then we have $f^{n+m} z = f^m z$. But then we see that Ψ and φ match, in particular we have:

$$\varphi(z) = \frac{\varphi(y)}{f^n} = x$$

showing surjectivity, and completing the proof.

Lemma 5.1.15: Flasque on II

Let I be an injective R-module over a Noetherian ring R. Then $\tilde{I}$ on $X = \operatorname{Spec} R$ is flasque Note that $\widetilde{I}$ is flasque even if R is not Noetherian, however this proof would be adds technicality without adding insight.

Proof **:**
We will use Noetherian induction on $Y = (\operatorname{Supp} \tilde{I})$ (see section 0.3 for an overview of Noetherian induction). If Y consists of a single closed point of X, then I is a skyscraper sheaf which is certainly flasque.
In the general case, it suffices to show that for any open set $U \subseteq X, \Gamma(X, \tilde{I}) \to \Gamma(U, \tilde{I})$ is surjective. If $Y \cap U = \emptyset$, we are done, so say $Y \cap U \neq \emptyset$. Then there is an $f \in A$ such that the open set $X_f = D(f)$ is contained in U and $X_f \cap Y \neq \emptyset$. Let $Z = X - X_f$, and consider the following diagram:

$$\begin{tikzcd}
\Gamma(X, \widetilde{I}) \arrow[r] & \Gamma_Z(U, \widetilde{I}) \arrow[r] & \Gamma(X_f, \widetilde{I}) \\
\Gamma_Z(X, \widetilde{I}) \arrow[r] \arrow[u, hook] & \Gamma_Z(U, \widetilde{I}) \arrow[u, hook] &
\end{tikzcd}$$

where Γ_Z denotes sections with support in Z. Now given a section $s \in \Gamma(U, \tilde{I})$, consider its image s' in $\Gamma\left(X_f, \tilde{I}\right)$. By proposition 4.1.11 $\Gamma\left(X_f, \tilde{I}\right) = I_f$, so by lemma 5.1.13 there is a $t \in I = \Gamma(X, \tilde{I})$ restricting to s'. Let t' be the restriction of t to $\Gamma(U, \tilde{I})$. Then $s - t'$ goes to 0 in $\Gamma\left(X_f, \tilde{I}\right)$, so it has support in Z. Thus to complete the proof, it will be sufficient to show that $\Gamma_Z(X, \tilde{I}) \to \Gamma_Z(U, \tilde{I})$ is surjective.
Now, let $J = \Gamma_Z(X, \tilde{I})$. If $\mathfrak{a}$ is the ideal generated by f, then $J = \Gamma_{\mathfrak{a}}(I)$ so by lemma 5.1.12, J is also an injective A-module. Furthermore, the support of $\tilde{J}$ is contained in $Y \cap Z$, which is strictly smaller than Y. Hence by our induction hypothesis, $\tilde{J}$ is flasque. Since $\Gamma(U, \tilde{J}) = \Gamma_Z(U, \tilde{I})$, we conclude that

$$\Gamma_Z(X, \tilde{I}) \to \Gamma_Z(U, \tilde{I})$$

is surjective, as we sought to show.

Theorem 5.1.16: Affine Noetherian Scheme: Trivial Cohomology

Let $X = \operatorname{Spec} R$ be the spectrum of a Noetherian ring R. Then for all quasi-coherent sheaves $\mathscr{F}$ on X,

$$H^i(X, \mathscr{F}) = 0 \quad \forall i > 0$$

***Proof* :**
Let $\mathscr{F}$ be a quasi-coherent sheaf on the $X = \operatorname{Spec} R$ so that we have a module $M = \Gamma(X, \mathscr{F})$ and by corollary 4.1.10 $\mathscr{F} = \widetilde{M}$. Then from [8], we know we can take the injective resolution $0 \to M \to I^*$. Then as $\widetilde{(-)}$ is exact, we have that $0 \to \widetilde{M} \to \widetilde{I}^*$ is exact. By lemma 5.1.14, each $\widetilde{I}^i$ is flasque, hence by the principles of homology we can use this sequence to compute the cohomology groups. But then, applying the (in this case exact) Γ we have the sequence of R-modules $0 \to M \to I^*$ and we immediately get:

$$H^0(X, \mathscr{F}) = M \quad H^i(X, \mathscr{F}) = 0 \quad i > 0$$

as we sought to show.

Corollary 5.1.17: Injective Embedding of Sheaves On Neotherian Schemes

Let X be a Noetherian scheme and $\mathscr{F}$ a quasi-coherent sheaf on X. Then $\mathscr{F}$ can embed in a quasi-coherent flasque sheaf.

This corollary shall in fact use the Noetherian condition in a meaningful way, as shall be pointed out in the proof

***Proof* :**
As X is Noetherian, let $X = \bigcup_i^n \operatorname{Spec} R_i = \bigcup_i^n U_i$ be a finite open affine cover, and let $\mathscr{F}|_U = \widetilde{M_i}$ for each i. Embed each M_i into an injective R_i-module I_i, and define:

$$\mathscr{G} = \bigoplus_{i}^{n} \iota_* \left(\widetilde{I}_i\right)$$

where ι is the inclusion of $U_i \to X$ for each U_i (where we are overloading notation letting ι represent each inclusion). As for each i we have the injective map $\mathscr{F}|_{U_i} \to \widetilde{I}_i$, we get a map the injective maps $\mathscr{F} \to f_*\left(\widetilde{I}_i\right)$. Then taking the direct sum, we get the injective map $\mathscr{F} \to \mathscr{G}$. Now, each $\widetilde{I}_i$ is flasque by lemma 5.1 .14 and by definition quasi-coherent on U_i. Then $f_*\left(\widetilde{I}_i\right)$ is also flasque (exercise) and quasi-coherent (proposition 4.1.14 this uses the Noetherian assumption!). Then certainly the direct sum is flasque, completing the proof.

The final theorem shall show that having trivial cohomology > 0 characterizes affine schemes (with the necessary Noetherian condition to be consistent with what we've proven):

Theorem 5.1.18: Characterizing affine schemes using Cohomology

Let X be a Noetherian scheme. Then the following are equivalent:

1. X is affine
2. $H^i(X, \mathscr{F}) = 0$ for $i > 0$ and all for all quasi-coherent sheaves $\mathscr{F}$
3. $H^1(X, \mathscr{I}) = 0$ for $i > 0$ and all for all coherent sheaves of ideals $\mathscr{I}$

***Proof* :**
(1) $\Rightarrow$ (2) Theorem 5.1 .15
(2) $\Rightarrow$ (3) A-fortiori
(3) $\Rightarrow$ (1) We shall prove X is affine using corollary 3.1.9. First we show that X can be covered by open affine subsets of the form X_f, with $f \in R = \Gamma(X, \mathcal{O}_X)$. Let $\mathfrak{p}$ be a closed point of X, let U be an open affine neighborhood of $\mathfrak{p}$, and let $Y = X - U$. Then we have an exact sequence

$$0 \to \mathscr{I}_{Y \cup \{\mathfrak{p}\}} \to \mathscr{I}_Y \to k(\mathfrak{p}) \to 0$$

where $\mathscr{I}_Y$ and $\mathscr{I}_{Y \cup \{\mathfrak{p}\}}$ are the ideal sheaves of the closed sets Y and $Y \cup \{\mathfrak{p}\}$, respectively. The quotient is the skyscraper sheaf $k(\mathfrak{p}) = \mathcal{O}_{\mathfrak{p}}/\mathfrak{m}_{\mathfrak{p}}$ at $\mathfrak{p}$. Now from the exact sequence of cohomology, by (3) we get an exact sequence

$$\Gamma(X, \mathscr{I}_Y) \to \Gamma(X, k(\mathfrak{p})) \to H^1\left(X, \mathscr{I}_{Y \cup \{\mathfrak{p}\}}\right) = 0.$$

So there is an element $f \in \Gamma(X, \mathscr{I}_Y)$ which goes to 1 in $k(\mathfrak{p})$, i.e., $f_{\mathfrak{p}} \equiv 1 \,(\mathrm{mod}\, \mathfrak{m}_{\mathfrak{p}})$. Since $\mathscr{I}_Y \subseteq \mathcal{O}_X$, we can consider f as an element of R. Then by construction, $\mathfrak{p} \in X_f \subseteq U$. Furthermore, $X_f = U_{\bar{f}}$, where $\bar{f}$ is the image of f in $\Gamma(U, \mathcal{O}_U)$, so X_f is affine.
Thus every closed point of X has an open affine neighborhood of the form X_f. By quasi-compactness, we can cover X with a finite number of these, corresponding to $f_1, \ldots, f_r \in R$.
Let us verify that $(f_1, \ldots, f_r) = R$. We use $f_1, \ldots, f_r$ to define a map $\alpha : \mathcal{O}_X^r \to \mathcal{O}_X$ by sending $\langle a_1, \ldots, a_r \rangle$ to $\sum f_i a_i$. Since the X_{f_i} cover X, this is a surjective map of sheaves. Let $\mathscr{F}$ be the kernel:

$$0 \to \mathscr{F} \to \mathcal{O}_X^r \xrightarrow{\alpha} \mathcal{O}_X \to 0$$

Filter $\mathscr{F}$ as follows:

$$\mathscr{F} = \mathscr{F} \cap \mathcal{O}_X^r \supseteq \mathscr{F} \cap \mathcal{O}_X^{r-1} \supseteq \cdots \supseteq \mathscr{F} \cap \mathcal{O}_X$$

for a suitable ordering of the factors of $\mathcal{O}_X^r$. Each of the quotients of this filtration is a coherent sheaf of ideals in $\mathcal{O}_X$. Thus by (3) and the long exact sequence of cohomology, we climb up the filtration and deduce that $H^1(X, \mathscr{F}) = 0$. But then $\Gamma(X, \mathcal{O}_X^r) \to \Gamma(X, \mathcal{O}_X)$ is surjective, which tells us that $f_1, \ldots, f_r$ generates R, as we sought to show.

5.2 Computation: Čech Cohomology

The Čech Cohomology computes to what degree a topological space X with a sheaf of abelian groups is covered. In the case where X is a Noetherian separated scheme with a quasi-coherent sheaf and the covering is an affine open covering, then the Čech cohomology groups shall be coincide with the cohomology groups defined by the Global section functor. Čech cohomology shall provide practical way to compute the cohomology. The following construction will be familiar to those who have seen the computation of group cohomology or have covered algebraic topology (see [2, chapter 4])

Let X be a topological space and let $\mathcal{U} = (U_i)_{i \in I}$ be an open covering. Give I a well-ordering[1]. Then for any finite set of indices $i_0, ..., i_p \in I$, let

$$U_{i_0,\ldots,i_p} = U_{i_0} \cap \cdots \cap U_{i_p}$$

Let $\mathscr{F}$ be a sheaf of abelian groups on X. Define $C^*(\mathcal{U}, \mathscr{F})$ to be:

$$C^p(\mathcal{U}, \mathscr{F}) = \prod_{i_0 < \cdots < i_p} \mathscr{F}(U_{i_0,\ldots,i_p})$$

where each element $\alpha \in C^p(\mathcal{U}, \mathscr{F})$ is given choosing an element $\alpha_{i_0,\ldots,i_p} \in \mathscr{F}(U_{i_0,\ldots,i_p})$. For each $(p+1)$-tuple $i_0 < \cdots < i_p$. Define the coboundary map $d : C^p \to C^{p+1}$ to be:

$$(d\alpha)_{i_0,\ldots,i_{p+1}} = \sum_{k=0}^{p+1} (-1)^k \alpha_{i_0,\ldots,\widehat{i_k},\ldots,i_{p+1}}|_{U_{i_0,\ldots,i_{p+1}}}$$

As usual, $d^2 = 0$, meaning we have a complex and hence a cohomology:

Definition 5.2.1: Čech Cohomology

Let X be a topological space and let $\mathcal{U}$ be an open covering of X. For any sheaf of abelian groups $\mathscr{F}$ on X, define the *Čech cohomology group* of $\mathscr{F}$ with respect to $\mathcal{U}$ to be:

$$\check{\mathrm{H}}^p(\mathcal{U}, \mathscr{F}) = h^p(C^*(\mathcal{U}, \mathscr{F})) = \frac{\ker\, d^{+1}}{\operatorname{im} d}$$

[1]Alternatively, make sure that if indices repeat then the element is 0 , and rearranging the indices gives multiplies the function by $(-1)^\sigma$

Note that $\check{\mathrm{H}}^p(\mathfrak{U}, -)$ is not in general a δ-functor, that is if $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$ is a short exact sequence of sheaves of abelian groups on $X, \check{H}^p(\mathfrak{U}, -)$ is not a δ-functor:

Example 5.2.2: Čech cohomology not δ-functor

Say $\mathfrak{U}$ consists of only X. Then $\check{\mathrm{H}}^p(\mathfrak{U}, \mathscr{F}) = 0$ for $p > 0$ because all intersection are trivial. Furthermore, $\check{\mathrm{H}}^0(\mathfrak{U}, \mathscr{F}) = \check{\mathrm{H}}^0(X, \mathscr{F}) = \Gamma(X, \mathscr{F})$. Hence, if $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$, then applying $\check{\mathrm{H}}^*(\mathfrak{U}, -)$ we get the "long exact sequence":

$$0 \to \Gamma(X, \mathscr{F}') \to \Gamma(X, \mathscr{F}) \to \Gamma(X, \mathscr{F}'') \to 0$$

But $\Gamma(X, -)$ need not be right-exact, and hence this is not a long exact sequence!

This is one of the few Cohomologies theories which does not work with derived functors, even though they are central in most of homological algebra. Čech cohomology is very useful for computing.

Example 5.2.3: Čech Cohomology

1. Let $X = \mathbb{P}^1_k, \mathscr{F}$ be sheaf of differentials Ω, and $\mathfrak{U}$ the usual open covering of two $\mathbb{A}^1_k$ with coordinates x and $1/x$; label these sets U, V. Then we get two terms:

$$\begin{aligned} C^0 &= \Gamma(U, \Omega) \times \Gamma(V, \Omega) \\ C^1 &= \Gamma(U \cap V, \Omega) \end{aligned}$$

   Then:

$$\begin{aligned} \Gamma(U, \Omega) &= k[x]dx \\ \Gamma(V, \Omega) &= k[y]dy \\ \Gamma(U \cap V, \Omega) &= k\left[x, \frac{1}{x}\right] dx \end{aligned}$$

   The differential map $d : C^0 \to C^1$ is given by:

$$\begin{aligned} x &\mapsto x \\ y &\mapsto \frac{1}{z} \\ dy &\mapsto \frac{1}{x^2}dx \end{aligned}$$

   Hence, $\ker d$ is all elements $(f(x)dx, g(y)dy) \in C^0$ such that

$$f(x) = \frac{-1}{x^2} g\left(\frac{1}{x}\right)$$

   But this is true if and only if $f = g = 0$. Hence:

$$\check{\mathrm{H}}^0(\mathfrak{U}, \Omega) = 0$$

Next, for $\check{\mathrm{H}}^1$, the image of d is all expression of the form:

$$\left(f(x) = \frac{1}{x^2} g\left(\frac{1}{x}\right)\right) dx$$

This is a sub-space of $k[x, 1/x]dx$ generated by all $x^n dx$ with $-1 \neq n \in \mathbb{Z}$. Hence

$$\check{\mathrm{H}}^1(\mathfrak{U}, \Omega) = k$$

which is generated by the image of $x^{-1}dx$.

2. Let us compute the Čech cohomology in a more classical setting: take S^1 and put the constant $\mathbb{Z}$ sheaf on it, $\underline{\mathbb{Z}}$. Let U, V be two open semi-circle where $U \cap V$ is the disjoint small open intervals.

Let us now build-up towards showing the cohomology given by the derived functor on Γ matches the Čech cohomology

Lemma 5.2.4: 0th Cohomology Matching

Let X be a topological space, $\mathfrak{U}$ an open covering, and $\mathscr{F}$ a sheaf of abelian groups on X. Then:

$$\check{\mathrm{H}}^0(\mathfrak{U}, \mathscr{F}) = \Gamma(X, \mathscr{F})$$

***Proof* :**
By definition, $\check{\mathrm{H}}^0(\mathfrak{U}, \mathscr{F}) = \ker\left(d : C^0(\mathfrak{U}, \mathscr{F}) \to C^1(\mathfrak{U}, \mathscr{F})\right)$. If $\alpha \in C^0$ is given by $\{\alpha_i \in \mathcal{F}(U_i)\}$, then for each $i < j$, $(d\alpha)_{ij} = \alpha_j - \alpha_i$. So $d\alpha = 0$ says the sections α_i and α_j agree on $U_i \cap U_j$. But then by the sheaf axioms $\ker d = \Gamma(X, \mathcal{F})$.

We next work towards fixing the lack of a δ-functor. Let $X, \mathfrak{U}, \mathscr{F}$ be as usual, $V \subseteq X$ an open subset, and $\iota : V \to X$ the natural inclusion map. Define the complex $\mathscr{C}(\mathfrak{U}, \mathscr{F})$ of sheaves of on X for each $p \geq 0$ as:

$$\mathscr{C}(\mathfrak{U}, \mathscr{F}) = \prod_{i_0 < \cdots < i_p} f_*\left(\mathscr{F}|_{U_{i_0, \ldots, i_p}}\right)$$

and define the differential map $d : \mathscr{C}^p \to \mathscr{C}^{p+1}$ in the same way. Importantly, this was defined so that:

$$\Gamma\left(X, \mathscr{C}^p(\mathfrak{U}, \mathscr{F})\right) = C^p(\mathfrak{U}, \mathscr{F})$$

Lemma 5.2.5: Čech Cohomology Resolution

Let X be a topological space, $\mathfrak{U}$ an open covering, and $\mathscr{F}$ a sheaf of abelian groups. Then the complex $\mathscr{C} * (\mathfrak{U}, \mathscr{F})$ is a resolution of $\mathscr{F}$, namely we have a long exact sequence:

$$0 \to \mathscr{F} \xrightarrow{\epsilon} \mathscr{C}^0(\mathfrak{U}, \mathscr{F}) \to \mathscr{C}^1(\mathfrak{U}, \mathscr{F}) \to \cdots$$

for appropriate natural map $\epsilon : \mathscr{F} \to \mathscr{C}^0$

Proof **:**
Define $\epsilon : \mathscr{F} \to \mathfrak{U}^0$ by taking the product of the natural maps $\mathscr{F} \to f_* \left(\mathscr{F}|_{U_i}\right)$ for $i \in I$. Then the exactness at the first step follows from the sheaf axioms for $\mathscr{F}$.
For the exactness of the complex $\mathscr{C}$ for $p \geq 1$, it is enough to check exactness on the stalks. Let $x \in X$, and suppose $x \in U_j$. For each $p \geq 1$, define the map $k : \mathscr{C}^p(\mathfrak{U}, \mathscr{F})_x \to \mathcal{C}^{p-1}(\mathfrak{U}, \mathscr{F})_x$ as follows: for $\alpha_x \in \mathscr{C}^p(\mathfrak{U}, \mathscr{F})_x$, it is represented by a section $\alpha \in \Gamma\left(V, \mathscr{C}^p(\mathfrak{U}, \mathscr{F})\right)$ over a neighborhood V of x, which we may choose so small that $V \subseteq U_j$. Now for any p-tuple $i_0 < \ldots < i_{p-1}$, set

$$(k\alpha)_{i_0,\ldots,i_{p-1}} = \alpha_{j,i_0,\ldots,i_{p-1}}$$

This makes sense because $V \cap U_{i_0,\ldots,i_{p-1}} = V \cap U_{j,i_0,\ldots,i_{p-1}}$. Then take the stalk of $k\alpha$ at x to get the required map k. Now one checks that for any $p \geq 1, \alpha \in \mathscr{C}_x^p$,

$$(dk + kd)(\alpha) = \alpha$$

Thus k is a homotopy operator for the complex $\mathscr{C}_x$, showing that the identity map is homotopic to the zero map! It follows that the cohomology groups $h^p\left(\mathscr{C}_x\right)$ of this complex are 0 for $p \geq 1$.

Lemma 5.2.6: Čech and Flasque Sheaves

Let X be a topological space, $\mathfrak{U}$ be an open covering, and let $\mathscr{F}$ be a flasque sheaf of abelian groups on X. Then for all $p > 0$ we have $\check{\mathrm{H}}^p(\mathfrak{U}, \mathscr{F}) = 0$.

Proof **:**
Consider the resolution $0 \to \mathscr{F} \to \mathcal{C}(\mathfrak{U}, \mathscr{F})$ given by lemma 5.2.3. As $\mathscr{F}$ is flasque, the sheaves $\mathcal{C}^p(\mathfrak{U}, \mathscr{F})$ are flasque for each $p \geq 0$. Indeed, for any $i_0, \ldots, i_p$, $\mathscr{F}|_{U_{i_0},\ldots,i_p}$ is a flasque sheaf on $U_{i_0,\ldots,i_p}$, f_* preserves flasque sheaves, and a product of flasque sheaves is flasque. Thus, we can use this resolution to compute the usual cohomology groups of $\mathscr{F}$. But $\mathscr{F}$ is flasque, so $H^p(X, \mathscr{F}) = 0$ for $p > 0$. On the other hand, the answer given by this resolution is

$$h^p(\Gamma(X, \mathcal{C} \cdot (\mathfrak{U}, \mathscr{F}))) = \check{\mathrm{H}}^p(\mathfrak{U}, \mathscr{F})$$

But then $\check{\mathrm{H}}^p(\mathfrak{U}, \mathscr{F}) = 0$ for $p > 0$, as we sought to show.

Lemma 5.2.7: Chech Cohomology Functorial Map

Let X be a topological space and $\mathfrak{U}$ an open covering. Then for each $p \geq 0$ there is a natural map, functorial in $\mathscr{F}$,
$$\check{\mathrm{H}}^p(\mathfrak{U}, \mathscr{F}) \to H^p(X, \mathscr{F})$$

Proof **:**
This follows from comparing two acyclic resolution (see [8]), namely:

$$0 \to \mathscr{F} \to \mathscr{I}^*$$
$$0 \to \mathscr{F} \to \mathscr{C}^*(\mathfrak{U}, \mathscr{F})$$

which gives a morphism of complexes $\mathscr{C}^*(\mathfrak{U}, \mathscr{F}) \to \mathscr{I}^*$ which induces the identity map on $\mathscr{F}$ and is unique up to homotopy. By applying the functors $\Gamma(X, -)$ and h^p, we extract the desired maps.

Theorem 5.2.8: Čech Cohomology Equivalence

Let X be a Noetherian separated scheme, $\mathfrak{U}$ an open affine cover of X, and let $\mathscr{F}$ be a quasi-coherent sheaf on X. Then for all $p \geq 0$, the natural maps

$$\check{\mathrm{H}}^p(\mathfrak{U}, \mathscr{F}) \to H^p(X, \mathscr{F})$$

are isomorphisms.

Proof **:**
For $p = 0$ we have an isomorphism by lemma 5.2.2, For $p > 0$, by corollary 5.1.16 embed $\mathscr{F}$ in a flasque, quasi-coherent sheaf $\mathcal{G}$, and let $\mathcal{Q}$ be the quotient:

$$0 \to \mathscr{F} \to \mathcal{G} \to \mathcal{Q} \to 0$$

For each $i_0 < \ldots < i_p$, the open set $U_{i_0,\ldots,i_p}$ is affine, since it is an intersection of affine open subsets of a separated scheme. Since $\mathscr{F}$ is quasi-coherent, we therefore have an exact sequence

$$0 \to \mathscr{F}\left(U_{i_0,\ldots,i_p}\right) \to \mathcal{G}\left(U_{i_0,\ldots,i_p}\right) \to \mathcal{Q}\left(U_{i_0,\ldots,i_p}\right) \to 0$$

of abelian groups by proposition 4.1.13. Taking products, we find that the corresponding sequence of Čech complexes

$$0 \to C^{\cdot}(\mathfrak{U}, \mathscr{F}) \to C^{\cdot}(\mathfrak{U}, \mathcal{G}) \to C^{\cdot}(\mathfrak{U}, \mathcal{Q}) \to 0$$

is exact. Therefore we get a long exact sequence of Čech cohomology groups. Since $\mathcal{G}$ is flasque, its Čech cohomology vanishes for $p > 0$, so we have an exact sequence

$$0 \to \check{\mathrm{H}}^0(\mathfrak{U}, \mathscr{F}) \to \check{\mathrm{H}}^0(\mathfrak{U}, \mathcal{G}) \to \check{\mathrm{H}}^0(\mathfrak{U}, \mathcal{Q}) \to \check{\mathrm{H}}^1(\mathfrak{U}, \mathscr{F}) \to 0$$

and isomorphisms

$$\check{\mathrm{H}}^p(\mathfrak{U}, \mathcal{Q}) \to \check{\mathrm{H}}^{p+1}(\mathfrak{U}, \mathscr{F})$$

for each $p \geq 1$. Now comparing with the long exact sequence of usual cohomology for the above short exact sequence, using the case $p = 0$, the fact it is flasque, we conclude that the natural map

$$\check{\mathrm{H}}^1(\mathfrak{U}, \mathscr{F}) \to H^1(X, \mathscr{F})$$

is an isomorphism. But $\mathcal{Q}$ is also quasi-coherent (proposition 4.1.14), so we obtain the result for all p by induction, completing the proof.

5.2.1 Application: Sheaf Cohomology of Projective Spaces

Let us now compute the cohomology for the twisted sheaves on projective space. Let R be a Noetherian ring, $S_\bullet = R[x_0, \ldots, x_n]$, and $X = \operatorname{Proj} S = \mathbb{P}^n_R$. Let $\mathcal{O}_X(1)$ be the twisted sheaf of Serre. If $\mathscr{F}$ is a sheaf of $\mathcal{O}_X$-module, let $\Gamma_*(\mathscr{F})$ be the graded S-module

Theorem 5.2.9: Twisted Sheaf Cohomology

Let R be a Noetherian ring, and let $X = \mathbb{P}^n_R$, with $n \geq 1$. Then:

1. $H^0(X, \mathcal{O}_X(-d)) \cong 0$ for $d \in \mathbb{N}$
2. $H^n(X, \mathcal{O}_X(-n-1)) \cong R$.
3. The natural map
$$H^0(X, \mathcal{O}_X(n)) \times H^n(X, \mathcal{O}_X(-d-n-1)) \to H^n(X, \mathcal{O}_X(-n-1)) \cong R$$
is a perfect pairing (isomorphic bilinear map) of finitely generated free R-modules, for each $d \in \mathbb{Z}$.
4. $H^i(X, \mathcal{O}_X(d)) = 0$ for $0 < i < n$ and all $d \in \mathbb{Z}$

Overall:

$$H^i\left(\mathbb{P}^n_R, \mathcal{O}_{\mathbb{P}^n_R}(d)\right) = \begin{cases} (R[x_0, \ldots, x_n])_d & i = 0, d \geq 0 \\ \left(\frac{1}{x_0 x_1 \cdots x_n} R[1/x_0, \ldots, 1/x_n]\right)_d & i = n, d < 0 \\ 0 & \text{otherwise} \end{cases}$$

Notice how this is different from the singular cohomology of projective space, namely the higher order cohomology groups when $R = \mathbb{C}$ and $d = 0$ are trivial! See [2, section 4.1] for computations of the singular cohomology of $\mathbb{P}^n_{\mathbb{C}}$.

***Proof* :**

Let us first setup the Čech cohomology. Let $\mathscr{F} = \bigoplus_{d \in \mathbb{Z}} \mathcal{O}_X(d)$. As cohomology commutes with direct sums on Noetherian space, the cohomology on $\mathscr{F}$ will be the direct sum of the cohomology on $\mathcal{O}_X(d)$ for $d \in \mathbb{Z}$. Hence, we can compute $\mathscr{F}$ to find the cohomologies.

Let $D(x_i)$ cover $\operatorname{Proj} S, 0 \leq i \leq n$: label this open cover $\mathcal{U}$. Then for any direct subset of indices, we have:

$$\mathscr{F}\left(U_{i_0,\ldots,i_p}\right) = S_{x_{i_0},\ldots,x_{i_p}}\Big)$$

Thus, the grading of $\mathscr{F}$ is given by the grading of S, and we get:

$$C^*(\mathcal{U},\mathscr{F}) : \prod S_{x_{i_0}} \to \prod S_{x_{i_0},x_{i_1}} \to \cdots \to S_{x_{i_0},\ldots,x_{i_n}}$$

1. $H^0(X,\mathscr{F})$ is the kernel of d^0 (the first map), which is just S by proposition 4.2.7.
2. $H^0\left(X,\mathcal{O}_X(-d)\right)$ must be 0 as we have the isomorphism above.
3. $H^n(X,\mathscr{F})$ is the cokernel of of d^{n-1} (the last map). To see what this is, notice that $S_{x_{i_0},\ldots,x_{i_n}}$ is a free R-module generated by monomial $\left\{x_0^{l_0}x_1^{l_1}\cdots x_n^{l_n}\middle| l_i \in \mathbb{Z}\right\}$. The image of d^{n-1} is generated by the sub-basis where at least one $l_i \geq 0$. Hence, $H^n(\mathscr{F},X)$ is a free R-module generated by the negative monomials:

   $$\left\{x_0^{l_0}x_1^{l_1}\cdots x_n^{l_n}\middle| l_i \in -\mathbb{N}\right\}$$

   Note that this is graded by $\sum_i l_i$. Now there is only one monomial of degree $-n-1$, namely:

   $$x_0^{-1}\cdots x_n^{-1}$$

   hence:

   $$H^n\left(X,\mathcal{O}_X(-n-1)\right) \cong R$$

4. Note that $H^n\left(X,\mathcal{O}_X(-d-n-1)\right) \cong 0$ since there are no negative monomials of degree bigger than 0 for $m < -n-1$. For $d \geq 0, H^0\left(X,\mathcal{O}_X(d)\right)$ has a basis consisting of the usual monomials of degree d, i.e., $\left\{x_0^{m_0}\cdots x_n^{m_n}\middle| m_i \geq 0 \text{ and } \sum m_i = d\right\}$. The natural pairing with $H^n\left(X,\mathcal{O}_X(-n-d-1)\right)$ into $H^n\left(X,\mathcal{O}_X(-n-1)\right)$ is determined by

   $$\left(x_0^{m_0}\cdots x_n^{m_n}\right)\cdot\left(x_0^{l_0}\cdots x_n^{l_n}\right) = x_0^{m_0+l_0}\cdots x_n^{m_n+l_n}$$

   where $\sum l_i = -d-n-1$, and the object on the right becomes 0 if any $m_i + l_i \geq 0$. So it is clear that there is a perfect pairing under which $x_0^{-m_0-1}\cdots x_n^{-m_n-1}$ is the dual basis element of $x_0^{m_0}\cdots x_n^{m_n}$.

5. Let us do induction on n. If $n=1$ there is nothing to prove, so let $n>1$. If we localize the complex $C^*(\mathfrak{U},\mathscr{F})$ with respect to x_n, as graded S-modules, we get the Čech complex for the sheaf $\mathscr{F}|_{U_n}$ on the space U_n, with respect to the open affine covering $\left\{U_i \cap U_n \middle| i = 0,\ldots,n\right\}$. By (4.5), this complex gives the cohomology of $\mathscr{F}|_{U_n}$ on U_n, which is 0 for $i>0$ by (3.5). Since localization is an exact functor, we conclude that $H^i(X,\mathscr{F})_{x_n} = 0$ for $i>0$. In other words, every element of $H^i(X,\mathscr{F})$, for $i>0$, is annihilated by some power of x_n.
   Next let's show that for $0<i<n$, multiplication by x_n induces a bijective map of $H^i(X,\mathscr{F})$ into itself. Then it will follow that this module is 0 . Consider the exact sequence of graded S-modules

   $$0 \to S(-1) \xrightarrow{x_n} S \to S/\left(x_n\right) \to 0$$

This gives the exact sequence of sheaves

$$0 \to \mathcal{O}_X(-1) \to \mathcal{O}_X \to \mathcal{O}_H \to 0$$

on X, where H is the hyperplane $x_n = 0$. Twisting by all $d \in \mathbb{Z}$ and taking the direct sum, we have

$$0 \to \mathscr{F}(-1) \to \mathscr{F} \to \mathscr{F}_H \to 0$$

where $\mathscr{F}_H = \bigoplus_{d\in\mathbb{Z}} \mathcal{O}_H(d)$. Taking cohomology, we get a long exact sequence

$$\cdots \to H^i(X, \mathscr{F}(-1)) \to H^i(X, \mathscr{F}) \to H^i(X, \mathscr{F}_H) \to \cdots$$

Considered as graded S-modules, $H^i(X, \mathscr{F}(-1))$ is just $H^i(X, \mathscr{F})$ shifted one place, and the map $H^i(X, \mathscr{F}(-1)) \to H^i(X, \mathscr{F})$ of the exact sequence is multiplication by x_n.
Now H is isomorphic to $\mathbb{P}^{n-1}_R$, and $H^i(X, \mathscr{F}_H) = H^i(H, \bigoplus \mathcal{O}_H(d))$ by lemma 5.1.10. So we can apply our induction hypothesis to $\mathscr{F}_H$, and find that $H^i(X, \mathscr{F}_H) = 0$ for $0 < i < n-1$. Furthermore, for $i = 0$ we have an exact sequence

$$0 \to H^0(X, \mathscr{F}(-1)) \to H^0(X, \mathscr{F}) \to H^0(X, \mathscr{F}_H) \to 0$$

by (1), since $H^0(X, \mathscr{F}_H)$ is just $S/(x_n)$. At the other end of the exact sequence we have

$$0 \to H^{n-1}(X, \mathscr{F}_H) \xrightarrow{\partial} H^n(X, \mathscr{F}(-1)) \xrightarrow{x_n} H^n(X, \mathscr{F}) \to 0$$

Indeed, we have described $H^n(X, \mathscr{F})$ above as the free R-module with basis formed by the negative monomials in $x_0, \ldots, x_n$. So it is clear that x_n is surjective. On the other hand, the kernel of x_n is the free submodule generated by those negative monomials $x_0^{l_0} \cdots x_n^{l_n}$ with $l_n = -1$. Since $H^{n-1}(X, \mathscr{F}_H)$ is the free R-module with basis consisting of the negative monomials in $x_0, \ldots, x_{n-1}$, and ∂ is division by x_n, the sequence is exact. In particular, ∂ is injective.
Putting these results all together, the long exact sequence of cohomology shows that the map multiplication by $x_n : H^i(X, \mathscr{F}(-1)) \to H^i(X, \mathscr{F})$ is bijective for $0 < i < n$, as we sought to show.

5.3 Serre Duality Theorem

Recall from [2, Chapter 4.3] that a compact n-dimensional smooth manifold satisfies *Poincaré Duality*:

$$H^i(M; \mathbb{Z}) \cong H_{n-i}(M; \mathbb{Z})$$

In other words, cycles on a manifold and cocycles are in fact dual concepts to each other (some concrete consequence of this is that looking at the "boundary" and the derivative are in fact two sides of the same coin). By using sheaves of differentials, we recover a similar duality for schemes. In this section, we shall start by building up some more cohomological theory which will

5.3.1 Ext Group on Sheaves

Recall from [8, chatper 27.3] that $\mathrm{Hom}(M,-)$ is a left-exact functor that gives rise to right-derived functors $\mathrm{Ext}^i(M,-)$. Let us translate these results to sheaves. First, recall that if $\mathscr{F},\mathscr{G}$ are sheaves on X, $\mathrm{Hom}(\mathscr{F},\mathscr{G})$ is the group of $\mathcal{O}_X$-module homomorphism, and $\mathcal{H}om(\mathscr{F},\mathscr{G})$ is the *hom-sheaf*. From these, we get the following:

Definition 5.3.1: Ext Group For Sheaves

Let $(X,\mathcal{O}_X)$ be a ringed space and $\mathscr{F}$ a $\mathcal{O}_X$-module. Then the *right-derived functors* of $\mathrm{Hom}(\mathscr{F},-)$ are denoted as $\mathrm{Ext}^i(\mathscr{F},-)$, and the right-derived functors of $\mathcal{H}om(\mathscr{F},-)$ is $\mathcal{E}xt^i(\mathscr{F},-)$.

For both Ext and $\mathcal{E}xt$, we have that $\mathrm{Ext}^0 = \mathrm{Hom}$, and if $\mathscr{G}$ is injective than Ext^i and $\mathcal{E}xt^i$ is trivial for $i>0$. One natural result we need to establish is that an in injective shear $\mathscr{I}$ on X restricts to an injective sheaf $\mathscr{I}|_U$ on U for any open subset:

Lemma 5.3.2: Injective Sheaves and Restriction

Let $\mathscr{I}$ be an injective sheaf on X. Then $\mathscr{I}|_U$ is an injective sheaf on U.

***Proof* :**
The key is to remember that injective naturally extend to global maps (see [13, p. III 6.1])

Proposition 5.3.3: Open Subset and Ext Group

Let $(X,\mathcal{O}_X)$ be a ringed space and $U\subseteq X$ an open subset. Then:

$$\mathcal{E}xt^i_X(\mathscr{F},\mathscr{G})|_U \cong \mathcal{E}xt^i_U(\mathscr{F}|_U,\mathscr{G}|_U)$$

***Proof* :**
To outline: first, by [8, Chapter 27] from $\mathscr{G}$ we get a δ-functor from X-**Mod** to U-**Mod**, which by the comments above agree at $i=0$. With $\mathscr{G}$ injective, both sides vanish on $i>0$, and the rest follows from lemma 5.3.2.

For computational purposes, we state the usual functor-representation result:

Proposition 5.3.4: Functor Representation and Ext

Let $\mathscr{G}$ be any $\mathcal{O}_X$-module on X. Then:

1. $\mathcal{E}xt^0(\mathcal{O}_X,\mathscr{G})\cong\mathscr{G}$
2. $\mathcal{E}xt^i(\mathcal{O}_X,\mathscr{G})=0$ for $i>0$
3. $\mathrm{Ext}^i(\mathcal{O}_X,\mathscr{G})=H^i(X,\mathscr{G})$ for $i>0$

Proof **:**
For (1) and (2), $\mathcal{H}om(\mathcal{O}_X, -)$ is the identity functor. Point (3) follows from $\Gamma(X, -)$ and $\mathrm{Hom}(\mathcal{O}_X, -)$ being the same (see corollary 5.1.7).

5.3.2 The Theorem

I will write down these results in more detail later. For now, I will put down the important theorems for reference:

Theorem 5.3.5: Duality for Projective Space

Let $X = \mathbb{P}^n_k$ over a field k. Then:

1. $H^n(X, \omega_X) \cong k$. Fix one such isomorphism;
2. for any coherent sheaf $\mathcal{F}$ on X, the natural pairing
$$\mathrm{Hom}(\mathcal{F}, \omega) \times H^n(X, \mathcal{F}) \to H^n(X, \omega) \cong k$$
is a perfect pairing of finite-dimensional vector spaces over k;
3. for every $i \geq 0$ there is a natural functorial isomorphism
$$\mathrm{Ext}^i(\mathcal{F}, \omega) \to H^{n-i}(X, \mathcal{F})',$$
where the tick, $'$, denotes the dual vector space, which for $i = 0$ is the one induced by the pairing of (2).

Proof **:**
Hartshorne III 7.1 (p.239)

Theorem 5.3.6: Duality for a Projective Scheme

Let X be a projective scheme of dimension n over an algebraically closed field k, ω_X be a dualizing sheaf on X, and let $\mathcal{O}(1)$ be a very ample sheaf on X. Then:

1. for all $i \geq 0$ and $\mathcal{F}$ coherent on X, there are natural functorial maps

$$\theta^i : \mathrm{Ext}^i(\mathcal{F}, \omega_X) \to H^{n-i}(X, \mathcal{F})',$$

   such that θ^0 is the map given in the definition of dualizing sheaf above;

2. the following conditions are equivalent:
   (a) X is Cohen-Macaulay and equidimensional (that is, all irreducible components have the same dimension);
   (b) for any $\mathcal{F}$ locally free on X, we have $H^i(X, \mathcal{F}(-q)) = 0$ for $i < n$ and $q \gg 0$;
   (c) the maps θ^i of (1) are isomorphisms for all $i \geq 0$ and all $\mathcal{F}$ coherent on X.

Proof **:**
Hartshorne III 7.6 (p.241)

Corollary 5.3.7: Serre Duality: Nonsingular Projective Variety

Let X be a nonsingular projective variety over an algebraically closed field k. Then the dualizing sheaf ω_X° is isomorphic to the canonical sheaf ω_X.

Proof **:**
As X is projective, embed it into $\mathbb{P}^n_k$. Then by theorem 4.5.21, X is a local complete intersection in $\mathbb{P}^n_k$, and so by proposition 4.5.26

$$\omega_X \cong \Omega_P \otimes \bigwedge^r \widetilde{\mathscr{I}/\mathscr{I}^2}$$

as we sought to show.

5.4 Families of Schemes

here

Recall that the fibers of a scheme morphism is a scheme. We may consider what type of families these fibers for.

Definition 5.4.1: Family of Schemes

Let X, B be schemes. Then a *Family of schemes* is given by a morphism $\pi : X \to B$. The scheme B is sometimes called the *parameter space*.

The notion of a family of schemes as it stands is much too general: consider any family $\pi : X \to B$, choose $b \in B$, and create a new family by taking the disjoint union of $X \setminus \pi^{-1}(b)$ and Y for some arbitrary scheme Y where Y is sent to b. From this, it can be deduced that there is no real properties that can be deduced, and further constrictions must be given.

Higher Direct Image Sheaves

here

5.4.1 Flat Morphisms

here

5.4.2 Smooth Morphisms

here

5.4.3 On Proper Morphisms: prelude to Zariski Main Theorem here

here

5.5 Dimension and Semi-continuity

here

5.6 *GAGA

I want to put here the most important GAGA results so that I may freely use them.

6

Application: Curves and Surfaces

In this chapter, we shall use the techniques we have taken the time to build to understand better low-dimensional algebraic-geometric objects, namely *curves* and *surfaces*.

6.1 Algebraic Curves

Intuitively, an algebraic curve ought to be a one-dimensional object given by a set of equations. In geometric settings, we shall want to work over algebraically closed fields, while in the number-theoretic setting we shall want to consider other fields and rings. For a moment let's consider curves over $\mathbb{C}$. As we saw in ref:HERE, all such curves can be projectivised and birationally into projective space. Furthermore, by the process of normalization, there is a birational map from the non-singular projective curves the projective curve. Hence, up birational equivalence we can consider algebraic curves as complete smooth objects; this can be thought of as the *geometric* way of looking at curves. If we go the analytic route, we shall see that curves correspond to compact Riemann surfaces due to GAGA. It is not too bad to show that smooth complete curves over $\mathbb{C}$ create compact Riemann surfaces, however the that all compact Riemann surface are given and characterized by smooth complete curves is a little harder to prove (it is non-trivial to show that a compact Riemann surface has a non-trivial meromorphic function defined on it; see [12]). On the other hand, looking at curves *algebraically*, we saw in [8] they correspond to fields $K/\mathbb{C}$ of transcendence degree 1. These three perspective shall constantly inspire each other to push forward the properties of we can define for curves.

Let us first state the definition of a curve in terms of scheme-theoretic language:

Definition 6.1.1: Curve

Let X be a k-scheme. Then if X is Noetherian, integral, separated dimension 1 scheme of finite type over k, then X is called a *curve*. If X is proper over k, it is called *complete*, and if all local rings are regular, it is called *regular* or *nonsingular*.

In this section, a curve will mean complete nonsingular curve over an algebraically closed field, or in terms of schemes it is an regular integral scheme of dimension 1 that is proper over k . When we say point, we shall mean closed point (the only other point is the generic point as the scheme is integral and dimension 1).

The following result will be important for being able to show that *nonsingular* curves that are rational are in fact isomorphic to the projective line. Note that this is *not* true for singular rational curves, and in general two rationally equivalent curves are *not* isomorphic[1].

Proposition 6.1.2: Proper Curve Equivalence

Let X be a nonsingular curve over k with function field K. Then the following are equivalent:

($\Rightarrow$) X is projective

($\Rightarrow$) X is proper

($\Rightarrow$) $X \cong t(C_K)$ where C_K is an abstract nonsingular curve and t is the functor from varieties to schemes (see theorem 3.1.19)

Proof **:**
(1) follows from (2) and (3) follows from (1) from classical algebraic geometry, see section 0.2. For (2) $\Rightarrow$ (3), as X is complete every DVR of K/k has a unique centre. Then as the closed points of X are all DVR are in one-to-one correspondence with the DVR's of K/k, we have a natural bijection $X \cong t(C_K)$, which is certainly an isomorphism.

We next generalize one more classical result about the image of projective varieties:

Proposition 6.1.3: Image of Complete Nonsingular Curve is Closed

Let X be a complete nonsingular curve over k, Y any curve over k, and $f : X \to Y$ a morphism. Then either the image of f is a point, or f is surjective, and so Y is complete. In the latter case, $\kappa(X)$ is a finite extension of $\kappa(Y)$, and so f is a finite morphism.

Proof **:**
As X is complete, $f(X)$ must be closed and proper over k, and $f(X)$ is irreducible. It follows that it either maps to a point, or to all of Y (recall we defined curves to be integral). Hence Y is complete as well.

To show f is finite, As f is dominant we have $\kappa(Y) \subseteq \kappa(X)$. As both have transcendence degree 1, $\kappa(X)$ is a finite extension of $\kappa(Y)$. Now, taking $\operatorname{Spec} S = V \subseteq Y$, and $U = f^{-1}(V)$, then it should

[1] See section 6.1.3 for more examples

be clear that $U = \operatorname{Spec} R$ where R is the integral closure of S, and as integral closure form finite S-modules, we have the desired result.

6.1.1 Divisors and Riemann-Roch

As all nontrivial maps between nonsingular complete curves must be finite, we have the following natural definition:

Definition 6.1.4: Degree of Morphism of Curves

Let $f : X \to Y$ be a finite morphism of curves. Then the *degree* of f is the degree of the finite extension of $[\kappa(X) : \kappa(Y)]$.

From this, can pull-back the divisor class group. Let $f : X \to Y$ be a finite morphism of curves. Define $f^* : \operatorname{WDiv} Y \to \operatorname{WDiv} X$ as follows: for each point $y \in Y$ choose $t \in \mathcal{O}_{Y,y}$ such that $\nu_y(t) = 1$ (where we choose the natural valuation on $\mathcal{O}_{Y,y}$). Then map it to:

$$f^*(y) = \sum_{f(x)=y} \nu_x(t) \cdot x$$

As f is a finite morphism, this is a finite sum. The reader should verify that this is indeed a well-defined function (namely, for another choice $t' = ut$ where u is unit, we get the same result). Then we see that f will map principal divisors to principal divisor, and hence we have:

$$f^* : \operatorname{Cl} Y \to \operatorname{Cl} X$$

Let us compute some examples of divisors on curves. To that end, we prove two useful results:

Proposition 6.1.5: Finite Map and Degrees

Let $f : X \to Y$ be a finite morphism of nonsingular curves. Then for any divisor D of Y,

$$\deg f^*(D) = \deg f \cdot \deg D$$

Proof **:**
As we are working with nonsingular curves, it suffice to show that for every (closed) point $y \in Y$, $\deg f^*(y) = \deg f$. Let $V = \operatorname{Spec} S \subseteq Y$ be an affine open subset containing y, and let R be the integral closure of S in $K(X)$. Then as we saw in proposition 6.1.3, $U = \operatorname{Spec} R$ is equal to $f^{-1}(V) \subseteq X$. Let $\mathfrak{m}_y$ be the maximal ideal of y in S. Then we can localize S and R with respect to $S = B \setminus \mathfrak{m}_y$, and we get a ring extension:

$$\mathcal{O}_{Y,y} \hookrightarrow R'$$

where R' is a finitely generated $\mathcal{O}_{Y,y}$-module. Now, R' is torsion free and has rank $r = [K(X) : K(Y)]$, and so $'$ i a free $\mathcal{O}_{Y,y}$-module of rank $r = \deg f$. If t is a local parameter of y, then R'/tR' is a K-vector pace of dimension r

On the other hand, the points $x_i \in X$ such that $f(x_i) = y$ are in one-to-one correspondence with the maximal ideals $\mathfrak{m}_i \subseteq R'$, and for each i $R'_{\mathfrak{m}_i} = \mathcal{O}_{X,x_i}$. Now, by construction we have:

$$tR' = \bigcap_i (tR'_{\mathfrak{m}_i} \cap R')$$

Then as these are co-prime, we can apply the Chinese remainder theorem and see that the dimension of the quotients are:

$$\dim_k R'/tR' = \sum_i \dim_k \frac{R'}{tR'_{\mathfrak{m}_i} \cap R'}$$

and as $R'/(tR'_{\mathfrak{m}_i} \cap R') \cong R'_{\mathfrak{m}_i}/tR'_{\mathfrak{m}_i} = \mathcal{O}_{X,x_i}/t\mathcal{O}_{X,x_i}$, we have the dimensions in the sum are equal to $\nu_{x_i}(t)$. Then as $f^*(y) = \sum \nu_{x_i}(t)x_i$, we have that

$$\deg f^*(q) = \deg f$$

as we sought to show.

Corollary 6.1.6: Degree Map and Integers

Lt X be a complete nonsingular curve. Then the principal divisors have degree 0, and the degree function induces a surjective morphism:

$$\deg : \operatorname{Cl} X \twoheadrightarrow \mathbb{Z}$$

***Proof* :**
Take $f \in \kappa(X)^*$. If $f \in k$, then $\operatorname{div} f = 0$, so assume $f \notin k$. Then we have the field extension $\kappa(X)/k(f)$ which induces a morphism $\varphi : X \to \mathbb{P}^1_k$ that is finite by proposition 6.1.3.Then $\operatorname{div} f = \varphi^*(\{0\} - \{\infty\})$, and as $\{0\} - \{\infty\}$ is a divisor of degree 0 in $\mathbb{P}^1_k$, we get that $\operatorname{div} f = 0$.

Thus, the degree map is independent of principal divisors, and so by proposition 4.3.10 we get the surjection, completing the proof.

Example 6.1.7: Divisors On Curves and Groups on Elliptic Curves

1. Recall that a (complete) curve is *rational* if it is birational to $\mathbb{P}^1_k$, and furthermore as X is a nonsingular curve, they are in fact *isomorphic* (by proposition 6.1.2). Then as all points correspond to principal divisors. I claim that X is a complete nonsingular curve is rational if and only if there exist two distinct points $x, y \in X$ such that $[x] = [y]$. Then there is a rational function $f \in \kappa(X)$ where $\operatorname{div} f = 1 \cdot x - 1 \cdot y$. Now, take the morphism $\varphi : X \to \mathbb{P}^1_k$ defined in corollary 6.1.6. Then $\varphi^*(\{0\}) = 1 \cdot x$, and so φ is of degree 1. But then it is invertible, that is it is birational, and so X is rational.

2. Let $X \subseteq \mathbb{P}^2_k$ be the nonsingular curve curve $y^2z = x^3 - xz^2$ where $\operatorname{char}(k) \neq 2$. Then as we saw in [8, chapter 21.3], this curve is not rational[a]. Take K be the kernel of $\operatorname{Cl} X \to \mathbb{Z}$, which by corollary 6.1.6 is nonempty. We shall show that there is a one-to-one correspondence between the closed points of X and the elements of K, which shows that this curve has a natural group structure on it! In fact *all* elliptic curves will have a natural group structure

on them, making them natural group varieties! The following visual illustrates how we shall define the multiplication structure:

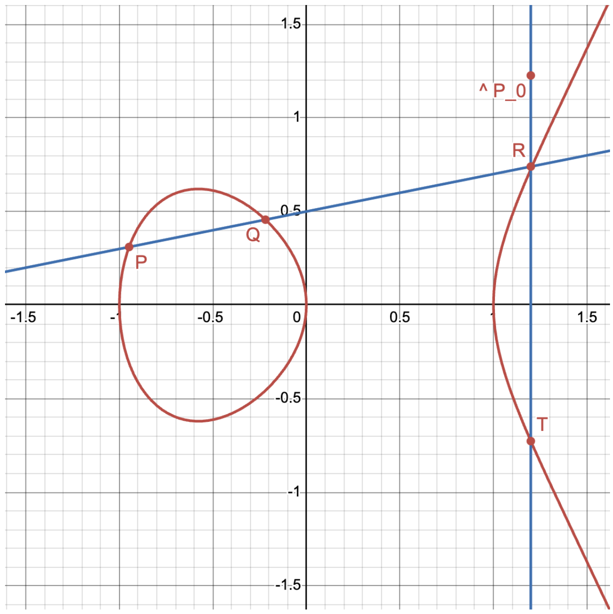


Figure 6.1: Group law on X in $z \neq 0$

Let $P_0 = [0 : 1 : 0]$ on X. This is an inflection point, hence the tangent line $z = 0$ at the point meets the curve in the divisor $3P_0$ (check this). If $L \subseteq \mathbb{P}^2_k$ is any other line meeting X in three points, say P, Q, R (which my coincide), then L is linearly equivalent to $z = 0$ in $\mathbb{P}^2_k$, that is we have

$$[P + Q + R] = [3P_0]$$

Now, to any closed point $P \in X$, associate to it the divisor $P - P_0 \in K$. his map is injective, since if $[P - P_0] = [Q - P_0]$ implies $[P] = [Q]$, and so X would be rational by example (1), a contradiction.

Then let's show that the map from the closed points of X to K is surjective. Take $D = \sum n_i P_i \in K$. Let us first take it where $\sum_i n_i = 0$. Then we can write

$$D = \sum_i n_i(P_i - P_0)$$

Now, for any point R, the line P_0R meet X also at the point T (where we always count intersection with multiplicity so that if $R = P_0$, then the line P_0R is the tangent line at P_0 and then $T = P_0$). Then $[P_0 + R + T] = [3P_0$, so $[R - P_0] = -[T - P_0]$. Now, if i is an index where $n_i < 0$, then take $P_i = R$. Then replacing P_i by T, we get divisors in the same cosets with the ith coefficient $-n_i > 0$. By repeating this process, we can assume that $D = \sum_i n_i(P_i - P_0)$ where all $n_i > 0$. If $\sum_i n_i = 1$, we are done, so suppose $\sum_i n_i \geq 2$. Let P, Q be two (not necessarily distinct) points P_i which occur in D. Let the line PQ meet X in R let the line P_0R meet X in T. Then:

$$[P + Q + R] = [3P_0] \qquad \text{and} \qquad [P_0 + R_0 + T] = [3P_0]$$

and so:

$$[P - P_0] + [Q - P_0] = [T - P_0]$$

By replacing P and Q by T, we get another equivalent divisor of the same form whose $\sum_i n_i$ is one less, and so by induction:

$$[D] = [P - P_0]$$

giving us the bijective correspondence. Finally, it is not too hard o show that this addition and inverse morphisms $+ : X \times X \to X$ and $(-)^{-1} : X \to X$, and hence we have a natural group variety!

More generally, if X is a complete nonsingular curve, then K is isomorphic to the group of closed points of an abelian variety known as the *Jacobian variety* of X. Then an *abelian variety*, is a complete group variety over a field k. The dimension of a Jacobean variety shall be shown in section ref:HERE to be related to the *genus* of the curve (see section 6.1). In general, the divisor class group of such a curve is an extension of $\mathbb{Z}$ by the group of closed points of the Jacobian variety of X Notice it has a continuous and a discrete component.

This process to higher dimensional nonsingular varieties: if X is a nonsingular projective variety of dimension ≥ 2 then we can define the subgroup$K \subseteq \operatorname{Cl} X$ such that $\operatorname{Cl} X/K$ is a finitely generated abelian group known as the *Néron-Severe group* of X, and K is isomorphic to the group of closed points of an abelian variety known as the *Picard Variety* (generalizing the Picard group from Number Theory!)

[a] We shall return a lot to this curve in section 6.1.3

If we are working over a characterize 0 curve then as we are smooth and proper, then the analytification of curve define a Riemann surface, and in fact a compact oriented surface as it is proper and it is complex (giving a natural orientation). Recall compact oriented surfaces are classified up to diffeomorphism (and since we are in low enough dimension, this is equivalent up to homeomorphism[2]) by their genus. These split up curves into a natural countable family of curves:

Definition 6.1.8: Genus of Curve

Let X be a curve. Then

1. The *arithmetic genus* of X is $1 - P_X(0)$ where P_X is the Hilbert polynomial of X. It is denoted $p_a(X)$
2. The *geometric genus* of X is $\dim_k \Gamma(X, \omega_X)$ where ω_X is the canonical sheaf. It is denoted $p_g(X)$.

These are in fact all equivalence, and naturally is also measured by the 1st cohomology group (see [2]). We first need a bit of build-up:

[2] Recall in real dimension $1, 2, 3$ that homeomorphic and diffeomorphic are essentially equivalent

Definition 6.1.9: Euler Characteristic

Let X be a projective scheme over k, $\mathscr{F}$ a coherent sheaf on X. Then the *Euler characteristic* of $\mathscr{F}$ is:

$$\chi(\mathscr{F}) = \sum_i (-1)^i \dim_k H^i(X, \mathscr{F})$$

The Euler Characteristic is additive, meaning that if $0 \to \mathscr{F}' \to \mathscr{F} \to \mathscr{F}'' \to 0$, then

$$\chi(\mathscr{F}) = \chi(\mathscr{F}') + \chi(\mathscr{F}'')$$

as the reader should verify.

Lemma 6.1.10: Hilbert Polynomial and Euler Characteristic

let X be a projective scheme over k, $\mathcal{O}_X(1)$ a very ample invertible sheaf on X over k, and $\mathscr{F}$ a coherent sheaf on X. Then there exists a polynomial $P(z) \in \mathbb{Q}[z]$ such that:

$$\chi(\mathscr{F}(n)) = P(n)$$

for all $n \in \mathbb{Z}$. This polynomial is called the *Hilbert polynomial* of $\mathscr{F}$.

To see the connection, if you let $X = \mathbb{P}^n_k$ and $M = \Gamma_*(\mathscr{F})$, then the Hilbert polynomial of $\mathscr{F}$ is just the Hilbert polynomial of M.

***Proof* :**
exercise

Lemma 6.1.11: Hilbert Polynomial and Cohomology

Let X be a projective scheme of dimension r over k. If X is integral and k algebraically closed, then:

$$p_a(X) = \sum_{i=1}^{r-1} (-1)^i \dim_k H^{r-1}(X, \mathcal{O}_X)$$

In particular, if X is a curve, then:

$$p_a(X) = \dim_k H^1(X, \mathcal{O}_X)$$

***Proof* :**
exercise

Proposition 6.1.12: Equivalence of Genus

Let X be a curve. Then:

$$p_a(X) = p_g(X) = \dim_k H^1(X, \mathcal{O}_X)$$

Hence, this number is simply called the *genus* of X and will be denoted g.

Proof **:**
We have $p_a(X) = \dim_k H^1(X, \mathcal{O}_X)$ by lemma 6.1.11. Next by Serre duality, we have that if X is a projective nonsingular curve, $H^1(X, \mathcal{O}_X)$ and $H^0(X, \omega_X)$ are dual vector spaces, and hence $p_a = \dim_k H^1(X, \mathcal{O}_X) = \dim_k \Gamma(X, \omega_X) = p_g$, completing the proof.

From this, we can deduce the genus must be non-negative (as we hope it would by our topological understanding of the genus). Furthermore, for each genus $g \geq 0$, we can associate to it a curve of genus g: Take a (primitive will) divisor of a quadratic nonsingular surface of type $g+1$ or $g+2$[3]. Then as we saw in section 5.2.1, we get a corresponding genus g curve.

One of the most important results for classifying curves comes from the Riemann-Roch theorem which relates the genus, the divisors, and the degree of a curve. As we are working over curves, the Weil and Cartier divisors coincide, and so for each divisor D we have a natural map $D \mapsto \mathcal{L}(D)$ to an invertible sheaf defined in a natural way using the principal parts of the covering of the curve. Recall that a divisor $D = \sum_i n_i P_i$ is called *effective* if $n_i \geq 0$. These don't form a group, however we still have the notion of "cosets": two divisors are said to be *linearly equivalent* and if they differ by a principal divisor, and all the divisors in an equivalent class is called a *complete linear system*; let $|D|$ denote this set.

As shown in (Harthsorne II 7.7, didn't put it down yet), if we have $\mathrm{Cl}\,(C)$, then if $|D| = |D'|$ we have $D = D' + \mathrm{div}\,(f)$ for some rational function f. Any other such principal divisor differs by a constant multiple (if we had $D = D' + \mathrm{div}\,(f) = D' + \mathrm{div}\,(f')$, then f/f' would be a polynomial and hence constant on projective space). We thus get that there is a one-to-one correspondence between the effective Cartier divisors and:

$$\frac{H^0(X, \mathcal{L}(D)) - \{0\}}{k^*}$$

Let $\ell(D) = \dim_k H^0(X, \mathcal{L}(D))$ (we may thus say that the dimension of $|D|$ is $\ell(D) - 1$). This number is finite by theorem 4.4.7.

Lemma 6.1.13: Divisors and Degree

Let D be a divisor on a curve X. Then if $\ell(D) \neq 0$, we must have

$$\deg D \geq 0.$$

Furthermore, if $\ell(D) \neq 0$ and $\deg D = 0$, then $D \sim 0$, that is,

$$\mathcal{L}(D) \cong \mathcal{O}_X$$

[3] see ref:HERE as to why

Proof **:**
If $\ell(D) \neq 0$, then the complete linear system $|D|$ is nonempty. Hence D is linearly equivalent to some effective divisor. Since the degree depends only on the linear equivalence class, and the degree of an effective divisor is nonnegative, we find

$$\deg D \geq 0.$$

If $\deg D = 0$, then D is linearly equivalent to an effective divisor of degree 0. But there is only one such, namely the zero divisor.

Finally, we establish one more convention: let $\Omega_{X/k}$ or Ω_X represent the sheaf of relative differentials of X over k Since X has dimension 1, it is an invertible sheaf on X, and so is equal to the canonical sheaf ω_X on X. We call any divisor in the corresponding linear equivalence class a *canonical divisor*, and denote it by K. Then we finally get:

Theorem 6.1.14: Riemann-Roch Theorem

Let D be a divisor on a curve X of genus g. Then:

$$\ell(D) - \ell(K - D) = \deg D + 1 - g$$

Proof **:**
The divisor $K - D$ corresponds to the invertible sheaf $\omega_X \otimes \mathscr{L}(D)^\vee$. Since X is projective (by proposition 6.1.2), we can apply Serre duality (namely, proposition 4.5.26) to conclude that the vector space $H^0(X, \omega_X \otimes \mathscr{L}(D)^\sim)$ is dual to $H^1(X, \mathscr{L}(D))$. Thus we have to show that for any D,

$$\chi(\mathscr{L}(D)) = \deg D + 1 - g,$$

where for any coherent sheaf $\mathscr{F}$ on X, $\chi(\mathscr{F})$ is the Euler characteristic

$$\chi(\mathscr{F}) = \dim H^0(X, \mathscr{F}) - \dim H^1(X, \mathscr{F}).$$

First, consider the case where $D = 0$. Then the formula gives

$$\dim H^0(X, \mathscr{O}_X) - \dim H^1(X, \mathscr{O}_X) = 0 + 1 - g.$$

since $H^0(X, \mathscr{O}_X) = k$ for any projective variety, and $\dim H^1(X, \mathscr{O}_X) = g$ by proposition 6.1.12.

Next, let D be any divisor, and let P be any point. Let's start by showing that the formula is true for D if and only if it is true for $D + P$. Since any divisor can be reached from 0 in a finite number of steps by adding or subtracting a point each time, this will show the result holds for all D.

Consider P as a closed subscheme of X. Its structure sheaf is a skyscraper sheaf k sitting at the point P, which we denote by $k(P)$, and its ideal sheaf is $\mathscr{L}(-P)$ by proposition 4.3.29. Therefore we get a short exact sequence

$$0 \to \mathscr{L}(-P) \to \mathscr{O}_X \to k(P) \to 0.$$

Tensoring with $\mathscr{L}(D + P)$ we get

$$0 \to \mathscr{L}(D) \to \mathscr{L}(D+P) \to k(P) \to 0.$$

Since $\mathscr{L}(D+P)$ is locally free of rank 1, tensoring by it does not affect the sheaf $k(P)$. Now, the Euler characteristic is additive on short exact sequences, and $\chi(k(P)) = 1$, so we have

$$\chi(\mathscr{L}(D+P)) = \chi(\mathscr{L}(D)) + 1.$$

On the other hand, $\deg(D+P) = \deg D + 1$, so our formula is true for D if and only if it is true for $D+P$. But, we get

$$\chi(\mathscr{L}(D)) = \deg D + 1 - g,$$

from which we can conclude that:

$$\ell(D) - \ell(K-D) = \deg D + 1 - g$$

as we sought to show.

Hartshorne goes through many examples here!

6.1.2 Hurwitz's Theorem

Recall that a finite morphism $f : X \to Y$ between schemes preserves dimension, has finite fibers, that normalization maps are finite morphisms, and that such maps are proper. In the case of curves, we can think of finite morphisms are *covering maps* (think of the Riemann surfaces defined to make square-roots maps well-defined). We endeavor to link the genus of Y to X given a finite morphism via *Hurwitz's Theorem*.

Recall that the *degree* of a finite morphism $f : X \to Y$ is $[K(X) : K(Y)]$. As we are working with smooth curves, each local ring is a DVR. Choose a point $p \in X$ and let $q = f(p)$. Take a uniformizer $t \in \mathcal{O}_Q$, and identify t with $f_q^{\#}(t)$ where $f_q^{\#} : \mathcal{O}_q \to \mathcal{O}_p$ is the natural map induces by f. This may no longer be a uniformizer (consider $f(x) = x^2$ on the affine line). This value is independent of choice of uniformizer, and hence we define the following:

Definition 6.1.15: Ramification Index

With the setup above, the *ramification index* of f at the point $p \in X$ is given by:

$$\nu_p(t) = e_p$$

where ν_p is the discrete valuation at p. If $e_p > 1$, we say that f is *ramified* p if $e_p > 1$, and unramified if $e_p = 1$.

If f is unramified everywhere, it is easy to see that it is étale. If char$k = 0$ or char$k = p$ and $p \nmid e_p$, we say that the ramification is *tame*, and it is *wild* otherwise.

6.1.3 Elliptic Curves

here

6.2 Surfaces

Here

Part II

Intersection Theory

6.2.1 Inspiration For now, this part will very heavily be inspired by Fulton [16] and Vakil [15]

Throughout most this chapter, schemes shall always be $\overline{k}$-schemes of finite type over an algebraically closed field. A point shall be a closed point unless stated otherwise. If needed, such a scheme shall be called an *algebraic scheme*.

7

Rational Equivalence

Recall from [8] that we had to define *projective plane curves* to be an equivalence relation on homogeneous polynomials of $k[x, y, z]$ up to a constant ($f \sim g$ if and only if $f = \alpha g$). This was specifically done keep track of multiplicity information. By the correspondence between homogeneous and affine curves, we can without loss of generality work with $[f(x, y)], [g(x, y)] \in k[x, y]$ and choose any representative; we shall simply call these *plane curves*. Given a choice of plane curves f, g let F, G denote the resulting variety. The intersection of two (affine or projective) plane curves F, G at P was defined as:

$$i(P, f \cdot g) = \dim_K \frac{\mathcal{O}\mathbb{A}^2_k, p}{(f, g)} = \dim_K \mathcal{O}_{Z,P}$$

where $Z = \mathcal{Z}(f, g)$. It was proven that:

1. (commutative on intersection): $i(P, F \cdot G) = i(P, G \cdot F)$

2. (linear on intersection): if $F_1 + F_2$ is the curve defined by $f_1 f_2$, then $i(P, (F_1 + F_2) \cdot G) = i(P, F_1 \cdot G) = i(P, F_2 \cdot G)$

3. (invariance of representation): if F' is given by $f + gh$ for some $h \in k[x, y]$:

$$i(P, F' \cdot G) = i(P, F \cdot G)$$

4. If F, G have have a common component through P (i.e. the dimension of the irreducible variety in $F \cap G$ containing p is greater than 0), then $i(P, F \cdot G) = \infty$. Otherwise, $i(P, F \cdot G) \in \mathbb{N}_{\geq 0}$.

5. If $P \notin F \cap G$:

$$i(P, F \cdot G) = 0$$

6. if $f(x,y) = x - a$ and $g(x,y) = y - b$ (more generally $\frac{\partial(f,g)}{\partial(x,y)} \neq 0$ at p) then:

$$i(P, F \cdot G) = 1$$

7. If F has no common component with G and H and P is a simple point of F:

$$i(P, G \cdot H) \geq \min i(P, F \cdot G), i(P, F \cdot H))$$

7.1 Basic Definitions and Examples

will define order here more precisely, just need defn of rational equivalence down for now

Definition 7.1.1: K-cycle

Let X be an algebraic scheme. Then a k-*cycle* is a finite formal sum:

$$\sum_i n_i[V_i]$$

where the V_i are k-dimensional subvarieties of X and $n_i \in \mathbb{Z}$.

The group of k-cycle is denoted $Z_k(X)$.

Let $W \subseteq X$ be a $(k+1)$-dimensional variety and $f \in k(W)^*$ be a nonzero rational function. Then:

$$[\operatorname{div}(r)] = \sum_{\substack{V \subseteq W \\ \operatorname{codim}(V)=1}} \operatorname{ord}_V(r)[V]$$

It is easy to check this is a finite sum (exercise ref:HERE).

Definition 7.1.2: Rational Equivalence

A k-cycle $\alpha \in Z_k(X)$ is *rationally equivalent to 0* if they exists a finite number of $(k+1)$-dimensional subvareities W_i and $r_i \in k(W_i)^*$ such that

$$\alpha = \sum_{W_i} [\operatorname{div}(r_i)]$$

This equivalence is denoted $\alpha \sim 0$. Two k-cycles $\alpha, \beta \in Z_k(X)$ are *rationally equivalent* if there exists their difference is rationally equivalent to 0:

$$\alpha \sim \beta \qquad \text{iff} \qquad \alpha - \beta \sim 0$$

As $[\operatorname{div}(r^{-1}] = -[\operatorname{div}(r)]$ for a rational function r, the quotient is a well-defined group:

Definition 7.1.3: Chow Group

Let $Z_k(X)$ denote the k-cycles of X and $\mathrm{Rat}_k(X)$ the k-cycles rationally equivalent to 0. Then the *Chow group* is:

$$A_k(X) = \frac{Z_k(X)}{\mathrm{Rat}_k(X)}$$

Elements of $A_k(X)$ are called a *k-cycle class*. In cohomological context, this group is often denoted $CH_i(X)$.

By taking the direction sum $Z_*(X) = \sum_i^\infty A_k(X)$ and $\mathrm{Rat}_*(X) \sum_i^\infty$, we call the elements of $Z_*(X)$ a *cycle* and define the group $A_*(X)$ to be the quotient of these two groups.

A cycle is *positive* if it is not zero and given $\alpha = \{\alpha_k\}_k$ each coefficient of α_k is positive. A *cycle class* is positive it there there is a cycle representative that is positive.

Example 7.1.4: Chow Group

1. $A_k(X) = A_k(X_{\mathrm{red}})$
2. disjoint unions become direct sums
3. If $X_1, X_2 \subseteq X$ are closed subschemes then we get an exact sequence (note the lack of left-exactness)

$$A_k(X_1 \cap X_2) \to A_k(X_1) \oplus A_k(X_2) \to A_k(X_1 \cup X_2)) \to 0$$

4. If X is n-dimensional, it has no $(n+1)$-dimensional irreducible components, so:

$$A_n(X) = Z_n(X)$$

Thus, it has two be that two rationally equivalent cycles irreducible components of A_n will have the same coefficients (including possibly 0). Thus, the *coefficient of V in α* for an irreducible component $V \subseteq X$ and cycle class α is the coefficient of $[V]$ in α.

Let us now look at how these classes change under the usual cohomological morphisms.

Recall that if $f : X \to Y$ is a scheme morphism, the image of a closed subvariety need not be a closed subvariety ($f : V(xy-1) \to \mathbb{A}^1_k$ where $(x, y) \mapsto x$ being the canonical counter-example, see example 3.4.11). If f is proper, this is the case as the image of an irreducible set is always irreducible, and by definition f is universally closed and hence the image of a closed set is closed. It thus makes sense to pushforward the Chow group. If $V \subseteq X$ is a subvariety and $W = f(X)$, there is an induced embedding $R(W) \hookrightarrow R(V)$. If $\dim_K V = \dim_K W$, this is a finite field extension. Let:

$$\deg(V/W) = \begin{cases} [R(V) : R(W)] & \dim_K(V) = \dim_K(W) \\ 0 & \text{otherwise} \end{cases}$$

and:

$$f_*[V] = \deg(V/W)[W]$$

Giving a map:

$$f_* : Z_k(X) \to Z_k(Y)$$

where if f, g are proper morphisms that compose then $(fg)_* = f_* g_*$. If $\dim_{\mathbb{C}}(V) = \dim_{\mathbb{C}}(W)$, then V is generically a covering of W with $\deg(V/W)$ sheets, and the pushforward agrees with the pushforward in topology. Importantly:

Theorem 7.1.5: Proper Morphism Preserves Rational Equivalence

Let $f : X \to Y$ be a proper morphism and $\alpha \in Z_k(X)$. Then if $\alpha \sim 0$, $f_*(\alpha) \sim 0 \sim f_*(0)$.

Thus, there is an induced homomorphism:

$$f_* A_k(X) \to A_k(Y)$$

making f_* a covariant functor

***Proof* :**

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